GEOMETRY · COMMON CORE HIGH SCHOOL MATH

# Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

96 question sets • 2915 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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# Precise definitions of angle, circle, and line segment

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.1** |
| **BACKWARD 1** | Identify points, lines, rays, segments, and angles | **4.G.A.1** |
| **BACKWARD 2** | See an angle as two rays with a common endpoint, measured by… | **4.MD.C.5.a** |
| **BACKWARD 3** | Use the center, radius, and diameter of a circle | **7.G.B.4** |
| **FORWARD 1** | Test proposed definitions with counterexamples | **HSG-CO.A.1** |
| **FORWARD 2** | Write precise definitions of new figures from earlier definitions | **HSG-CO.A.1** |

SET 1 • HSG-CO.A.1

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Use only the words point, plane, and distance, and words built from them.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

10

**QUESTION ID**

## S1-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**A circle is the set of all points in a plane that are a given distance (the radius) from a given point in that plane (the center).**

**GRADING NOTE** Any equivalent wording. Essential: (1) ALL points, (2) in one plane, (3) at one fixed distance, (4) from one fixed point, the center. Not accepted: "a round shape" (not precise); leaving out the plane (that describes a sphere); "points within a distance" (that is a disk).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Points A and B lie on line $\ell$. Which is a precise definition of segment AB?

**A.** Every point of line $\ell$

**B.** Points A and B together with every point C of line $\ell$ for which AC + CB = AB

**C.** Point A and every point of line $\ell$ on the same side of A as B

**D.** Every point of the plane that is closer to A than to B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5**

11

**QUESTION ID**

## S1-M-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. Points A and B together with every point C of line $\ell$ for which AC + CB = AB**

**GRADING NOTE** A segment is its two endpoints and every point between them; "C is between A and B" means C is on the line and AC + CB = AB (distance along a line). The other choices describe line $\ell$, ray AB, and a half-plane.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5**

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

12

**QUESTION ID**

## S1-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**An angle is the figure formed by two rays that have the same endpoint. The common endpoint is the vertex, and the two rays are the sides.**

**GRADING NOTE** Any equivalent wording. Essential: two rays (not lines), one common endpoint, that endpoint named as the vertex. Not accepted: "the space between two lines", "a turn", or "two lines that cross" (two crossing lines form four angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5**

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

13

QUESTION ID

**S1-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**She must add "in a plane" (the plane that contains O). Without it, all points 5 cm from O form a sphere, not a circle.**

**GRADING NOTE** Any equivalent wording that adds the plane condition and explains that the points in space would form a sphere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

**True or false?**

An angle with its vertex at the center of a circle that cuts off $\frac{1}{360}$ of the circle measures 1°, whatever the radius of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

14

QUESTION ID

**S1-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Degree measure is defined from distance around a circular arc: a 1° angle cuts off 1/360 of any circle centered at its vertex. The fraction of the circle, not the radius, sets the measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

# BACKWARD 1

Identify points, lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

What is shown in the figure?

**A.** Ray AB

**B.** Ray BA

**C.** Line AB

**D.** Segment AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q1/5

16

QUESTION ID

**S1-B1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A. Ray AB**

**GRADING NOTE** It starts at A and goes through B without end, so it is ray AB (endpoint named first).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q1/5

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q2/5

17

QUESTION ID

## S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

**A segment has 2 endpoints; a ray has 1.**

**GRADING NOTE** Any equivalent form. Both numbers are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q2/5

**True or false?**

A line segment has exactly two endpoints.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q3/5

18

QUESTION ID

**S1-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A segment is two endpoints and the points between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q3/5

Which figure goes on without end in exactly one direction?

**A.** A line

**B.** A point

**C.** A line segment

**D.** A ray

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q4/5

19

QUESTION ID

**S1-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. A ray**

**GRADING NOTE** A ray has one endpoint and continues in one direction; a line continues in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q4/5

**True or false?**

Ray AB and ray BA are the same ray.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

**Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q5/5**

20

QUESTION ID

## S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Ray AB starts at A and goes through B; ray BA starts at B and goes through A, in the opposite direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

**Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q5/5**

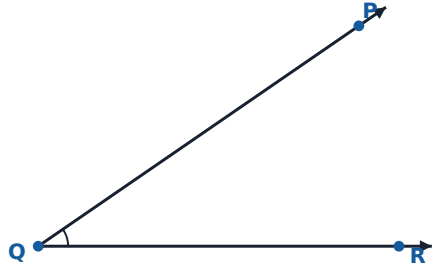
SET 1 • HSG-CO.A.1

# BACKWARD 2

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q1/5

22

QUESTION ID

## S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

**Vertex Q; sides ray QP and ray QR.**

**GRADING NOTE** Any equivalent form. The middle letter names the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q2/5

23

**QUESTION ID**

## S1-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form. 1/4 of 360° is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q3/5

24

**QUESTION ID**

## S1-B2-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

$\frac{1}{6}$

**GRADING NOTE** Any equivalent form. 60/360 = 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q3/5

**True or false?**

The measure of an angle depends on how long its sides are drawn.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q4/5

25

QUESTION ID

## S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The sides are rays; drawing them longer does not change the opening between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q4/5

**True or false?**

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q5/5

26

QUESTION ID

**S1-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** That is the meaning of one degree.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSG-CO.A.1

# BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

28

QUESTION ID

## S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

**Yes. Every point of the plane that is 7 cm from O is on the circle.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5**

29

**QUESTION ID**

## S1-B3-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**9 in.**

**GRADING NOTE** Any equivalent form. The radius is half the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5**

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

**A.** Inside the circle

**B.** Outside the circle

**C.** On the circle

**D.** There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

30

QUESTION ID

**S1-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. Inside the circle**

**GRADING NOTE** D is closer to the center than the radius, so it is inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

**True or false?**

Every radius of a circle has the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

**Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5**

31

QUESTION ID

## S1-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** All points of a circle are the same distance from its center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

**Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5**

**True or false?**

A point 10 units from the center of a circle of radius 5 lies on the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

32

QUESTION ID

**S1-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Points on the circle are exactly 5 units from the center; this point is outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

# FORWARD 1

Test proposed definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

34

**QUESTION ID**

## S1-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**For example, two points that are each 3 cm from O (or an arc of a circle, or a sphere). The definition must say ALL points at that distance, in one plane.**

**GRADING NOTE** Any set that fits the wording but is not a whole circle: finitely many points, an arc, or a sphere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

35

**QUESTION ID**

## S1-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**The top vertex of an isosceles triangle with base AB has AM = MB but is not on AB. Repair: M is the point ON segment AB with AM = MB.**

**GRADING NOTE** Any counterexample with M off segment AB and AM = MB, plus a repair that puts M on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

**A.** A circle of radius 2 centered at O in a plane

**B.** A sphere of radius 2 centered at O

**C.** A disk of radius 2

**D.** A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

36

QUESTION ID

**S1-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. A sphere of radius 2 centered at O**

**GRADING NOTE** In space, the points 2 units from O form a sphere. A disk also contains points closer than 2 units, so it does not fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

**True or false?**

"Two lines are parallel if they never meet" is a complete definition, even in three-dimensional space.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

37

QUESTION ID

## S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Skew lines in space never meet but are not parallel. The definition needs "in the same plane".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

38

**QUESTION ID**

## S1-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**A short segment through the center whose endpoints are not on the circle. Repair: a diameter is a segment through the center whose endpoints are both on the circle.**

**GRADING NOTE** Any counterexample plus a repair that requires both endpoints on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

# FORWARD 2

Write precise definitions of new figures from earlier definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

40

QUESTION ID

**S1-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Ray AB is point A, point B, every point between A and B, and every point P of line AB such that B is between A and P.**

**GRADING NOTE** Any equivalent definition built from A, B, line AB, and betweenness.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

41

QUESTION ID

**S1-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**The midpoint of segment AB is the point M on segment AB such that AM = MB.**

**GRADING NOTE** Any equivalent wording. M must be on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

42

**QUESTION ID**

## S1-F2-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**A semicircle is P, Q, and all points of the circle that lie on one side of line PQ.**

**GRADING NOTE** Any equivalent wording. Essential: the endpoints of a diameter and the points of the circle on one side of the diameter line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the bisector of ∠ABC?

**A.** A ray from B that lies outside ∠ABC

**B.** The segment that joins A and C

**C.** Any line through B

**D.** Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

43

**QUESTION ID**

## S1-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC**

**GRADING NOTE** The bisector is the ray from the vertex that splits the angle into two angles of equal measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

**True or false?**

The definition "the midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

44

QUESTION ID

## S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Every point on the perpendicular bisector of AB satisfies AM = MB; only one of them is on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

# Precise definitions of perpendicular lines and parallel lines

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.1** |
| **BACKWARD 1** | Recognize perpendicular and parallel lines and right angles | **4.G.A.1** |
| **BACKWARD 2** | Find the other angles when two lines cross | **7.G.B.5** |
| **FORWARD 1** | Prove consequences of the definition of perpendicular lines | **HSG-CO.C.9** |
| **FORWARD 2** | Reason about three lines in a plane and in space | **HSG-CO.C.9** |

SET 2 • HSG-CO.A.1

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

47

QUESTION ID

## S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

**Two lines are parallel if they lie in the same plane and do not intersect.**

**GRADING NOTE** Any equivalent wording. Essential: in the same plane, and no point in common. Not accepted: "lines that never meet" without the plane (skew lines in space never meet), or "lines that point the same way" (not precise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

48

QUESTION ID

## S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

**Two lines are perpendicular if they intersect and form a right angle.**

**GRADING NOTE** Any equivalent wording, such as "intersect to form four right angles" or "intersect to form congruent adjacent angles". Not accepted: "lines that cross" or "lines that make a T" (not precise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines $\ell$ and m do not intersect. Which additional fact makes them parallel?

**A.** They lie in the same plane

**B.** They both look horizontal in a drawing

**C.** They are drawn the same length

**D.** They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

49

QUESTION ID

## S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

## A. They lie in the same plane

**GRADING NOTE** Parallel lines are coplanar lines that do not meet. Lines in space can miss each other without being parallel (skew lines), even if both are perpendicular to one segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

**True or false?**

If two lines meet and form one right angle, they form four right angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5**

50

QUESTION ID

## S2-M-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The angle next to a right angle is its supplement, 90°, and vertical angles are equal, so all four angles are 90°.

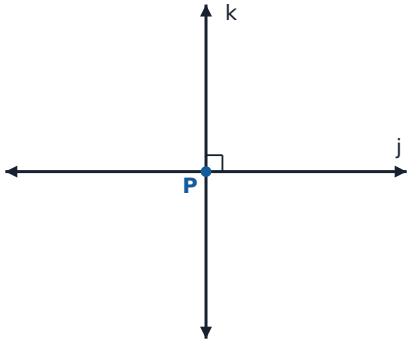
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5**

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.

Answer: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

51

QUESTION ID

## S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

**Yes. The lines intersect and form a right angle, which is the definition of perpendicular lines.**

**GRADING NOTE** Any equivalent explanation that names both parts of the definition: they intersect, and an angle is a right angle (given by the mark).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

# BACKWARD 1

Recognize perpendicular and parallel lines and right angles • Grade 4 • Geometry • 4.G.A.1

**True or false?**

Lines that meet at a 90° angle are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

**Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5**

53

QUESTION ID

## S2-B1-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** That is the definition of perpendicular lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

**Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5**

**True or false?**

Any two lines that cross are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

**Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5**

54

QUESTION ID

## S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Lines can cross at angles other than 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

**Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5**

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

55

**QUESTION ID**

## S2-B1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of the opposite sides of a rectangle?

**A.** They are parallel

**B.** They are perpendicular

**C.** They meet at one point

**D.** They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

56

QUESTION ID

## S2-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

## A. They are parallel

**GRADING NOTE** Opposite sides of a rectangle never meet and lie in one plane, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

**A.** Two straight streets that never meet

**B.** Two streets that curve around a park

**C.** Two streets that meet at a sharp point

**D.** Two streets that cross at a square corner

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

57

QUESTION ID

**S2-B1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**D. Two streets that cross at a square corner**

**GRADING NOTE** Perpendicular lines meet at a right (square) angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

# BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

59

**QUESTION ID**

## S2-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**90°, 90°, 90°**

**GRADING NOTE** Any order. The adjacent angles are 180° − 90°; the vertical angle is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

60

**QUESTION ID**

## S2-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**110°, 70°, 110°**

**GRADING NOTE** Any order. Adjacent angles: 180° − 70° = 110°; the vertical angle is 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

**True or false?**

Vertical angles formed by two crossing lines are equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

61

QUESTION ID

**S2-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Vertical angles have equal measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

**A.** 180°

**B.** 45°

**C.** 90°

**D.** 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

62

**QUESTION ID**

**S2-B2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 90°**

**GRADING NOTE** A linear pair adds to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

**True or false?**

Two lines cross and one angle measures 100°.
The angle next to it also measures 100°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

63

QUESTION ID
**S2-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** The angle next to it forms a linear pair with it, so it measures 80°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

# FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell$ and m are perpendicular and meet at point P.
Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5**

65

**QUESTION ID**

## S2-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**All four angles measure 90°.**

**GRADING NOTE** REQUIRED WORK: Start from the definition (one angle at P is a right angle). Use the linear pair (supplementary) to get the adjacent angles and vertical angles (or another linear pair) to get the fourth, with a reason for each step.

Any valid proof. Essential: (1) definition gives one 90° angle; (2) each adjacent angle forms a linear pair with it, so it is 180° − 90° = 90°; (3) the fourth angle is vertical to the first (or forms a linear pair with a 90° angle). Invalid: "they look like right angles".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5**

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

66

**QUESTION ID**

## S2-F1-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**Each angle of the pair measures 90°, so the lines are perpendicular.**

**GRADING NOTE** REQUIRED WORK: Linear pair angles add to 180°; congruent means equal measures, so each is 90°; then apply the definition.

Any valid proof. Essential: x + x = 180° from the linear pair, so x = 90°; the lines meet and form a right angle, which is the definition of perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

67

**QUESTION ID**

**S2-F1-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**25**

**GRADING NOTE** Any equivalent form. Perpendicular means the angle is 90°: 3x + 15 = 90, so x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

**True or false?**

If two intersecting lines form a pair of vertical angles that are supplementary, the lines are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

68

QUESTION ID

## S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Vertical angles are equal, so each is 180° ÷ 2 = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at point O. Which statement is enough to prove that p ⊥ q?

**A.** Two of the angles at O are vertical angles

**B.** Two adjacent angles at O add to 180°

**C.** Two of the angles at O are acute

**D.** One of the four angles at O measures 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

69

**QUESTION ID**

## S2-F1-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. One of the four angles at O measures 90°**

**GRADING NOTE** The definition needs a right angle. Every pair of crossing lines has vertical angles and linear pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

# FORWARD 2

Reason about three lines in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5**

71

**QUESTION ID**

## S2-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**a ∥ b**

**GRADING NOTE** REQUIRED WORK: Use the corresponding angles that a and b make with t (both right angles, so congruent) and the converse of the corresponding angles theorem; or argue by contradiction with the triangle angle sum.

Any valid proof. Essential: both corresponding angles at t are 90°, so they are congruent; congruent corresponding angles make the lines parallel. Alternative: if a and b met, they would form a triangle with t having two 90° angles, so its angle sum would exceed 180°. Invalid: "they look parallel".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5**

**True or false?**

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

72

QUESTION ID

## S2-F2-Q2

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Counterexample: in a corner of a room, the two floor edges are both perpendicular to the vertical edge, but they meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

**A.** t and b do not meet

**B.** t is perpendicular to b

**C.** t is parallel to b

**D.** The angle between t and b cannot be known

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

73

QUESTION ID

**S2-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. t is perpendicular to b**

**GRADING NOTE** t crosses a, so it crosses b. The corresponding angle at b equals the right angle at a, so t ⊥ b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

74

**QUESTION ID**

## S2-F2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**x = 5**

**GRADING NOTE** Form required: an equation of the line. A line perpendicular to the x-axis is vertical, so every point has x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

**True or false?**

In a plane, through a point not on a line there is exactly one line parallel to the given line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

75

QUESTION ID

**S2-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** This is the parallel postulate of Euclidean geometry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

# Transformations as functions: inputs, outputs, and preimages

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.2** |
| **BACKWARD 1** | Evaluate a rule for a given input | **8.F.A.1** |
| **BACKWARD 2** | Apply coordinate rules for translations, reflections, and rotations | **8.G.A.3** |
| **BACKWARD 3** | Read and locate ordered pairs in all four quadrants | **6.NS.C.6.c** |
| **FORWARD 1** | Compose two transformations and write one rule | **HSG-CO.A.2** |
| **FORWARD 2** | Find the inverse of a transformation | **HSG-CO.A.2** |

SET 3 • HSG-CO.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

78

**QUESTION ID**

## S3-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(6, 2)**

**GRADING NOTE** Form required: an ordered pair. Input (2, 5): x + 4 = 6 and y − 3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by R(x, y) = (−y, x).
Find R(3, −1).

R(3, −1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

79

QUESTION ID

**S3-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**(1, 3)**

**GRADING NOTE** Form required: an ordered pair. −y = −(−1) = 1 and x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5**

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

80

QUESTION ID

**S3-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**(4, −7)**

**GRADING NOTE** Form required: an ordered pair. F(4, −7) = (4, 7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

**A.** T(x, y) = (x − 5, y − 2)

**B.** T(x, y) = (5x, 2y)

**C.** T(x, y) = (x + 2, y + 5)

**D.** T(x, y) = (x + 5, y + 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

81

**QUESTION ID**

## S3-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. T(x, y) = (x + 5, y + 2)**

**GRADING NOTE** Right adds to x and up adds to y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

**True or false?**

Under the transformation D(x, y) = (2x, 2y), the point (0, 0) is its own output.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

82

QUESTION ID

**S3-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** D(0, 0) = (0, 0): the origin is a fixed point of D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

# BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

f(x) = 3x − 2. Find f(4).

f(4) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

84

QUESTION ID

**S3-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**10**

**GRADING NOTE** Any equivalent form. 3(4) − 2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

85

**QUESTION ID**

## S3-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

g(x) = x + 5. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

86

**QUESTION ID**

## S3-B1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**7**

**GRADING NOTE** Any equivalent form. 7 + 5 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

**True or false?**

A function assigns exactly one output to each input.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

**Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5**

87

**QUESTION ID**

## S3-B1-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** That is the definition of a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

**Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5**

**True or false?**

The rule "the input 3 gives the outputs 4 and 5" describes a function.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

88

QUESTION ID

## S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** A function gives exactly one output for each input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

# BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

90

**QUESTION ID**

**S3-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**(−1, 3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

91

QUESTION ID

**S3-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**(−5, 2)**

**GRADING NOTE** Form required: an ordered pair. (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

**A.** (1, 4)

**B.** (−1, 4)

**C.** (4, −1)

**D.** (−4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

92

**QUESTION ID**

## S3-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (−4, −1)**

**GRADING NOTE** A 180° rotation about the origin sends (x, y) to (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

**True or false?**

Reflecting the point (3, 6) across the x-axis gives (3, −6).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

**Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5**

93

QUESTION ID

## S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** (x, y) → (x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

**Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5**

**True or false?**

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

94

QUESTION ID

**S3-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** (x, y) → (−y, x) gives (−5, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

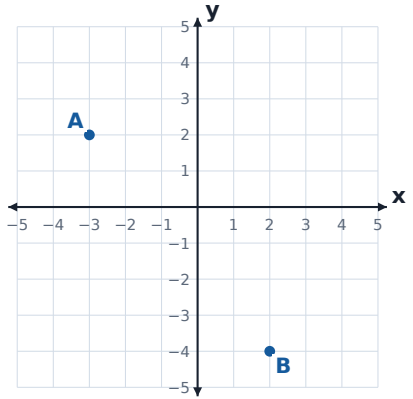
SET 3 • HSG-CO.A.2

# BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

**Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5**

96

**QUESTION ID**

## S3-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(−3, 2)**

**GRADING NOTE** Form required: an ordered pair (x, y).

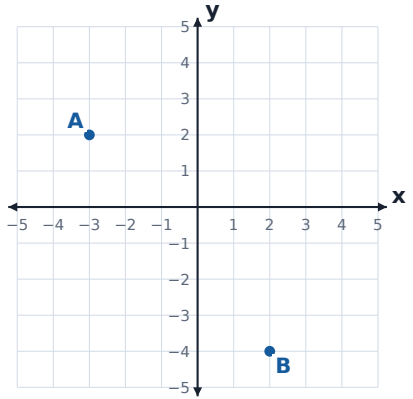
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

**Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

What are the coordinates of point B?

B = _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

97

QUESTION ID
**S3-B3-Q2**

TYPE
Short answer

CORRECT ANSWER
**(2, −4)**

GRADING NOTE Form required: an ordered pair (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

**A.** (−2, −5)

**B.** (−2, 5)

**C.** (2, 5)

**D.** (2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

98

QUESTION ID

**S3-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. (−2, −5)**

**GRADING NOTE** Quadrant III has $x < 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

**True or false?**

The point (0, −3) lies on the y-axis.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

99

QUESTION ID

**S3-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Its x-coordinate is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

**True or false?**

The point (−2, 5) is 2 units to the right of the y-axis.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

100

QUESTION ID

**S3-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** x = −2 is 2 units to the LEFT of the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

# FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

T(x, y) = (x + 2, y) and R(x, y) = (−x, y). Find R(T(1, 4)).

R(T(1, 4)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

102

**QUESTION ID**

## S3-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(−3, 4)**

**GRADING NOTE** Form required: an ordered pair. T(1, 4) = (3, 4); R(3, 4) = (−3, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

103

QUESTION ID

**S3-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**(−x − 2, y)**

**GRADING NOTE** Any equivalent form, such as $(-(x + 2), y)$. Apply T first, then R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

T(x, y) = (x + 2, y) and R(x, y) = (−x, y). Find T(R(1, 4)).

T(R(1, 4)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

104

**QUESTION ID**

## S3-F1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**(1, 4)**

**GRADING NOTE** Form required: an ordered pair. R(1, 4) = (−1, 4); T(−1, 4) = (1, 4). The order changes the answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

**True or false?**

For any two transformations, doing them in either order gives the same result.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

105

QUESTION ID

**S3-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A translation and a reflection give different results in different orders (for example, R(T(1, 4)) ≠ T(R(1, 4)) above).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$.
Which rule is $S(T(x, y))$?

**A.** $(x + 2, y + 4)$

**B.** $(x - 3, y - 5)$

**C.** $(x + 2, y - 4)$

**D.** $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

106

**QUESTION ID**
**S3-F1-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. (x + 2, y + 4)**

**GRADING NOTE** $S(x + 3, y - 1) = (x + 3 - 1, y - 1 + 5) = (x + 2, y + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

# FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of T(x, y) = (x − 6, y + 2).

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

108

QUESTION ID

**S3-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**(x + 6, y − 2)**

**GRADING NOTE** Any equivalent form. The inverse undoes T: T(x + 6, y − 2) = (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

R(x, y) = (−y, x) is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

109

QUESTION ID

**S3-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**(y, −x)**

**GRADING NOTE** Any equivalent form: a 90° clockwise rotation. Check: R(y, −x) = (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

**True or false?**

Every rotation is its own inverse.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5



QUESTION ID

## S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** A 90° rotation is undone by a 270° (or −90°) rotation; only a 180° rotation undoes itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

**A.** (x − 3, y − 3)

**B.** (3y, 3x)

**C.** (−3x, −3y)

**D.** $(\frac{x}{3}, \frac{y}{3})$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

111

QUESTION ID

**S3-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. (x/3, y/3)**

**GRADING NOTE** Dividing each coordinate by 3 undoes multiplying by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

**True or false?**

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

112

QUESTION ID

**S3-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** P is not one-to-one: (2, 1) and (2, 5) have the same output (2, 0), so no rule can undo it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

# Compare transformations that preserve distance and angle with those that do not

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.2** |
| **BACKWARD 1** | Find the distance between two points | **8.G.B.8** |
| **BACKWARD 2** | Know that rigid motions keep lengths and angle measures | **8.G.A.1** |
| **FORWARD 1** | Classify transformations as rigid motions, similarity transformatio... | **HSG-SRT.A.2** |
| **FORWARD 2** | Prove with variables that a transformation preserves or scales ev... | **HSG-SRT.A.2** |

SET 4 • HSG-CO.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

**A.** R(x, y) = (−x, y)

**B.** T(x, y) = (x + 4, y − 1)

**C.** H(x, y) = (2x, y)

**D.** Q(x, y) = (−y, x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

115

QUESTION ID

**S4-M-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**C. H(x, y) = (2x, y)**

**GRADING NOTE** H stretches horizontally: (0, 0) and (1, 0) are 1 unit apart, but their images (0, 0) and (2, 0) are 2 units apart. The others are a translation, a reflection, and a rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

116

QUESTION ID

**S4-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**H(A) = (0, 0), H(B) = (6, 4); the distance is √52 = 2√13 ≈ 7.21; no.**

**GRADING NOTE** Form required: both images, the distance (exact or to 0.01), and the conclusion. $\sqrt{(6^2 + 4^2)} = \sqrt{52} \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation T(x, y) = (x − 2, y + 5) maps A(1, 1) and B(4, 5) to A′ and B′. Show that A′B′ = AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

117

QUESTION ID

**S4-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**A′(−1, 6), B′(2, 10).**
**AB = √(3² + 4²) = 5**
**and**
**A′B′ = √(3² + 4²) = 5,**
**so A′B′ = AB.**

GRADING NOTE Form required: both images and both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

**True or false?**

The transformation V(x, y) = (x, 3y) preserves the measure of every angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

118

QUESTION ID

**S4-M-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The right angle at (0, 0) between the rays through (1, 1) and (1, −1) maps to the angle between the rays through (1, 3) and (1, −3), which is about 143°, not 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

119

QUESTION ID

**S4-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**Distance: no; every distance is multiplied by 3. Angle measure: yes; S is a dilation, which keeps every angle measure.**

**GRADING NOTE** Form required: both answers with reasons. Example: (0, 0) and (1, 0) are 1 apart; their images are 3 apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

# BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

121

**QUESTION ID**

## S4-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**5**

**GRADING NOTE** Any equivalent form. $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between (−2, 3) and (4, 3).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

122

**QUESTION ID**

**S4-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form. Same y, so |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

123

QUESTION ID

**S4-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**13**

**GRADING NOTE** Any equivalent form. √(25 + 144) = 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

**True or false?**

The distance from (0, 0) to (1, 1) is 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5**

124

QUESTION ID

## S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** $\sqrt{(1^2 + 1^2)} = \sqrt{2} \approx 1.41$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5**

Find the distance between (2, −1) and (5, 3).

**A.** 25

**B.** √7

**C.** 5

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

125

QUESTION ID

**S4-B1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. 5**

**GRADING NOTE** $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

# BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

**True or false?**

A rotation keeps every segment the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

**Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5**

127

QUESTION ID

## S4-B2-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Rotations are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

**Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5**

**True or false?**

A reflection can change the measure of an angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

**Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5**

128

QUESTION ID

## S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Reflections keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

**Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5**

A triangle is translated. What changes?

**A.** Its angle measures

**B.** Its side lengths

**C.** The positions of its vertices

**D.** Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

129

**QUESTION ID**

**S4-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The positions of its vertices**

**GRADING NOTE** A translation moves the figure but keeps its lengths, angles, and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

130

**QUESTION ID**

## S4-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**7 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

131

**QUESTION ID**

**S4-B2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**35°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

# FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

**A.** (x, y) → (4x, y)

**B.** (x, y) → (x + 4, y)

**C.** (x, y) → (x, y + 4)

**D.** (x, y) → (4x, 4y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

133

**QUESTION ID**

## S4-F1-Q1

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (x, y) → (4x, 4y)**

**GRADING NOTE** Multiplying both coordinates by 4 is a dilation: angles stay, distances are multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify (x, y) → (−2x, −2y) as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

134

**QUESTION ID**

## S4-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**A similarity transformation that is not rigid: every distance is doubled and angles are kept (a dilation by 2 followed by a 180° rotation).**

**GRADING NOTE** Any equivalent explanation that shows the factor 2 for distances and that angles are kept.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5**

Classify (x, y) → (x + y, y) as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

135

**QUESTION ID**

## S4-F1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**Neither. (0, 0) and (0, 1) are 1 apart but their images (0, 0) and (1, 1) are √2 apart, while (0, 0) and (1, 0) stay 1 apart: distances are not all multiplied by one number, and the right angle at the origin becomes 45°.**

**GRADING NOTE** Any correct counterexample showing that distances are not scaled by a single factor (or an angle changes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

**True or false?**

Every similarity transformation is a rigid motion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

136

QUESTION ID

## S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Only similarity transformations with scale factor 1 are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

**True or false?**

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

137

QUESTION ID

## S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Multiplying all distances by k keeps triangles similar (SSS), so angle measures stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

# FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that T(x, y) = (x + a, y + b) preserves the distance between ANY two points $(x_1, y_1)$ and $(x_2, y_2)$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

139

QUESTION ID

**S4-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**The image distance equals $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$, the original distance.**

GRADING NOTE

REQUIRED WORK: Write both images, apply the distance formula to the images, and show that a and b cancel.

Any valid algebraic proof. Essential: images $(x_1 + a, y_1 + b)$ and $(x_2 + a, y_2 + b)$; the differences are $x_2 - x_1$ and $y_2 - y_1$, so the distance formula gives the same value. Invalid: checking one or two numerical examples only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

140

**QUESTION ID**

## S4-F2-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The image distance is $3\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.**

**GRADING NOTE** REQUIRED WORK: Write both images, apply the distance formula, and factor out 9 under the square root.

Any valid proof. Essential: $\sqrt{((3x_2 - 3x_1)^2 + (3y_2 - 3y_1)^2)} = \sqrt{(9[(x_2 - x_1)^2 + (y_2 - y_1)^2])} = 3\cdot$(original distance).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of $(x_1, y_1)$ and $(x_2, y_2)$ is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

141

**QUESTION ID**

## S4-F2-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**Exactly the pairs with $x_1 = x_2$ (points on the same vertical line).**

**GRADING NOTE** Any equivalent condition. The distance is unchanged only if $4(x_2 - x_1)^2 = (x_2 - x_1)^2$, that is, $x_2 = x_1$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F2-Q3**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5**

**True or false?**

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

142

**QUESTION ID**

## S4-F2-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** H(x, y) = (2x, y) keeps vertical distances but changes others. A general claim needs a proof for all points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

**True or false?**

One example in which a distance is preserved is enough to prove that a transformation preserves distance.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

143

**QUESTION ID**

## S4-F2-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Every pair of points must keep its distance; one example proves nothing about all pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

# Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.3** |
| **BACKWARD 1** | Identify lines of symmetry of figures | **4.G.A.3** |
| **BACKWARD 2** | Know the side and angle properties of parallelograms, rectangles,… | **5.G.B.3** |
| **BACKWARD 3** | Rotate points by 90° and 180° about the origin | **8.G.A.3** |
| **FORWARD 1** | Use a symmetry to deduce congruent parts of the figure | **HSG-CO.B.6** |
| **FORWARD 2** | Combine symmetries and identify a figure from its symmetries | **HSG-CO.A.5** |

SET 5 • HSG-CO.A.3

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

146

**QUESTION ID**

## S5-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Rotations: 180° and 360° about the center O. Reflections: across the line through the midpoints of AB and CD, and across the line through the midpoints of BC and DA.**

**GRADING NOTE** Form required: both rotations and both reflections, with the center and the lines named. The diagonals are NOT lines of symmetry of a non-square rectangle, and a 90° turn does not work because the sides have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Which transformation carries it onto itself?

**A.** A 90° rotation about the point where its diagonals meet

**B.** A reflection across diagonal PR

**C.** A reflection across the line through the midpoints of PQ and RS

**D.** A 180° rotation about the point where its diagonals meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

147

QUESTION ID

## S5-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

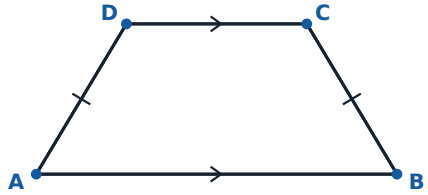
**D. A 180° rotation about the point where its diagonals meet**

**GRADING NOTE** The diagonals of a parallelogram bisect each other, so a 180° turn about their intersection swaps P with R and Q with S. A general parallelogram has no line of symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5**

Trapezoid ABCD is isosceles, with bases AB and DC and legs AD = BC. Describe every rotation and reflection that carries it onto itself.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

148

QUESTION ID

**S5-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Only the reflection across the line through the midpoints of the two bases, and the 360° rotation.**

GRADING NOTE Form required: the one reflection (named by its line) and the 360° rotation. No rotation of less than 360° works because the bases have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

**True or false?**

A 90° rotation about its center carries a rectangle that is not a square onto itself.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

149

**QUESTION ID**

## S5-M-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** A 90° turn would send a long side onto a short side, so the image does not match the rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has EF ∥ HG, EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations and reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

150

**QUESTION ID**

## S5-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Only the 360° rotation (the identity). It has no line of symmetry and no other rotation symmetry.**

**GRADING NOTE** Form required: the conclusion with a reason. A symmetry would have to send the 3-unit leg to a congruent leg, but the only other leg is 5 units long, and the two bases have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

# BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

**Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5**

152

**QUESTION ID**

## S5-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**4**

**GRADING NOTE** Any equivalent form. Two through opposite vertices and two through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

**Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5**

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

153

**QUESTION ID**

## S5-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**2**

**GRADING NOTE** Any equivalent form. Only the lines through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

**True or false?**

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

154

**QUESTION ID**

## S5-B1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Folding along a diagonal does not match the sides of a non-square rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

**A.** An isosceles triangle that is not equilateral

**B.** A square

**C.** A circle

**D.** A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

155

**QUESTION ID**

## S5-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. An isosceles triangle that is not equilateral**

**GRADING NOTE** The line through the vertex angle and the midpoint of the base is its only line of symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

**True or false?**

An isosceles trapezoid has a line of symmetry.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

156

QUESTION ID

**S5-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The line through the midpoints of its two bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

# BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

**True or false?**

In every parallelogram, the diagonals have equal lengths.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

158

QUESTION ID

**S5-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only rectangles (including squares) have equal diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

**True or false?**

Every rectangle is a parallelogram.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

**Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5**

159

QUESTION ID

## S5-B2-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Both pairs of opposite sides of a rectangle are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

**Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5**

Which property do all rectangles have?

**A.** Diagonals that are perpendicular

**B.** Four sides of equal length

**C.** Four right angles

**D.** Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

160

QUESTION ID

**S5-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. Four right angles**

**GRADING NOTE** A rectangle has four right angles; only a square has all sides equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

161

QUESTION ID

**S5-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**110°, 70°, 110°**

**GRADING NOTE** Any order. Consecutive angles are supplementary; opposite angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

162

**QUESTION ID**

**S5-B2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**65°**

**GRADING NOTE** Any equivalent form. Base angles of an isosceles trapezoid are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

# BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

164

QUESTION ID

**S5-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**(−3, −1)**

**GRADING NOTE** Form required: an ordered pair. (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

165

QUESTION ID

**S5-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**(0, 2)**

**GRADING NOTE** Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

**True or false?**

A 180° rotation about the origin sends (x, y) to (−x, −y).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

**Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5**

166

QUESTION ID

**S5-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** That is the 180° rotation rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

**Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5**

Rotate the point (−4, 2) by 90° counterclockwise about the origin.

**A.** (4, −2)

**B.** (−4, −2)

**C.** (2, 4)

**D.** (−2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

167

**QUESTION ID**

**S5-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (−2, −4)**

**GRADING NOTE** (x, y) → (−y, x) gives (−2, −4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

**True or false?**

A rotation of 360° about a point moves every point to a new position.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

168

QUESTION ID

**S5-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A 360° rotation returns every point to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

# FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

170

QUESTION ID

**S5-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**A goes to B (and D goes to C), so ∠DAB ≅ ∠CBA.**

**GRADING NOTE** Any equivalent answer naming A → B and a pair of base angles (∠D ≅ ∠C also accepted). A reflection keeps angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

171

QUESTION ID

**S5-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**PQ ≅ RS and QR ≅ SP.**

**GRADING NOTE** Any equivalent form. The rotation sends segment PQ onto segment RS and QR onto SP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

**True or false?**

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

172

QUESTION ID

**S5-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The reflection that swaps A with B and D with C sends diagonal AC onto diagonal BD, and reflections keep lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

**A.** AB = BC

**B.** AC = AB

**C.** AD = AB

**D.** AD = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

173

**QUESTION ID**

## S5-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. AD = BC

**GRADING NOTE** The reflection sends side AD onto side BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

**True or false?**

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

174

QUESTION ID

## S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Every parallelogram has 180° rotation symmetry; most are not rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

# FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

176

**QUESTION ID**

## S5-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**A 180° rotation about O.**

**GRADING NOTE** Any equivalent form. Reflections across perpendicular lines through O combine to a 180° rotation about O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

**A.** A parallelogram that is not a rectangle or a rhombus

**B.** A square

**C.** A rectangle that is not a square

**D.** A kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

177

QUESTION ID

**S5-F2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**D. A kite that is not a rhombus**

**GRADING NOTE** A kite has one line of symmetry, through two opposite vertices. A rectangle has two lines and a 180° turn; a general parallelogram has a 180° turn and no line; a square has eight symmetries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

**True or false?**

A quadrilateral that is carried onto itself by a 180° rotation about a point must be a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

178

QUESTION ID

**S5-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Every parallelogram has 180° rotation symmetry about the crossing point of its diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

179

QUESTION ID

**S5-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**A parallelogram that is not a rectangle or a rhombus.**

**GRADING NOTE** Any equivalent answer. A rectangle or a rhombus would also have reflection symmetries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

**True or false?**

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

180

QUESTION ID

**S5-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Each point goes to its image and back again.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

# Rotations and reflections that carry a regular polygon onto itself

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.3** |
| **BACKWARD 1** | Find the angle of a fraction of a full turn | **4.MD.C.5.b** |
| **BACKWARD 2** | Recognize regular polygons | **5.G.B.3** |
| **BACKWARD 3** | Find lines of symmetry of regular polygons | **4.G.A.3** |
| **FORWARD 1** | Work backward from a symmetry to the regular polygon | **HSG-CO.A.3** |
| **FORWARD 2** | Combine symmetries of a regular polygon | **HSG-CO.A.5** |

SET 6 • HSG-CO.A.3

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

183

QUESTION ID
**S6-M-Q1**

TYPE
Short answer

CORRECT ANSWER
**72°, 144°, 216°, 288°, and 360°**

**GRADING NOTE** Form required: all five angles. Each is a multiple of 360° ÷ 5 = 72°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

184

QUESTION ID

**S6-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**6 reflections: 3 across lines through pairs of opposite vertices and 3 across lines through the midpoints of pairs of opposite sides.**

**GRADING NOTE** Form required: the number and both kinds of lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Which rotation about its center carries a regular octagon onto itself?

**A.** 100°

**B.** 135°

**C.** 150°

**D.** 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

185

QUESTION ID

**S6-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 135°**

**GRADING NOTE** The rotations are multiples of 360° ÷ 8 = 45°; 135° = 3 × 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

**True or false?**

Each line of symmetry of a regular nonagon (9 sides) passes through one vertex and the midpoint of the opposite side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

186

QUESTION ID

**S6-M-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A regular polygon with an odd number of sides has n lines of symmetry, each through a vertex and the midpoint of the opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

187

QUESTION ID

**S6-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**3 reflections: across each line through a vertex and the midpoint of the opposite side.**

**GRADING NOTE** Form required: the number and the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

# BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

189

QUESTION ID

**S6-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**72°**

**GRADING NOTE** Any equivalent form. 360 ÷ 5 = 72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

190

**QUESTION ID**

## S6-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**45°**

**GRADING NOTE** Any equivalent form. 360 ÷ 8 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

191

**QUESTION ID**

## S6-B1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**180°**

**GRADING NOTE** Any equivalent form. 3/6 = 1/2 and 1/2 of 360 is 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

**True or false?**

$\frac{1}{12}$ of a full turn is 30°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

**Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5**

192

QUESTION ID

## S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** 360 ÷ 12 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

**Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5**

**True or false?**

$\frac{1}{9}$ of a full turn is 36°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

**Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5**

193

QUESTION ID

## S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** 360 ÷ 9 = 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

**Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5**

SET 6 • HSG-CO.A.3

# BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

**A.** A right triangle

**B.** A rhombus that is not a square

**C.** A rectangle that is not a square

**D.** An equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

195

QUESTION ID

**S6-B2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**D. An equilateral triangle**

**GRADING NOTE** A regular polygon has all sides equal and all angles equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

**True or false?**

A polygon with all sides equal must be regular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

**Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5**

196

QUESTION ID

## S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** A rhombus that is not a square has equal sides but unequal angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

**Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5**

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

197

**QUESTION ID**

**S6-B2-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

198

QUESTION ID

**S6-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**A square**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

**True or false?**

A regular hexagon has 6 equal angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

**Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5**

199

QUESTION ID

## S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Regular means all sides and all angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

**Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5**

SET 6 • HSG-CO.A.3

# BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

**Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5**

201

**QUESTION ID**

## S6-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

**Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5**

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

**Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5**

202

**QUESTION ID**

**S6-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

**Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5**

**True or false?**

Every line of symmetry of a regular polygon passes through its center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

**Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5**

203

QUESTION ID

**S6-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A reflection that carries the polygon onto itself keeps its center fixed, so the center is on the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

**Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5**

A line of symmetry of a square passes through

**A.** two adjacent vertices

**B.** exactly one vertex and no midpoint

**C.** no point of the square

**D.** two opposite vertices, or the midpoints of two opposite sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

204

**QUESTION ID**

## S6-B3-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. two opposite vertices, or the midpoints of two opposite sides**

**GRADING NOTE** A square has 2 diagonal lines and 2 lines through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

**True or false?**

A regular hexagon has exactly 3 lines of symmetry.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

205

QUESTION ID

**S6-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It has 6: 3 through opposite vertices and 3 through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

# FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

207

**QUESTION ID**

**S6-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Any equivalent form. 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

208

**QUESTION ID**

## S6-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**18°**

**GRADING NOTE** Any equivalent form. A regular n-gon has n lines of symmetry, so n = 20 and 360 ÷ 20 = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

**True or false?**

A regular polygon can have 50° as its smallest positive rotation onto itself.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

209

QUESTION ID

## S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** 360 ÷ 50 = 7.2 is not a whole number of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

**A.** 35°

**B.** 70°

**C.** 80°

**D.** 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

210

QUESTION ID

**S6-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 40°**

**GRADING NOTE** The angle must be 360° ÷ n for a whole number n ≥ 3; 360 ÷ 40 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

211

QUESTION ID

**S6-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**24**

**GRADING NOTE** Any equivalent form. 12 rotations (30°, 60°, ..., 360°) and 12 reflections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

# FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

213

**QUESTION ID**

## S6-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**The position of C**

**GRADING NOTE** Any equivalent wording. Each 60° turn moves a vertex one place counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

214

QUESTION ID

**S6-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**The position of E**

**GRADING NOTE** Any equivalent wording. The rotation moves B to the position of C; the reflection across AD swaps C and E.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

**True or false?**

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

215

QUESTION ID

## S6-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Two reflections across lines through the center combine into a rotation about the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

**A.** The position of C

**B.** The position of A

**C.** The position of D

**D.** The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

216

QUESTION ID

**S6-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. The position of A**

**GRADING NOTE** The reflection across AC sends B to the position of D; the 90° turn moves D to the position of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

217

**QUESTION ID**

**S6-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**A 144° rotation about the center**

**GRADING NOTE** Any equivalent form. Rotation angles about the same center add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

# Definitions of reflections and translations

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.4** |
| **BACKWARD 1** | Find the midpoint of a horizontal or vertical segment | **6.NS.C.8** |
| **BACKWARD 2** | Recognize perpendicular and parallel segments on a grid | **4.G.A.1** |
| **BACKWARD 3** | Reflect and translate points with coordinate rules | **8.G.A.3** |
| **FORWARD 1** | Find lines of reflection and images in slanted, non-axis cases | **HSG-CO.A.4** |
| **FORWARD 2** | Prove from the definitions that reflections and translations keep l… | **HSG-CO.A.4** |

SET 7 • HSG-CO.A.4

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A reflection across line $\ell$ maps each point P not on $\ell$ to the point P′ such that _____. Each point on $\ell$ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

220

QUESTION ID

**S7-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**$\ell$ is the perpendicular bisector of segment PP′; itself.**

**GRADING NOTE** Any equivalent wording. Both blanks are needed. "P′ is the same distance from $\ell$" alone is not enough: it must be on the perpendicular through P, on the other side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Points A and B are different. Write a precise definition of the translation that maps A to B.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

221

QUESTION ID

**S7-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**It maps each point P to the point P′ such that PP′ has the same length as AB, is parallel to AB (or on line AB), and points in the same direction.**

**GRADING NOTE** Any equivalent wording. Essential: same length, parallel (or collinear), same direction. Accept "ABP′P is a parallelogram" for points P not on line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

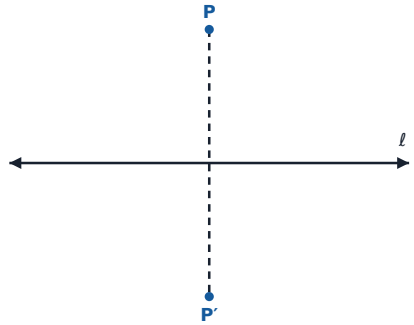
Point P′ is the image of point P under a reflection across line $\ell$, and P is not on $\ell$. Which must be true?

**A.** $\ell$ is perpendicular to PP′ and passes through its midpoint

**B.** $\ell$ passes through P

**C.** PP′ is parallel to $\ell$

**D.** P and P′ are on the same side of $\ell$




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

222

QUESTION ID

**S7-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. $\ell$ is perpendicular to PP′ and passes through its midpoint**

**GRADING NOTE** By the definition of a reflection, $\ell$ is the perpendicular bisector of PP′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

**True or false?**

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

223

QUESTION ID

**S7-M-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** That is what the definition of a translation says about every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

224

QUESTION ID

**S7-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**x = 3**

**GRADING NOTE** Form required: an equation of the line. The line is the perpendicular bisector of PP′: PP′ is horizontal with midpoint (3, 2), so the line is the vertical line x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

# BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

226

QUESTION ID

**S7-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(5, 4)**

**GRADING NOTE** Form required: an ordered pair. Halfway from x = 2 to x = 8 is x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

227

**QUESTION ID**

## S7-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(1, 5)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

228

QUESTION ID

**S7-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form. (−3 + 7) ÷ 2 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

**True or false?**

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

229

QUESTION ID

**S7-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE (−4 + 6) ÷ 2 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

**True or false?**

The midpoint of a segment is closer to one endpoint than to the other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

230

QUESTION ID

**S7-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is the same distance from both endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

# BACKWARD 2

Recognize perpendicular and parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

**True or false?**

A horizontal segment and a vertical segment are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

232

QUESTION ID

**S7-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Horizontal and vertical directions meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

**True or false?**

Any two segments of the same length are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

**Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5**

233

QUESTION ID

**S7-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Length does not decide direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

**Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5**

Which pair of segments is parallel?

**A.** From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

**B.** From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

**C.** From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

**D.** From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

234

QUESTION ID

**S7-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)**

**GRADING NOTE** Both segments in the first pair go 3 right and 1 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) that is perpendicular to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

235

**QUESTION ID**

## S7-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Any horizontal segment through (5, 3), for example from (4, 3) to (8, 3).**

**GRADING NOTE** Any horizontal segment containing (5, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

**A.** They are parallel

**B.** They are perpendicular to each other

**C.** They meet at the origin

**D.** They are the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

236

QUESTION ID

**S7-B2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. They are parallel**

**GRADING NOTE** Both are vertical lines, so they are parallel (or the same line if they have the same x-value).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

# BACKWARD 3

Reflect and translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

238

**QUESTION ID**

## S7-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(4, 3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

Translate the point (−1, 5) 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

239

**QUESTION ID**

**S7-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(5, 3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

**A.** (7, 2)

**B.** (−7, −2)

**C.** (−2, 7)

**D.** (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

240

QUESTION ID

**S7-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. (7, 2)**

**GRADING NOTE** Reflecting across y = x swaps the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

**True or false?**

Reflecting the point (−6, 1) across the y-axis gives (6, 1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

241

QUESTION ID

**S7-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

**True or false?**

Translating (3, 3) by 2 units left gives (3, 1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

**Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5**

242

QUESTION ID

## S7-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Left changes x: (1, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

**Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5**

SET 7 • HSG-CO.A.4

# FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

244

**QUESTION ID**

## S7-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -2x + 5$**

**GRADING NOTE** Any equivalent equation. The midpoint of PP′ is (2, 1); PP′ has slope 1/2, so the perpendicular bisector has slope −2: $y - 1 = -2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A translation maps A(2, −1) to A′(−3, 4). Where does it map B(0, 6)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

245

QUESTION ID

**S7-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**(−5, 11)**

**GRADING NOTE** Form required: an ordered pair. The translation moves every point 5 left and 5 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

**True or false?**

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

246

QUESTION ID
**S7-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Both segments are perpendicular to the line of reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

247

QUESTION ID

**S7-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**Yes. The midpoint (2, 2) is on y = x, and AA′ (slope −1) is perpendicular to y = x (slope 1).**

**GRADING NOTE** Form required: the conclusion with both conditions of the definition checked (midpoint and perpendicularity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

**A.** (−1, 0)

**B.** (5, 0)

**C.** (2, −3)

**D.** (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

248

**QUESTION ID**

**S7-F1-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (2, −3)**

**GRADING NOTE** The line is x = 2, the vertical perpendicular bisector of the horizontal segment PP′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

# FORWARD 2

Prove from the definitions that reflections and translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line $\ell$, and line PQ is not perpendicular to $\ell$. A reflection across $\ell$ maps them to P′ and Q′. Let PP′ meet $\ell$ at M and QQ′ meet $\ell$ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5**

250

**QUESTION ID**

**S7-F2-Q1**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**PQ = P′Q′**

**GRADING NOTE** REQUIRED WORK: Use the definition ($\ell$ is the perpendicular bisector of PP′ and of QQ′), then two pairs of congruent triangles (SAS): PMN ≅ P′MN, then PNQ ≅ P′NQ′.

Any valid proof. Essential: PM = P′M and QN = Q′N with right angles at M and N (definition); triangles PMN and P′MN are congruent by SAS, so PN = P′N and ∠PNM ≅ ∠P′NM; subtracting from the right angles gives ∠PNQ ≅ ∠P′NQ′; then triangles PNQ and P′NQ′ are congruent by SAS, so PQ = P′Q′. Alternative: coordinates with $\ell$ as the x-axis. Invalid: "reflections preserve distance" (that is what is being proved).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5**

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

251

QUESTION ID

**S7-F2-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**PQ = P′Q′**

**GRADING NOTE** REQUIRED WORK: Use the definition (PP′ and QQ′ are parallel, the same length, and the same direction) to show PP′Q′Q is a parallelogram.

Any valid proof. Essential: PP′ ∥ QQ′ and PP′ = QQ′ in the same direction, so PP′Q′Q is a parallelogram (one pair of opposite sides parallel and congruent); opposite sides of a parallelogram are congruent, so PQ = P′Q′. Invalid: assuming translations preserve length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5**

**True or false?**

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

252

QUESTION ID

**S7-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** If Q is on line PP′, all four points lie on one line, so there is no parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

**True or false?**

Under a reflection, a point not on the line of reflection and its image are on the same side of the line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

253

QUESTION ID

**S7-F2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are on opposite sides; the line is the perpendicular bisector of the segment joining them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across $\ell$ keeps the length of segment PQ, which fact is used first?

**A.** $\ell$ is the perpendicular bisector of PP′ and of QQ′

**B.** $\ell$ bisects ∠PQP′

**C.** $\ell$ is parallel to PQ

**D.** $\ell$ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

254

**QUESTION ID**
**S7-F2-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. $\ell$ is the perpendicular bisector of PP′ and of QQ′**

**GRADING NOTE** The proof starts from the definition of a reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

# Definition of a rotation

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.4** |
| **BACKWARD 1** | Describe turns in degrees, clockwise and counterclockwise | **4.MD.C.5.b** |
| **BACKWARD 2** | Points on a circle are the same distance from its center | **7.G.B.4** |
| **BACKWARD 3** | Rotate points 90°, 180°, and 270° about the origin | **8.G.A.3** |
| **FORWARD 1** | Rotate about a center that is not the origin | **HSG-CO.A.4** |
| **FORWARD 2** | Find the center and angle of a rotation from points and their imag.. | **HSG-CO.A.4** |

SET 8 • HSG-CO.A.4

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point $P \neq O$ to the point $P'$ such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

257

**QUESTION ID**

## S8-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$OP' = OP$ ($P'$ is on the circle with center O through P); $m\angle POP' = \theta$, measured counterclockwise from ray OP; itself.**

**GRADING NOTE** Any equivalent wording. All three blanks are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

**A.** O is the midpoint of PP′

**B.** O is on the perpendicular bisector of PP′

**C.** OP′ = 2 · OP

**D.** PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

258

QUESTION ID

**S8-M-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B. O is on the perpendicular bisector of PP′**

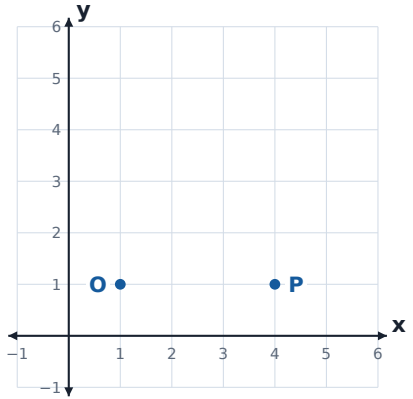
**GRADING NOTE** OP = OP′, so O is equidistant from P and P′ and lies on the perpendicular bisector of PP′. O is the midpoint only for a 180° rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.




P′ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

259

QUESTION ID
**S8-M-Q3**

TYPE
Short answer

CORRECT ANSWER
**(1, 4)**

GRADING NOTE Form required: an ordered pair. P is 3 units right of O; after a 90° counterclockwise turn it is 3 units above O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

**True or false?**

Under a rotation about point O, each point P moves along the circle with center O that passes through P.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5**

260

**QUESTION ID**

## S8-M-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** The definition keeps OP′ = OP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5**

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

261

QUESTION ID

## S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

**The center is the same distance from P and from P′, so it is on the perpendicular bisector of PP′. That bisector passes through (1, 1) with slope 1, which is the line y = x.**

**GRADING NOTE** Any equivalent explanation that uses equal distances from the center (the definition) and the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

# BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

263

**QUESTION ID**

## S8-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Up**

**GRADING NOTE** Any equivalent wording (north).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

264

QUESTION ID

**S8-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Down**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

**True or false?**

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

265

QUESTION ID

## S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** 270° clockwise and 90° counterclockwise add to a full 360° turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

**A.** Down

**B.** Left

**C.** Right

**D.** Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

266

QUESTION ID

**S8-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. Right**

**GRADING NOTE** Clockwise from up is toward the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

267

**QUESTION ID**

## S8-B1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**270°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

# BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

269

**QUESTION ID**

## S8-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

**True or false?**

If OA = OB, then A and B lie on one circle with center O.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

270

QUESTION ID

**S8-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Both are on the circle with center O and radius OA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

**A.** (4, 4)

**B.** (3, 4)

**C.** (2, 3)

**D.** (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

271

**QUESTION ID**

## S8-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. (3, 4)**

**GRADING NOTE** $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

272

**QUESTION ID**

## S8-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**5 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

**True or false?**

Two points the same distance from O are always the same point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

273

QUESTION ID

**S8-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Many points lie on the circle of that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

# BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

275

QUESTION ID

**S8-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**(−2, 5)**

**GRADING NOTE** Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate (−3, 4) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

276

QUESTION ID

**S8-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**(3, −4)**

**GRADING NOTE** Form required: an ordered pair. (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

**A.** (−6, 1)

**B.** (1, −6)

**C.** (−1, −6)

**D.** (6, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

277

**QUESTION ID**

## S8-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (6, −1)**

**GRADING NOTE** (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

**True or false?**

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

278

QUESTION ID

**S8-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** 270° counterclockwise = 90° clockwise: (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

**True or false?**

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

**Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5**

279

QUESTION ID

**S8-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It gives (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

**Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5**

SET 8 • HSG-CO.A.4

# FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

281

QUESTION ID

**S8-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(3, 0)**

**GRADING NOTE** Form required: an ordered pair. P is (3, 1) from C; a 90° clockwise turn gives (1, −3) from C, which is (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

282

QUESTION ID

**S8-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**(−3, −1)**

**GRADING NOTE** Form required: an ordered pair. P is (2, 3) from C; the image is (−2, −3) from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5**

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

283

QUESTION ID

**S8-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**(0, 1)**

**GRADING NOTE** Form required: an ordered pair. P is (3, 0) from C; the image is (0, 3) from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

**A.** (4, 4)

**B.** (4, 1)

**C.** (1, −2)

**D.** (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

284

QUESTION ID

**S8-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. (4, 1)**

**GRADING NOTE** P is (0, 3) from C; 270° counterclockwise (= 90° clockwise) gives (3, 0) from C, which is (4, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

**True or false?**

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

285

QUESTION ID
**S8-F1-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** About C the image is (2, 2); about the origin it is (0, 4). The center matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

# FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

287

QUESTION ID

**S8-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Center (0, 0); 90° counterclockwise.**

**GRADING NOTE** Form required: the center and the angle with direction. The bisector of BB′ is y = x; the bisector of AA′ passes through (1, 2) with slope 2, which is y = 2x. They meet at (0, 0), and (1, 0) → (0, 1) is a 90° counterclockwise turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

288

QUESTION ID

**S8-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Center (2, 1); 90° counterclockwise.**

**GRADING NOTE** Form required: the center and the angle. Check: P is (2, 0) from (2, 1) and P′ is (0, 2) from it; Q is (2, 2) from it and Q′ is (−2, 2) from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

**True or false?**

The center of a nonzero rotation moves to a new position under the rotation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

289

QUESTION ID

**S8-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The center is the one fixed point of the rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

A rotation about C maps P to P′. CP = 5 and PP′ = 5√2. Which could be the angle of rotation?

**A.** 60°

**B.** 90°

**C.** 45°

**D.** 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

290

QUESTION ID

**S8-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 90°**

**GRADING NOTE** CP = CP′ = 5 and $PP'^2 = 50 = 5^2 + 5^2$, so triangle PCP′ has a right angle at C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

291

QUESTION ID

**S8-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**53°**

**GRADING NOTE** Rounding to the nearest degree required. The angle at the origin has tangent 4/3, so it is about 53.13°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

# Draw the image of a figure under a rotation, reflection, or translation

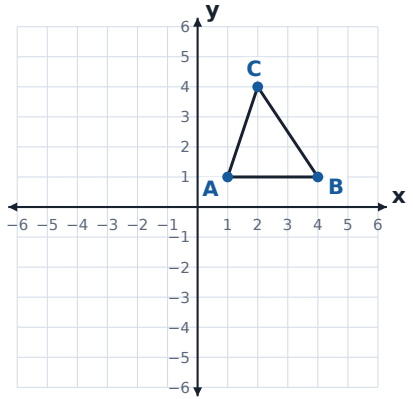
| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.5** |
| **BACKWARD 1** | Plot and read points in all four quadrants | **6.NS.C.6.c** |
| **BACKWARD 2** | Find image points with coordinate rules | **8.G.A.3** |
| **BACKWARD 3** | Count the distance from a point to a horizontal or vertical line | **6.NS.C.8** |
| **FORWARD 1** | Draw the image under a sequence of two transformations | **HSG-CO.A.5** |
| **FORWARD 2** | Draw rotations and reflections off the grid with compass, protract… | **HSG-CO.A.5** |

SET 9 • HSG-CO.A.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5**

294

**QUESTION ID**

## S9-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**A′(−4, −1), B′(−1, −1), C′(−3, 2), joined to form triangle A′B′C′.**

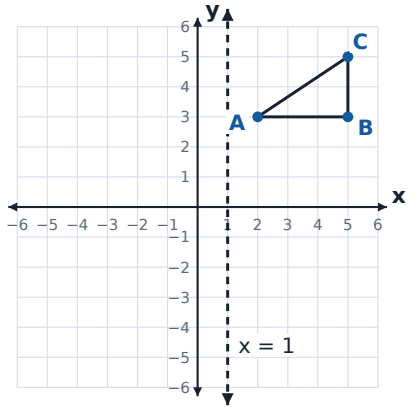
**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (x − 5, y − 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5**

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

295

**QUESTION ID**

## S9-M-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

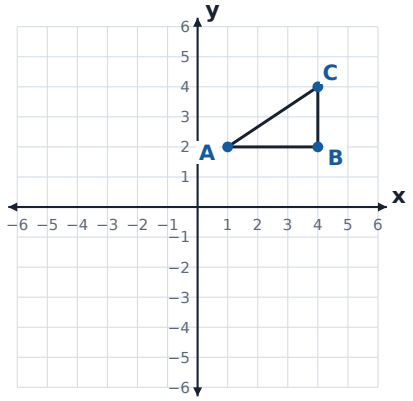
**A′(0, 3), B′(−3, 3), C′(−3, 5).**

**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Each image is the same distance from x = 1 on the other side: x → 2 − x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5**

296

**QUESTION ID**

## S9-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

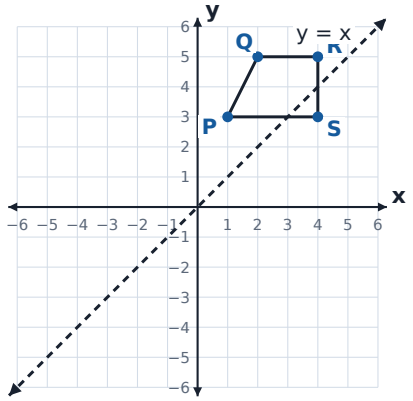
**A′(−2, 1), B′(−2, 4), C′(−4, 4).**

**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5**

Draw the image of quadrilateral PQRS under the reflection across the line y = x. Label the image P′Q′R′S′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

297

**QUESTION ID**

## S9-M-Q4

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

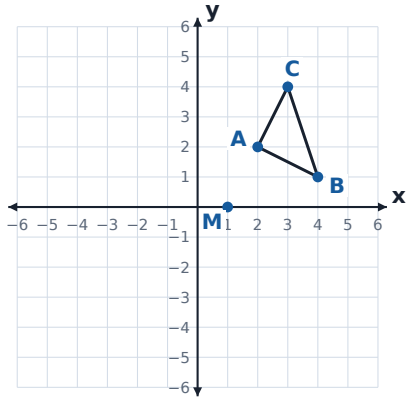
**P′(3, 1), Q′(5, 2), R′(5, 4), S′(3, 4).**

**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Reflection across y = x swaps the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5**

298

**QUESTION ID**

## S9-M-Q5

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**A′(0, −2), B′(−2, −1), C′(−1, −4).**

**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Each image is directly opposite its point through M: (x, y) → (2 − x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5**

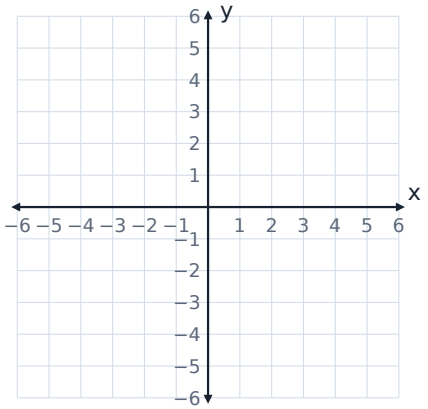
SET 9 • HSG-CO.A.5

# BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

**ANSWER KEY** Reduced copy of the question page, with its answer

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

300

**QUESTION ID**

## S9-B1-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**A at (−3, 2), B at (0, −4), C at (5, 1).**

**GRADING NOTE** Each point within about a quarter of a grid square of its location.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

301

**QUESTION ID**

## S9-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(−4, −1)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

**A.** (2, 2)

**B.** (0, −2)

**C.** (0, 2)

**D.** (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

302

**QUESTION ID**

**S9-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (0, 2)**

**GRADING NOTE** On the y-axis, x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

**True or false?**

The point (3, −2) is in Quadrant IV.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

303

QUESTION ID

**S9-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** $x > 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

**True or false?**

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

304

QUESTION ID

**S9-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The x-coordinate (2) is the horizontal move.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

# BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of (3, −2) under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

306

**QUESTION ID**

## S9-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(−1, −1)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of (−2, 5) under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

307

**QUESTION ID**

**S9-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(−2, −5)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

**A.** (−3, −1)

**B.** (3, −1)

**C.** (1, −3)

**D.** (−1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

308

**QUESTION ID**

**S9-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (1, −3)**

**GRADING NOTE** (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

**True or false?**

The image of (4, −1) under a reflection across the y-axis is (−4, −1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

**Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5**

309

QUESTION ID

**S9-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

**Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5**

**True or false?**

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

310

QUESTION ID

**S9-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is (−2, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

# BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

312

QUESTION ID

**S9-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form. |5 − 1| = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point (−2, −1) from the horizontal line y = 3?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

313

QUESTION ID

**S9-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form. |−1 − 3| = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

**A.** (−3, 4)

**B.** (−1, 4)

**C.** (2, 1)

**D.** (5, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

314

**QUESTION ID**

**S9-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. (−1, 4)**

**GRADING NOTE** 2 − 3 = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

**True or false?**

The point (0, 4) is 4 units from the x-axis.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

**Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5**

315

QUESTION ID

## S9-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Its distance from y = 0 is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

**Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5**

**True or false?**

The point (6, 2) is 6 units from the line x = 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

316

QUESTION ID

**S9-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is |6 − 2| = 4 units away.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

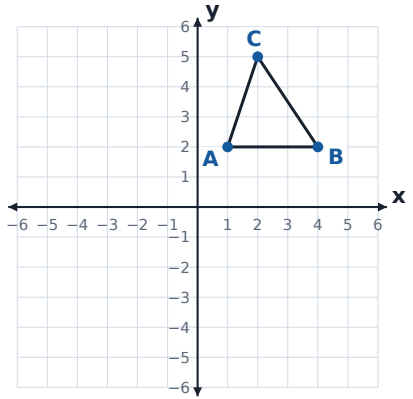
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

# FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5**

318

**QUESTION ID**

## S9-F1-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

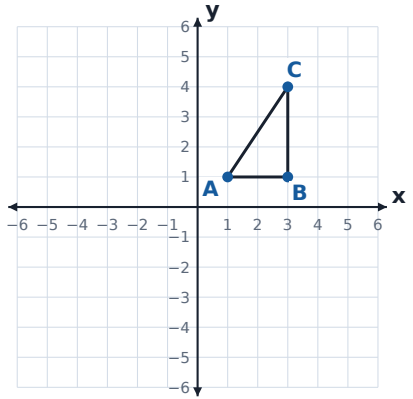
**A″(−1, −1), B″(−4, −1), C″(−2, 2).**

**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−x, y) → (−x, y − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5**

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5**

319

**QUESTION ID**

## S9-F1-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**A″(−1, −1), B″(−1, −3), C″(−4, −3).**

**GRADING NOTE** Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−y, x) → (−y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5**

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

320

QUESTION ID

**S9-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**(−4, −1)**

**GRADING NOTE** Form required: an ordered pair. (2, 1) → (4, 1) → (−4, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

**True or false?**

Reflecting a figure across the x-axis and then across the y-axis gives the same image as reflecting it across the line y = x.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

321

QUESTION ID

**S9-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The two reflections give (x, y) → (−x, −y), a 180° rotation; reflecting across y = x gives (y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

**A.** (3, 1)

**B.** (−1, 3)

**C.** (1, −3)

**D.** (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

322

**QUESTION ID**

## S9-F1-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (3, −1)**

**GRADING NOTE** (1, 3) → (−1, 3) → (3, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

# FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5



**QUESTION ID**
## S9-F2-Q1

**TYPE**
Drawing / plotting

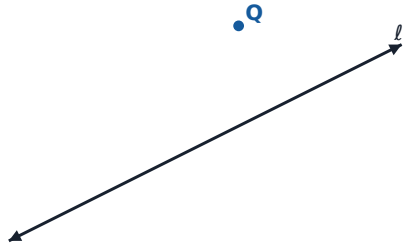
**CORRECT ANSWER**

**P′ is on the circle with center O through P, with m∠POP′ = 60° measured counterclockwise from ray OP.**

**GRADING NOTE** Accept within about 2° and 2 mm. Evidence: the arc (or equal compass distance) from O and the 60° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line $\ell$. Mark and label the image Q′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

325

**QUESTION ID**

## S9-F2-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

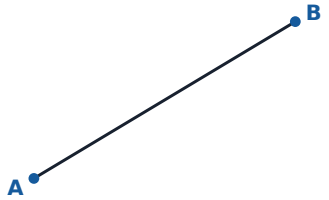
**Q′ is on the perpendicular from Q to $\ell$, on the other side of $\ell$, the same distance from $\ell$ as Q ($\ell$ is the perpendicular bisector of QQ′).**

**GRADING NOTE** Accept within about 2 mm. Evidence: the construction of the perpendicular from Q to $\ell$ and equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

326

QUESTION ID

## S9-F2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**AB′ has the same length as AB, with m∠BAB′ = 90° measured clockwise from ray AB; A stays fixed.**

**GRADING NOTE** Accept within about 2° and 2 mm. Evidence: equal lengths and the 90° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

**True or false?**

To rotate point P about point O by 45°, it is enough to draw a 45° angle at O; the distance from O does not matter.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

327

QUESTION ID

## S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The image must also be the same distance from O as P (on the circle centered at O through P).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Describe how to reflect point P across a slanted line $\ell$ using only a compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

328

QUESTION ID

## S9-F2-Q5

TYPE

Short answer

CORRECT ANSWER

**Construct the line through P perpendicular to $\ell$; mark where it meets $\ell$ (M); with the compass set to MP, mark the point P′ on that perpendicular, on the other side of $\ell$, so that MP′ = MP.**

**GRADING NOTE** Any equivalent construction (for example, two circles centered at two points of $\ell$ through P meet again at P′).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

# Specify a sequence of transformations that carries one figure onto another

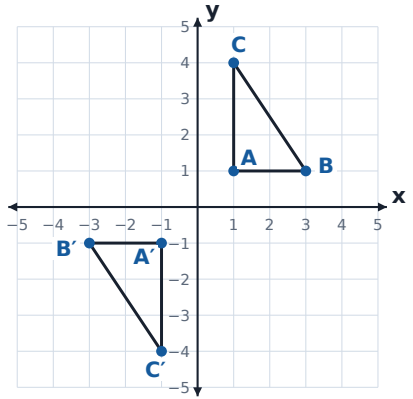
| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.A.5** |
| **BACKWARD 1** | Understand congruence as the result of one rigid motion | **8.G.A.2** |
| **BACKWARD 2** | Find images with coordinate rules for sequences | **8.G.A.3** |
| **BACKWARD 3** | Read the orientation (clockwise or counterclockwise order) of a tr… | **8.G.A.1** |
| **FORWARD 1** | Use the fewest transformations and justify | **HSG-CO.A.5** |
| **FORWARD 2** | Specify a similarity sequence: a dilation and rigid motions | **HSG-SRT.A.2** |

SET 10 • HSG-CO.A.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).




Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

331

QUESTION ID
**S10-M-Q1**

TYPE
Short answer

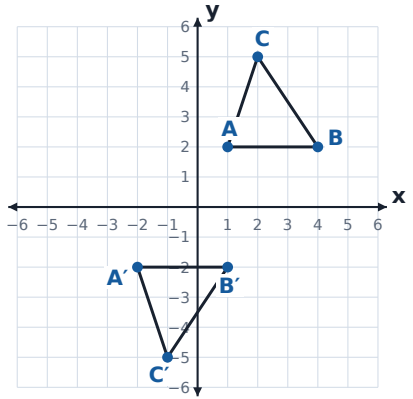
CORRECT ANSWER
**A 180° rotation about the origin.**

**GRADING NOTE** Any valid sequence, for example a reflection across the x-axis followed by a reflection across the y-axis. Check: every vertex must land on its partner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.




Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

332

**QUESTION ID**

## S10-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

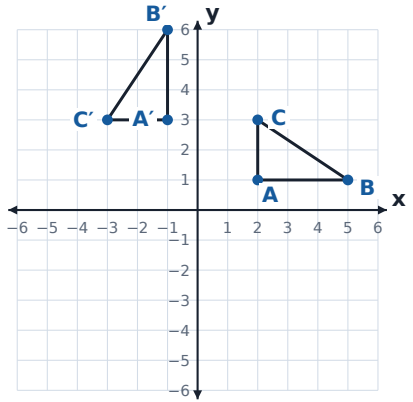
**Reflect across the x-axis, then translate 3 units left.**

**GRADING NOTE** Any valid sequence (for example, translate 3 units left, then reflect across the x-axis). The orientation is reversed, so the sequence must include a reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.




Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

333

QUESTION ID

**S10-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Rotate 90° counterclockwise about the origin, then translate 1 unit up.**

GRADING NOTE Any valid sequence. Check: (x, y) → (−y, x) → (−y, x + 1) sends A(2, 1) to (−1, 3), B to (−1, 6), C to (−3, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

**A.** Reflect across the y-axis, then translate 4 units right

**B.** Rotate 90° counterclockwise about the origin, then translate 4 units right

**C.** Rotate 90° clockwise about the origin, then translate 4 units right

**D.** Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5**

334

QUESTION ID

## S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

**B. Rotate 90° counterclockwise about the origin, then translate 4 units right**

**GRADING NOTE** The rotation sends B to (0, 2) and C to (−3, 0); the translation then gives B′(4, 2) and C′(1, 0). The other orders or turns send B or C elsewhere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5**

**True or false?**

Triangles PQR and P′Q′R′ are congruent. P, Q, R are in clockwise order and P′, Q′, R′ are in counterclockwise order. Some sequence of rotations and translations alone carries P to P′, Q to Q′, and R to R′.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

335

QUESTION ID

**S10-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Rotations and translations keep the order (orientation) of the vertices; a reflection is needed to reverse it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

# BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

**A.** A reflection

**B.** A translation

**C.** A rotation

**D.** A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

337

QUESTION ID

**S10-B1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**B. A translation**

**GRADING NOTE** Sliding every point the same way is a translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

338

QUESTION ID

**S10-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**A reflection**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

**True or false?**

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

339

QUESTION ID

## S10-B1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Rigid motions keep all lengths and angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

**True or false?**

A figure and its image under a rigid motion can have different areas.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

340

QUESTION ID

## S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Rigid motions keep area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

**A.** A translation 2 units up

**B.** A reflection across the x-axis

**C.** A 90° counterclockwise rotation about the origin

**D.** A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

341

**QUESTION ID**
**S10-B1-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**C. A 90° counterclockwise rotation about the origin**

**GRADING NOTE** (2, 0) → (0, 2) under (x, y) → (−y, x), and the origin stays fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

# BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

343

QUESTION ID

**S10-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**(−2, −3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point (−1, 4) is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

344

QUESTION ID

**S10-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**(−4, −1)**

**GRADING NOTE** Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

**A.** (3, −2)

**B.** (7, 4)

**C.** (7, −2)

**D.** (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

345

**QUESTION ID**

## S10-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. (3, −2)**

**GRADING NOTE** (5 − 2, 1 − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

**True or false?**

Point (0, 3) rotated 180° about the origin is (0, −3).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

346

QUESTION ID

**S10-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

**True or false?**

Point (4, 1) reflected across the line y = x is (4, −1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

347

QUESTION ID

**S10-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is (1, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

# BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

349

QUESTION ID

**S10-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**Counterclockwise**

**GRADING NOTE** Any equivalent wording. From A go right to B, then up-left to C: a counterclockwise turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

**True or false?**

A reflection reverses the orientation of a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

350

QUESTION ID

**S10-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A mirror image runs the other way around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

**True or false?**

A rotation reverses the orientation of a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

**Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5**

351

**QUESTION ID**

## S10-B3-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Rotations keep the order of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

**Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5**

Triangle PQR is clockwise. After two reflections, its image is

**A.** counterclockwise

**B.** either, depending on the lines

**C.** clockwise

**D.** not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

352

QUESTION ID

**S10-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. clockwise**

**GRADING NOTE** Each reflection reverses the order, so two reflections restore it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

353

**QUESTION ID**

## S10-B3-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

## Clockwise

**GRADING NOTE** Any equivalent wording. The translation keeps the order; the reflection reverses it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

# FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

355

QUESTION ID

**S10-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**A translation of 6 units, perpendicular to the lines (in the direction from the first line toward the second).**

**GRADING NOTE** Any equivalent wording with the distance (twice the gap) and the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

356

QUESTION ID

**S10-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**A rotation of 80° about O (in the direction from the first line toward the second).**

**GRADING NOTE** Any equivalent wording with the center and twice the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

357

**QUESTION ID**

## S10-F1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**Rotations and translations keep orientation, but these triangles have opposite orientations, so a reflection is needed.**

**GRADING NOTE** Any equivalent explanation based on orientation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

**True or false?**

Every rigid motion of the plane can be written as a sequence of at most three reflections.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

358

**QUESTION ID**

## S10-F1-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** This is a known theorem: a reflection fixes one vertex, a second fixes two, and a third (if needed) fixes the third vertex of a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

**A.** 2

**B.** 4

**C.** 3

**D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

359

**QUESTION ID**

## S10-F1-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 2**

**GRADING NOTE** The orientation is kept, so an even number is needed; two reflections give any rotation or translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

# FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

361

**QUESTION ID**

## S10-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Dilate by a factor of 3 about the origin, then rotate 90° counterclockwise about the origin.**

**GRADING NOTE** Any valid sequence. Check: (2, 0) → (6, 0) → (0, 6) and (0, 1) → (0, 3) → (−3, 0).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F2-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5**

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

362

QUESTION ID

**S10-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Dilate by 2 about the origin, then translate 2 units right and 2 units up.**

**GRADING NOTE** Any valid sequence. (x, y) → (2x + 2, 2y + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

**True or false?**

Any two figures that a sequence of rigid motions and a dilation carries one onto the other are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

363

QUESTION ID

## S10-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** They are similar; they are congruent only when the scale factor is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

**A.** Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

**B.** Dilate by 2 about the origin, then translate 1 right and 1 up

**C.** Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

**D.** Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

364

QUESTION ID

**S10-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up**

**GRADING NOTE** (3, 0) → (6, 0) → (0, 6) → (1, 7), and (0, 0) → (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

365

**QUESTION ID**

## S10-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**2**

**GRADING NOTE** Any equivalent form. Each side of XYZ is 2 times the matching side of PQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

# Predict the effect of a rigid motion on a figure

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.B.6** |
| **BACKWARD 1** | Know what rigid motions keep: lines, lengths, angles, and parallel l… | **8.G.A.1** |
| **BACKWARD 2** | Match the vertices and sides of a figure with its image | **8.G.A.2** |
| **FORWARD 1** | Predict the effect of a sequence of rigid motions | **HSG-CO.B.6** |
| **FORWARD 2** | Use a rigid motion to conclude that parts of a figure are congruent | **HSG-CO.B.7** |

SET 11 • HSG-CO.B.6

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line $\ell$. What are A′B′ and m∠B′?

**A.** 14 and 48°

**B.** 7 and 132°

**C.** 7 and 48°

**D.** They depend on where $\ell$ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

368

**QUESTION ID**

**S11-M-Q1**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 7 and 48°**

**GRADING NOTE** A reflection is a rigid motion: it keeps every length and every angle measure, wherever $\ell$ is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

369

QUESTION ID

**S11-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**P′Q′ ∥ S′R′. A rotation maps lines to lines and keeps them from meeting if they did not meet before, so parallel lines stay parallel.**

**GRADING NOTE** Any equivalent explanation. Essential: the images are parallel because rigid motions take parallel lines to parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

370

**QUESTION ID**

## S11-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**4 cm, on the other side of m.**

**GRADING NOTE** Any equivalent wording. m is the perpendicular bisector of AA′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

**True or false?**

A translation can reverse the orientation (the clockwise or counterclockwise order) of a triangle's vertices.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

371

QUESTION ID

**S11-M-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A translation slides every point the same way, so the order of the vertices around the triangle is kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line $\ell$. A 180° rotation about a point O on $\ell$ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

372

**QUESTION ID**

## S11-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**On $\ell$. A 180° rotation about a point of $\ell$ sends every point of $\ell$ to the point of $\ell$ on the opposite side of O, so $\ell$ maps onto itself.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

# BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

**True or false?**

Under a rotation, the image of a line is a line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

374

QUESTION ID

**S11-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Rigid motions map lines to lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

**True or false?**

A reflection can turn two parallel lines into two lines that meet.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

375

QUESTION ID

**S11-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Rigid motions keep parallel lines parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

376

**QUESTION ID**

## S11-B1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**52°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

**A.** 30 cm

**B.** 4.5 cm

**C.** 39 cm

**D.** 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

377

**QUESTION ID**

## S11-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 9 cm**

**GRADING NOTE** Rotations keep lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

378

QUESTION ID

**S11-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**Any two of: side lengths, angle measures, area, parallel sides, perimeter.**

GRADING NOTE Any two correct properties.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

# BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

380

QUESTION ID

**S11-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**B′C′**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

**A.** ∠R′

**B.** ∠P′

**C.** ∠Q′

**D.** ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

381

**QUESTION ID**

## S11-B2-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. ∠Q′**

**GRADING NOTE** The prime marks the image of each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

**True or false?**

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

382

QUESTION ID

**S11-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Corresponding sides of an image under a rigid motion are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

383

**QUESTION ID**

**S11-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**D′E′ = 6, E′F′ = 9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

**True or false?**

After a rotation, side A′B′ of the image always lies on top of side AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

**Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5**

384

QUESTION ID

## S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The image is usually in a new position; it only has the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

**Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5**

SET 11 • HSG-CO.B.6

# FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line $\ell$ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

386

QUESTION ID

**S11-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Counterclockwise. Each reflection reverses the orientation, so two reflections restore it.**

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° about point O and then rotated 50° about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

387

QUESTION ID

**S11-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**A rotation of 120° about O (in the same direction).**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

**True or false?**

A translation that is not the identity has exactly one fixed point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

388

QUESTION ID

**S11-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Every point moves by the same nonzero amount, so no point is fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

**A.** A translation 3 units right

**B.** A reflection across line $\ell$

**C.** A rotation of 90° about point O

**D.** The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

389

QUESTION ID

**S11-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. A rotation of 90° about point O**

GRADING NOTE A 90° rotation fixes only its center; a reflection fixes every point of its line; a translation fixes none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line $\ell$ is reflected across line m, and $\ell$ is perpendicular to m. What is the image of $\ell$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

390

QUESTION ID

**S11-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**$\ell$ itself. Each point of $\ell$ goes to the point of $\ell$ on the other side of m, the same distance away.**

GRADING NOTE Any equivalent explanation (a line perpendicular to the mirror maps onto itself).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

# FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

392

QUESTION ID

**S11-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**OC = 5 and CD = 7.**

**GRADING NOTE** Any equivalent form. The rotation sends OA to OC and AB to CD, keeping lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line $\ell$ maps triangle PQR onto triangle PSR (Q → S, and P and R are on $\ell$). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

393

**QUESTION ID**
**S11-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**64°**

**GRADING NOTE** Any equivalent form. ∠PQR maps to ∠PSR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

**True or false?**

If a rigid motion maps segment AB onto segment CD, then AB could be longer than CD.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

394

QUESTION ID

**S11-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Rigid motions keep lengths, so AB = CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

**A.** KL ∥ K′L′ or KL is on line K′L′

**B.** ∠L ≅ ∠L′

**C.** KL is on the same line as K′L′

**D.** KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

395

QUESTION ID

**S11-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. KL is on the same line as K′L′**

**GRADING NOTE** A translation keeps length and angle and keeps lines parallel to themselves, but the image is usually on a different line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

396

QUESTION ID

**S11-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**The two equal sides map to two sides of the same lengths, so the image has two equal sides.**

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

# Decide whether two figures are congruent using rigid motions

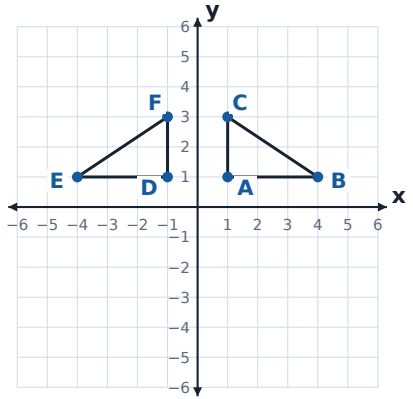
| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.B.6** |
| **BACKWARD 1** | Recognize congruent figures as images under rigid motions | **8.G.A.2** |
| **BACKWARD 2** | Find lengths on a grid with the Pythagorean Theorem | **8.G.B.8** |
| **FORWARD 1** | Decide congruence for figures given by measurements, including… | **HSG-CO.B.6** |
| **FORWARD 2** | Use an established congruence to find unknown parts | **HSG-CO.B.7** |

SET 12 • HSG-CO.B.6

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

399

**QUESTION ID**

## S12-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

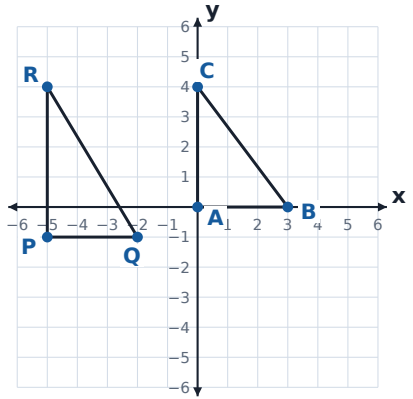
**Yes. The reflection across the y-axis carries A to D, B to E, and C to F.**

**GRADING NOTE** Any valid rigid motion or sequence. The conclusion and a motion that works are both required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

400

**QUESTION ID**

## S12-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**No. ABC has legs 3 and 4, but PQR has legs 3 and 5. A rigid motion keeps every length, so none can carry one onto the other.**

**GRADING NOTE** Form required: "No" with a length (or area) that differs. Areas 6 and 7.5 also show it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Which fact proves that two figures are NOT congruent?

**A.** One is above the other

**B.** One has area 12 and the other has area 15

**C.** They face opposite directions

**D.** They are in different quadrants

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

401

QUESTION ID

**S12-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. One has area 12 and the other has area 15**

**GRADING NOTE** Rigid motions keep area. Direction and position can be changed by rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

**True or false?**

Two figures are congruent if some sequence of rotations, reflections, and translations carries one onto the other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

402

QUESTION ID

**S12-M-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE That is the definition of congruence in terms of rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

403

QUESTION ID

**S12-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**Yes. A 90° rotation (followed by a translation if needed) carries R onto S.**

GRADING NOTE Any valid rigid motion. Both are 4 by 6 rectangles; turning one a quarter turn matches the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

# BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

**True or false?**

If one figure is a translation of another, the figures are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

405

QUESTION ID

**S12-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A translation is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

**True or false?**

Two figures with the same shape but different sizes are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

406

**QUESTION ID**

**S12-B1-Q2**

**TYPE**

True or false

**CORRECT ANSWER**

**B. False**

**GRADING NOTE** They are similar, not congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

**A.** A square and a larger square

**B.** A triangle and its reflection

**C.** A circle of radius 2 and a circle of radius 3

**D.** A triangle and a dilation of it by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

407

QUESTION ID

**S12-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. A triangle and its reflection**

**GRADING NOTE** A reflection keeps size and shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

408

**QUESTION ID**

## S12-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

## 5, 6, and 7

**GRADING NOTE** Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

409

QUESTION ID

**S12-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**Yes**

**GRADING NOTE** Any equivalent answer: both are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

# BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

411

QUESTION ID

**S12-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

412

QUESTION ID

**S12-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from (−2, 1) to (1, 5).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

413

QUESTION ID

**S12-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form. $\sqrt{(3^2 + 4^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

**True or false?**

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

414

QUESTION ID

## S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Both are √5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

**True or false?**

The segment from (0, 0) to (3, 3) has length 3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

415

QUESTION ID

**S12-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Its length is $\sqrt{18} = 3\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

# FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

417

**QUESTION ID**

## S12-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Translate P's center to Q's center. Every point 3 units from P's center goes to a point 3 units from Q's center, so circle P lands on circle Q.**

**GRADING NOTE** Any equivalent explanation using a translation that matches the centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

418

QUESTION ID

**S12-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**No. A rigid motion keeps angle measures, but one has 90° angles and the other a 60° angle.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

**True or false?**

Two triangles with the same perimeter must be congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

419

**QUESTION ID**

## S12-F1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** A 3-4-5 triangle and a 4-4-4 triangle both have perimeter 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Which information is enough to show two segments are congruent?

**A.** They have the same length

**B.** They have the same slope

**C.** They have the same midpoint

**D.** They both start at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

420

QUESTION ID

**S12-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. They have the same length**

GRADING NOTE A rigid motion can carry any segment onto any other segment of the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

421

QUESTION ID

## S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**Translate the vertex of angle 1 to the vertex of angle 2, rotate so one side lies along a side of angle 2, and reflect across that side if the other side is on the wrong side.**

**GRADING NOTE** Any valid description that matches vertex and both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

# FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

423

QUESTION ID

**S12-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**EF = 6, m∠G = 110°, CD = 4.**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

424

**QUESTION ID**

## S12-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form. RS = XY, so 2x + 3 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

**True or false?**

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

425

QUESTION ID

**S12-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE ∠B corresponds to ∠E.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

**A.** ∠S and ∠S′

**B.** PQ and RS

**C.** QR and S′T′

**D.** ∠S and ∠T′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

426

QUESTION ID

**S12-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. ∠S and ∠S′**

**GRADING NOTE** Each part corresponds to its own image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

427

**QUESTION ID**

**S12-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**14**

**GRADING NOTE** Any equivalent form. ∠BAC corresponds to ∠DCA, so 3y = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

# Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.B.7** |
| **BACKWARD 1** | Read a congruence statement to match corresponding parts | **8.G.A.2** |
| **BACKWARD 2** | Rigid motions keep lengths and angle measures | **8.G.A.1** |
| **BACKWARD 3** | Move a point onto another and a ray onto another with rigid motio… | **8.G.A.3** |
| **FORWARD 1** | Find which three corresponding parts are enough | **HSG-CO.B.8** |
| **FORWARD 2** | Use congruent triangles to prove a relationship in a figure | **HSG-SRT.B.5** |

SET 13 • HSG-CO.B.7

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries △ABC onto △DEF with A → D, B → E, C → F. Explain why AB = DE, BC = EF, CA = FD, ∠A ≅ ∠D, ∠B ≅ ∠E, and ∠C ≅ ∠F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5**

430

**QUESTION ID**

## S13-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**Each side and angle of ABC is carried onto the corresponding side and angle of DEF, and rigid motions keep lengths and angle measures.**

**GRADING NOTE**

REQUIRED WORK: Use the definition: the motion sends each vertex to its partner, so it sends each side and angle onto the corresponding one, and rigid motions keep length and angle measure.

Any valid explanation. Essential: (1) the motion maps segment AB onto segment DE (endpoints to endpoints), and similarly for the other sides and angles; (2) rigid motions preserve distance and angle measure. Invalid: "they look the same".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5**

In △ABC and △DEF, AB = DE, BC = EF, CA = FD, ∠A ≅ ∠D, ∠B ≅ ∠E, and ∠C ≅ ∠F. Explain how to build a rigid motion that carries △ABC onto △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5**

431

**QUESTION ID**

## S13-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**A translation, a rotation, and (if needed) a reflection carry A to D, B to E, and C to F.**

**GRADING NOTE** REQUIRED WORK: Translate A to D; rotate about D so ray AB lies on ray DE (B lands on E because AB = DE); if C is on the other side of line DE from F, reflect across DE. Explain why C then lands on F.

Any valid explanation. Essential: the translation and rotation match A and B; the reflection (if needed) puts C on F's side; C lands on F because ∠A ≅ ∠D puts C on ray DF and AC = DF fixes its distance from D. Accept using ∠B and ∠E instead. Invalid: stating the result without explaining why C lands on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5**

A rigid motion maps △PQR onto △XYZ with P → X, Q → Y, R → Z. Which must be true?

**A.** QR = YZ

**B.** PQ = YZ

**C.** ∠P ≅ ∠Y

**D.** QR = XZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5



**QUESTION ID**

**S13-M-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. QR = YZ**

**GRADING NOTE** Side QR corresponds to side YZ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

△JKL ≅ △MNO with J ↔ M, K ↔ N, L ↔ O. JK = 8, KL = 11, and m∠L = 37°. Find MN, NO, and m∠O.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

433

QUESTION ID

**S13-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**MN = 8, NO = 11, m∠O = 37°**

**GRADING NOTE** Any equivalent form. Corresponding parts of congruent triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

**True or false?**

Two triangles whose three pairs of corresponding angles are congruent must be congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

434

QUESTION ID

## S13-M-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** A small and a large equilateral triangle have the same angles but are not congruent: the sides must match too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

# BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

436

QUESTION ID

**S13-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**∠Y**

GRADING NOTE Any equivalent form. The order of the letters gives the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

437

**QUESTION ID**

## S13-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**XZ**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

**A.** QR and MN

**B.** QR and LN

**C.** PQ and MN

**D.** PR and LM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

438

QUESTION ID

**S13-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. QR and MN**

**GRADING NOTE** Q ↔ M and R ↔ N.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

**True or false?**

△DEF ≅ △GHI means that ∠F ≅ ∠I.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

439

QUESTION ID

**S13-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The third letters correspond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

**True or false?**

△DEF ≅ △GHI means that DE = HI.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

440

QUESTION ID

**S13-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** DE corresponds to GH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

# BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

**True or false?**

A rotation keeps the measure of every angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

442

QUESTION ID

**S13-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Rotations are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

443

**QUESTION ID**

## S13-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

## 12 units

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

**A.** 289°

**B.** 109°

**C.** 71°

**D.** 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

444

QUESTION ID

**S13-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 71°**

GRADING NOTE Reflections keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

**True or false?**

A translation can change the length of a segment if it moves the segment far enough.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

445

**QUESTION ID**

## S13-B2-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Translations keep every length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

446

QUESTION ID

**S13-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**Translations, rotations, and reflections**

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

# BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

448

QUESTION ID

**S13-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**3 units right and 4 units down**

**GRADING NOTE** Any equivalent form: (x, y) → (x + 3, y − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

449

**QUESTION ID**

**S13-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**90° counterclockwise**

**GRADING NOTE** Any equivalent form (270° clockwise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

**A.** A rotation about D

**B.** A translation along DE

**C.** A rotation about the midpoint of DE

**D.** A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

450

QUESTION ID

**S13-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. A reflection across line DE**

**GRADING NOTE** A reflection across DE keeps every point of the line fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

**True or false?**

Reflecting across a line keeps every point of that line fixed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

451

QUESTION ID

**S13-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Points on the mirror line are their own images.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

**True or false?**

Translating a triangle can keep exactly one of its vertices fixed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

452

QUESTION ID

**S13-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A nonzero translation moves every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

# FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

454

QUESTION ID

**S13-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Yes. Three congruent sides force the angles to be congruent too (SSS), so all six parts match and a rigid motion exists.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

455

**QUESTION ID**

## S13-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Triangles with sides 6 and 8 and a 30° angle opposite the 6-unit side: the 8-unit side can make the third side either long or short (SSA).**

**GRADING NOTE** Any valid SSA counterexample with the angle not between the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

**True or false?**

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

456

QUESTION ID
**S13-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** AAA gives similar triangles, not necessarily congruent ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

**A.** Three sides

**B.** Two sides and an angle not between them

**C.** Two sides and the angle between them

**D.** Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

457

QUESTION ID

**S13-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. Two sides and an angle not between them**

**GRADING NOTE** SSA can give two different triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

458

QUESTION ID

**S13-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**Yes. The angle sum is 180° in both triangles, so ∠C = 180° − ∠A − ∠B = ∠F.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

# FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

460

QUESTION ID

**S13-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**∠ABD ≅ ∠CBD, so BD bisects ∠ABC.**

**GRADING NOTE** REQUIRED WORK: Prove △ABD ≅ △CBD (SSS, with BD shared), then use corresponding angles.

Any valid proof. Essential: AB = CB, AD = CD, BD = BD (reflexive); SSS gives the congruence; corresponding angles ∠ABD and ∠CBD are congruent, which is the definition of a bisector. Invalid: assuming the angles are equal from the picture.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

461

QUESTION ID

**S13-F2-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**AB = CD**

**GRADING NOTE** REQUIRED WORK: Use vertical angles and the bisected segments to prove △AEB ≅ △CED by SAS.

Any valid proof. Essential: AE = CE and BE = DE (definition of bisect); ∠AEB ≅ ∠CED (vertical angles); SAS; corresponding sides AB = CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In △PQR and △STU, PQ = ST, QR = TU, and PR = SU. The triangles are congruent. What must be true of ∠R and ∠U? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

462

QUESTION ID

**S13-F2-Q3**

TYPE

Short answer

CORRECT ANSWER

**∠R ≅ ∠U, because corresponding angles of congruent triangles are congruent.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

**True or false?**

Once two triangles are proved congruent, only the parts used in the proof are known to be congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

463

QUESTION ID

## S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Congruence gives all six pairs of corresponding parts (CPCTC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

**A.** ∠M and ∠X

**B.** ∠X and ∠Y

**C.** XM and YM

**D.** XY and XM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

464

**QUESTION ID**

## S13-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. XM and YM

**GRADING NOTE** Midpoint means XM = MY.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

# Explain SAS congruence with rigid motions

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.B.8** |
| **BACKWARD 1** | Identify the included angle and included side | **7.G.A.2** |
| **BACKWARD 2** | Rigid motions map rays, lengths, and angles to equal ones | **8.G.A.1** |
| **BACKWARD 3** | Choose a translation, rotation, or reflection to move a point or ray… | **8.G.A.2** |
| **FORWARD 1** | Prove triangles congruent by SAS inside a larger figure | **HSG-SRT.B.5** |
| **FORWARD 2** | Decide whether given information is enough for SAS, and name a… | **HSG-SRT.B.5** |

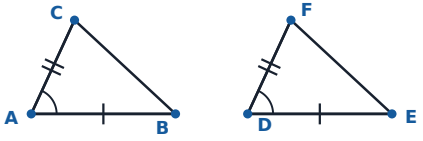
SET 14 • HSG-CO.B.8

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, AB = DE, AC = DF, and ∠A ≅ ∠D. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5**

467

QUESTION ID

## S14-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

**A translation, a rotation, and possibly a reflection carry A to D, B to E, and C to F, so the triangles are congruent.**

**GRADING NOTE** REQUIRED WORK: Translate A to D; rotate so ray AB lies on ray DE; reflect across DE if C is on the other side from F. Then explain why B lands on E and C lands on F.

Any valid explanation. Essential: (1) translate A onto D; (2) rotate about D so ray AB lies on ray DE; B lands on E because AB = DE; (3) reflect across DE if needed so C is on F's side; (4) ray AC lands on ray DF because ∠A ≅ ∠D; (5) C lands on F because AC = DF. Invalid: "the triangles look the same"; citing SAS to prove SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5**

In the rigid-motion proof of SAS, A has been moved to D and ray AB lies on ray DE, with C on the same side of line DE as F. Why must ray AC lie on ray DF?

**A.** BC = EF

**B.** The triangles have equal areas

**C.** AC = DF

**D.** ∠A ≅ ∠D, so both rays make the same angle with ray DE on the same side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5**

468

**QUESTION ID**

## S14-M-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. ∠A ≅ ∠D, so both rays make the same angle with ray DE on the same side**

**GRADING NOTE** There is only one ray from D on that side of line DE making that angle with ray DE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5**

In the rigid-motion proof of SAS, ray AC now lies on ray DF. Why does C land exactly on F?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

469

**QUESTION ID**

## S14-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**AC = DF and rigid motions keep length, so C is the point of ray DF at distance DF from D, which is F.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

Explain why a reflection is sometimes needed in the rigid-motion proof of SAS, and why it does not move D or E.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

470

**QUESTION ID**

## S14-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**After the rotation, C may be on the opposite side of line DE from F. Reflecting across line DE moves it to F's side; D and E are on the line, so they stay fixed.**

**GRADING NOTE** Any equivalent explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

**True or false?**

The rigid-motion argument for SAS still works if the congruent angle is not between the two congruent sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

471

QUESTION ID

## S14-M-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Then the angle does not fix where the third vertex goes; SSA can give two different triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

# BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

473

**QUESTION ID**

## S14-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**∠A**

**GRADING NOTE** Any equivalent form. The included angle is at the shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

474

QUESTION ID

**S14-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**PQ**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

In △XYZ, which angle is NOT included between sides XY and YZ?

**A.** Both ∠X and ∠Y

**B.** ∠X

**C.** ∠Y

**D.** None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

475

QUESTION ID

**S14-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. ∠X**

**GRADING NOTE** Sides XY and YZ share vertex Y, so ∠Y is included; ∠X is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

**True or false?**

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

476

QUESTION ID

## S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Two sides and the included angle fix the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

**True or false?**

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

477

QUESTION ID

**S14-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** When the angle is not between the sides, there may be two triangles or none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

# BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

**True or false?**

Under a rigid motion, the image of a ray is a ray.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

479

QUESTION ID

**S14-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Rigid motions map rays to rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

480

**QUESTION ID**

## S14-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**30°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

**A.** closer to D

**B.** not on the image ray

**C.** 5 units from D on the image ray

**D.** farther from D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

481

**QUESTION ID**

**S14-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 5 units from D on the image ray**

**GRADING NOTE** A rotation about D keeps distances from D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

**True or false?**

A reflection across line DE can move point D.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

482

QUESTION ID

## S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Points on the mirror line stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

483

QUESTION ID

**S14-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

# BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

485

QUESTION ID

**S14-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**A translation**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

486

QUESTION ID

**S14-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**A rotation about D**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

**A.** A translation along DE

**B.** A rotation of 90° about D

**C.** A rotation of 180° about E

**D.** A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5



**QUESTION ID**

## S14-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. A reflection across line DE**

**GRADING NOTE** Only the reflection keeps both D and E fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

**True or false?**

A rotation about point D keeps D fixed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

488

QUESTION ID

**S14-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE The center of a rotation is its own image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

**True or false?**

A translation that moves A onto D also keeps A fixed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

489

QUESTION ID

**S14-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It moves A onto D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

# FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

491

**QUESTION ID**

## S14-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**△ABC ≅ △CDA by SAS.**

**GRADING NOTE**

REQUIRED WORK: Use alternate interior angles with the parallel sides and the shared diagonal to apply SAS.

Any valid proof. Essential: AB = CD (given); ∠BAC ≅ ∠DCA (alternate interior angles, AB ∥ DC); AC = CA (reflexive); SAS with the angle between the sides. Invalid: using ∠B ≅ ∠D without proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

Point M is the midpoint of segment AB, and line $\ell$ is perpendicular to AB at M. Point P is on $\ell$. Prove that $\triangle PMA \cong \triangle PMB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

492

**QUESTION ID**

## S14-F1-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$\triangle PMA \cong \triangle PMB$ by SAS, so PA = PB.**

**GRADING NOTE** REQUIRED WORK: Use the midpoint, the right angles, and the shared side PM with SAS.

Any valid proof. Essential: AM = BM (midpoint); $\angle PMA \cong \angle PMB$ (both right angles); PM = PM; SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

Segments AC and BD cross at E, with AE = CE and BE = DE. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

493

QUESTION ID

**S14-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**△AEB ≅ △CED by SAS (the vertical angles at E are the included angles).**

**GRADING NOTE** Any equivalent correct pair (also △AED ≅ △CEB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

**True or false?**

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

494

QUESTION ID

## S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The shared side is congruent to itself (reflexive property).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

**A.** The angles are both right angles

**B.** Vertical angles are congruent

**C.** Base angles are congruent

**D.** Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

495

**QUESTION ID**

## S14-F1-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. Vertical angles are congruent

**GRADING NOTE** ∠PTR and ∠QTS (or the other pair) are vertical angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

# FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

497

**QUESTION ID**

**S14-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠B ≅ ∠E**

**GRADING NOTE** Form required: the included angle pair. (AC = DF would give SSS, not SAS.)

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F2-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5**

In △PQR and △XYZ, PQ = XY, PR = XZ, and ∠Q ≅ ∠Y. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

498

**QUESTION ID**

## S14-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**No. The congruent angle is not between the congruent sides (SSA), which does not guarantee congruence.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

**True or false?**

In △ABC and △ADC (sharing AC), AB = AD and ∠BCA ≅ ∠DCA. These facts prove △ABC ≅ △ADC by SAS.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

499

QUESTION ID

**S14-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The congruent angles at C are not included between AB and AC (this is SSA), so SAS does not apply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

**A.** △BCD and △FGH

**B.** △ABC and △EFG

**C.** △ABD and △EFH

**D.** △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

500

**QUESTION ID**

**S14-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. △ABC and △EFG**

**GRADING NOTE** The given parts are the two sides and the included angle of △ABC and △EFG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

501

**QUESTION ID**

## S14-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**No. SAS guarantees congruence; the third side is then determined (and equal).**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

# Explain ASA congruence with rigid motions

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.B.8** |
| **BACKWARD 1** | Two angles and the included side determine a triangle | **7.G.A.2** |
| **BACKWARD 2** | Two different lines meet in at most one point | **8.EE.C.8.a** |
| **BACKWARD 3** | Rigid motions keep rays and angle measures | **8.G.A.1** |
| **FORWARD 1** | Explain AAS from ASA and the triangle angle sum | **HSG-CO.C.10** |
| **FORWARD 2** | Prove congruence with ASA or AAS in overlapping triangles | **HSG-SRT.B.5** |

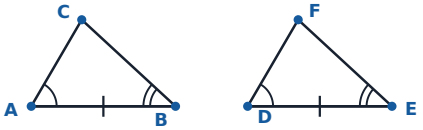
SET 15 • HSG-CO.B.8

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, ∠A ≅ ∠D, AB = DE, and ∠B ≅ ∠E. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5**

504

**QUESTION ID**

## S15-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**A translation, a rotation, and possibly a reflection carry A to D, B to E, and C to F.**

**GRADING NOTE** REQUIRED WORK: Translate A to D; rotate so ray AB lies on ray DE (B lands on E); reflect across DE if needed; then use the two angles to place rays AC and BC on rays DF and EF.

Any valid explanation. Essential: (1) A → D and B → E using AB = DE; (2) C on F's side of DE (reflect if needed); (3) ray AC lands on ray DF because ∠A ≅ ∠D; (4) ray BC lands on ray EF because ∠B ≅ ∠E; (5) C is where rays AC and BC meet, so it lands where DF and EF meet, which is F. Invalid: using side lengths AC or BC (not given); citing ASA to prove ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5**

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on F's side of line DE. Why does C land on F?

**A.** C and F are both the highest points

**B.** BC = EF

**C.** C is where rays AC and BC meet; those rays now lie on rays DF and EF, which meet only at F

**D.** AC = DF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

505

**QUESTION ID**
**S15-M-Q2**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**C. C is where rays AC and BC meet; those rays now lie on rays DF and EF, which meet only at F**

**GRADING NOTE** The two angles place the two rays; the intersection point is unique.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on the same side of line DE as F. Explain why ray BC lies on ray EF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

506

**QUESTION ID**

## S15-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**∠B ≅ ∠E, and on that side of line DE there is only one ray from E making that angle with ray ED, namely ray EF.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In the rigid-motion proof of ASA, why can rays DF and EF meet in only one point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

507

QUESTION ID

**S15-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**They lie on two different lines, and two different lines meet in at most one point.**

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

**True or false?**

In the rigid-motion proof of ASA, a reflection across line DE leaves D and E where they are.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

508

QUESTION ID

**S15-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Points on the line of reflection are fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

# BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

**True or false?**

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

510

QUESTION ID

**S15-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The side and the two angles at its ends fix where the other two sides meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

**True or false?**

Any two angles with a side between them make a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

511

QUESTION ID

**S15-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE The two angles must add to less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

512

**QUESTION ID**

**S15-B1-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**60°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

**A.** Angles 30° and 80° with a 4 cm side between them

**B.** One angle of 45° and nothing else

**C.** Angles 30°, 60°, and 90°

**D.** Angles 100° and 90° with a 3 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

513

QUESTION ID

**S15-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. Angles 30° and 80° with a 4 cm side between them**

**GRADING NOTE** Two angles with the included side fix the triangle; three angles alone fix only its shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

514

**QUESTION ID**

**S15-B1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**No. 95° + 90° = 185° is more than 180°.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

# BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines y = x + 1 and y = −x + 3 have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

516

QUESTION ID

**S15-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**1**

**GRADING NOTE** Any equivalent form. They meet at (1, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

**True or false?**

Two different lines can meet in exactly two points.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

517

QUESTION ID

**S15-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Two different lines meet in at most one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines y = 2x + 1 and y = 2x − 4

**A.** never meet

**B.** are the same line

**C.** meet at one point

**D.** meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

518

QUESTION ID

**S15-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. never meet**

GRADING NOTE Same slope, different intercepts: parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

**True or false?**

If two lines share two points, they are the same line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

519

QUESTION ID

**S15-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Two points determine one line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Where do the lines x = 2 and y = −3 meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

520

QUESTION ID

**S15-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**(2, −3)**

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

# BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

522

**QUESTION ID**

**S15-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**64°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

**True or false?**

A rigid motion maps an angle to an angle with the same measure.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

523

QUESTION ID

**S15-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Rigid motions keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

**A.** Exactly one

**B.** Two

**C.** Infinitely many

**D.** None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

524

**QUESTION ID**

## S15-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. Exactly one

**GRADING NOTE** On one side of a line there is one ray at a given angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

**True or false?**

After a rotation, an angle can become larger if its sides are longer.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

525

QUESTION ID

**S15-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Angle measure does not depend on side length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

526

**QUESTION ID**

**S15-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**4 cm from D (on the image ray).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

# FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC and △DEF, ∠A ≅ ∠D, ∠B ≅ ∠E, and BC = EF (the side is NOT between the angles). Prove △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

528

QUESTION ID

**S15-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**△ABC ≅ △DEF by ASA after showing ∠C ≅ ∠F.**

GRADING NOTE REQUIRED WORK: Use the triangle angle sum to show ∠C ≅ ∠F, then apply ASA with ∠B, BC, ∠C.

Any valid proof. Essential: m∠C = 180° − m∠A − m∠B = 180° − m∠D − m∠E = m∠F; then ∠B, side BC, ∠C match ∠E, EF, ∠F (ASA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

529

QUESTION ID

**S15-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Yes, by AAS: the third angles are both 65°, so ASA applies.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

**True or false?**

AAS is a valid congruence criterion only because the angles of a triangle add to 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

530

QUESTION ID

**S15-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The angle sum gives the third pair of angles, turning AAS into ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In △PQR and △XYZ, ∠P ≅ ∠X, ∠Q ≅ ∠Y, and PR = XZ. Which criterion proves congruence?

**A.** ASA

**B.** AAS

**C.** SAS

**D.** SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

531

QUESTION ID

**S15-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. AAS**

**GRADING NOTE** PR is not between ∠P and ∠Q, so it is AAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if ∠A ≅ ∠D and ∠C ≅ ∠F?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

532

QUESTION ID

**S15-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**AB = DE or BC = EF**

**GRADING NOTE** Any side not between the two angles (AC = DF would be ASA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

# FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

534

QUESTION ID

**S15-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**△ABE ≅ △DCE by ASA.**

GRADING NOTE REQUIRED WORK: Use alternate interior angles, the vertical angles at E, and AE = DE with ASA.

Any valid proof. Essential: ∠BAE ≅ ∠CDE (alternate interior, AB ∥ CD); AE = DE (midpoint); ∠AEB ≅ ∠DEC (vertical); ASA. AAS with ∠B ≅ ∠C also accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

In △ABC, the bisector of ∠A is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

535

**QUESTION ID**

## S15-F2-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AB = AC**

**GRADING NOTE** REQUIRED WORK: Prove △ABD ≅ △ACD by ASA using the bisected angle, the shared side AD, and the right angles.

Any valid proof. Essential: ∠BAD ≅ ∠CAD (bisector); AD = AD; ∠ADB ≅ ∠ADC (right angles); ASA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, ∠A is shared, AB = AC, and ∠B ≅ ∠C. Which criterion proves △ABE ≅ △ACD?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

536

**QUESTION ID**

**S15-F2-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**ASA (∠A, AB = AC, ∠B ≅ ∠C)**

**GRADING NOTE** Form required: the criterion with the matching parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

**True or false?**

An angle shared by two overlapping triangles cannot be used in a congruence proof, because it is the same angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

537

QUESTION ID

## S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** By the reflexive property, the shared angle is congruent to itself and can be used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In △JKL and △JML, ∠KJL ≅ ∠MJL and ∠JLK ≅ ∠JLM. Which side completes ASA?

**A.** JL = JL

**B.** JK = JM

**C.** KL = ML

**D.** JK = KL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

538

QUESTION ID

**S15-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. JL = JL**

**GRADING NOTE** JL is between the two pairs of angles and is shared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

# Explain SSS congruence with rigid motions

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.B.8** |
| **BACKWARD 1** | Three side lengths determine a triangle (triangle inequality) | **7.G.A.2** |
| **BACKWARD 2** | Find the points at given distances from two points (two circles) | **7.G.B.4** |
| **BACKWARD 3** | Rigid motions keep every length | **8.G.A.1** |
| **FORWARD 1** | Explain the Hypotenuse-Leg criterion for right triangles | **HSG-SRT.B.5** |
| **FORWARD 2** | Decide which criteria guarantee congruence, with counterexampl… | **HSG-SRT.B.5** |

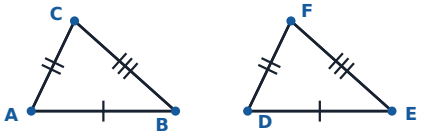
SET 16 • HSG-CO.B.8

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, AB = DE, BC = EF, and CA = FD. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5**

541

**QUESTION ID**

## S16-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**A translation and a rotation carry A to D and B to E; if needed, the reflection across DE carries C′ to F.**

**GRADING NOTE** REQUIRED WORK: Move A to D and B to E (translation and rotation, using AB = DE). If the image C′ of C is not F, show that line DE is the perpendicular bisector of C′F, so the reflection across DE carries C′ to F.

Any valid explanation. Essential: (1) A → D, B → E (AB = DE); (2) DC′ = DF and EC′ = EF (rigid motions keep lengths); (3) if C′ ≠ F, D and E are each equidistant from C′ and F, so line DE is the perpendicular bisector of C′F; (4) the reflection across DE fixes D and E and maps C′ to F. Alternative: C′ and F both lie on the circles centered at D and E with these radii, which meet in at most one point on each side of DE. Invalid: using angles (none are given); citing SSS to prove SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5**

In the rigid-motion proof of SSS, A is on D, B is on E, and C′ (the image of C) is not F. Why is line DE the perpendicular bisector of C′F?

**A.** DE bisects ∠D

**B.** DE is parallel to C′F

**C.** D and E are each the same distance from C′ as from F

**D.** C′F is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

542

**QUESTION ID**

**S16-M-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. D and E are each the same distance from C′ as from F**

**GRADING NOTE** A point equidistant from C′ and F lies on the perpendicular bisector of C′F; two such points determine that line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

In the rigid-motion proof of SSS, line DE is the perpendicular bisector of C′F. Why does the reflection across DE carry C′ onto F, and why is that enough?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

543

**QUESTION ID**

## S16-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**By the definition of a reflection, it maps C′ to F because DE is the perpendicular bisector of C′F; it keeps D and E fixed, so the whole triangle lands on DEF.**

**GRADING NOTE** Any equivalent explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

The rigid-motion proof of SSS uses no angle information. Explain how the three side lengths alone locate the third vertex.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

544

QUESTION ID

## S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

**Once D and E are matched, the third vertex must be DF from D and EF from E: it lies on two circles, which meet in at most one point on each side of DE.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

**True or false?**

A triangle with sides 5, 7, and 9 is congruent to a triangle with sides 9, 5, and 7.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

545

QUESTION ID

**S16-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Matching the sides by length gives three pairs of congruent sides (SSS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

# BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

**True or false?**

Side lengths 3, 4, and 8 can form a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

547

QUESTION ID

**S16-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 3 + 4 = 7 is less than 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

**True or false?**

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

548

QUESTION ID

**S16-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** 5 + 6 > 10, and three sides fix the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

**A.** 5, 5, 11

**B.** 4, 5, 7

**C.** 2, 3, 6

**D.** 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

549

QUESTION ID

**S16-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 4, 5, 7**

**GRADING NOTE** Each pair must add to more than the third side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

550

**QUESTION ID**

## S16-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Between 4 and 16**

**(4 < x < 16)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

551

**QUESTION ID**

## S16-B1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**No. Three side lengths determine one triangle shape and size.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

# BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

553

QUESTION ID

**S16-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form. The circles of radius 5 about A and B meet twice, once on each side of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

**Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5**

**True or false?**

All points 3 cm from a point A form a circle with center A and radius 3 cm.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

554

QUESTION ID

**S16-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** That is the definition of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

**A.** Both on line AB

**B.** One on each side of line AB

**C.** At A and B

**D.** Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

555

QUESTION ID

**S16-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. One on each side of line AB**

**GRADING NOTE** The picture is symmetric across line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

**True or false?**

Two circles always meet in exactly two points.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

**Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5**

556

QUESTION ID

## S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** They can meet in 0, 1, or 2 points (or be the same circle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

**Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5**

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

557

**QUESTION ID**

## S16-B2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**No. 3 + 4 = 7 < 10, so the circles do not meet.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

# BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

559

**QUESTION ID**

## S16-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**3, 5, and 7**

**GRADING NOTE** Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

**True or false?**

Under a rigid motion, the distance from the image of A to the image of C equals AC.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

560

QUESTION ID

**S16-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Rigid motions keep all distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which can a rigid motion change?

**A.** The perimeter

**B.** The length of a side

**C.** The area

**D.** The position of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

561

QUESTION ID

**S16-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. The position of a triangle**

**GRADING NOTE** Only the position changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

**True or false?**

A reflection doubles the distance from a point to the line of reflection.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

562

QUESTION ID

**S16-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The image is the same distance from the line, on the other side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

563

QUESTION ID

**S16-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**6 units**

GRADING NOTE Any equivalent form. D is on the line, so it is fixed, and distances are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

# FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

565

**QUESTION ID**

## S16-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**△ABC ≅ △DEF by SSS (BC = EF from the Pythagorean Theorem).**

**GRADING NOTE** REQUIRED WORK: Use the Pythagorean Theorem to show BC = EF, then apply SSS.

Any valid proof. Essential: BC = √(AB² − AC²) = √(DE² − DF²) = EF; then SSS.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F1-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5**

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

566

QUESTION ID

**S16-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Yes; the other leg is 12 in both.**

**GRADING NOTE** Form required: the conclusion and the leg. $\sqrt{(169 - 25)} = 12$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5**

**True or false?**

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

567

QUESTION ID

## S16-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The right angle is opposite the hypotenuse; the Pythagorean Theorem fixes the third side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

**A.** One pair of acute angles is equal

**B.** Their right angles are equal

**C.** Their hypotenuses are equal and one pair of legs is equal

**D.** Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

568

QUESTION ID

**S16-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. Their hypotenuses are equal and one pair of legs is equal**

GRADING NOTE HL needs the hypotenuse and a leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

569

**QUESTION ID**

## S16-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**PR = XZ = 8**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

# FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

571

QUESTION ID
**S16-F2-Q1**

TYPE
Short answer

CORRECT ANSWER
**Equilateral triangles with sides 2 and 5 have all angles 60° but are not congruent.**

**GRADING NOTE** Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

572

**QUESTION ID**

## S16-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**With sides 6 and 8 and a 30° angle opposite the 6, the third vertex can be in two places, giving two different triangles.**

**GRADING NOTE** Any valid SSA counterexample (two non-congruent triangles with the same SSA data).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F2-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5**

Which list contains only valid triangle congruence criteria?

**A.** SSS, SAS, ASA, AAS

**B.** SSS, SSA, ASA, AAS

**C.** SSA, AAA, SAS, HL

**D.** SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

573

**QUESTION ID**

## S16-F2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. SSS, SAS, ASA, AAS

**GRADING NOTE** SSA and AAA are not criteria (HL is a special right-triangle case).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

**True or false?**

If two triangles have the same three side lengths, they have the same three angle measures.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

574

QUESTION ID

## S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** SSS gives congruence, and corresponding angles of congruent triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

**True or false?**

Two triangles with two pairs of congruent sides and equal areas must be congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

575

QUESTION ID
**S16-F2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** The included angles could be θ and 180° − θ: equal areas (same sine), different third sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

# Prove that vertical angles are congruent

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.9** |
| **BACKWARD 1** | Angles in a linear pair add to 180° | **7.G.B.5** |
| **BACKWARD 2** | Use properties of equality as reasons | **HSA-REI.A.1** |
| **BACKWARD 3** | Identify vertical angles and linear pairs in a figure | **7.G.B.5** |
| **FORWARD 1** | Use the vertical angle theorem in multistep proofs | **HSG-CO.C.9** |
| **FORWARD 2** | Prove statements about the bisectors of angles formed by two lin… | **HSG-CO.C.9** |

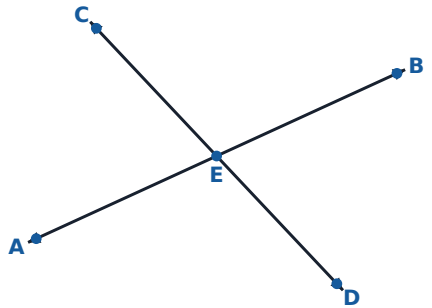
SET 17 • HSG-CO.C.9

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5**

578

**QUESTION ID**

## S17-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**∠AEC ≅ ∠BED**

**GRADING NOTE** REQUIRED WORK: Use the two linear pairs that share ∠CEB, and the subtraction property of equality.

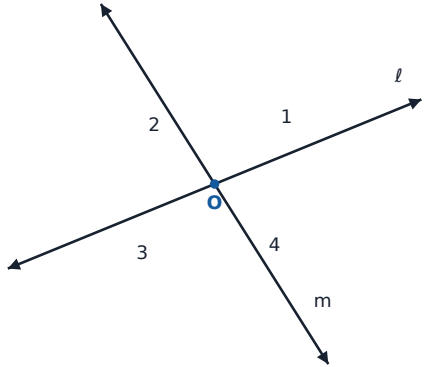
Any valid proof (paragraph, two-column, or flow). Essential: (1) m∠AEC + m∠CEB = 180° (linear pair: A, E, B are collinear); (2) m∠CEB + m∠BED = 180° (linear pair: C, E, D are collinear); (3) so m∠AEC + m∠CEB = m∠CEB + m∠BED (substitution/transitive); (4) subtract m∠CEB. Alternative: the 180° rotation about E maps ray EA to ray EB and ray EC to ray ED. Invalid: "they look equal", measuring, or citing the theorem itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5**

Lines $\ell$ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5**

579

**QUESTION ID**

## S17-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$\angle 2 \cong \angle 4$**

**GRADING NOTE** REQUIRED WORK: Use the linear pairs $\angle 1$ and $\angle 2$, and $\angle 1$ and $\angle 4$, then the subtraction property of equality.

Any valid proof (paragraph, two-column, or flow). Essential: $m\angle 1 + m\angle 2 = 180°$ and $m\angle 1 + m\angle 4 = 180°$ (linear pairs, each with a reason); so $m\angle 2 = 180° - m\angle 1 = m\angle 4$. Using $\angle 3$ as the shared angle is also valid. Invalid: using $\angle 2 \cong \angle 4$ as a step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A proof that ∠1 ≅ ∠3 for vertical angles 1 and 3 (with ∠2 next to both):

1. m∠1 + m∠2 = 180° (linear pair)
2. m∠2 + m∠3 = 180° (linear pair)
3. m∠1 + m∠2 = m∠2 + m∠3 (?)
4. m∠1 = m∠3 (?)

Give the reasons for steps 3 and 4.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

580

**QUESTION ID**
**S17-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**Step 3: substitution (or the transitive property of equality); step 4: subtraction property of equality.**

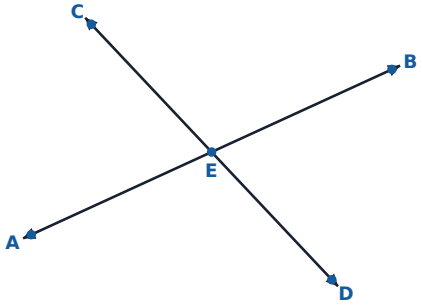
**GRADING NOTE** Any equivalent names for both reasons. Both are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why ∠AEC ≅ ∠BED.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

581

QUESTION ID

## S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

**The 180° rotation about E maps ∠AEC onto ∠BED, so the angles are congruent.**

GRADING NOTE

REQUIRED WORK: Use the 180° rotation about E and explain where it sends rays EA and EC.

Any valid explanation. Essential: a 180° rotation about E maps each line through E onto itself, sending ray EA to the opposite ray EB and ray EC to ray ED; so it maps ∠AEC onto ∠BED; rotations keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

A student measures two vertical angles in one drawing, finds that both are 47°, and writes "vertical angles are congruent." Why is this not a proof?

**A.** Protractors are never accurate

**B.** Vertical angles are not always congruent

**C.** It checks one example; a proof must show the result for every pair of vertical angles

**D.** The student should have measured all four angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

582

**QUESTION ID**

## S17-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. It checks one example; a proof must show the result for every pair of vertical angles**

**GRADING NOTE** Measuring can suggest a theorem but cannot prove it for all cases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

# BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

584

**QUESTION ID**

## S17-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**52°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

585

**QUESTION ID**

**S17-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**45°**

**GRADING NOTE** Any equivalent form. 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

**True or false?**

The two angles of a linear pair are always supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

586

QUESTION ID

**S17-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Together they form a straight angle, 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

**A.** ∠DBC and ∠ABC

**B.** ∠BAD and ∠BCD

**C.** ∠ABD and ∠DBC

**D.** ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

587

**QUESTION ID**

## S17-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. ∠ABD and ∠DBC

**GRADING NOTE** They share ray BD and their other sides are opposite rays BA and BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

**True or false?**

Two angles that add to 180° must form a linear pair.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

588

QUESTION ID

**S17-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Supplementary angles need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

# BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From m∠1 + 40° = m∠2 + 40°, which property gives m∠1 = m∠2?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

590

QUESTION ID

**S17-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Subtraction property of equality**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If x + y = 180 and y + z = 180, which property lets you write x + y = y + z?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

591

QUESTION ID
**S17-B2-Q2**

TYPE
Short answer

CORRECT ANSWER
**Substitution (or the transitive property of equality)**

**GRADING NOTE** Any equivalent wording.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-B2-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5**

Which property justifies: if a = b, then b = a?

**A.** Reflexive property

**B.** Symmetric property

**C.** Distributive property

**D.** Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

592

QUESTION ID

**S17-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. Symmetric property**

**GRADING NOTE** Swapping the sides is the symmetric property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

**True or false?**

The reflexive property says that any quantity equals itself, such as m∠A = m∠A.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

593

QUESTION ID

**S17-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE a = a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

**True or false?**

From 2x = 2y you may conclude x = y by the subtraction property of equality.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

594

**QUESTION ID**

## S17-B2-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** That step divides by 2 (division property).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

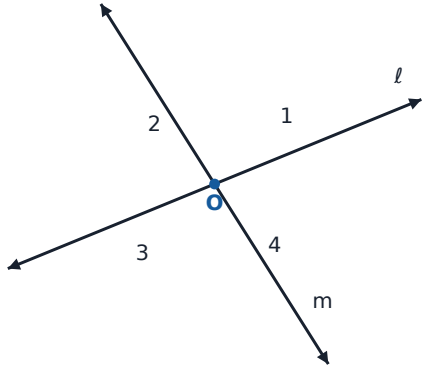
# BACKWARD 3

Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

Lines $\ell$ and m meet at O, forming ∠1, ∠2, ∠3, ∠4 in order. Which angle is vertical to ∠1?

**A.** ∠3 **B.** ∠1

**C.** ∠4 **D.** ∠2




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

596

**QUESTION ID**

## S17-B3-Q1

**TYPE**

Multiple choice

**CORRECT ANSWER**

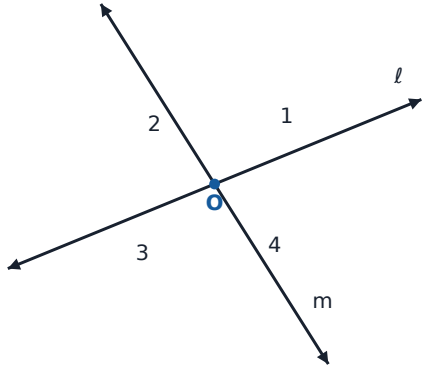
**A. ∠3**

**GRADING NOTE** Vertical angles are opposite each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

Lines $\ell$ and m meet at O, forming ∠1, ∠2, ∠3, ∠4 in order. Name the two angles that form linear pairs with ∠2.




Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

597

**QUESTION ID**
**S17-B3-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**∠1 and ∠3**

**GRADING NOTE** Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

**True or false?**

Vertical angles share a side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

598

QUESTION ID

**S17-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They share only the vertex; their sides are opposite rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

**True or false?**

When two lines cross, they form two pairs of vertical angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

599

**GRADING NOTE** ∠1 with ∠3, and ∠2 with ∠4.

**QUESTION ID**

## S17-B3-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

600

QUESTION ID

**S17-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**Vertical: 63°; adjacent: 117° and 117°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

# FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

602

**QUESTION ID**

**S17-F1-Q1**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠BEG = m∠GED, so EG bisects ∠BED.**

**GRADING NOTE** REQUIRED WORK: Combine the bisector definition with vertical angles (∠AEF and ∠BEG; ∠FEC and ∠GED).

Any valid proof (paragraph, two-column, or flow). Essential: m∠AEF = m∠FEC (bisector); ∠AEF ≅ ∠BEG and ∠FEC ≅ ∠DEG (vertical angles, since EG is opposite EF, EB opposite EA, ED opposite EC); so m∠BEG = m∠GED.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5**

603

**QUESTION ID**

## S17-F1-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The angles in order are 50°, 70°, 60°, 50°, 70°, 60°.**

**GRADING NOTE** REQUIRED WORK: Use the straight angle (180°) to find the third angle on one side, then vertical angles for the other three.

Any valid proof (paragraph, two-column, or flow). Essential: three adjacent angles on one side of a line add to 180°, so the third is 60°; each of the other three is vertical to one of these.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5**

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

604

QUESTION ID

**S17-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**x = 20; angles 90°, 90°, 90°, 90°**

GRADING NOTE Form required: x and the four angles. 4x + 10 = 6x − 30 gives x = 20, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

**True or false?**

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

605

QUESTION ID

**S17-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Its vertical angle is 90° and its neighbors are 180° − 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

**A.** 72°

**B.** 144°

**C.** 36°

**D.** 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

606

QUESTION ID

**S17-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. 72°**

GRADING NOTE 2x + 3x = 180, so x = 36 and m∠AEC = 72°; ∠BED is vertical to ∠AEC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

# FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

608

QUESTION ID

**S17-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**m∠EBF = 90°**

GRADING NOTE REQUIRED WORK: Express m∠EBF as half of m∠ABD plus half of m∠DBC.

Any valid proof (paragraph, two-column, or flow). Essential: m∠EBD = ½m∠ABD, m∠DBF = ½m∠DBC (bisectors); m∠EBF = ½(m∠ABD + m∠DBC) = ½(180°) = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

609

**QUESTION ID**
**S17-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**No. Two 30° angles in different places in a figure are congruent but not vertical angles.**

**GRADING NOTE** Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

610

**QUESTION ID**

## S17-F2-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The bisector rays are opposite rays, so they lie on one line.**

**GRADING NOTE** REQUIRED WORK: Show that the angle between the two bisector rays is 180°.

Any valid proof (paragraph, two-column, or flow). Essential: if the vertical angles each measure 2a and the other two each 180° − 2a, the angle from one bisector to the other is a + (180° − 2a) + a = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

**True or false?**

The bisectors of a pair of vertical angles are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

611

QUESTION ID

**S17-F2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They form one straight line: each bisector is the extension of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

**A.** 45°

**B.** It depends on the lines

**C.** 180°

**D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

612

**QUESTION ID**

**S17-F2-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 90°**

**GRADING NOTE** Adjacent angles formed by two lines are a linear pair, so their bisectors meet at 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

# Prove the angle theorems for parallel lines cut by a transversal

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.9** |
| **BACKWARD 1** | Name and use angles formed by parallel lines and a transversal | **8.G.A.5** |
| **BACKWARD 2** | Vertical angles and linear pairs as reasons | **HSG-CO.C.9** |
| **BACKWARD 3** | The parallel postulate and the definition of parallel lines | **HSG-CO.A.1** |
| **FORWARD 1** | Prove the converse: congruent alternate interior angles make line... | **HSG-CO.C.9** |
| **FORWARD 2** | Use parallel-line theorems to prove the triangle angle sum | **HSG-CO.C.10** |

SET 18 • HSG-CO.C.9

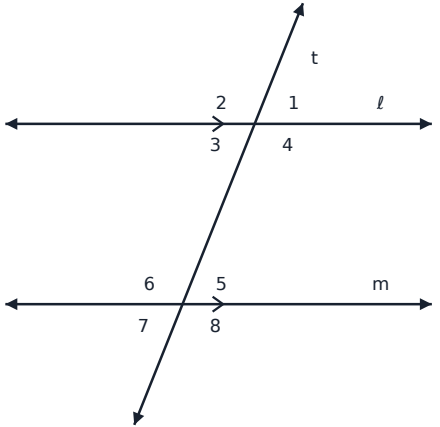
# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

Lines $\ell$ and m are parallel and are cut by transversal t. Assume the corresponding angles postulate (corresponding angles formed by a transversal of parallel lines are congruent). Prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5**

615

**QUESTION ID**

## S18-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$\angle 3 \cong \angle 5$**

**GRADING NOTE** REQUIRED WORK: Use the corresponding angle of $\angle 5$ at the top crossing and vertical angles.

Any valid proof (paragraph, two-column, or flow). Essential: $\angle 1 \cong \angle 5$ (corresponding angles postulate, $\ell \parallel$ m); $\angle 1 \cong \angle 3$ (vertical angles); so $\angle 3 \cong \angle 5$ (transitive). Invalid: "they look equal"; assuming alternate interior angles are congruent.

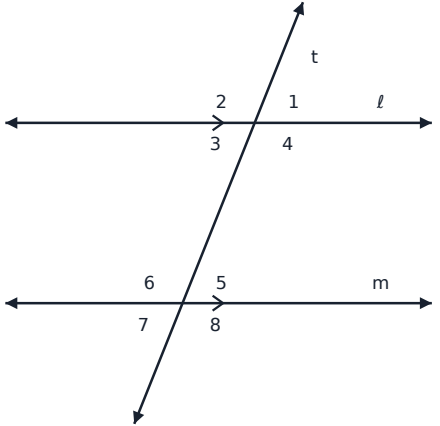
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Lines $\ell$ and m are parallel and are cut by transversal t. Assume that alternate interior angles are congruent. Prove that corresponding angles $\angle 2$ and $\angle 6$ are congruent.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5**

616

**QUESTION ID**

## S18-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$\angle 2 \cong \angle 6$**

**GRADING NOTE** REQUIRED WORK: Use an alternate interior angle and vertical angles.

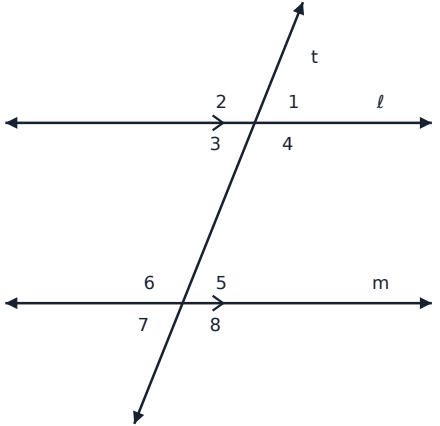
Any valid proof (paragraph, two-column, or flow). Essential: $\angle 4 \cong \angle 6$ (alternate interior angles); $\angle 2 \cong \angle 4$ (vertical angles); so $\angle 2 \cong \angle 6$ (transitive).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5**

Lines $\ell$ and m are parallel and are cut by transversal t, crossing $\ell$ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5**

617

**QUESTION ID**

## S18-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The translation that moves Q to P maps m onto $\ell$ and maps t onto itself, so it maps $\angle 5$ onto $\angle 1$; translations keep angle measure.**

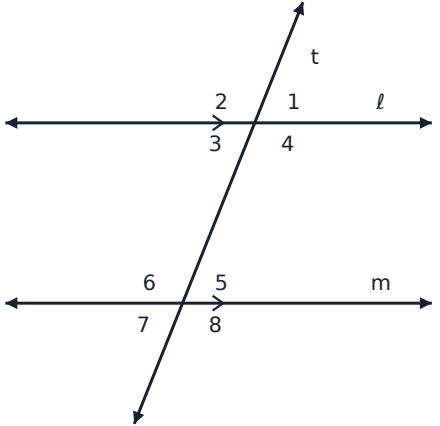
**GRADING NOTE** REQUIRED WORK: Translate along the transversal from Q to P. Explain why the image of m is $\ell$ and why the image of $\angle 5$ is $\angle 1$.

Any valid explanation. Essential: (1) the translation maps t onto itself (it moves along t); (2) it maps m to a line through P parallel to m; (3) by the parallel postulate that line is $\ell$; (4) so the angle formed by t and m at Q maps to the corresponding angle at P; (5) angle measure is kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5**

Lines $\ell \parallel m$ are cut by transversal t, and $m\angle 1 = 115°$. Find $m\angle 5$, $m\angle 3$, and $m\angle 6$. Give a reason for each.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

618

QUESTION ID

## S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

**$m\angle 5 = 115°$ (corresponding to $\angle 1$); $m\angle 3 = 115°$ (vertical to $\angle 1$); $m\angle 6 = 65°$ (linear pair with $\angle 5$).**

GRADING NOTE Form required: each measure with a valid reason. Other valid reason chains are accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

A student "proves" that alternate interior angles are congruent by saying: "The lines are parallel because the alternate interior angles are congruent." What is wrong?

**A.** Nothing; the argument is correct

**B.** The argument is circular: it assumes the conclusion

**C.** Alternate interior angles are supplementary

**D.** The lines must be perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

619

QUESTION ID

**S18-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. The argument is circular: it assumes the conclusion**

**GRADING NOTE** The statement uses the result it is trying to prove (and states a converse instead).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

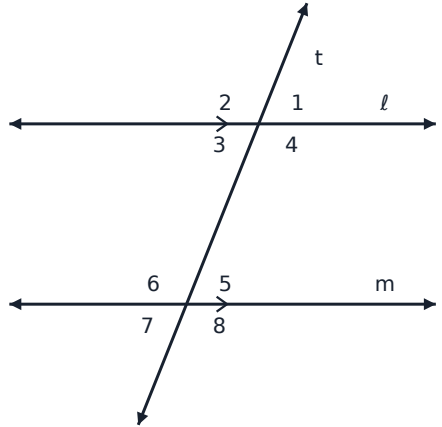
SET 18 • HSG-CO.C.9

# BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.




Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

621

**QUESTION ID**

**S18-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**70°**

**GRADING NOTE** Any equivalent form. $\angle 4$ and $\angle 6$ are alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

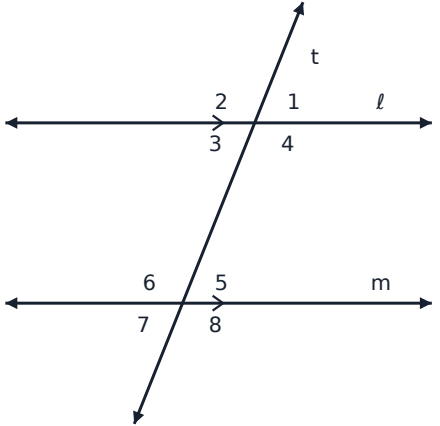
Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?

**A.** $\angle 7$

**B.** $\angle 8$

**C.** $\angle 6$

**D.** $\angle 5$




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

622

QUESTION ID
**S18-B1-Q2**

TYPE
Multiple choice

CORRECT ANSWER
**C. $\angle 6$**

**GRADING NOTE** Both are in the upper-left position at their crossings.

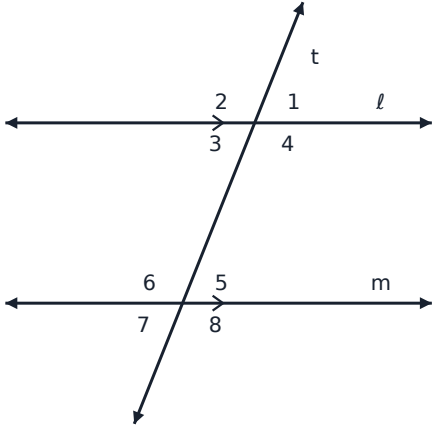
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

**True or false?**

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.

**A.** True **B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

623

**QUESTION ID**
**S18-B1-Q3**

**TYPE**
True or false

**CORRECT ANSWER**
**A. True**

**GRADING NOTE** They are on the same side of t, between the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

**True or false?**

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

624

QUESTION ID

**S18-B1-Q4**

TYPE

True or false

CORRECT ANSWER

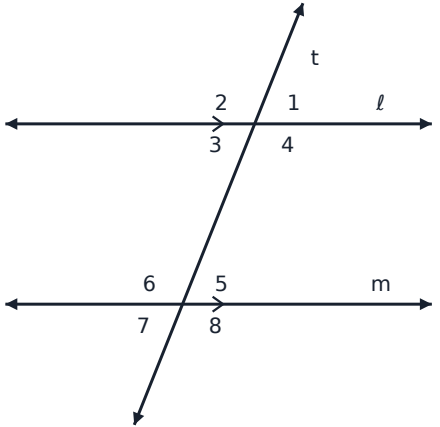
**B. False**

**GRADING NOTE** They are congruent only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures $112°$. Find the measure of $\angle 2$.




Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

625

**QUESTION ID**

**S18-B1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**112°**

**GRADING NOTE** Any equivalent form. $\angle 8$ and $\angle 2$ are alternate exterior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

# BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

627

QUESTION ID

**S18-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Vertical angles are congruent**

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

628

QUESTION ID

**S18-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**They form a linear pair (supplementary)**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

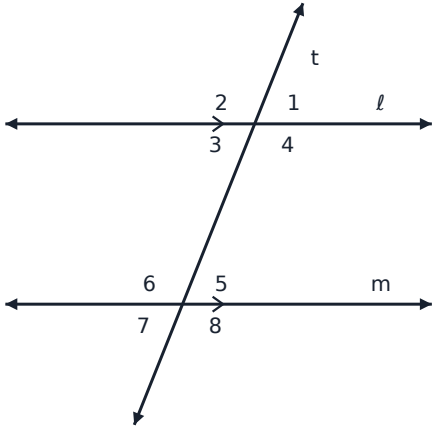
At the top crossing, ∠1 = 110°. What is ∠2 and why?

**A.** 110°, linear pair

**B.** 70°, linear pair

**C.** 110°, vertical angles

**D.** 70°, vertical angles




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

629

**QUESTION ID**

## S18-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 70°, linear pair**

**GRADING NOTE** ∠1 and ∠2 are next to each other on line ℓ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

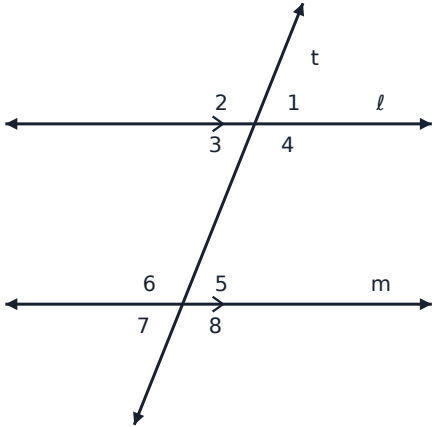
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

**True or false?**

Angles 1 and 3 are vertical angles in the figure.

**A.** True

**B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

630

QUESTION ID
**S18-B2-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** They are opposite each other at the top crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

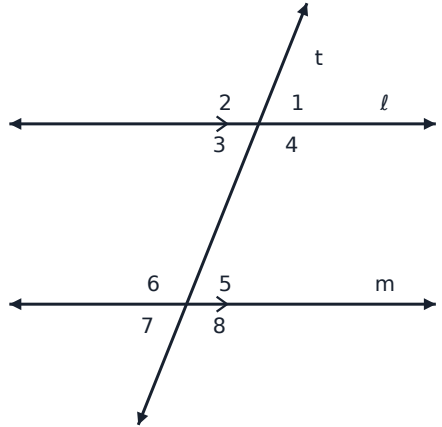
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

**True or false?**

Angles 1 and 4 are vertical angles in the figure.

**A.** True

**B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

631

QUESTION ID

**S18-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are adjacent: a linear pair along t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

# BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**True or false?**

Through a point not on a line, there is exactly one line parallel to it (in a plane).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

633

QUESTION ID
**S18-B3-Q1**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** The parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

Line $\ell$ is parallel to line m. Line n passes through a point of $\ell$ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

634

**QUESTION ID**

**S18-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**n is the same line as $\ell$.**

**GRADING NOTE** Any equivalent answer: by the parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

**A.** $x = 2$ and $y = 2$

**B.** $y = x$ and $y = -x$

**C.** $y = 2x + 1$ and $y = -2x + 1$

**D.** $y = 2x + 1$ and $y = 2x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

635

QUESTION ID

**S18-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. $y = 2x + 1$ and $y = 2x - 5$**

**GRADING NOTE** Same slope, different intercepts: they never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

**True or false?**

Two lines in a plane that never meet are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

636

QUESTION ID

**S18-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition of parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

Line $\ell \parallel$ line m. A translation maps m to a line through a point of $\ell$. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

637

**QUESTION ID**

**S18-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**Parallel to m (a translation maps a line to a parallel line or itself); so it is $\ell$.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSG-CO.C.9

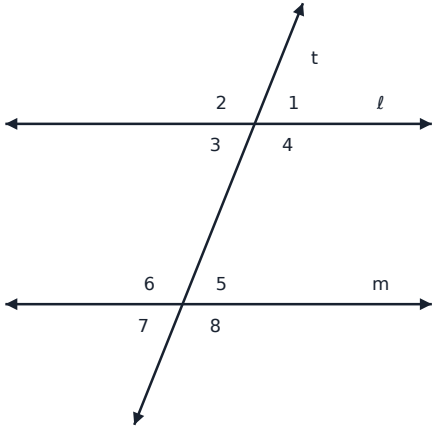
# FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

Lines $\ell$ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5**

639

**QUESTION ID**

## S18-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$\ell \parallel$ m**

**GRADING NOTE**

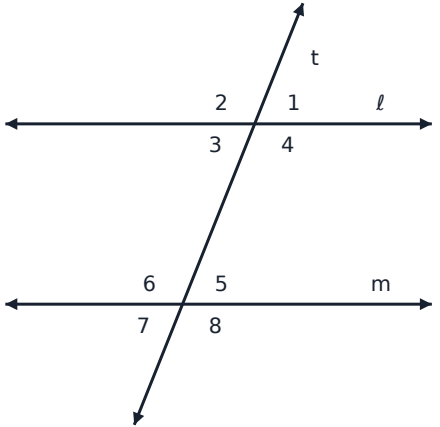
REQUIRED WORK: Indirect proof: suppose $\ell$ and m meet; they would form a triangle with t whose angles contradict a known result. Or use the 180° rotation about the midpoint of PQ.

Any valid proof (paragraph, two-column, or flow). Essential (indirect): if $\ell$ and m met at a point R, triangle PQR would have an exterior angle equal to a remote interior angle, which is impossible (an exterior angle is greater than each remote interior angle); so they do not meet. Alternative: the 180° rotation about the midpoint of PQ swaps P and Q and maps $\ell$ onto m, so if they met on one side they would also meet on the other, giving two common points. Invalid: assuming $\ell \parallel$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5**

Lines $\ell$ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel m$? Explain.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

640

**QUESTION ID**

## S18-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**x = 25, because then the alternate interior angles are congruent (105°).**

**GRADING NOTE** Form required: x with the converse as the reason. $5x - 20 = 3x + 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

**True or false?**

If same-side interior angles are supplementary, the lines are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

641

QUESTION ID

## S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Then the alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines $\ell$ and m are parallel?

**A.** A pair of corresponding angles is congruent

**B.** A pair of vertical angles is congruent

**C.** A linear pair adds to 180°

**D.** Two angles at the same crossing are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

642

**QUESTION ID**

## S18-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. A pair of corresponding angles is congruent**

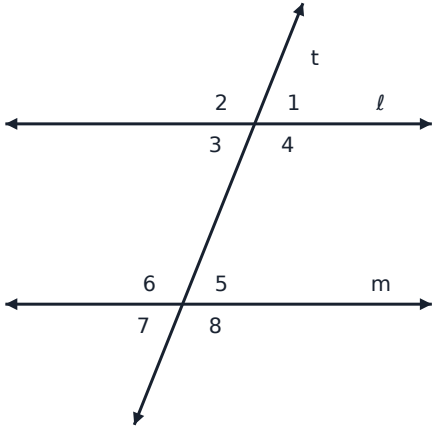
**GRADING NOTE** Vertical angles and linear pairs hold for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Lines $\ell$ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are $\ell$ and m parallel? Explain.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

643

QUESTION ID

**S18-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**No. ∠1 and ∠7 are alternate exterior angles; if $\ell \parallel$ m they would be congruent, but 72° ≠ 108°.**

GRADING NOTE Form required: the conclusion with a reason. Equivalently, ∠5 is vertical to ∠7, so $m\angle 5 = 108°$, but its corresponding angle ∠1 is 72°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

# FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

645

QUESTION ID

**S18-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**m∠A + m∠B + m∠C = 180°**

**GRADING NOTE** REQUIRED WORK: Name the two new angles at B, use alternate interior angles, and the straight angle at B.

Any valid proof (paragraph, two-column, or flow). Essential: the line through B parallel to AC exists (parallel postulate); the angles it makes with BA and BC are congruent to ∠A and ∠C (alternate interior angles); those two angles and ∠ABC form a straight angle, 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5**

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

646

QUESTION ID

**S18-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**∠A = 50°, ∠C = 65°, ∠B = 65°**

GRADING NOTE Form required: all three. Alternate interior angles give ∠A and ∠C; 180 − 50 − 65 = 65.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

**True or false?**

The proof of the triangle angle sum works without any fact about parallel lines.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

647

QUESTION ID

**S18-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It uses the parallel through one vertex and the alternate interior angle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

**A.** Alternate interior angles theorem

**B.** Isosceles triangle theorem

**C.** Triangle inequality

**D.** Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5



**QUESTION ID**

## S18-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. Alternate interior angles theorem

**GRADING NOTE** Sides BA and BC are transversals of the parallel lines (AC and the line through B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

649

**QUESTION ID**

## S18-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The exterior angle forms a linear pair with ∠C, so it is 180° − m∠C, which equals m∠A + m∠B by the angle sum.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

# Prove the perpendicular bisector theorem and its converse

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.9** |
| **BACKWARD 1** | Use the definitions of midpoint and perpendicular bisector | **HSG-CO.A.1** |
| **BACKWARD 2** | Use SAS and SSS to prove triangles congruent | **HSG-CO.B.8** |
| **BACKWARD 3** | Corresponding parts of congruent triangles and right angles from… | **HSG-CO.B.7** |
| **FORWARD 1** | Use the perpendicular bisector to find points equidistant from two… | **HSG-GPE.B.5** |
| **FORWARD 2** | Prove that the perpendicular bisectors of a triangle meet at one p… | **HSG-C.A.3** |

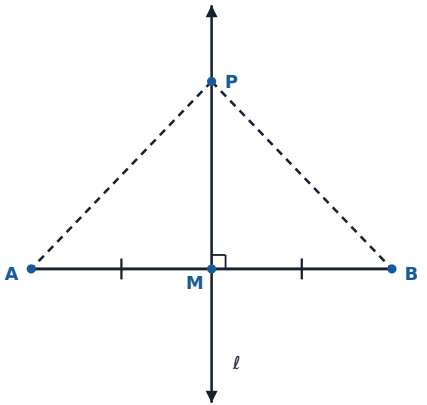
SET 19 • HSG-CO.C.9

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Line $\ell$ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on $\ell$. Prove that PA = PB.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5**

652

**QUESTION ID**

## S19-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**PA = PB**

**GRADING NOTE** REQUIRED WORK: Use SAS on triangles PMA and PMB.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM (M is the midpoint); ∠PMA ≅ ∠PMB (both right angles, $\ell \perp$ AB); PM = PM (reflexive); △PMA ≅ △PMB (SAS); PA = PB (corresponding parts). If P = M, PA = MA = MB = PB directly. Alternative: the reflection across $\ell$ fixes P and swaps A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

653

**QUESTION ID**

## S19-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**PM ⊥ AB, so P is on the perpendicular bisector of AB.**

**GRADING NOTE** REQUIRED WORK: Use SSS on triangles PMA and PMB, then the linear pair of congruent angles at M.

Any valid proof (paragraph, two-column, or flow). Essential: PA = PB (given), AM = BM (midpoint), PM = PM; △PMA ≅ △PMB (SSS); ∠PMA ≅ ∠PMB; they form a linear pair, so each is 90°; so PM is perpendicular to AB through its midpoint. Invalid: drawing the perpendicular bisector through P and assuming it passes through P (that is the claim).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

The theorem says the points on the perpendicular bisector of AB are EXACTLY the points equidistant from A and B. Which two statements must be proved?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

654

**QUESTION ID**

## S19-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**(1) Every point on the perpendicular bisector is equidistant from A and B. (2) Every point equidistant from A and B is on the perpendicular bisector.**

**GRADING NOTE** Any equivalent wording of both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

655

QUESTION ID

**S19-M-Q4**

TYPE

Answer with required work

CORRECT ANSWER

**PA = PB = √(a² + y²) for P(0, y); and PA = PB gives (x + a)² = (x − a)², so x = 0.**

GRADING NOTE REQUIRED WORK: Use the distance formula in both directions.

Any valid proof (paragraph, two-column, or flow). Essential: both directions. (1) √(a² + y²) both ways. (2) (x + a)² + y² = (x − a)² + y² simplifies to 4ax = 0, and a ≠ 0, so x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A student proves only that every point on the perpendicular bisector of AB is equidistant from A and B. What is missing?

**A.** A proof that the bisector is a line

**B.** Nothing; one direction proves "exactly"

**C.** The converse: every point equidistant from A and B is on the perpendicular bisector

**D.** A measurement of PA and PB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

656

**QUESTION ID**

## S19-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The converse: every point equidistant from A and B is on the perpendicular bisector**

**GRADING NOTE** "Exactly those points" needs both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

# BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

658

QUESTION ID

**S19-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

**True or false?**

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

659

QUESTION ID

**S19-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** That is its definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line $\ell$ meets AB at M with AM = MB, but $\ell$ is not perpendicular to AB. What is $\ell$?

**A.** Not related to AB

**B.** Parallel to AB

**C.** The perpendicular bisector of AB

**D.** A bisector of AB that is not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

660

**QUESTION ID**

## S19-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. A bisector of AB that is not the perpendicular bisector**

**GRADING NOTE** It bisects the segment but is not perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line $\ell$ is perpendicular to AB at M, and AM = 3, MB = 5. Is $\ell$ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

661

QUESTION ID

**S19-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**No. M is not the midpoint (3 ≠ 5).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

**True or false?**

A segment has exactly one perpendicular bisector in a plane.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

662

QUESTION ID

**S19-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Exactly one line is perpendicular to AB at its midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

# BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △PMA and △PMB, PM is shared, AM = BM, and ∠PMA ≅ ∠PMB. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

664

**QUESTION ID**

**S19-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**SAS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

In △PMA and △PMB, PA = PB, AM = BM, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

665

QUESTION ID

**S19-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**SSS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

**A.** SAS

**B.** ASA

**C.** SSS

**D.** AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

666

**QUESTION ID**

**S19-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. SSS**

**GRADING NOTE** Three pairs of congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

**True or false?**

A side shared by two triangles counts as a pair of congruent sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

667

QUESTION ID

**S19-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

**True or false?**

SSA is a valid congruence criterion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

668

QUESTION ID

**S19-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Two sides and a non-included angle do not always determine a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

# BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△PMA ≅ △PMB. Which angle corresponds to ∠PMA?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

670

**QUESTION ID**

**S19-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠PMB**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

671

QUESTION ID

**S19-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

**True or false?**

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

672

QUESTION ID

**S19-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

△XYZ ≅ △XWZ. Which pair of segments must be congruent?

**A.** YZ and XW

**B.** YZ and WZ

**C.** XY and XZ

**D.** XZ and YW

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

673

**QUESTION ID**

**S19-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. YZ and WZ**

**GRADING NOTE** Y ↔ W and Z ↔ Z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

**True or false?**

Two angles in a linear pair are always congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

674

QUESTION ID

**S19-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are supplementary; they are congruent only when both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

# FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

676

**QUESTION ID**

## S19-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -x + 7$**

**GRADING NOTE** Any equivalent equation. Midpoint (4, 3); slope of AB is 1, so the bisector has slope −1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S19-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5**

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

677

**QUESTION ID**

## S19-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(7, 0)**

**GRADING NOTE** Form required: an ordered pair. It is where y = −x + 7 meets y = 0.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S19-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5**

**True or false?**

The point (4, 4) is equidistant from (0, 0) and (8, 6).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

678

QUESTION ID
**S19-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Distances are $\sqrt{32} \approx 5.66$ and $\sqrt{20} \approx 4.47$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

**A.** (3, 1)

**B.** (4, 4)

**C.** (2, 7)

**D.** (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

679

**QUESTION ID**

## S19-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (2, 7)**

**GRADING NOTE** Points on x = 2.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S19-F1-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5**

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

680

QUESTION ID
**S19-F1-Q5**

TYPE
Short answer

CORRECT ANSWER
**The vertical line x = 1.**

**GRADING NOTE** Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

# FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

682

QUESTION ID

**S19-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**The three perpendicular bisectors meet at O, and OA = OB = OC.**

**GRADING NOTE** REQUIRED WORK: Let the bisectors of AB and BC meet at O; use the theorem and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of AB and BC are not parallel (AB and BC are not parallel), so they meet at O; OA = OB and OB = OC (theorem); so OA = OC; by the converse O is on the bisector of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

683

QUESTION ID

**S19-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**(3, 4)**

**GRADING NOTE** Form required: an ordered pair. Bisectors x = 3 and y = 4 meet at (3, 4); it is 5 from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

**True or false?**

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

684

QUESTION ID

**S19-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** For an obtuse triangle it is outside; for a right triangle it is on the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns are to share a well equally distant from all three. Where should it go?

**A.** Where the perpendicular bisectors of the triangle's sides meet

**B.** Where the angle bisectors meet

**C.** At the midpoint of the longest side

**D.** Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

685

**QUESTION ID**

## S19-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. Where the perpendicular bisectors of the triangle's sides meet**

**GRADING NOTE** Points equidistant from two towns are on the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

686

**QUESTION ID**
**S19-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**OA = OB and OB = OC give OA = OC, and by the converse O is on the perpendicular bisector of AC.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

# Prove that the angles of a triangle add to 180°

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.10** |
| **BACKWARD 1** | Use the angle sum and parallel-line angles informally | **8.G.A.5** |
| **BACKWARD 2** | Angles that form a straight angle add to 180° | **4.MD.C.7** |
| **BACKWARD 3** | Alternate interior angles of parallel lines are congruent | **HSG-CO.C.9** |
| **FORWARD 1** | Prove the exterior angle theorem | **HSG-CO.C.10** |
| **FORWARD 2** | Generalize the angle sum to polygons | **HSG-CO.C.10** |

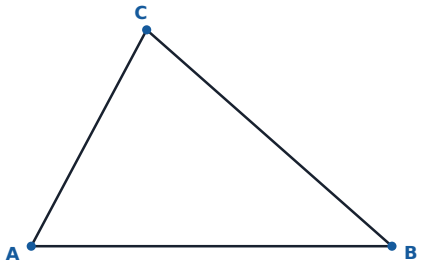
SET 20 • HSG-CO.C.10

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
m∠A + m∠B + m∠C = 180° in triangle ABC. Draw the line through C parallel to AB.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5**

689

**QUESTION ID**

## S20-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠A + m∠B + m∠C = 180°**

**GRADING NOTE** REQUIRED WORK: Name the two new angles at C, use alternate interior angles with AB, and the straight angle at C.

Any valid proof (paragraph, two-column, or flow). Essential: (1) through C there is exactly one line parallel to AB (parallel postulate); (2) the angles it makes with CA and CB are congruent to ∠A and ∠B (alternate interior angles, transversals CA and CB); (3) those two angles and ∠ACB form a straight angle, 180°; (4) substitute. Invalid: measuring angles; citing the exterior angle theorem (it depends on this result).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5**

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

690

**QUESTION ID**

## S20-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠P + m∠Q + m∠R = 180°**

**GRADING NOTE** REQUIRED WORK: Same strategy with vertex P: alternate interior angles and a straight angle at P.

Any valid proof (paragraph, two-column, or flow). Essential: unique parallel through P; the two angles at P next to ∠QPR are congruent to ∠Q and ∠R (alternate interior); the three angles at P make 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

691

QUESTION ID

**S20-M-Q3**

TYPE

Answer with required work

CORRECT ANSWER

**m∠A + m∠B + m∠ACB = 180°**

**GRADING NOTE** REQUIRED WORK: Use alternate interior angles (∠A and ∠ACE) and corresponding angles (∠B and ∠ECD), then the straight angle BCD.

Any valid proof (paragraph, two-column, or flow). Essential: ∠ACE ≅ ∠A (alternate interior, transversal AC); ∠ECD ≅ ∠B (corresponding, transversal BD); ∠ACB + ∠ACE + ∠ECD = 180° (straight angle at C).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

A proof of the triangle angle sum (line $\ell$ through C parallel to AB, making $\angle 1$ next to CA and $\angle 2$ next to CB):

1. $m\angle 1 + m\angle ACB + m\angle 2 = 180°$ (?)
2. $\angle 1 \cong \angle A$ and $\angle 2 \cong \angle B$ (?)
3. $m\angle A + m\angle ACB + m\angle B = 180°$ (?)

Give the three reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

692

**QUESTION ID**

## S20-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1: the angles form a straight angle (angle addition); 2: alternate interior angles are congruent ($\ell \parallel$ AB); 3: substitution.**

**GRADING NOTE** Any equivalent names for all three reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

A student proves the angle sum theorem using "an exterior angle equals the sum of the two remote interior angles." What is wrong?

**A.** Exterior angles do not exist in triangles

**B.** Nothing is wrong

**C.** That theorem is usually proved FROM the angle sum, so the argument is circular

**D.** The exterior angle theorem is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

693

QUESTION ID

**S20-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. That theorem is usually proved FROM the angle sum, so the argument is circular**

**GRADING NOTE** Using a result that depends on the theorem being proved is circular reasoning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

# BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

695

QUESTION ID

**S20-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**65°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

**True or false?**

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

696

QUESTION ID

**S20-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

697

QUESTION ID

**S20-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**30**

**GRADING NOTE** Any equivalent form. 6x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at $\ell$ is 72°. Which angle at m must also be 72°?

**A.** Its alternate interior angle

**B.** Its same-side interior angle

**C.** Every angle at m

**D.** The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

698

**QUESTION ID**

**S20-B1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. Its alternate interior angle**

**GRADING NOTE** Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

**True or false?**

A triangle can have two obtuse angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

699

QUESTION ID

## S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Two angles over 90° would already add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

# BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

701

QUESTION ID

**S20-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**65°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

**True or false?**

The angles around a point on one side of a straight line add to 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

702

**QUESTION ID**

## S20-B2-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** They make a straight angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

∠1, ∠2, and ∠3 form a straight angle, with ∠1 = ∠3 = 50°. Find ∠2.

**A.** 50°

**B.** 100°

**C.** 80°

**D.** 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

703

**QUESTION ID**

**S20-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 80°**

**GRADING NOTE** 180 − 100 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

704

QUESTION ID

**S20-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**36**

**GRADING NOTE** Any equivalent form. 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

**True or false?**

Two angles that make up a straight angle must be equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

705

QUESTION ID

**S20-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They add to 180° but can be any pair, such as 30° and 150°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

# BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

707

**QUESTION ID**

## S20-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**58°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

**True or false?**

Alternate interior angles formed by a transversal of parallel lines are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

708

QUESTION ID

**S20-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** This is the theorem proved in an earlier set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Line $\ell$ through C is parallel to AB. Which angle at C is congruent to $\angle A$?

**A.** The angle between CA and CB

**B.** The angle between $\ell$ and CB

**C.** The angle between $\ell$ and CA

**D.** $\angle ACB$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

709

**QUESTION ID**

## S20-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The angle between $\ell$ and CA**

**GRADING NOTE** CA is a transversal of $\ell$ and AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

710

QUESTION ID

**S20-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**The parallel postulate: through a point not on a line there is exactly one parallel line.**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

**True or false?**

Alternate interior angles are congruent for any two lines cut by a transversal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

711

QUESTION ID

**S20-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

# FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that m∠ACD = m∠A + m∠B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

713

QUESTION ID

## S20-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

**m∠ACD = m∠A + m∠B**

GRADING NOTE

REQUIRED WORK: Use the linear pair at C and the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ACD + m∠ACB = 180° (linear pair); m∠A + m∠B + m∠ACB = 180° (angle sum); substitute and subtract m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

714

**QUESTION ID**

## S20-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Remote interior: 77°; adjacent interior: 55°.**

**GRADING NOTE** Form required: both angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

**True or false?**

An exterior angle of a triangle is always greater than 90°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

715

QUESTION ID

**S20-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE The exterior angle at an obtuse vertex is acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

**A.** 5

**B.** 35

**C.** 25

**D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

716

QUESTION ID

**S20-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 25**

GRADING NOTE 5x + 10 = 4x + 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

717

**QUESTION ID**
**S20-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**

**Each exterior angle is 180° minus its interior angle, so the sum is 540° − 180° = 360°.**

**GRADING NOTE** Any valid argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

# FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

719

QUESTION ID

**S20-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**$(n - 2) \cdot 180°$**

**GRADING NOTE** REQUIRED WORK: Draw the diagonals from one vertex to divide the polygon into triangles and use the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: from one vertex there are n − 3 diagonals, making n − 2 triangles; their angles together make up exactly the polygon's angles; each triangle contributes 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

720

QUESTION ID

**S20-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**135°**

**GRADING NOTE** Any equivalent form. $(8 - 2) \cdot 180 \div 8 = 135$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

721

**QUESTION ID**

**S20-F2-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Any equivalent form. Each exterior angle is 24°; 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

**True or false?**

The interior angles of a hexagon add to 1,080°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

722

QUESTION ID

**S20-F2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $(6 - 2) \cdot 180 = 720°$; 1,080° is the sum for an octagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

**A.** 125

**B.** 115

**C.** 135

**D.** 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

723

QUESTION ID

**S20-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. 115**

GRADING NOTE 540 − 425 = 115.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

# Prove that the base angles of an isosceles triangle are congruent

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.10** |
| **BACKWARD 1** | Prove triangles congruent with SAS or SSS | **HSG-CO.B.8** |
| **BACKWARD 2** | Use the definitions of angle bisector and midpoint | **HSG-CO.A.1** |
| **BACKWARD 3** | Classify triangles and name legs, base, and base angles | **4.G.A.2** |
| **FORWARD 1** | Prove the converse: congruent base angles give congruent sides | **HSG-CO.C.10** |
| **FORWARD 2** | Use isosceles triangle theorems in multistep problems and proofs | **HSG-CO.C.10** |

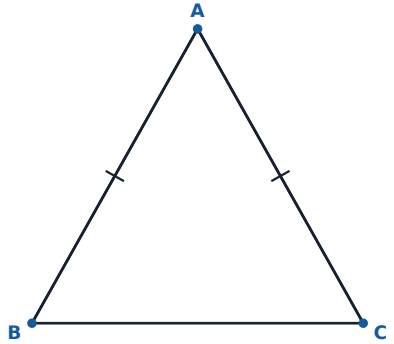
SET 21 • HSG-CO.C.10

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5**

726

**QUESTION ID**

## S21-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**∠B ≅ ∠C**

**GRADING NOTE** REQUIRED WORK: Prove △ABD ≅ △ACD by SAS, then use corresponding angles.

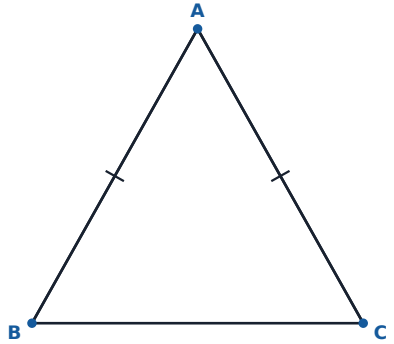
Any valid proof (paragraph, two-column, or flow). Essential: AB = AC (given); ∠BAD ≅ ∠CAD (definition of angle bisector); AD = AD; △ABD ≅ △ACD (SAS); ∠B ≅ ∠C (corresponding angles). Invalid: "∠B and ∠C look equal"; using the fact that AD ⊥ BC (not yet known).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5**

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5**

727

**QUESTION ID**

## S21-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**∠B ≅ ∠C**

**GRADING NOTE**

REQUIRED WORK: Prove △ABM ≅ △ACM by SSS.

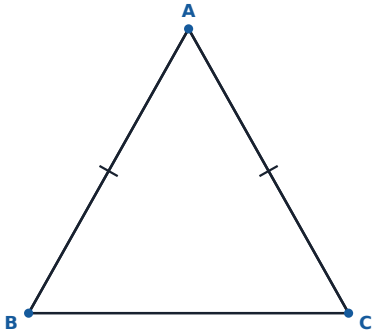
Any valid proof (paragraph, two-column, or flow). Essential: AB = AC; BM = CM (midpoint); AM = AM; SSS; corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5**

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

728

QUESTION ID

## S21-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

**The reflection maps ∠ABC onto ∠ACB, so ∠B ≅ ∠C.**

**GRADING NOTE** REQUIRED WORK: Explain where the reflection sends ray AB, point B, and point C.

Any valid explanation. Essential: the reflection across the bisector of ∠A swaps rays AB and AC (the bisector makes equal angles with them); B goes to the point of ray AC at distance AB = AC, which is C; so B ↔ C and the triangle maps onto itself with ∠B ↔ ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5**

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5**

729

**QUESTION ID**

## S21-M-Q4

**TYPE**

Answer with required work

**CORRECT ANSWER**

**All three angles are congruent (each is 60°).**

**GRADING NOTE** REQUIRED WORK: Apply the isosceles base angle theorem twice, with two different pairs of equal sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC gives ∠B ≅ ∠C; AB = BC gives ∠A ≅ ∠C; so all three are congruent (and each is 60° by the angle sum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5**

In the proof that base angles of isosceles triangle ABC (AB = AC) are congruent, the bisector of ∠A meets BC at D. Which criterion proves △ABD ≅ △ACD?

**A.** SAS

**B.** SSS

**C.** AAS

**D.** ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

730

QUESTION ID

**S21-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. SAS**

**GRADING NOTE** AB = AC, ∠BAD ≅ ∠CAD, and AD is shared: two sides and the included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

# BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

732

QUESTION ID

**S21-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**SAS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM.
Which criterion shows △ABM ≅ △ACM?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

733

QUESTION ID
**S21-B1-Q2**

TYPE
Short answer

CORRECT ANSWER
**SSS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

**True or false?**

A shared side counts as a pair of congruent sides in a congruence proof.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

734

QUESTION ID

**S21-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

**A.** Three sides

**B.** Two sides and the included angle

**C.** Two angles and the included side

**D.** Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

735

QUESTION ID

**S21-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. Two sides and a non-included angle**

**GRADING NOTE** SSA is not a criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

**True or false?**

If △ABD ≅ △ACD, then ∠ABD ≅ ∠ACD.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

736

QUESTION ID

**S21-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

# BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

738

**QUESTION ID**

**S21-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**32°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

739

QUESTION ID

**S21-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

**True or false?**

An angle bisector divides an angle into two congruent angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

740

QUESTION ID

**S21-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

**A.** m∠BAD = m∠BAC

**B.** m∠BAD = m∠DAC

**C.** AD = AB

**D.** BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

741

QUESTION ID

**S21-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. m∠BAD = m∠DAC**

**GRADING NOTE** Equal angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

**True or false?**

The bisector of ∠A of triangle ABC always passes through the midpoint of BC.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

742

QUESTION ID

**S21-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only when AB = AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

# BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

744

**QUESTION ID**

## S21-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Base QR; base angles ∠Q and ∠R**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

**True or false?**

An isosceles triangle has at least two congruent sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

745

**QUESTION ID**

## S21-B3-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Definition used in this course.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

**A.** Sides 3, 3, 3

**B.** Sides 5, 5, 5

**C.** Sides 4, 5, 6

**D.** Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

746

**QUESTION ID**

## S21-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. Sides 4, 5, 6

**GRADING NOTE** No two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

747

**QUESTION ID**

**S21-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠X**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

**True or false?**

Every equilateral triangle is isosceles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

748

QUESTION ID

**S21-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE It has at least two congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

# FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, ∠B ≅ ∠C. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

750

QUESTION ID

**S21-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**AB = AC**

GRADING NOTE REQUIRED WORK: Draw the bisector of ∠A meeting BC at D and prove △ABD ≅ △ACD by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (given); ∠BAD ≅ ∠CAD (bisector); AD = AD; AAS; AB = AC. Invalid: using AB = AC to choose the auxiliary line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

751

**QUESTION ID**

## S21-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**x = 4; PQ = 14**

**GRADING NOTE** Form required: x and PQ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

**True or false?**

A triangle with two congruent angles must have two congruent sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

752

QUESTION ID

**S21-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE This is the converse of the isosceles triangle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has m∠A = 50° and m∠B = 65°. Which two sides are congruent?

**A.** No two sides

**B.** AB and BC

**C.** AB and AC

**D.** AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

753

QUESTION ID

**S21-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. AB and AC**

**GRADING NOTE** m∠C = 65° = m∠B, so the sides opposite them, AC and AB, are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

754

QUESTION ID

**S21-F1-Q5**

TYPE

Answer with required work

CORRECT ANSWER

**All three sides are congruent.**

GRADING NOTE

REQUIRED WORK: Apply the converse twice.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠B gives BC = AC; ∠B ≅ ∠C gives AC = AB; so all three sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

# FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

756

**QUESTION ID**

**S21-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**71°**

**GRADING NOTE** Any equivalent form. (180 − 38) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

757

**QUESTION ID**

**S21-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form. Triangles ABD and ACD are isosceles, so ∠BAD = ∠B and ∠CAD = ∠C; then 2(∠B + ∠C) = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

758

QUESTION ID

**S21-F2-Q3**

TYPE

Answer with required work

CORRECT ANSWER

**AD = AE**

GRADING NOTE REQUIRED WORK: Use the base angles and SAS on triangles ABD and ACE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (base angles); AB = AC; BD = CE; △ABD ≅ △ACE (SAS); AD = AE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

**True or false?**

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

759

QUESTION ID

**S21-F2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE (180 − 100) ÷ 2 = 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

**A.** 56°

**B.** 112°

**C.** 68°

**D.** 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

760

**QUESTION ID**

**S21-F2-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 44°**

**GRADING NOTE** ∠PQR = 68° = ∠R, so ∠P = 180 − 136 = 44.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

# Prove the midsegment theorem

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.10** |
| **BACKWARD 1** | Find midpoints of segments in the coordinate plane | **6.G.A.3** |
| **BACKWARD 2** | Find and compare slopes to decide whether lines are parallel | **8.EE.B.6** |
| **BACKWARD 3** | Find lengths with the distance formula | **8.G.B.8** |
| **BACKWARD 4** | Prove lines parallel from congruent alternate interior angles | **HSG-CO.C.9** |
| **FORWARD 1** | Prove that the midpoints of a quadrilateral's sides form a parallelo | **HSG-CO.C.11** |
| **FORWARD 2** | Midsegment triangles: similarity, scale factor, and area | **HSG-SRT.A.2** |

SET 22 • HSG-CO.C.10

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

763

QUESTION ID

**S22-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**D(b, c), E(a + b, c); DE is horizontal like AB, and DE = a = ½(2a) = ½AB.**

GRADING NOTE REQUIRED WORK: Find D and E with the midpoint formula, then compare slopes and lengths.

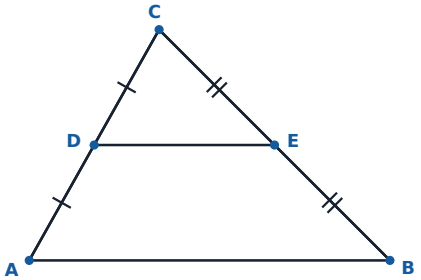
Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; DE has slope 0 (same y) and AB has slope 0, so DE ∥ AB (and DE ≠ AB since c ≠ 0); DE = |a + b − b| = a and AB = 2a. Invalid: checking one numerical triangle only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5**

764

**QUESTION ID**

## S22-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**DE ∥ AB and DE = ½AB**

**GRADING NOTE** REQUIRED WORK: Prove △DEC ≅ △FEB (SAS), deduce BF = DC = AD and BF ∥ AD, so ABFD is a parallelogram; then use its opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: CE = BE, DE = FE, vertical angles at E, so △DEC ≅ △FEB (SAS); BF = DC = AD; ∠DCE ≅ ∠FBE gives BF ∥ AC (alternate interior converse); one pair of opposite sides of ABFD is parallel and congruent, so ABFD is a parallelogram; so DF ∥ AB and DF = AB; DE = ½DF = ½AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5**

In the coordinate proof of the midsegment theorem, the vertices are A(0, 0), B(2a, 0), C(2b, 2c). Why are the coordinates written with 2a, 2b, and 2c?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

765

QUESTION ID

**S22-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**So the midpoints have whole-letter coordinates (a, b, c) without fractions, which keeps the algebra simple; any triangle can still be placed this way.**

**GRADING NOTE** Any equivalent explanation that mentions avoiding fractions in the midpoint formula and that the placement is general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A proof of the midsegment theorem with A(0, 0), B(2a, 0), C(2b, 2c):

1. D = (b, c), E = (a + b, c) (?)
2. slope DE = 0 = slope AB (?)
3. DE ∥ AB (?)
4. DE = a and AB = 2a, so DE = ½AB (?)

Give the four reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

766

**QUESTION ID**

## S22-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1: midpoint formula; 2: slope formula (equal y-coordinates); 3: lines with equal slopes are parallel; 4: distance formula (horizontal distances).**

**GRADING NOTE** Any equivalent names for all four reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

A student checks the triangle A(0, 0), B(6, 0), C(2, 4), finds the midsegment is parallel to AB and 3 units long, and concludes the theorem is proved. What is the flaw?

**A.** One example cannot prove a statement about every triangle

**B.** The midsegment should be 6 units long

**C.** Midpoints cannot be found with coordinates

**D.** The midsegment is not parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

767

QUESTION ID

**S22-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. One example cannot prove a statement about every triangle**

GRADING NOTE A general proof must work for every triangle, for example with variable coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

# BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

769

**QUESTION ID**

## S22-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(5, 4)**

**GRADING NOTE** Form required: an ordered pair. Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the midpoint of the segment from (−3, 1) to (5, −7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

770

**QUESTION ID**

**S22-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(1, −3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

771

**QUESTION ID**

## S22-B1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**(a, c)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

**True or false?**

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

772

QUESTION ID

**S22-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE ((1 + 7) ÷ 2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

**True or false?**

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

773

QUESTION ID

**S22-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is (3, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

# BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

775

QUESTION ID

**S22-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form. (8 − 2) ÷ (4 − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the slope of the line through (b, c) and (a + b, c).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

776

QUESTION ID

**S22-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0**

GRADING NOTE Any equivalent form. The y-coordinates are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

**True or false?**

Two different nonvertical lines with the same slope are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

777

QUESTION ID

**S22-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Same steepness, different positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

**A.** From (1, 3) to (5, 5)

**B.** From (2, 2) to (2, 6)

**C.** From (1, 3) to (3, 7)

**D.** From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

778

QUESTION ID

**S22-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. From (1, 3) to (5, 5)**

**GRADING NOTE** Both have slope 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

**True or false?**

A horizontal segment and a vertical segment have the same slope.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

779

QUESTION ID

**S22-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Horizontal has slope 0; vertical is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

# BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

781

QUESTION ID

**S22-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**a (that is, |a|, with a > 0)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

782

QUESTION ID

**S22-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from (−2, 5) to (4, 5).

**A.** 3

**B.** 10

**C.** 6

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

783

**QUESTION ID**

**S22-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 6**

**GRADING NOTE** Horizontal: |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

**True or false?**

The distance from (1, 2) to (4, 6) is 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

784

QUESTION ID

**S22-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE $\sqrt{(9 + 16)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

**True or false?**

The distance from (0, 0) to (2a, 0) is a, when a > 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

785

QUESTION ID

**S22-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It is 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

# BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell$ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

787

**QUESTION ID**

## S22-B4-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

$\ell \parallel m$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

**True or false?**

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

788

QUESTION ID

**S22-B4-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Converse of the alternate interior angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In △ABC, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that EF = DE, and △DEC ≅ △FEB. Which fact shows BF ∥ AC?

**A.** DE = EF

**B.** ∠DCE ≅ ∠FBE are alternate interior angles

**C.** BF = DC

**D.** E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

789

QUESTION ID
**S22-B4-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**B. ∠DCE ≅ ∠FBE are alternate interior angles**

**GRADING NOTE** Congruent alternate interior angles (transversal BC) give parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

790

**QUESTION ID**

## S22-B4-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

**True or false?**

Congruent vertical angles prove that two lines are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

791

QUESTION ID

**S22-B4-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Vertical angles are congruent for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

# FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

793

**QUESTION ID**

## S22-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**PQ ∥ AC ∥ SR and PQ = ½AC = SR, so PQRS is a parallelogram.**

**GRADING NOTE** REQUIRED WORK: Draw diagonal AC and apply the midsegment theorem in triangles ABC and ADC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ABC, PQ ∥ AC and PQ = ½AC; in △ADC, SR ∥ AC and SR = ½AC; so PQ ∥ SR and PQ = SR; a quadrilateral with one pair of opposite sides parallel and congruent is a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

794

**QUESTION ID**

**S22-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**24**

**GRADING NOTE** Any equivalent form. Its sides are ½AC, ½BD, ½AC, ½BD: 5 + 7 + 5 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

**True or false?**

The midpoint quadrilateral of any quadrilateral is a parallelogram.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

795

QUESTION ID

## S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** This is Varignon's theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

**A.** rectangle

**B.** trapezoid that is not a parallelogram

**C.** rhombus

**D.** kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

796

**QUESTION ID**

## S22-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. rectangle

**GRADING NOTE** Its sides are parallel to the diagonals, so adjacent sides are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

797

**QUESTION ID**

**S22-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**19**

**GRADING NOTE** Any equivalent form. Each midsegment is half the parallel side: 6 + 8 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

# FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

799

**QUESTION ID**

## S22-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Each side of DEF is half of a side of ABC (midsegment theorem), so the triangles are similar by SSS similarity with scale factor ½.**

**GRADING NOTE** Any equivalent explanation with the factor ½ (or 2 from DEF to ABC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

800

**QUESTION ID**

## S22-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**12**

**GRADING NOTE** Any equivalent form. Scale factor ½ gives area factor ¼; also the 4 small triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

**True or false?**

The four triangles formed by drawing all three midsegments of a triangle are similar to it with scale factor ⅓.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

801

QUESTION ID
**S22-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** The scale factor is ½: each side is half a side of the original.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

The midsegment parallel to AB in triangle ABC is 9. A line segment connects the midpoints of that midsegment's triangle the same way. How long is the new segment parallel to AB?

**A.** 9

**B.** 18

**C.** 3

**D.** 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

802

**QUESTION ID**

**S22-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 4.5**

**GRADING NOTE** Each repetition halves the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

Triangle ABC has vertices A(0, 0), B(8, 0), C(2, 6). Find the vertices of its midsegment triangle and verify the ratio of areas.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

803

QUESTION ID
**S22-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**(4, 0), (5, 3), (1, 3); area of ABC = 24 and area of the midsegment triangle = 6, a ratio of 1 : 4.**

**GRADING NOTE** Form required: the three midpoints and both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

# Prove that the medians of a triangle meet at one point

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.10** |
| **BACKWARD 1** | Find midpoints, including with variables | **6.G.A.3** |
| **BACKWARD 2** | Use the midsegment theorem | **HSG-CO.C.10** |
| **BACKWARD 3** | Find the point a given fraction of the way along a segment | **7.RP.A.2** |
| **BACKWARD 4** | Write the equation of a line through two points | **8.F.B.4** |
| **FORWARD 1** | Locate the centroid and use the 2 : 1 ratio | **HSG-GPE.B.6** |
| **FORWARD 2** | Prove that the angle bisectors of a triangle meet at one point | **HSG-C.A.3** |

SET 23 • HSG-CO.C.10

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2b, 0), and C(2c, 2d), with b ≠ 0 and d ≠ 0. Prove that the point G((2b + 2c)/3, 2d/3) lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5**

806

**QUESTION ID**

## S23-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**G is on the median from A (to (b + c, d)), from B (to (c, d)), and from C (to (b, 0)), so the medians meet at G.**

**GRADING NOTE** REQUIRED WORK: Find the midpoint of each side, then show G is two-thirds of the way from each vertex to the opposite midpoint (or on each median line).

Any valid proof (paragraph, two-column, or flow). Essential: the three midpoints (b + c, d), (c, d), (b, 0); for each median, G = vertex + ⅔(midpoint − vertex): A + ⅔((b + c, d) − A) = G; B + ⅔((c, d) − (2b, 0)) = G; C + ⅔((b, 0) − (2c, 2d)) = G. Alternative: show G satisfies each median's equation. Invalid: checking one triangle with numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5**

In triangle ABC, medians BE and CF meet at G. Let H and K be the midpoints of BG and CG. Prove that BG = 2GE. Then explain why the third median also passes through G.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

807

QUESTION ID

## S23-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

**BG = 2GE; the same argument for medians AD and BE gives the point that is two-thirds along BE, which is G, so AD passes through G.**

**GRADING NOTE** REQUIRED WORK: Use the midsegment theorem in triangles ABC and GBC to show FEKH is a parallelogram, then its diagonals.

Any valid proof (paragraph, two-column, or flow). Essential: FE ∥ BC and FE = ½BC (midsegment in △ABC); HK ∥ BC and HK = ½BC (midsegment in △GBC); so FEKH is a parallelogram; its diagonals bisect each other, so HG = GE; with BH = HG, BG = 2GE; any two medians meet at the point two-thirds along each, so all three share it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

Explain why proving that the point two-thirds of the way along EACH median is the same point proves that the three medians meet at one point.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

808

**QUESTION ID**

## S23-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**Each median contains its own two-thirds point. If those three points are one point, that point lies on all three medians, so they meet there.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

Triangle PQR has P(0, 0), Q(6, 0), and R(0, 6). Find the three medians' equations and prove that they meet at one point.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5**

809

**QUESTION ID**

## S23-M-Q4

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The medians meet at (2, 2).**

**GRADING NOTE** REQUIRED WORK: Find each median as a line through a vertex and a midpoint, find where two meet, and check that the third passes through that point.

Any valid proof (paragraph, two-column, or flow). Essential: midpoints (3, 3), (0, 3), (3, 0); medians y = x, y = −½x + 3 (from Q), y = −2x + 6 (from R); y = x and y = −2x + 6 meet at (2, 2); (2, 2) satisfies y = −½x + 3. (This proves the claim for this triangle; the general proof needs variables.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5**

**True or false?**

Drawing medians carefully in three different triangles and seeing that they meet proves that the medians of every triangle meet at one point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

810

QUESTION ID
**S23-M-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Examples suggest a theorem but cannot prove it for every triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

SET 23 • HSG-CO.C.10

# BACKWARD 1

Find midpoints, including with variables • Grade 6 • Geometry • 6.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the midpoint of the segment from (2b, 0) to (2c, 2d).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

812

QUESTION ID

**S23-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(b + c, d)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (0, 6) to (6, 0).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

813

**QUESTION ID**

## S23-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(3, 3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

A median of a triangle joins a vertex to

**A.** the midpoint of the opposite side

**B.** another vertex

**C.** the foot of the perpendicular from that vertex

**D.** the midpoint of an adjacent side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

814

QUESTION ID

**S23-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. the midpoint of the opposite side**

**GRADING NOTE** Definition of a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

**True or false?**

A triangle has exactly three medians.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

815

QUESTION ID

**S23-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** One from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

**True or false?**

A median always bisects the angle at its vertex.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

816

QUESTION ID

## S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Only in special cases (for example, the vertex angle of an isosceles triangle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

# BACKWARD 2

Use the midsegment theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, F and E are the midpoints of AB and AC. BC = 14. Find FE.

FE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

818

**QUESTION ID**

## S23-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

**True or false?**

The segment joining the midpoints of two sides of a triangle is parallel to the third side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

819

**QUESTION ID**

## S23-B2-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Midsegment theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

In triangle GBC, H and K are midpoints of GB and GC. Which statement is true?

**A.** HK ⊥ BC

**B.** HK ∥ GB

**C.** HK ∥ BC and HK = ½BC

**D.** HK = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

820

QUESTION ID

**S23-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. HK ∥ BC and HK = ½BC**

GRADING NOTE Midsegment theorem in triangle GBC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

A midsegment of a triangle is 11 cm. How long is the side parallel to it?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

821

**QUESTION ID**

## S23-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**22 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

**True or false?**

A midsegment of a triangle is always half as long as each of the other two sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

822

QUESTION ID

**S23-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only half as long as the side it is parallel to.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSG-CO.C.10

# BACKWARD 3

Find the point a given fraction of the way along a segment • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

On a number line, what point is ⅔ of the way from 0 to 9?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

824

**QUESTION ID**
**S23-B3-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

On a number line, what point is ⅔ of the way from 3 to 12?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

825

QUESTION ID

**S23-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**9**

GRADING NOTE Any equivalent form. 3 + ⅔(9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

A point is ⅔ of the way from A to D. What is the ratio of its distances from A and from D?

**A.** 2 : 3

**B.** 3 : 2

**C.** 1 : 2

**D.** 2 : 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

826

**QUESTION ID**

## S23-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 2 : 1**

**GRADING NOTE** ⅔ and ⅓ of the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

**True or false?**

The point ⅔ of the way from (0, 0) to (6, 3) is (4, 2).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

827

QUESTION ID

**S23-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** ⅔ of each coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

**True or false?**

The point ⅔ of the way from (0, 0) to (3, 6) is (2, 2).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

828

QUESTION ID

**S23-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSG-CO.C.10

# BACKWARD 4

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 0) and (3, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

830

QUESTION ID

**S23-B4-Q1**

TYPE

Short answer

CORRECT ANSWER

**y = x**

**GRADING NOTE** Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

Write an equation of the line through (6, 0) and (0, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5**

831

**QUESTION ID**

## S23-B4-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -\frac{1}{2}x + 3$**

**GRADING NOTE** Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5**

Which point is on the line y = −2x + 6?

**A.** (1, 2)

**B.** (3, 3)

**C.** (0, 2)

**D.** (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

832

**QUESTION ID**

**S23-B4-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (2, 2)**

**GRADING NOTE** −2(2) + 6 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

**True or false?**

The line through (0, 6) and (3, 0) has slope −2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5**

833

QUESTION ID

## S23-B4-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

GRADING NOTE (0 − 6) ÷ (3 − 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5**

**True or false?**

The line y = x passes through (2, 3).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

834

QUESTION ID

**S23-B4-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $2 \neq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

SET 23 • HSG-CO.C.10

# FORWARD 1

Locate the centroid and use the 2 : 1 ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point where the medians of the triangle with vertices (1, 2), (7, 4), and (4, 9) meet.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

836

**QUESTION ID**

**S23-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**(4, 5)**

**GRADING NOTE** Form required: an ordered pair. Average the vertices.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5**

Median AD of a triangle is 18 cm long. How far is the centroid from A and from D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

837

**QUESTION ID**

**S23-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**12 cm from A and 6 cm from D**

**GRADING NOTE** Form required: both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

**True or false?**

The centroid divides each median so that the part from the vertex is half the part from the midpoint.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

838

QUESTION ID
**S23-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** The part from the vertex is twice the other part (2 : 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

G is the centroid of triangle ABC, and GE = 5 on median BE. Find BE.

**A.** 20

**B.** 10

**C.** 7.5

**D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

839

QUESTION ID

**S23-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 15**

**GRADING NOTE** BG = 10, so BE = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

Explain why the centroid of a triangle is the average of its three vertices.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

840

QUESTION ID

## S23-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**The centroid is ⅔ of the way from A to the midpoint of BC: A + ⅔((B + C)/2 − A) = (A + B + C)/3.**

**GRADING NOTE** Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

# FORWARD 2

Prove that the angle bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the bisectors of the three angles of triangle ABC meet at one point that is equidistant from the three sides.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

842

**QUESTION ID**

## S23-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The three angle bisectors meet at one point I, equidistant from all three sides.**

**GRADING NOTE** REQUIRED WORK: Use the fact that a point on an angle bisector is equidistant from the angle's sides, and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of ∠A and ∠B meet at I; I is equidistant from AB and AC, and from AB and BC; so it is equidistant from AC and BC; by the converse it lies on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

Why must the bisectors of ∠A and ∠B of a triangle meet inside the triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

843

QUESTION ID

**S23-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Each bisector goes from its vertex into the triangle's interior and reaches the opposite side, so the two bisectors cross inside the triangle.**

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

**True or false?**

A point on the bisector of an angle is equidistant from the two sides of the angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

844

QUESTION ID

**S23-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Proved with congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

The point where the angle bisectors of a triangle meet is the center of the

**A.** circumscribed circle

**B.** nine-point circle

**C.** inscribed circle

**D.** largest square inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

845

QUESTION ID

**S23-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. inscribed circle**

**GRADING NOTE** It is equidistant from the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

**True or false?**

The angle bisectors and the medians of every triangle meet at the same point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

846

QUESTION ID

**S23-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only in an equilateral triangle in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

# Prove that opposite sides and opposite angles of a parallelogram are congruent

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.11** |
| **BACKWARD 1** | Alternate interior and same-side interior angles of parallel lines | **HSG-CO.C.9** |
| **BACKWARD 2** | Prove triangles congruent with ASA | **HSG-CO.B.8** |
| **BACKWARD 3** | Know the definition of a parallelogram | **5.G.B.3** |
| **FORWARD 1** | Prove that the diagonals of a rhombus are perpendicular | **HSG-CO.C.11** |
| **FORWARD 2** | Prove the parallelogram properties with coordinates | **HSG-GPE.B.4** |

SET 24 • HSG-CO.C.11

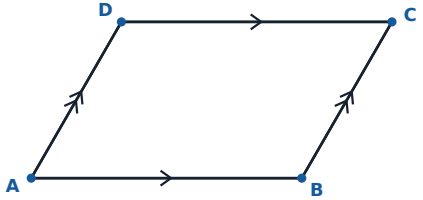
# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

ABCD is a parallelogram (AB ∥ DC and AD ∥ BC). Prove that AB = CD and AD = CB. (Draw diagonal AC.)

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

849

QUESTION ID

**S24-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**AB = CD and AD = CB**

**GRADING NOTE** REQUIRED WORK: Prove △ABC ≅ △CDA by ASA, using alternate interior angles from both pairs of parallel sides and the shared diagonal.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA (alternate interior, AB ∥ DC); ∠BCA ≅ ∠DAC (alternate interior, AD ∥ BC); AC = CA (reflexive); △ABC ≅ △CDA (ASA); corresponding sides. Invalid: "opposite sides look equal"; using the theorem being proved.

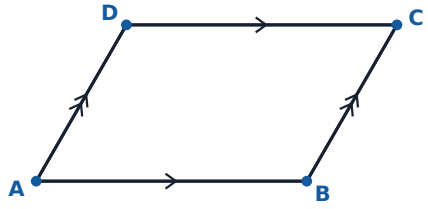
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

ABCD is a parallelogram.
Prove that ∠B ≅ ∠D.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5**

850

**QUESTION ID**
**S24-M-Q2**

**TYPE**
Answer with required work

**CORRECT ANSWER**
**∠B ≅ ∠D**

**GRADING NOTE** REQUIRED WORK: Use diagonal AC and the congruent triangles ABC and CDA (ASA), then corresponding angles.

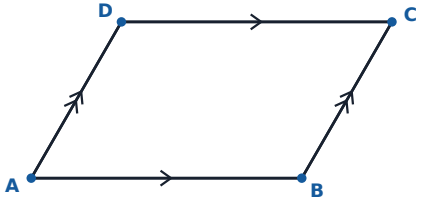
Any valid proof (paragraph, two-column, or flow). Essential: the same ASA congruence △ABC ≅ △CDA; ∠B and ∠D are corresponding angles. Alternative: consecutive angles are supplementary (same-side interior), so ∠B = 180° − ∠A = ∠D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5**

ABCD is a parallelogram. Prove that ∠A ≅ ∠C using same-side interior angles (no diagonal).

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

851

**QUESTION ID**

## S24-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**∠A ≅ ∠C**

**GRADING NOTE** REQUIRED WORK: Show ∠A and ∠B are supplementary and ∠B and ∠C are supplementary, then subtract.

Any valid proof (paragraph, two-column, or flow). Essential: AD ∥ BC with transversal AB gives m∠A + m∠B = 180°; AB ∥ DC with transversal BC gives m∠B + m∠C = 180°; so m∠A = m∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

A proof that opposite sides of parallelogram ABCD are congruent (diagonal AC drawn):

1. ∠BAC ≅ ∠DCA (?)
2. ∠BCA ≅ ∠DAC (?)
3. AC = CA (?)
4. △ABC ≅ △CDA (?)
5. AB = CD and BC = DA (?)

Give the five reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5**

852

**QUESTION ID**

## S24-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1: alternate interior angles (AB ∥ DC); 2: alternate interior angles (AD ∥ BC); 3: reflexive property; 4: ASA; 5: corresponding parts of congruent triangles.**

**GRADING NOTE** Any equivalent names for all five reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5**

In the proof that opposite sides of parallelogram ABCD are congruent, which congruence criterion is used for triangles ABC and CDA?

**A.** SSA

**B.** ASA

**C.** SAS

**D.** SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

853

**QUESTION ID**

## S24-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. ASA

**GRADING NOTE** Two pairs of alternate interior angles and the shared side between them. SSS would assume what is being proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

SET 24 • HSG-CO.C.11

# BACKWARD 1

Alternate interior and same-side interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

AB ∥ DC and diagonal AC is drawn. Which angle at C is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

855

**QUESTION ID**
**S24-B1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**∠DCA**

**GRADING NOTE** Any equivalent form. Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

**True or false?**

Same-side interior angles formed by a transversal of parallel lines are supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

856

QUESTION ID

**S24-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

In parallelogram ABCD, side AB is a transversal of AD and BC. What is m∠A + m∠B?

**A.** It depends on the figure

**B.** 90°

**C.** 360°

**D.** 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

857

**QUESTION ID**

## S24-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 180°**

**GRADING NOTE** Same-side interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

In parallelogram ABCD, m∠A = 65°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

858

QUESTION ID

**S24-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**115°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

**True or false?**

Alternate interior angles of parallel lines are supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

859

QUESTION ID

**S24-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

# BACKWARD 2

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Two triangles have two pairs of congruent angles and the congruent sides between them. Which criterion applies?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

861

QUESTION ID

**S24-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**ASA**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

**True or false?**

A diagonal shared by two triangles can be the included side in ASA.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

862

QUESTION ID

**S24-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

In △ABC and △CDA, ∠BAC ≅ ∠DCA, ∠BCA ≅ ∠DAC, and AC is shared. Which correspondence is correct?

**A.** A ↔ D, B ↔ C, C ↔ A

**B.** A ↔ A, B ↔ D, C ↔ C

**C.** A ↔ C, B ↔ D, C ↔ A

**D.** A ↔ C, B ↔ A, C ↔ D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

863

QUESTION ID

**S24-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. A ↔ C, B ↔ D, C ↔ A**

**GRADING NOTE** Match the equal angles: A ↔ C and C ↔ A, so B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

After proving △ABC ≅ △CDA, which side corresponds to AB?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5



**QUESTION ID**

**S24-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**CD**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

**True or false?**

ASA requires the side to be between the two angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

865

QUESTION ID

**S24-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Otherwise it is AAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSG-CO.C.11

# BACKWARD 3

Know the definition of a parallelogram • Grade 5 • Geometry • 5.G.B.3

**True or false?**

A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

867

QUESTION ID

**S24-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

Which quadrilateral is always a parallelogram?

**A.** A trapezoid

**B.** A kite

**C.** Any quadrilateral with one right angle

**D.** A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

868

**QUESTION ID**

## S24-B3-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. A rectangle

**GRADING NOTE** A rectangle has both pairs of opposite sides parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

Name the two pairs of opposite sides of parallelogram PQRS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

869

QUESTION ID

**S24-B3-Q3**

TYPE

Short answer

CORRECT ANSWER

**PQ and RS; QR and SP**

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

**True or false?**

In parallelogram ABCD, sides AB and BC are opposite sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

870

QUESTION ID

**S24-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are consecutive sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

Name the two pairs of opposite angles of parallelogram WXYZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

871

**QUESTION ID**

**S24-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠W and ∠Y; ∠X and ∠Z**

**GRADING NOTE** Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSG-CO.C.11

# FORWARD 1

Prove that the diagonals of a rhombus are perpendicular • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

ABCD is a rhombus (a parallelogram with all four sides congruent), with diagonals meeting at E. Prove that AC ⊥ BD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5**

873

**QUESTION ID**

## S24-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

## AC ⊥ BD

**GRADING NOTE** REQUIRED WORK: Use that the diagonals of a parallelogram bisect each other, then SSS on triangles AEB and AED (or the perpendicular bisector converse).

Any valid proof (paragraph, two-column, or flow). Essential: BE = ED (diagonals of a parallelogram bisect each other); AB = AD; AE = AE; △AEB ≅ △AED (SSS); ∠AEB ≅ ∠AED form a linear pair, so both are 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5**

In rhombus ABCD, diagonals AC = 16 and BD = 12. Find the side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

874

**QUESTION ID**

**S24-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** Any equivalent form. Half-diagonals 8 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

**True or false?**

Each diagonal of a rhombus bisects two of its angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5**

875

QUESTION ID

## S24-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** SSS on the triangles formed by a diagonal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5**

Which property does every rhombus have but not every parallelogram?

**A.** Opposite sides parallel

**B.** Perpendicular diagonals

**C.** Opposite angles congruent

**D.** Diagonals that bisect each other

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

876

QUESTION ID

**S24-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. Perpendicular diagonals**

**GRADING NOTE** The other three hold for every parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

In rhombus PQRS, m∠PQR = 110°. Find m∠PQS, where QS is a diagonal.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

877

**QUESTION ID**

## S24-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**55°**

**GRADING NOTE** Any equivalent form. The diagonal bisects ∠PQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

# FORWARD 2

Prove the parallelogram properties with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), with a > 0 and c > 0. Prove that AB = DC and AD = BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

879

**QUESTION ID**

**S24-F2-Q1**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AB = DC = a and AD = BC = √(b² + c²)**

**GRADING NOTE** REQUIRED WORK: Use the distance formula on each pair of opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = a; DC = |a + b − b| = a; AD = √(b² + c²); BC = √((a + b − a)² + c²) = √(b² + c²).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-F2-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5**

Explain why the vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c) describe a general parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

880

**QUESTION ID**

## S24-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Any parallelogram can be moved (rigid motion) so that A is at the origin and AB lies on the positive x-axis; then D = (b, c) and C = B + (D − A) = (a + b, c).**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

**True or false?**

With vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), slopes show AC ⊥ BD for every choice of a, b, and c.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

881

QUESTION ID
**S24-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** The product of the slopes is $c^2/(b^2 - a^2)$, which is $-1$ only when $a^2 = b^2 + c^2$ (a rhombus).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

For the parallelogram A(0, 0), B(6, 0), C(8, 4), D(2, 4), what is AD?

**A.** √20 + 2

**B.** 4

**C.** 2√5

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

882

QUESTION ID

**S24-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 2√5**

**GRADING NOTE** √(4 + 16) = √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

Parallelogram ABCD has A(1, 1), B(5, 1), and D(2, 4). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

883

**QUESTION ID**

## S24-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**(6, 4)**

**GRADING NOTE** Form required: an ordered pair. C = B + (D − A).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-F2-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5**

SET 25 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

# Prove that the diagonals of a parallelogram bisect each other

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.11** |
| **BACKWARD 1** | Opposite sides of a parallelogram are congruent | **HSG-CO.C.11** |
| **BACKWARD 2** | Vertical and alternate interior angles at the crossing of the diago… | **HSG-CO.C.9** |
| **BACKWARD 3** | Find the midpoint of a segment and use the definition of bisect | **6.G.A.3** |
| **FORWARD 1** | Prove that a 180° rotation carries a parallelogram onto itself | **HSG-CO.A.3** |
| **FORWARD 2** | Use diagonal bisection with coordinates and algebra | **HSG-GPE.B.4** |

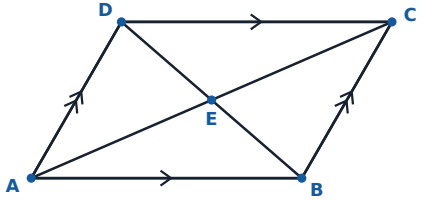
SET 25 • HSG-CO.C.11

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram with diagonals AC and BD meeting at E. Prove that AE = CE and BE = DE.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5**

886

**QUESTION ID**

## S25-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AE = CE and BE = DE**

**GRADING NOTE** REQUIRED WORK: Prove △ABE ≅ △CDE by ASA (or AAS), using alternate interior angles and AB = CD.

Any valid proof (paragraph, two-column, or flow). Essential: AB = CD (opposite sides of a parallelogram); ∠BAE ≅ ∠DCE and ∠ABE ≅ ∠CDE (alternate interior, AB ∥ DC); △ABE ≅ △CDE (ASA); corresponding sides. Vertical angles at E may replace one angle pair (AAS). Invalid: assuming E is a midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

PQRS is a parallelogram with diagonals PR and QS meeting at T. Prove that T is the midpoint of both diagonals.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

887

QUESTION ID

**S25-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**PT = TR and QT = TS, so T is the midpoint of both diagonals.**

GRADING NOTE REQUIRED WORK: Prove △PTQ ≅ △RTS, using alternate interior angles and PQ = RS.

Any valid proof (paragraph, two-column, or flow). Essential: PQ = RS; two pairs of alternate interior angles (PQ ∥ SR); ASA; corresponding sides; definition of midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c). Prove that its diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5**

888

QUESTION ID

## S25-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

**Both diagonals have midpoint ((a + b)/2, c/2).**

GRADING NOTE

REQUIRED WORK: Show the two diagonals have the same midpoint.

Any valid proof (paragraph, two-column, or flow). Essential: midpoint of AC = ((a + b)/2, c/2); midpoint of BD = ((a + b)/2, c/2); the same point lies on both diagonals and is the midpoint of each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A proof that the diagonals of parallelogram ABCD bisect each other (they meet at E):

1. AB = CD (?)
2. ∠BAE ≅ ∠DCE (?)
3. ∠AEB ≅ ∠CED (?)
4. △ABE ≅ △CDE (?)
5. AE = CE, BE = DE (?)

Give the five reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5**

889

**QUESTION ID**

## S25-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1: opposite sides of a parallelogram are congruent; 2: alternate interior angles (AB ∥ DC); 3: vertical angles; 4: AAS; 5: corresponding parts.**

**GRADING NOTE** Any equivalent names for all five reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5**

Which previously proved fact does the proof that a parallelogram's diagonals bisect each other use?

**A.** Consecutive sides are congruent

**B.** Diagonals are perpendicular

**C.** Diagonals of a parallelogram are congruent

**D.** Opposite sides of a parallelogram are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

890

**QUESTION ID**

## S25-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Opposite sides of a parallelogram are congruent**

**GRADING NOTE** AB = CD is needed for the congruence. The other facts are false for general parallelograms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

SET 25 • HSG-CO.C.11

# BACKWARD 1

Opposite sides of a parallelogram are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

In parallelogram ABCD, AB = 9 and BC = 5. Find CD and DA.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

892

QUESTION ID

**S25-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**CD = 9, DA = 5**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

**True or false?**

Opposite sides of every parallelogram are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

893

QUESTION ID

**S25-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Proved in the previous set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In parallelogram WXYZ, WX = 2x + 1 and YZ = x + 6. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

894

QUESTION ID

**S25-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**5**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

In parallelogram PQRS, which side is congruent to QR?

**A.** PQ

**B.** PR

**C.** SP

**D.** RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

895

QUESTION ID

**S25-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. SP**

GRADING NOTE Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

**True or false?**

Consecutive sides of every parallelogram are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

896

QUESTION ID

**S25-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only in a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

# BACKWARD 2

Vertical and alternate interior angles at the crossing of the diagonals • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Diagonals of parallelogram ABCD meet at E. Which angle is vertical to ∠AEB?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

898

QUESTION ID
**S25-B2-Q1**

TYPE
Short answer

CORRECT ANSWER
**∠CED**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

AB ∥ DC with transversal AC. Which angle is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

899

**QUESTION ID**

**S25-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠DCA**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

**True or false?**

Vertical angles are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

900

QUESTION ID

**S25-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Proved earlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

AB ∥ DC with transversal BD. Which angle is congruent to ∠ABD?

**A.** ∠CDB

**B.** ∠ABC

**C.** ∠ADB

**D.** ∠CBD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

901

QUESTION ID

**S25-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. ∠CDB**

GRADING NOTE Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

**True or false?**

∠AEB and ∠BEC are vertical angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

902

QUESTION ID

**S25-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They form a linear pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSG-CO.C.11

# BACKWARD 3

Find the midpoint of a segment and use the definition of bisect • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (0, 0) to (a + b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

904

**QUESTION ID**

**S25-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**((a + b)/2, c/2)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

Find the midpoint of the segment from (a, 0) to (b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

905

**QUESTION ID**

## S25-B3-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**((a + b)/2, c/2)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

**True or false?**

A segment bisects another segment if it passes through its midpoint.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

**Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5**

906

**QUESTION ID**

## S25-B3-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Definition of bisect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

**Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Segment AC = 18 is bisected at E. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

907

**QUESTION ID**

## S25-B3-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

**True or false?**

If two segments have the same midpoint, each bisects the other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

**Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5**

908

QUESTION ID

**S25-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Their common point is the midpoint of both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

**Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5**

SET 25 • HSG-CO.C.11

# FORWARD 1

Prove that a 180° rotation carries a parallelogram onto itself • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

ABCD is a parallelogram whose diagonals meet at E. Prove that the 180° rotation about E maps ABCD onto itself.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

910

**QUESTION ID**

## S25-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**A ↔ C and B ↔ D, so the parallelogram maps onto itself.**

**GRADING NOTE** REQUIRED WORK: Use the fact that E is the midpoint of both diagonals to find the images of the vertices.

Any valid proof (paragraph, two-column, or flow). Essential: a 180° rotation about E maps each point P to the point P′ with E the midpoint of PP′; E is the midpoint of AC and BD, so A → C, C → A, B → D, D → B; segments map to segments, so each side maps onto its opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

Parallelogram ABCD has diagonals meeting at E(2, 3), and A(−1, 1). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

911

**QUESTION ID**

## S25-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(5, 5)**

**GRADING NOTE** Form required: an ordered pair. E is the midpoint of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

**True or false?**

A quadrilateral whose diagonals bisect each other has 180° rotation symmetry about their crossing point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

912

QUESTION ID

**S25-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The rotation swaps opposite vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

The 180° rotation about the center of parallelogram PQRS maps side PQ onto

**A.** side SP

**B.** side RS

**C.** side QR

**D.** diagonal PR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

913

QUESTION ID

**S25-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. side RS**

GRADING NOTE P → R and Q → S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

A 180° rotation about the intersection of its diagonals carries a parallelogram onto itself. Explain why this implies that opposite angles of a parallelogram are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

914

**QUESTION ID**

## S25-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The rotation maps each angle onto the opposite angle, and rotations keep angle measures.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

# FORWARD 2

Use diagonal bisection with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(7, 2), C(9, 6). Find D and the point where the diagonals meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

916

**QUESTION ID**

## S25-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**D(3, 6); diagonals meet at (5, 4)**

**GRADING NOTE** Form required: both points. Midpoint of AC = (5, 4) = midpoint of BD.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F2-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5**

In parallelogram ABCD, diagonals meet at E.
AE = 3x − 2 and CE = x + 8. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

917

**QUESTION ID**
**S25-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**26**

**GRADING NOTE** Any equivalent form. 3x − 2 = x + 8 gives x = 5; AE = 13.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F2-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5**

**True or false?**

The diagonals of every parallelogram are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5



QUESTION ID

## S25-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** They bisect each other but are congruent only in a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

Three vertices of a parallelogram are (0, 0), (4, 0), (1, 3). Which could be the fourth vertex?

**A.** (4, 3)

**B.** (3, 3)

**C.** (5, 3)

**D.** (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

919

**QUESTION ID**

**S25-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (5, 3)**

**GRADING NOTE** (0, 0), (4, 0), (5, 3), (1, 3) is a parallelogram (others are possible: (3, −3), (−3, 3)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

In parallelogram PQRS, diagonals meet at T. PT = y + 4, TR = 2y − 1, QT = 3y. Find QS.

QS = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

920

**QUESTION ID**

**S25-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**30**

**GRADING NOTE** Any equivalent form. y = 5, QT = 15.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F2-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5**

SET 26 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

# Prove the converses: conditions that make a quadrilateral a parallelogram

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.11** |
| **BACKWARD 1** | Converses of the parallel-line angle theorems | **HSG-CO.C.9** |
| **BACKWARD 2** | Prove triangles congruent with SSS and SAS | **HSG-CO.B.8** |
| **BACKWARD 3** | Angle sum of a quadrilateral | **8.G.A.5** |
| **FORWARD 1** | Prove that a quadrilateral is a parallelogram using coordinates | **HSG-GPE.B.4** |
| **FORWARD 2** | Decide which conditions are enough to guarantee a parallelogram | **HSG-CO.C.11** |

SET 26 • HSG-CO.C.11

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB = CD and AD = CB. Prove that ABCD is a parallelogram. (Draw diagonal AC.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

923

**QUESTION ID**

## S26-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AB ∥ DC and AD ∥ BC, so ABCD is a parallelogram.**

**GRADING NOTE** REQUIRED WORK: Prove △ABC ≅ △CDA by SSS, then use the converse of the alternate interior angles theorem twice.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = CD, BC = DA, AC shared); ∠BAC ≅ ∠DCA, so AB ∥ DC (converse); ∠BCA ≅ ∠DAC, so BC ∥ AD; definition. Invalid: using properties of parallelograms before proving it is one.

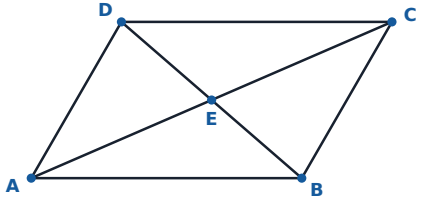
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In quadrilateral ABCD, diagonals AC and BD bisect each other at E. Prove that ABCD is a parallelogram.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

924

QUESTION ID

## S26-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

**AB ∥ DC and AD ∥ BC, so ABCD is a parallelogram.**

**GRADING NOTE** REQUIRED WORK: Prove △AEB ≅ △CED and △AED ≅ △CEB by SAS (vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AE = CE, BE = DE (bisect); vertical angles; SAS twice; ∠EAB ≅ ∠ECD gives AB ∥ DC; ∠EAD ≅ ∠ECB gives AD ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

925

**QUESTION ID**

## S26-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AD ∥ BC, so with AB ∥ DC, ABCD is a parallelogram.**

**GRADING NOTE** REQUIRED WORK: Draw diagonal AC; prove △ABC ≅ △CDA by SAS, then show AD ∥ BC.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA (alternate interior, AB ∥ DC); AB = CD; AC shared; SAS; ∠BCA ≅ ∠DAC; AD ∥ BC (converse).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In quadrilateral ABCD, ∠A ≅ ∠C and ∠B ≅ ∠D. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

926

**QUESTION ID**

## S26-M-Q4

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AD ∥ BC and AB ∥ DC.**

**GRADING NOTE** REQUIRED WORK: Use the quadrilateral angle sum to show consecutive angles are supplementary, then the converse of the same-side interior angles theorem.

Any valid proof (paragraph, two-column, or flow). Essential: $2m\angle A + 2m\angle B = 360°$, so $m\angle A + m\angle B = 180°$; same-side interior angles supplementary gives AD ∥ BC; similarly $m\angle A + m\angle D = 180°$ gives AB ∥ DC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

Quadrilateral ABCD has AB ∥ DC and AD = BC. Which conclusion is justified?

**A.** ABCD must be a rhombus

**B.** ABCD may not be a parallelogram (it could be an isosceles trapezoid)

**C.** ABCD must be a rectangle

**D.** ABCD must be a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

927

QUESTION ID

**S26-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. ABCD may not be a parallelogram (it could be an isosceles trapezoid)**

**GRADING NOTE** An isosceles trapezoid has one pair of parallel sides and the other pair congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

SET 26 • HSG-CO.C.11

# BACKWARD 1

Converses of the parallel-line angle theorems • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

Two lines are cut by a transversal so that alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

929

QUESTION ID

**S26-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**The lines are parallel.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

**True or false?**

If same-side interior angles are supplementary, the lines are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

930

QUESTION ID

## S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

Which fact proves two lines are parallel?

**A.** Congruent corresponding angles

**B.** A linear pair

**C.** Two acute angles

**D.** Congruent vertical angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

931

QUESTION ID

**S26-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. Congruent corresponding angles**

GRADING NOTE Corresponding angles converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

Same-side interior angles measure 104° and 76°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

932

**QUESTION ID**

## S26-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes (they add to 180°).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

**True or false?**

If alternate interior angles are 40° and 50°, the lines are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

933

QUESTION ID

**S26-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They must be congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

# BACKWARD 2

Prove triangles congruent with SSS and SAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

△ABC and △CDA share AC, and AB = CD, BC = DA.
Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

935

QUESTION ID

**S26-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**SSS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△AEB and △CED: AE = CE, BE = DE, and ∠AEB ≅ ∠CED. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

936

**QUESTION ID**
**S26-B2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**SAS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

**True or false?**

Vertical angles can serve as the included angle in SAS.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

937

QUESTION ID

**S26-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

After △ABC ≅ △CDA, which angle is congruent to ∠BAC?

**A.** ∠BCA

**B.** ∠ABC

**C.** ∠DAC

**D.** ∠DCA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

938

QUESTION ID

**S26-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. ∠DCA**

**GRADING NOTE** A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

**True or false?**

SSA proves two triangles congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

939

QUESTION ID

## S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

GRADING NOTE Not a valid criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSG-CO.C.11

# BACKWARD 3

Angle sum of a quadrilateral • Grade 8 • Geometry • 8.G.A.5

Three angles of a quadrilateral are 80°, 95°, and 110°. Find the fourth.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

941

QUESTION ID

**S26-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**75°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

**True or false?**

The angles of every quadrilateral add to 360°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

942

QUESTION ID

**S26-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE A diagonal splits it into two triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

A quadrilateral has angles x, x, 2x, 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

943

**QUESTION ID**

**S26-B3-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**60**

**GRADING NOTE** Any equivalent form. 6x = 360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

Opposite angles of a quadrilateral are 70° and 70°, and another is 110°. Find the fourth.

**A.** 90°

**B.** 70°

**C.** 110°

**D.** 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

944

**QUESTION ID**

**S26-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 110°**

**GRADING NOTE** 360 − 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

**True or false?**

A quadrilateral can have four acute angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

945

QUESTION ID

**S26-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Four angles under 90° add to less than 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSG-CO.C.11

# FORWARD 1

Prove that a quadrilateral is a parallelogram using coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(6, 2), C(8, 6), D(3, 5) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

947

QUESTION ID

## S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**It is a parallelogram: the diagonals AC and BD have the same midpoint (4.5, 3.5), so they bisect each other.**

**GRADING NOTE** Any valid method (slopes of opposite sides, or opposite side lengths, or diagonal midpoints) with the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 1), R(5, 4), S(0, 3) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

948

QUESTION ID

## S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**Not a parallelogram: PQ has slope 1/4 but SR has slope 1/5, so PQ is not parallel to SR.**

**GRADING NOTE** Any valid evidence and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

**True or false?**

If the midpoints of both diagonals of a quadrilateral are the same point, the quadrilateral is a parallelogram.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

949

QUESTION ID
**S26-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

Which test alone proves that ABCD is a parallelogram?

**A.** AC and BD have equal lengths

**B.** AB ⊥ BC

**C.** AB and DC have equal slopes and equal lengths

**D.** AB and DC have equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

950

**QUESTION ID**

## S26-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. AB and DC have equal slopes and equal lengths**

**GRADING NOTE** One pair of opposite sides parallel and congruent.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5**

A(0, 0), B(5, 0), C(7, 3), D(x, 3) form a parallelogram ABCD. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

951

**QUESTION ID**

## S26-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**2**

**GRADING NOTE** Any equivalent form. DC must equal AB = 5.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5**

SET 26 • HSG-CO.C.11

# FORWARD 2

Decide which conditions are enough to guarantee a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Is a quadrilateral with one pair of opposite sides congruent and the other pair parallel always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

953

QUESTION ID

**S26-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**No. An isosceles trapezoid is a counterexample.**

**GRADING NOTE** Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

Is a quadrilateral with both pairs of opposite angles congruent always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

954

**QUESTION ID**

## S26-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes. The angle sum makes consecutive angles supplementary, so both pairs of opposite sides are parallel.**

**GRADING NOTE** Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

**True or false?**

A quadrilateral with one diagonal that bisects the other must be a parallelogram.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

955

QUESTION ID

**S26-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A kite has one diagonal bisecting the other but is not a parallelogram in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

Which condition is NOT enough to prove a quadrilateral is a parallelogram?

**A.** One pair of opposite sides congruent

**B.** Both pairs of opposite sides congruent

**C.** Diagonals bisect each other

**D.** One pair of opposite sides parallel and congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

956

**QUESTION ID**

## S26-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. One pair of opposite sides congruent**

**GRADING NOTE** One pair of congruent sides alone allows many shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

Is a quadrilateral with perpendicular diagonals always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

957

**QUESTION ID**

## S26-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**No. A kite (not a rhombus) has perpendicular diagonals but is not a parallelogram.**

**GRADING NOTE** Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

# Prove that rectangles are parallelograms with congruent diagonals

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.C.11** |
| **BACKWARD 1** | Parallelogram properties: opposite sides and consecutive angles | **HSG-CO.C.11** |
| **BACKWARD 2** | SAS and SSS with overlapping triangles | **HSG-CO.B.8** |
| **BACKWARD 3** | Use the Pythagorean Theorem for diagonals | **8.G.B.7** |
| **FORWARD 1** | Decide whether four points form a rectangle | **HSG-GPE.B.4** |
| **FORWARD 2** | Prove diagonal properties of rhombuses and squares | **HSG-CO.C.11** |

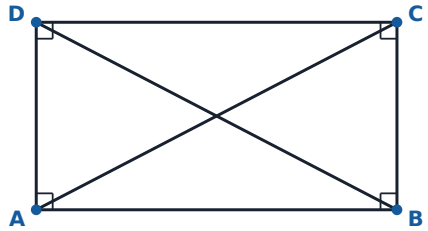
SET 27 • HSG-CO.C.11

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rectangle. Prove that AC = BD.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

960

**QUESTION ID**

## S27-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AC = BD**

**GRADING NOTE** REQUIRED WORK: Prove △ABC ≅ △BAD by SAS (right angles, shared side AB, and AD = BC).

Any valid proof (paragraph, two-column, or flow). Essential: a rectangle is a parallelogram, so AD = BC; ∠ABC ≅ ∠BAD (right angles); AB shared; SAS; corresponding sides AC = BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5**

ABCD is a parallelogram with AC = BD. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

961

QUESTION ID

**S27-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**All four angles are right angles, so ABCD is a rectangle.**

**GRADING NOTE** REQUIRED WORK: Prove △ABC ≅ △BAD by SSS, show ∠A ≅ ∠B, then use that consecutive angles are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: BC = AD (opposite sides); AC = BD (given); AB shared; SSS; ∠DAB ≅ ∠CBA; they are supplementary (consecutive angles), so each is 90°; opposite angles then are 90° too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

Rectangle ABCD has A(0, 0), B(a, 0), C(a, b), D(0, b), with a, b > 0. Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5**

962

**QUESTION ID**

## S27-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$AC = BD = \sqrt{(a^2 + b^2)}$**

**GRADING NOTE** REQUIRED WORK: Use the distance formula on both diagonals.

Any valid proof (paragraph, two-column, or flow). Essential: $AC = \sqrt{(a^2 + b^2)}$ and $BD = \sqrt{(a^2 + b^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5**

Explain why a rectangle is a parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

963

**QUESTION ID**

## S27-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Its opposite angles are congruent (all 90°), so it is a parallelogram; or: both pairs of opposite sides are perpendicular to the same side, so they are parallel.**

**GRADING NOTE** Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

Which statement is the converse proved in this set?

**A.** Every rectangle has congruent diagonals

**B.** Every quadrilateral with congruent diagonals is a rectangle

**C.** A parallelogram with congruent diagonals is a rectangle

**D.** A rectangle has perpendicular diagonals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

964

**QUESTION ID**

## S27-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. A parallelogram with congruent diagonals is a rectangle**

**GRADING NOTE** The converse needs the quadrilateral to be a parallelogram; an isosceles trapezoid also has congruent diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

SET 27 • HSG-CO.C.11

# BACKWARD 1

Parallelogram properties: opposite sides and consecutive angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = 90°. Find the other three angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

966

QUESTION ID

**S27-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**90°, 90°, 90°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

**True or false?**

Consecutive angles of a parallelogram are supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

967

QUESTION ID

**S27-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Same-side interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

In parallelogram PQRS, m∠P = m∠Q. Find m∠P.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

968

**QUESTION ID**

## S27-B1-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form. They are also supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

In parallelogram ABCD, which side is congruent to AD?

**A.** AC

**B.** CD

**C.** AB

**D.** BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

969

QUESTION ID

**S27-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. BC**

GRADING NOTE Opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

**True or false?**

In every parallelogram, all angles are equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

970

QUESTION ID

**S27-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only in rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

# BACKWARD 2

SAS and SSS with overlapping triangles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △BAD overlap and share AB. Which corresponding angle is at A in △BAD for ∠B in △ABC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5



**QUESTION ID**

**S27-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠BAD**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

**True or false?**

Overlapping triangles can share a side used in a congruence proof.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

973

QUESTION ID

**S27-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

△ABC and △BAD: AB shared, BC = AD, AC = BD. Which criterion?

**A.** SSS

**B.** ASA

**C.** SAS

**D.** AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

974

**QUESTION ID**

**S27-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. SSS**

**GRADING NOTE** Three pairs of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ABC and △BAD: AB shared, BC = AD, ∠ABC ≅ ∠BAD. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

975

**QUESTION ID**
**S27-B2-Q4**

**TYPE**
Short answer

**CORRECT ANSWER**
**SAS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

**True or false?**

In △ABC ≅ △BAD, vertex C corresponds to vertex C.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

976

QUESTION ID
**S27-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** C ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSG-CO.C.11

# BACKWARD 3

Use the Pythagorean Theorem for diagonals • Grade 8 • Geometry • 8.G.B.7

A rectangle is 6 by 8. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

978

**QUESTION ID**

## S27-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

A rectangle is 5 by 12. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

979

**QUESTION ID**

## S27-B3-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

**True or false?**

A rectangle with sides a and b has diagonal $\sqrt{(a^2 + b^2)}$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

980

QUESTION ID

**S27-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

A square has side 4. Its diagonal is

**A.** 8

**B.** 4√2

**C.** 2√2

**D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

981

QUESTION ID

**S27-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 4√2**

GRADING NOTE √32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

**True or false?**

The two diagonals of a 3-by-7 rectangle have different lengths.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

982

QUESTION ID

**S27-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Both are √58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSG-CO.C.11

# FORWARD 1

Decide whether four points form a rectangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(5, 3), C(4, 5), D(0, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

984

QUESTION ID

**S27-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Rectangle: it is a parallelogram (AB ∥ DC, slope ½; AD ∥ BC, slope −2) and AB ⊥ AD (½ · −2 = −1).**

**GRADING NOTE** Any valid method (or: diagonals bisect each other and are congruent, both 5 units long).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 0), R(5, 3), S(1, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

985

**QUESTION ID**

## S27-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Not a rectangle: it is a parallelogram, but its diagonals PR = √34 and QS = √18 are not congruent.**

**GRADING NOTE** Any valid evidence and the conclusion.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5**

**True or false?**

A quadrilateral whose diagonals are congruent and bisect each other is a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

986

QUESTION ID
**S27-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Parallelogram with congruent diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

Which set of facts proves ABCD is a rectangle?

**A.** Diagonals bisect each other and are congruent

**B.** Diagonals are congruent

**C.** One angle is a right angle

**D.** Opposite sides are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

987

**QUESTION ID**

## S27-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. Diagonals bisect each other and are congruent**

**GRADING NOTE** Congruent diagonals alone fit an isosceles trapezoid.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F1-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5**

A(0, 0), B(6, 0), and D(0, 4) are three vertices of rectangle ABCD. Find C and verify the diagonals are congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

988

**QUESTION ID**

**S27-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**C(6, 4);**
**AC = BD = √52**
**= 2√13.**

**GRADING NOTE** Form required: C and both diagonal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

# FORWARD 2

Prove diagonal properties of rhombuses and squares • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Prove that a parallelogram whose diagonals are perpendicular is a rhombus.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

990

QUESTION ID

**S27-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**All four sides are congruent.**

GRADING NOTE REQUIRED WORK: Use that the diagonals bisect each other, then SAS on two adjacent triangles.

Any valid proof (paragraph, two-column, or flow). Essential: diagonals bisect at E; right angles at E; shared segment; SAS gives AB = AD; opposite sides give the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

Which property must a quadrilateral have to be a square, using diagonals only?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

991

QUESTION ID

**S27-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Its diagonals bisect each other, are congruent, and are perpendicular.**

**GRADING NOTE** Any equivalent complete list.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

**True or false?**

A rectangle with congruent diagonals is a square.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

992

QUESTION ID

**S27-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Every rectangle has congruent diagonals; a square needs perpendicular diagonals too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

A parallelogram has congruent, perpendicular diagonals. It is a

**A.** rectangle that is not a square

**B.** rhombus that is not a square

**C.** square

**D.** trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

993

**QUESTION ID**

**S27-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. square**

**GRADING NOTE** Rectangle (congruent) and rhombus (perpendicular).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

A square has diagonal 10. Find its side length (exact) and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

994

**QUESTION ID**

## S27-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Side $5\sqrt{2}$; area 50**

**GRADING NOTE** Form required: exact side and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

# Construct a copy of a segment

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.12** |
| **BACKWARD 1** | Compare and measure lengths | **2.MD.A.1** |
| **BACKWARD 2** | A compass draws all points at one distance from its center | **7.G.B.4** |
| **BACKWARD 3** | Know rays, segments, and their endpoints | **HSG-CO.A.1** |
| **FORWARD 1** | Construct a triangle from three given segments | **HSG-CO.D.12** |
| **FORWARD 2** | Decide whether segments can form a triangle, and construct sums | **HSG-CO.D.12** |

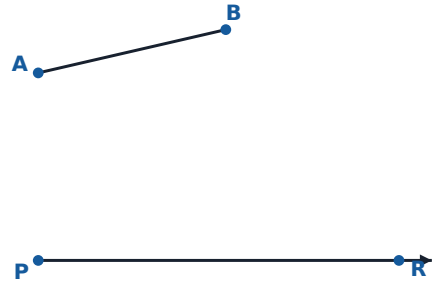
SET 28 • HSG-CO.D.12

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = AB.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

997

QUESTION ID

## S28-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**A compass arc centered at P with radius AB crosses ray PR at Q; PQ = AB.**

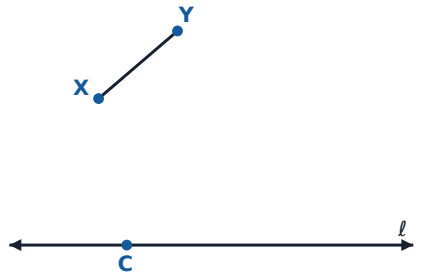
**GRADING NOTE** Grade the construction marks, not only the final figure: the compass set to AB (an arc or tick at A and B is allowed) and an arc centered at P meeting the ray at Q, which is labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

On line $\ell$, starting at point C, construct a segment CD with CD = XY.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

998

**QUESTION ID**

## S28-M-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**An arc centered at C with radius XY crosses $\ell$ at D (either side of C is correct).**

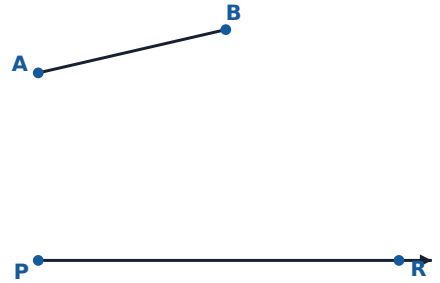
**GRADING NOTE** Grade the construction marks, not only the final figure: compass opening equal to XY, arc centered at C, D marked on $\ell$. Either intersection point is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = 2 · AB.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3
High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5



**QUESTION ID**
## S28-M-Q3

**TYPE**
Drawing / plotting

**CORRECT ANSWER**

**Copy AB from P to get M on the ray, then copy AB again from M (away from P) to get Q; PQ = 2AB.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two arcs of radius AB, one centered at P and one centered at the first point, with Q labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3
High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the steps for copying segment AB onto ray PR, starting at P, with a compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

1000

QUESTION ID

**S28-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**1. Put the compass point on A and open it to B. 2. Without changing the opening, put the compass point on P. 3. Draw an arc that crosses ray PR; label the crossing Q. Then PQ = AB.**

GRADING NOTE Any equivalent steps. Essential: the compass is set to AB and is not changed, the arc is centered at P, and the crossing is labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

1000

**ANSWER KEY** Reduced copy of the question page, with its answer

You have no compass, only a straightedge and a strip of paper. Describe how to copy segment AB onto line $\ell$ starting at point C.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

1001

**QUESTION ID**

## S28-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Lay the strip along AB and mark the positions of A and B on its edge. Move the strip to $\ell$, put the A-mark on C along $\ell$, and mark the point at the B-mark: that is D, with CD = AB.**

**GRADING NOTE** Any valid method using the paper strip (or string) to transfer the length without measuring it with a ruler.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

SET 28 • HSG-CO.D.12

# BACKWARD 1

Compare and measure lengths • Grade 2 • Measurement & Data • 2.MD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Segment AB is 6 cm and segment CD is 6 cm. Are they congruent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

1003

QUESTION ID

**S28-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Yes**

GRADING NOTE Any equivalent answer: equal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

**True or false?**

Two segments are congruent if they have the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

1004

QUESTION ID

## S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Definition of congruent segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

Which tool transfers a length without reading numbers?

**A.** A compass

**B.** A calculator

**C.** A protractor

**D.** A thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

1005

QUESTION ID

**S28-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. A compass**

**GRADING NOTE** The compass keeps a fixed opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A segment 4.5 cm long is copied twice end to end on a line. How long is the result?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

1006

QUESTION ID

**S28-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**9 cm**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A segment drawn longer on the page than another is congruent to it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

1007

QUESTION ID

**S28-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Congruent segments have equal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

# BACKWARD 2

A compass draws all points at one distance from its center • Grade 7 • Geometry • 7.G.B.4

A compass is opened to 5 cm with its point on P. How far from P is every point on the arc it draws?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

1009

**QUESTION ID**

## S28-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**5 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

**True or false?**

Changing the compass opening between steps of a construction keeps the copy accurate.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5**

1010

**QUESTION ID**

## S28-B2-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** The opening must stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5**

An arc centered at P crosses a ray from P at Q. What is PQ?

**A.** It cannot be known

**B.** Half the opening

**C.** Twice the opening

**D.** The compass opening

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

1011

**QUESTION ID**

## S28-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. The compass opening

**GRADING NOTE** Q is on the arc, so it is one radius from P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

**True or false?**

Every point on a circle is the same distance from the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

1012

QUESTION ID

**S28-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An arc centered at C with radius 3 cm crosses a line through C. How many crossing points are there?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

1013

QUESTION ID

**S28-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSG-CO.D.12

# BACKWARD 3

Know rays, segments, and their endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**True or false?**

Ray PR starts at P and continues through R without end.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

1015

QUESTION ID

**S28-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition of a ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Q is on ray PR with PQ = 5. Must Q be between P and R?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

1016

QUESTION ID

**S28-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**No. It is between P and R only if PR > 5; otherwise R is between P and Q.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

Segment AB has endpoints

**A.** none

**B.** only B

**C.** only A

**D.** A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

1017

**QUESTION ID**

**S28-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. A and B**

**GRADING NOTE** A segment has two endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

**True or false?**

A point on ray PR can be on the opposite side of P from R.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

1018

QUESTION ID
**S28-B3-Q4**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** The ray goes only from P toward R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name the endpoint of ray PR.

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

1019

QUESTION ID

**S28-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**P**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

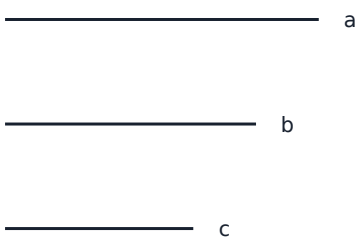
SET 28 • HSG-CO.D.12

# FORWARD 1

Construct a triangle from three given segments • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a triangle whose sides are congruent to the three segments a, b, and c.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

1021

**QUESTION ID**

## S28-F1-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Copy a as base XY; arcs centered at X with radius b and at Y with radius c meet at Z; triangle XYZ has sides a, b, c.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the copied base, two arcs with radii b and c, and their intersection joined to the base endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the two arcs in the triangle construction locate the third vertex.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

1022

QUESTION ID

**S28-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**The third vertex must be b from one end of the base and c from the other, so it lies on both circles, where the arcs cross.**

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

**True or false?**

Any three segments can be used to construct a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

1023

QUESTION ID

**S28-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The two arcs meet only if each length is less than the sum of the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

To construct a triangle from three given segments, arcs are drawn from the two ends of the base, and they cross at two points, one on each side of the base. The two triangles formed are

**A.** similar but not congruent

**B.** different in size

**C.** not triangles

**D.** congruent (mirror images)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

1024

**QUESTION ID**

## S28-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. congruent (mirror images)**

**GRADING NOTE** Reflection across the base swaps them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

Segments of length 3, 4, and 6 are given. Can a triangle be constructed? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

1025

**QUESTION ID**

## S28-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes. 3 + 4 > 6, so the arcs meet.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

# FORWARD 2

Decide whether segments can form a triangle, and construct sums and differences • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

**ANSWER KEY** Reduced copy of the question page, with its answer

Segments of length 2, 3, and 6 are given. Can a triangle be constructed? Explain what happens to the arcs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

1027

**QUESTION ID**

## S28-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**No. 2 + 3 < 6, so the arcs of radii 2 and 3 centered at the ends of the 6 base do not meet.**

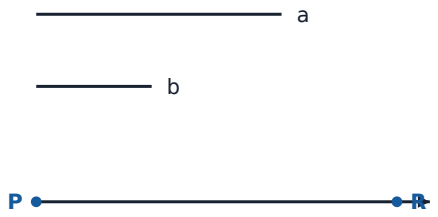
**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

On the ray from P, construct a segment of length a − b (a is the longer segment).




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

1028

**QUESTION ID**

## S28-F2-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Copy a from P to Q on the ray; copy b from Q back toward P to S; PS = a − b.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two compass arcs (radius a at P, radius b at Q toward P), with the result labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

**True or false?**

If the arcs in a triangle construction just touch at one point on the base, the "triangle" is flat.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

1029

QUESTION ID

**S28-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Then one length equals the sum of the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

Which lengths CANNOT form a triangle?

**A.** 1, 2, 3

**B.** 3, 4, 5

**C.** 4, 4, 7

**D.** 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

1030

QUESTION ID

**S28-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 1, 2, 3**

GRADING NOTE 1 + 2 = 3, so the figure is flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

Two sides of a constructed triangle are 5 and 8. For what third lengths will the arcs meet in two points?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

1031

**QUESTION ID**
**S28-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**Any length between 3 and 13 (3 < x < 13).**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

SET 29 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

# Construct a copy of an angle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.12** |
| **BACKWARD 1** | Measure and draw angles with a protractor | **4.MD.C.6** |
| **BACKWARD 2** | Copy a segment with a compass | **HSG-CO.D.12** |
| **BACKWARD 3** | Arcs and circles in constructions | **7.G.B.4** |
| **FORWARD 1** | Justify why the angle-copy construction works | **HSG-CO.B.8** |
| **FORWARD 2** | Use angle copying to construct a parallel line | **HSG-CO.D.12** |

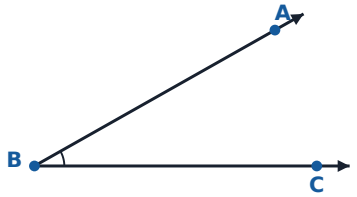
SET 29 • HSG-CO.D.12

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

1034

**QUESTION ID**

## S29-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Arc centered at B meets the sides at D and E; same arc centered at P meets ray PR at F; arc centered at F with radius DE meets the first arc at G; ray PG.**

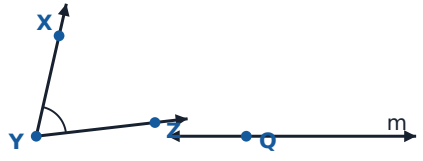
**GRADING NOTE** Grade the construction marks, not only the final figure: the arc at B, the same-radius arc at P, the arc with radius DE centered on ray PR, and ray PG drawn. The copy may open above or below ray PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an angle at point Q on line m congruent to ∠XYZ, opening upward.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

1035

QUESTION ID

## S29-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Same steps as copying an angle: arcs of the same radius at Y and Q, then the chord length transferred; the new ray from Q opens upward.**

**GRADING NOTE** Grade the construction marks, not only the final figure: equal-radius arcs at Y and Q and the transferred chord arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

Write the steps to copy ∠ABC so that its vertex is at P and one side is ray PR.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

1036

QUESTION ID

**S29-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**1. Arc centered at B crossing BA at D and BC at E. 2. Same arc centered at P crossing ray PR at F. 3. Set the compass to DE. 4. Arc centered at F crossing the second arc at G. 5. Draw ray PG.**

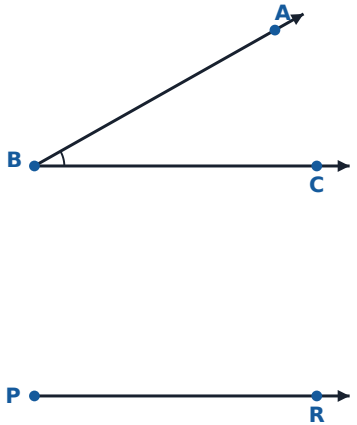
**GRADING NOTE** Any equivalent steps with all five actions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an angle at P on ray PR whose measure is twice m∠ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

1037

QUESTION ID

## S29-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**Copy ∠ABC onto ray PR to get ray PG, then copy ∠ABC again onto ray PG (on the other side from R) to get ray PH; ∠RPH = 2∠ABC.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two complete angle copies sharing ray PG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

**True or false?**

When copying an angle, the first arcs at B and at P must have the same radius.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

1038

QUESTION ID

**S29-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The chord length DE transfers the angle only if the two arcs have equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

SET 29 • HSG-CO.D.12

# BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor shows a ray at 0° and the other ray at 65°. What is the angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

1040

**QUESTION ID**

**S29-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**65°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

**True or false?**

An angle's measure does not change when its sides are drawn longer.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

**Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5**

1041

QUESTION ID

## S29-B1-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Measure depends only on the opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

**Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Which angle is obtuse?

**A.** 90°

**B.** 45°

**C.** 130°

**D.** 20°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

1042

QUESTION ID

**S29-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 130°**

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

Two angles of 35° and 35° are placed side by side. What is the combined angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

1043

**QUESTION ID**

## S29-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**70°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

**True or false?**

Two angles with sides of different lengths cannot be congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

1044

**QUESTION ID**

## S29-B1-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Only the measures matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSG-CO.D.12

# BACKWARD 2

Copy a segment with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which length is transferred with the compass in the fourth step?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

1046

QUESTION ID

**S29-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**The chord DE between the arc crossings on the two sides**

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

**True or false?**

To copy a length, set the compass to the length and do not change it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

1047

QUESTION ID

**S29-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Fixed opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

An arc centered at F with radius DE crosses another arc at G. What is FG?

**A.** Half of DE

**B.** BD

**C.** DE

**D.** PF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

1048

QUESTION ID

**S29-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. DE**

GRADING NOTE G is on the circle about F with radius DE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Arcs centered at B and at P are both drawn with radius 3 cm. How far is F (on the arc at P) from P?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

1049

**QUESTION ID**
**S29-B2-Q4**

**TYPE**
Short answer

**CORRECT ANSWER**
**3 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

**True or false?**

A segment copy can be checked only by measuring with a ruler.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

1050

QUESTION ID

**S29-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Compass marks are the evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSG-CO.D.12

# BACKWARD 3

Arcs and circles in constructions • Grade 7 • Geometry • 7.G.B.4

**True or false?**

Two arcs centered at different points can cross at two points.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

1052

QUESTION ID

**S29-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Two circles can meet twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

Points D and E are both on an arc centered at B with radius 4. Find BD and BE.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

1053

**QUESTION ID**

**S29-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**BD = BE = 4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

Which describes an arc drawn with a compass?

**A.** Part of a circle

**B.** A ray

**C.** A straight segment

**D.** A polygon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

1054

**QUESTION ID**

## S29-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. Part of a circle

**GRADING NOTE** The compass draws circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

**True or false?**

An arc centered at P passes through P.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

1055

QUESTION ID

**S29-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Points on it are one radius away from P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What does the point of the compass mark in a construction?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

1056

QUESTION ID

**S29-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**The center of the arc**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSG-CO.D.12

# FORWARD 1

Justify why the angle-copy construction works • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

In copying ∠ABC to ∠FPG, BD = BE = PF = PG (same arc radius) and DE = FG (same compass setting). Prove that ∠DBE ≅ ∠FPG.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

1058

**QUESTION ID**

## S29-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**∠DBE ≅ ∠FPG**

**GRADING NOTE** REQUIRED WORK: Prove △DBE ≅ △FPG by SSS, then use corresponding angles.

Any valid proof. Essential: BD = PF, BE = PG, DE = FG (construction); SSS; corresponding angles ∠DBE and ∠FPG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

Which congruence criterion explains the angle-copy construction?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

1059

QUESTION ID

**S29-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**SSS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

**True or false?**

The angle copy would fail if the second arc used a different radius than the first.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5**

1060

QUESTION ID

## S29-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Then the triangles would not have matching sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5**

Which three pairs of segments are congruent in the angle-copy construction?

**A.** BA and PR, BC and PG, AC and RG

**B.** Only DE and FG

**C.** BD and DE, PF and FG, BE and PG

**D.** BD and PF, BE and PG, DE and FG

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

1061

**QUESTION ID**

## S29-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. BD and PF, BE and PG, DE and FG**

**GRADING NOTE** Two radii and the chord.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

Why does the construction not need to measure the angle with a protractor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

1062

**QUESTION ID**

## S29-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**SSS guarantees the triangles, and so the angles, are congruent.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

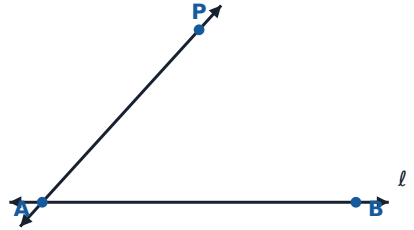
SET 29 • HSG-CO.D.12

# FORWARD 2

Use angle copying to construct a parallel line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line $\ell$ by copying an angle.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

1064

QUESTION ID

## S29-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Copy the angle that line AP makes with $\ell$ at A to vertex P, in the corresponding position; the new side through P is parallel to $\ell$.**

**GRADING NOTE** Grade the construction marks, not only the final figure: a complete angle copy at P (equal-radius arcs and chord) in the corresponding position, and the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

Why does copying a corresponding angle at P give a line parallel to $\ell$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

1065

**QUESTION ID**

## S29-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Congruent corresponding angles make the lines parallel (converse of the corresponding angles theorem).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

**True or false?**

Copying the angle so the two angles are congruent same-side interior angles also produces a parallel line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

1066

QUESTION ID

**S29-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Same-side interior angles must be supplementary, not congruent (unless both are 90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

To construct the line through P parallel to $\ell$ by copying an angle, a transversal is drawn through P meeting $\ell$. Which angle is copied?

**A.** The angle between $\ell$ and itself

**B.** The angle the transversal makes with $\ell$

**C.** A right angle

**D.** Any 45° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

1067

**QUESTION ID**
**S29-F2-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. The angle the transversal makes with $\ell$**

**GRADING NOTE** The transversal through P and a point of $\ell$ creates the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

Describe a different way to construct a parallel through P using two perpendicular constructions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

1068

**QUESTION ID**

## S29-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Construct the perpendicular from P to $\ell$, then construct the perpendicular to that line at P; it is parallel to $\ell$.**

**GRADING NOTE** Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

SET 30 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

# Construct the perpendicular bisector of a segment (bisect a segment)

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.12** |
| **BACKWARD 1** | Points equidistant from A and B lie on the perpendicular bisector | **HSG-CO.C.9** |
| **BACKWARD 2** | Know midpoint and perpendicular bisector definitions | **HSG-CO.A.1** |
| **BACKWARD 3** | Arcs with one radius from two centers | **7.G.B.4** |
| **FORWARD 1** | Justify the perpendicular bisector construction | **HSG-CO.C.9** |
| **FORWARD 2** | Construct the circumcenter and circumscribed circle of a triangle | **HSG-C.A.3** |

SET 30 • HSG-CO.D.12

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of segment AB. Label the midpoint M.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

1071

**QUESTION ID**

## S30-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Arcs of equal radius (more than half of AB) centered at A and at B meet at two points; the line through them is the perpendicular bisector, meeting AB at M.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two equal-radius arcs from A and B crossing on both sides of AB, the line through the crossings, and M labeled.

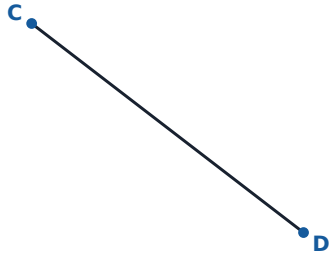
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the midpoint M of segment CD.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

1072

QUESTION ID

## S30-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**The perpendicular bisector construction locates M where the line through the two arc crossings meets CD.**

GRADING NOTE Grade the construction marks, not only the final figure: the two arc crossings and the line through them; M labeled on CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the point P on line $\ell$ that is the same distance from A as from B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

1073

**QUESTION ID**

## S30-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the perpendicular bisector of AB; the point P where it meets $\ell$ is equidistant from A and B.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the full perpendicular bisector construction for AB and its intersection with $\ell$ marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

Write the steps for constructing the perpendicular bisector of segment AB.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

1074

QUESTION ID

## S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

**1. Open the compass to more than half of AB. 2. Draw arcs centered at A above and below AB. 3. With the same opening, draw arcs centered at B crossing the first arcs. 4. Draw the line through the two crossing points.**

GRADING NOTE Any equivalent steps. Essential: the same opening for both centers, more than half of AB, and the line through both crossings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

Describe how to find the perpendicular bisector of segment AB by folding paper.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

1075

QUESTION ID

**S30-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**Fold the paper so that A lands exactly on B and crease it; the crease is the perpendicular bisector of AB.**

**GRADING NOTE** Any equivalent description (the fold matches A onto B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

SET 30 • HSG-CO.D.12

# BACKWARD 1

Points equidistant from A and B lie on the perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

A point equidistant from A and B lies on the perpendicular bisector of AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

1077

QUESTION ID

**S30-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Converse of the perpendicular bisector theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

Points P and Q are each equidistant from A and B. What line do P and Q determine?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

1078

QUESTION ID

**S30-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**The perpendicular bisector of AB**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

In the perpendicular bisector construction for segment AB, the two arc crossings are equidistant from A and B because

**A.** the crossings are on AB

**B.** the arcs have the same radius

**C.** the compass is opened to AB

**D.** the arcs are drawn above AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

1079

QUESTION ID

**S30-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. the arcs have the same radius**

**GRADING NOTE** Each crossing is one radius from A and one radius from B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

**True or false?**

Every point on the perpendicular bisector of AB is equidistant from A and B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

1080

QUESTION ID

**S30-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Two points determine more than one line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

1081

QUESTION ID

**S30-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Exactly one line passes through two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

# BACKWARD 2

Know midpoint and perpendicular bisector definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

M is the midpoint of AB, and AM = 3.5. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

1083

QUESTION ID

**S30-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

**True or false?**

The perpendicular bisector of a segment passes through its midpoint.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

1084

**QUESTION ID**

**S30-B2-Q2**

**TYPE**

True or false

**CORRECT ANSWER**

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

A line crosses AB at its midpoint at a 60° angle. It is

**A.** the perpendicular bisector

**B.** not a bisector

**C.** parallel to AB

**D.** a bisector but not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

1085

**QUESTION ID**

**S30-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. a bisector but not the perpendicular bisector**

**GRADING NOTE** It bisects but is not perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line k is perpendicular to AB and passes through its midpoint. What is k called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

1086

**QUESTION ID**

**S30-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**The perpendicular bisector of AB**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

**True or false?**

A segment has many perpendicular bisectors in one plane.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

1087

QUESTION ID

**S30-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSG-CO.D.12

# BACKWARD 3

Arcs with one radius from two centers • Grade 7 • Geometry • 7.G.B.4

Two arcs of radius 5 are centered at A and B, and AB = 6. Do they cross?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

1089

**QUESTION ID**

## S30-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes, at two points (5 + 5 > 6).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

**True or false?**

If the compass opening is less than half of AB, the arcs from A and B do not meet.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

1090

QUESTION ID

**S30-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The two radii add to less than AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

Why must the opening be more than half of AB?

**A.** It does not matter

**B.** So the midpoint is on the arc

**C.** So the arcs from A and B cross

**D.** So the line is longer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

1091

**QUESTION ID**

## S30-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. So the arcs from A and B cross**

**GRADING NOTE** Equal radii must add to more than AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

A point X is on both arcs (radius r) centered at A and B. Find XA and XB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

1092

**QUESTION ID**

## S30-B3-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**XA = XB = r**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

**True or false?**

The two arc crossings are on the same side of AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

1093

QUESTION ID

**S30-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** One is on each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSG-CO.D.12

# FORWARD 1

Justify the perpendicular bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

To construct the perpendicular bisector of AB, arcs of radius r (more than half of AB) centered at A and B meet at P and Q. Prove that line PQ is the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

1095

QUESTION ID

**S30-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**PQ is the perpendicular bisector of AB.**

GRADING NOTE REQUIRED WORK: Use the converse of the perpendicular bisector theorem for P and for Q.

Any valid proof. Essential: PA = PB = r and QA = QB = r; each of P and Q is on the perpendicular bisector (converse); two points determine that line. Alternative: SSS gives congruent triangles APQ and BPQ, then SAS/linear pair at the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the perpendicular bisector construction for AB would still work with a larger compass opening.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

1096

QUESTION ID

**S30-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Any equal radius greater than half of AB gives two points equidistant from A and B.**

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

**True or false?**

The construction works only if the arcs are drawn with radius exactly AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

1097

QUESTION ID

**S30-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Any equal radius over half of AB works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

Which theorem justifies that the arc crossings lie on the perpendicular bisector?

**A.** Midsegment theorem

**B.** Triangle angle sum

**C.** Converse of the perpendicular bisector theorem

**D.** Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

1098

**QUESTION ID**

## S30-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. Converse of the perpendicular bisector theorem**

**GRADING NOTE** Equidistant points lie on the bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

Why does folding A onto B produce the perpendicular bisector?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

1099

QUESTION ID

**S30-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**The fold is a reflection swapping A and B, so the crease is the perpendicular bisector of AB (definition of reflection).**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

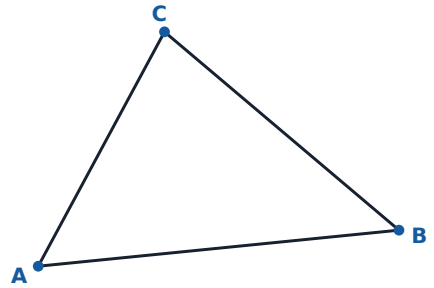
SET 30 • HSG-CO.D.12

# FORWARD 2

Construct the circumcenter and circumscribed circle of a triangle • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the point O equidistant from A, B, and C, and draw the circle through A, B, and C.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

1101

**QUESTION ID**

## S30-F2-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the perpendicular bisectors of two sides; they meet at O; the circle centered at O through A also passes through B and C.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two complete perpendicular bisector constructions, O labeled, and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

Why is it enough to construct only two perpendicular bisectors to find the circumcenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

1102

**QUESTION ID**

## S30-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Their crossing is equidistant from all three vertices, so it is on the third bisector too.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

**True or false?**

The circumcenter of an obtuse triangle lies on its longest side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

1103

QUESTION ID

**S30-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It lies outside, beyond the longest side; it is on the longest side only for a right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

Where is the circumcenter of a right triangle?

**A.** Inside, near the centroid

**B.** At the midpoint of the hypotenuse

**C.** Outside the triangle

**D.** At the right-angle vertex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

1104

QUESTION ID

**S30-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. At the midpoint of the hypotenuse**

GRADING NOTE The hypotenuse is a diameter of the circumcircle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

A triangle has A(0, 0), B(8, 0), C(0, 6). Where is its circumcenter, and what is the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

1105

QUESTION ID

**S30-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**(4, 3); radius 5**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

# Construct the bisector of an angle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.12** |
| **BACKWARD 1** | Know the definition of an angle bisector | **HSG-CO.A.1** |
| **BACKWARD 2** | Prove triangles congruent with SSS | **HSG-CO.B.8** |
| **BACKWARD 3** | Measure angles with a protractor | **4.MD.C.6** |
| **FORWARD 1** | Justify the angle bisector construction | **HSG-CO.B.8** |
| **FORWARD 2** | Construct special angles and the incenter by bisecting | **HSG-CO.D.12** |

SET 31 • HSG-CO.D.12

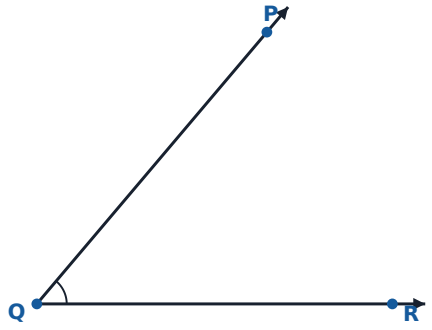
# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the bisector of ∠PQR.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

1108

**QUESTION ID**

## S31-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Arc centered at Q crosses the sides at X and Y; equal arcs centered at X and Y meet at Z inside the angle; ray QZ is the bisector.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the arc at Q, two equal-radius arcs from X and Y crossing inside the angle, and ray QZ.

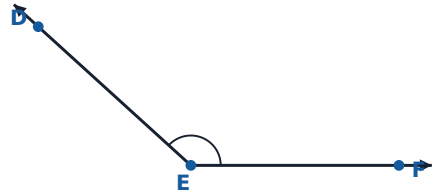
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the bisector of obtuse angle ∠DEF.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

1109

**QUESTION ID**

## S31-M-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Same steps: arc at E, equal arcs from the two crossing points, ray from E through their intersection.**

**GRADING NOTE** Grade the construction marks, not only the final figure: all three arcs and the bisector ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a 45° angle with vertex A on line $\ell$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

1110

**QUESTION ID**

## S31-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the perpendicular to $\ell$ at A (bisect the straight angle), then bisect one of the right angles.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the perpendicular construction at A and the angle bisector construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

Write the steps for bisecting ∠ABC with compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

1111

**QUESTION ID**

## S31-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1. Arc centered at B crossing BA at X and BC at Y. 2. With a fixed opening, arcs centered at X and at Y crossing at Z. 3. Draw ray BZ.**

**GRADING NOTE** Any equivalent steps. Essential: equal radii from X and Y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Describe how to bisect an angle drawn on paper by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

1112

QUESTION ID

**S31-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**Fold so that one side of the angle lies exactly on the other side, with the fold through the vertex; the crease is the bisector.**

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

SET 31 • HSG-CO.D.12

# BACKWARD 1

Know the definition of an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Ray BD bisects ∠ABC, and m∠ABD = 27°. Find m∠ABC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

1114

QUESTION ID

**S31-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**54°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

**True or false?**

An angle bisector divides the angle into two congruent angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

1115

QUESTION ID

**S31-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

Ray BD bisects ∠ABC with m∠ABC = 3x + 6 and m∠ABD = 24. Find x.

**A.** 6

**B.** 10

**C.** 18

**D.** 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

1116

QUESTION ID

**S31-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 14**

GRADING NOTE 3x + 6 = 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 90° angle is bisected. What is each part?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

1117

QUESTION ID

**S31-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**45°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

**True or false?**

An angle bisector is a segment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

1118

QUESTION ID

**S31-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It is a ray from the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

# BACKWARD 2

Prove triangles congruent with SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share a side and have two more pairs of congruent sides. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

1120

**QUESTION ID**

**S31-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**SSS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

**True or false?**

SSS guarantees that corresponding angles are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

1121

QUESTION ID

**S31-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

In the bisector construction, which triangles are congruent?

**A.** △ABC and △XYZ

**B.** △BXZ and △BYZ

**C.** △BXY and △XYZ

**D.** None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

1122

QUESTION ID

**S31-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. △BXZ and △BYZ**

GRADING NOTE BX = BY, XZ = YZ, BZ shared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

After △BXZ ≅ △BYZ, which angle is congruent to ∠XBZ?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

1123

**QUESTION ID**

**S31-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**∠YBZ**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

**True or false?**

SSS requires knowing one angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

1124

QUESTION ID

**S31-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only three sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSG-CO.D.12

# BACKWARD 3

Measure angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

A protractor reads 0° on one side and 112° on the other. Half of this angle is

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

1126

**QUESTION ID**

**S31-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**56°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

**True or false?**

Measuring with a protractor is a construction.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

1127

QUESTION ID

**S31-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Constructions use compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An angle of 150° is bisected. Each part is

**A.** 60°

**B.** 150°

**C.** 75°

**D.** 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

1128

**QUESTION ID**

**S31-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 75°**

**GRADING NOTE** 150 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

After bisecting, a student measures 38° and 38°. What was the original angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

1129

**QUESTION ID**

## S31-B3-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**76°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

**True or false?**

A protractor can check a construction.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

1130

QUESTION ID

**S31-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Measuring can check, though it does not replace the construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSG-CO.D.12

# FORWARD 1

Justify the angle bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In bisecting ∠ABC, an arc centered at B crosses the sides at X and Y, and equal arcs centered at X and Y meet at Z. Prove that ray BZ bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5**

1132

**QUESTION ID**

## S31-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**∠XBZ ≅ ∠YBZ**

**GRADING NOTE** REQUIRED WORK: Prove △BXZ ≅ △BYZ by SSS.

Any valid proof. Essential: BX = BY (same arc), XZ = YZ (equal arcs), BZ shared; SSS; corresponding angles; definition of bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

**High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5**

In the angle bisector construction, an arc centered at the vertex crosses the sides at X and Y, and then arcs from X and Y cross at Z. Why must the arcs from X and Y have the same radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

1133

**QUESTION ID**

## S31-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**So that XZ = YZ, which SSS needs.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

**True or false?**

The bisector construction works for obtuse angles too.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

1134

QUESTION ID

**S31-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE The SSS argument does not depend on the size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

Which criterion justifies the angle bisector construction?

**A.** AAS

**B.** ASA

**C.** SSS

**D.** SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

1135

QUESTION ID

**S31-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. SSS**

GRADING NOTE Three pairs of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

Explain why folding one side onto the other gives the bisector.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

1136

QUESTION ID

**S31-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**The fold is a reflection that maps one side onto the other, so it makes two congruent angles at the vertex.**

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

# FORWARD 2

Construct special angles and the incenter by bisecting • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a 30° angle. (Hint: start from an equilateral triangle.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

1138

QUESTION ID

## S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Construct an equilateral triangle on AB (60° angle at A), then bisect the angle at A to get 30°.**

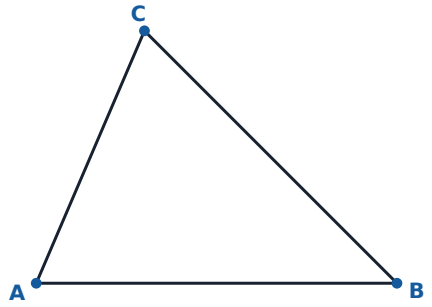
**GRADING NOTE** Grade the construction marks, not only the final figure: arcs of radius AB from A and B, the 60° angle, and its bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the point inside triangle ABC that is equidistant from its three sides.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

1139

**QUESTION ID**

## S31-F2-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Bisect two angles of the triangle; their intersection is the incenter.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two complete angle bisector constructions and the crossing labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

**True or false?**

Bisecting an angle and then bisecting each half divides the angle into three equal parts.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

1140

QUESTION ID

## S31-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It gives four equal parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Starting from 60° and 90° angles, which angle CANNOT be constructed by bisecting angles and adding angles together?

**A.** 15°

**B.** 75°

**C.** 20°

**D.** 22.5°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

1141

QUESTION ID

**S31-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 20°**

GRADING NOTE Bisecting gives 30°, 15°, 45°, 22.5°, ... and 75° = 60° + 15°. 20° would require trisecting 60°, which these steps cannot do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

1141

Explain how to construct a 135° angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

1142

**QUESTION ID**

## S31-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Construct a right angle, bisect it to get 45°, and add the 45° angle to the right angle (90° + 45°).**

**GRADING NOTE** Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

# Construct a perpendicular to a line through a point on or off the line

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.12** |
| **BACKWARD 1** | Know the definition of perpendicular lines | **HSG-CO.A.1** |
| **BACKWARD 2** | Construct the perpendicular bisector of a segment | **HSG-CO.D.12** |
| **BACKWARD 3** | Recognize perpendicular lines in figures | **4.G.A.1** |
| **FORWARD 1** | Construct an altitude of a triangle and use it for area | **HSG-CO.D.12** |
| **FORWARD 2** | Construct a square on a given segment | **HSG-CO.D.12** |

SET 32 • HSG-CO.D.12

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line $\ell$. (P is on $\ell$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

1145

**QUESTION ID**

## S32-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Arcs centered at P mark A and B on $\ell$ equally far from P; then construct the perpendicular bisector of AB, which passes through P.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the two equal marks on $\ell$ and the perpendicular bisector arcs, with the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line $\ell$. (P is not on $\ell$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

1146

QUESTION ID

## S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**An arc centered at P crosses $\ell$ at A and B; equal arcs from A and B meet at Q below $\ell$; line PQ is perpendicular to $\ell$.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the arc from P meeting $\ell$ twice, equal arcs from A and B, and line PQ.

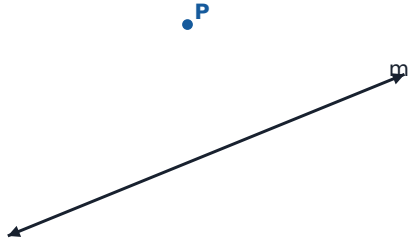
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the perpendicular from point P to the slanted line m.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

1147

**QUESTION ID**

## S32-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Same method as for a point not on the line: arc from P meets m twice, equal arcs from those points meet, and the line through P and the crossing.**

**GRADING NOTE** Grade the construction marks, not only the final figure: all arcs and the perpendicular line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

Write the steps for constructing the perpendicular from a point P not on line $\ell$.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

1148

**QUESTION ID**

## S32-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1. Arc centered at P crossing $\ell$ at A and B. 2. Equal arcs centered at A and B (radius more than half of AB) meeting at Q on the other side. 3. Draw line PQ.**

**GRADING NOTE** Any equivalent steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

Explain why the perpendicular construction from a point P not on $\ell$ is really a perpendicular bisector construction.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

1149

**QUESTION ID**

## S32-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**P is equidistant from A and B (same arc), and so is Q; so line PQ is the perpendicular bisector of AB, which is perpendicular to $\ell$.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSG-CO.D.12

# BACKWARD 1

Know the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**True or false?**

Perpendicular lines meet at a right angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

1151

QUESTION ID

**S32-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line k ⊥ line ℓ. What is the measure of each angle they form?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

1152

QUESTION ID

**S32-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**90°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which pair of lines is perpendicular?

**A.** y = 3 and y = −3

**B.** y = 2x and y = 2x + 1

**C.** y = x and y = 2x

**D.** y = 2x and y = −½x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

1153

**QUESTION ID**

**S32-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. y = 2x and y = −½x**

**GRADING NOTE** Slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

**True or false?**

Through a point, there are many lines perpendicular to a given line in a plane.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

1154

QUESTION ID

**S32-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

The shortest path from a point to a line follows which kind of line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

1155

**QUESTION ID**

## S32-B1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The perpendicular from the point to the line**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

# BACKWARD 2

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

After marking A and B on $\ell$, which construction finishes the perpendicular?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

1157

QUESTION ID

**S32-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**The perpendicular bisector of AB**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

**True or false?**

The perpendicular bisector construction needs arcs of equal radius from both endpoints.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

1158

QUESTION ID

**S32-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Equidistant crossings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

The arc crossings of the perpendicular bisector construction are

**A.** equidistant from A and B

**B.** on segment AB

**C.** on line $\ell$

**D.** at A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

1159

**QUESTION ID**

**S32-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. equidistant from A and B**

**GRADING NOTE** Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

P is on $\ell$, and A and B are marked on $\ell$ with PA = PB.
Where does the perpendicular bisector of AB cross $\ell$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

1160

**QUESTION ID**
**S32-B2-Q4**

**TYPE**
Short answer

**CORRECT ANSWER**
**At P**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The perpendicular bisector of AB passes through A.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

1161

QUESTION ID

**S32-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It passes through the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSG-CO.D.12

# BACKWARD 3

Recognize perpendicular lines in figures • Grade 4 • Geometry • 4.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A square's adjacent sides are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

1163

QUESTION ID

**S32-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE They meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

Which letter is made of perpendicular segments?

**A.** V

**B.** T

**C.** O

**D.** S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

1164

**QUESTION ID**

## S32-B3-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. T**

**GRADING NOTE** T has a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

How many right angles do two perpendicular lines make?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

1165

**QUESTION ID**

## S32-B3-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

**True or false?**

Two lines that look almost perpendicular must be perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

1166

QUESTION ID

**S32-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Appearance is not proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

Name a pair of perpendicular edges of a rectangular door.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

1167

**QUESTION ID**

## S32-B3-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**A vertical side and the top (or bottom) edge**

**GRADING NOTE** Any correct pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

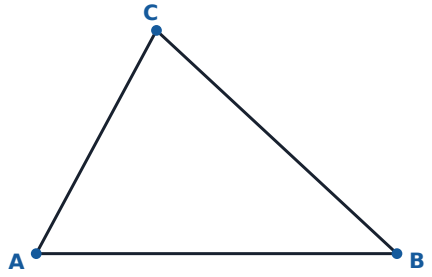
SET 32 • HSG-CO.D.12

# FORWARD 1

Construct an altitude of a triangle and use it for area • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the altitude from C to side AB of triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

1169

**QUESTION ID**

## S32-F1-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the perpendicular from C to line AB; the segment from C to the foot is the altitude.**

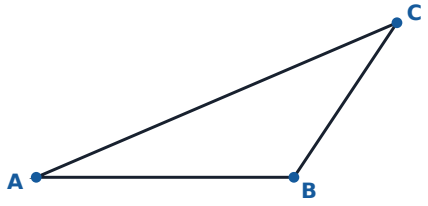
**GRADING NOTE** Grade the construction marks, not only the final figure: the perpendicular construction from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the altitude from C in obtuse triangle ABC (the foot is outside AB). Extend AB as needed.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

1170

**QUESTION ID**

## S32-F1-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Extend line AB past B and construct the perpendicular from C to that line.**

**GRADING NOTE** Grade the construction marks, not only the final figure: AB extended and the perpendicular construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has base 10 and its constructed altitude measures 6. Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

1171

**QUESTION ID**

**S32-F1-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**30**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

**True or false?**

Every altitude of a triangle lies inside the triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

1172

QUESTION ID

**S32-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE An obtuse triangle has two altitudes outside it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

In a right triangle, the altitude from the right-angle vertex to the hypotenuse is

**A.** the hypotenuse

**B.** inside the triangle

**C.** outside the triangle

**D.** a leg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

1173

QUESTION ID

**S32-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. inside the triangle**

**GRADING NOTE** It meets the hypotenuse between its endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

# FORWARD 2

Construct a square on a given segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a square with side AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

1175

QUESTION ID

## S32-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Construct perpendiculars to AB at A and at B; copy AB onto each to get D and C; join C and D.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two perpendicular constructions and two segment copies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

A square on segment AB is constructed by drawing perpendiculars to AB at A and at B and copying AB onto each. Explain why the figure is a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

1176

**QUESTION ID**

## S32-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**It has four right angles (perpendiculars, and the fourth by the angle sum) and four sides equal to AB.**

**GRADING NOTE** Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

**True or false?**

A quadrilateral with four congruent sides is always a square.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

1177

QUESTION ID

**S32-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A rhombus need not have right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

To construct a rectangle 2AB long and AB wide, which additional step is needed?

**A.** Bisect AB

**B.** Construct a 60° angle

**C.** Draw a circle through A and B

**D.** Copy AB twice along the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

1178

QUESTION ID

**S32-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. Copy AB twice along the base**

**GRADING NOTE** The base must be doubled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

A square is constructed on a segment of length 5. What is the length of its diagonal?

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

1179

**QUESTION ID**

**S32-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**$5\sqrt{2}$**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

SET 33 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

# Construct a line parallel to a given line through a point not on it

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.12** |
| **BACKWARD 1** | Copy an angle with compass and straightedge | **HSG-CO.D.12** |
| **BACKWARD 2** | Converse: congruent corresponding angles give parallel lines | **HSG-CO.C.9** |
| **BACKWARD 3** | Recognize parallel lines | **4.G.A.1** |
| **FORWARD 1** | Justify the parallel-line constructions | **HSG-CO.C.9** |
| **FORWARD 2** | Divide a segment into equal parts with parallel lines | **HSG-SRT.B.4** |

SET 33 • HSG-CO.D.12

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line $\ell$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

1182

**QUESTION ID**

## S33-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Draw a transversal through P meeting $\ell$ at A; copy the angle it makes with $\ell$ at A to P in the corresponding position; the new side is parallel to $\ell$.**

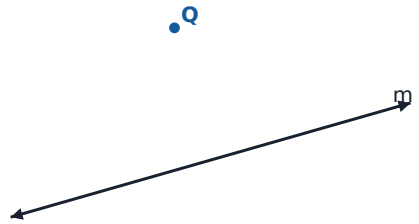
**GRADING NOTE** Grade the construction marks, not only the final figure: a transversal, a complete angle copy at P (equal-radius arcs and transferred chord), and the parallel line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through Q parallel to the slanted line m.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

1183

**QUESTION ID**

## S33-M-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Same method: a transversal through Q, then copy the corresponding (or alternate interior) angle at Q.**

**GRADING NOTE** Grade the construction marks, not only the final figure: all arcs of the angle copy and the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to $\ell$ using two perpendiculars.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

1184

**QUESTION ID**

## S33-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the perpendicular k from P to $\ell$, then the perpendicular to k at P; it is parallel to $\ell$.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two complete perpendicular constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

Write the steps for constructing the line through P parallel to line $\ell$ by copying an angle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

1185

**QUESTION ID**

## S33-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1. Draw a line through P crossing $\ell$ at A. 2. Copy the angle between $\ell$ and line AP at A to vertex P, on the same side, using AP as a side. 3. Extend the new side into a line.**

**GRADING NOTE** Any equivalent steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

Why is the line constructed by copying the corresponding angle parallel to $\ell$?

**A.** It is perpendicular to the transversal

**B.** It passes through P

**C.** Congruent corresponding angles make the lines parallel

**D.** It looks parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

1186

**QUESTION ID**
**S33-M-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**C. Congruent corresponding angles make the lines parallel**

**GRADING NOTE** Converse of the corresponding angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

SET 33 • HSG-CO.D.12

# BACKWARD 1

Copy an angle with compass and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which three lengths are transferred?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

1188

QUESTION ID

**S33-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**The radius of the first arc (twice) and the chord between the side crossings**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

**True or false?**

An angle copy uses arcs of the same radius at both vertices.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

1189

QUESTION ID

**S33-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

Which step completes an angle copy?

**A.** Draw a circle

**B.** Measure with a protractor

**C.** Draw the ray from the new vertex through the last arc crossing

**D.** Bisect the angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

1190

QUESTION ID

**S33-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. Draw the ray from the new vertex through the last arc crossing**

GRADING NOTE The ray is the new side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

Why does the angle copy work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

1191

QUESTION ID

**S33-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**SSS makes the two triangles congruent.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

**True or false?**

An angle copy needs a protractor.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

1192

QUESTION ID

**S33-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

# BACKWARD 2

Converse: congruent corresponding angles give parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

Congruent corresponding angles mean the lines are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

1194

QUESTION ID

**S33-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A transversal makes 62° angles with two lines in corresponding positions. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

1195

QUESTION ID

**S33-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Yes**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

Which also proves the lines parallel?

**A.** Congruent vertical angles

**B.** Supplementary linear pairs

**C.** Congruent alternate interior angles

**D.** Two acute angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

1196

QUESTION ID

**S33-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. Congruent alternate interior angles**

GRADING NOTE Alternate interior converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

**True or false?**

Same-side interior angles of 70° and 70° prove the lines parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

1197

QUESTION ID

**S33-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They must add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Corresponding angles are (2x + 10)° and 50°. For which x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

1198

QUESTION ID

**S33-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**20**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSG-CO.D.12

# BACKWARD 3

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The opposite edges of a ruler are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

1200

QUESTION ID

**S33-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** They never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

Which letter has parallel segments?

**A.** V

**B.** A

**C.** H

**D.** X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

1201

**QUESTION ID**

**S33-B3-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. H**

**GRADING NOTE** H's two uprights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two railroad rails stay the same distance apart. What are they modeling?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

1202

**QUESTION ID**

**S33-B3-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**Parallel lines**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Lines that cross are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

1203

QUESTION ID

**S33-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Parallel lines never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How many lines through a point are parallel to a given line (in a plane)?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

1204

**QUESTION ID**

**S33-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**1**

**GRADING NOTE** Parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSG-CO.D.12

# FORWARD 1

Justify the parallel-line constructions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

To construct a line through P parallel to $\ell$, a transversal through P meets $\ell$ at A, and the angle at A is copied at P in the corresponding position. Prove that the new line is parallel to $\ell$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

1206

**QUESTION ID**

## S33-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The new line $\parallel \ell$.**

**GRADING NOTE** REQUIRED WORK: Use the congruence of the copied angle (SSS) and the converse of the corresponding angles theorem.

Any valid proof. Essential: the copy is congruent (SSS from the construction); the angles are corresponding angles along AP; the converse gives parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In the two-perpendicular construction, k ⊥ ℓ and n ⊥ k at P. Prove that n ∥ ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5**

1207

**QUESTION ID**

## S33-F1-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**n ∥ ℓ**

**GRADING NOTE** REQUIRED WORK: Both n and ℓ make right angles with k (corresponding angles congruent), or two lines perpendicular to the same line in a plane are parallel.

Any valid proof (paragraph, two-column, or flow). Essential: n ⊥ k and ℓ ⊥ k give a pair of congruent (90°) corresponding angles where k crosses n and ℓ; congruent corresponding angles give n ∥ ℓ. Alternative: two lines perpendicular to the same line are parallel (with that theorem cited). Invalid: "they look parallel".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

**High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5**

**True or false?**

Copying the alternate interior angle at P also gives a parallel line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

1208

QUESTION ID

**S33-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Alternate interior converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

Which construction gives a line through P parallel to $\ell$?

**A.** Copy, at P, the angle that a transversal through P makes with $\ell$, in the corresponding position

**B.** Bisect the angle at P

**C.** Draw a circle centered at P

**D.** Draw the perpendicular bisector of the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

1209

QUESTION ID

**S33-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. Copy, at P, the angle that a transversal through P makes with $\ell$, in the corresponding position**

**GRADING NOTE** Only the angle copy gives congruent corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

By the parallel postulate, how many lines could the construction produce?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

1210

QUESTION ID

**S33-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**Exactly one (the unique parallel through P).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

# FORWARD 2

Divide a segment into equal parts with parallel lines • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Divide segment AB into three congruent parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

1212

**QUESTION ID**

## S33-F2-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Draw a ray from A; mark three equal steps C, D, E on it with the compass; join E to B; construct parallels to EB through D and C; they cut AB into three equal parts.**

**GRADING NOTE** Grade the construction marks, not only the final figure: three equal compass steps on the auxiliary ray and two parallel constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

To divide AB into three equal parts, three equal steps are marked on a ray from A, the last mark is joined to B, and parallels to that segment are drawn through the other marks. Why do the parallel lines divide AB into equal parts?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

1213

**QUESTION ID**
**S33-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**Parallel lines cut the two sides of the triangle proportionally (triangle proportionality), so equal steps on the ray give equal parts of AB.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

**True or false?**

The same method can divide a segment into 5 equal parts only if its length is a multiple of 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

1214

QUESTION ID

## S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Equal compass steps on the auxiliary ray work for any segment; nothing is measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

To divide AB in the ratio 2 : 3, how many equal steps are marked on the auxiliary ray?

**A.** 2

**B.** 5

**C.** 3

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

1215

**QUESTION ID**

## S33-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 5**

**GRADING NOTE** 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A segment of length 12 is divided into 4 parts this way. How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

1216

QUESTION ID

**S33-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**3**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

SET 34 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

# Construct an equilateral triangle inscribed in a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.13** |
| **BACKWARD 1** | Central angles of a regular polygon | **HSG-CO.A.3** |
| **BACKWARD 2** | Copy a length with a compass | **HSG-CO.D.12** |
| **BACKWARD 3** | Circle vocabulary: center, radius, chord, diameter | **7.G.B.4** |
| **FORWARD 1** | Justify that the inscribed triangle is equilateral | **HSG-C.A.2** |
| **FORWARD 2** | Construct a regular dodecagon by bisecting arcs | **HSG-CO.D.13** |

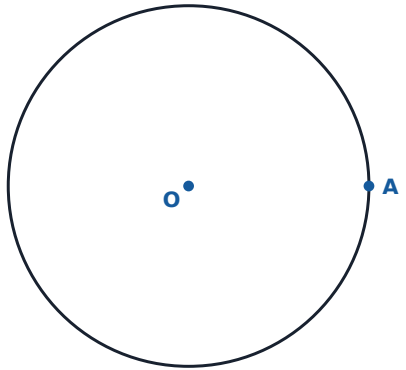
SET 34 • HSG-CO.D.13

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle, with one vertex at A.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

1219

QUESTION ID

## S34-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**With the compass set to the radius, step around the circle from A marking six points; join every other point (A, the 2nd, the 4th).**

**GRADING NOTE** Grade the construction marks, not only the final figure: radius-length arcs stepped around the circle (or the diameter-and-arcs method) and the triangle joining alternate points.

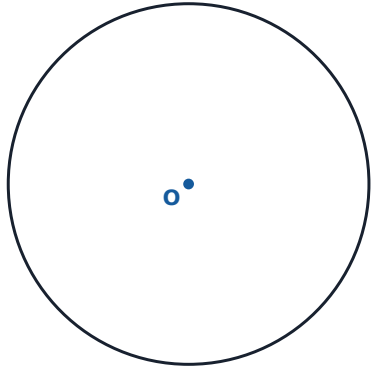
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle (choose any starting vertex).




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

1220

QUESTION ID

## S34-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Mark a point on the circle, step the radius around the circle to get six points, and join alternate points.**

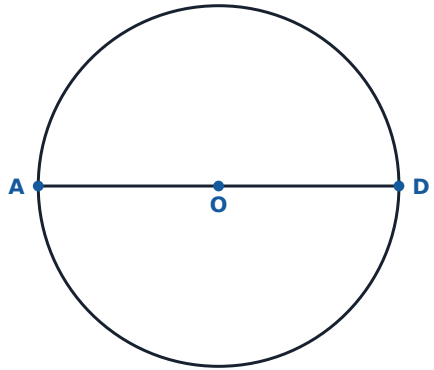
GRADING NOTE Grade the construction marks, not only the final figure: stepping arcs and the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle using a diameter: draw diameter AD, then an arc centered at D with radius DO.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

1221

QUESTION ID

**S34-M-Q3**

TYPE

Drawing / plotting

CORRECT ANSWER

**The arc centered at D with radius DO meets the circle at B and C; triangle ABC is equilateral.**

GRADING NOTE Grade the construction marks, not only the final figure: the arc at D and the triangle ABC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

Write the steps for inscribing an equilateral triangle in a circle with center O.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

1222

**QUESTION ID**

## S34-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1. Mark A on the circle. 2. With the compass set to the radius, mark successive points around the circle (6 in all). 3. Join A to the 2nd and 4th points, and those to each other.**

**GRADING NOTE** Any valid set of steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

When a compass set to the radius of a circle is stepped around the circle, why does the radius fit exactly six times?

**A.** Any length fits six times

**B.** The compass is 6 cm

**C.** The circle has six parts

**D.** Each step makes an equilateral triangle with the center, a 60° central angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

1223

QUESTION ID
**S34-M-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**D. Each step makes an equilateral triangle with the center, a 60° central angle**

**GRADING NOTE** Six 60° angles make 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

SET 34 • HSG-CO.D.13

# BACKWARD 1

Central angles of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

What rotation carries an equilateral triangle onto itself, about its center (smallest positive)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

1225

**QUESTION ID**

**S34-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**120°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Into how many equal central angles does a regular hexagon divide 360°?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

1226

**QUESTION ID**
**S34-B1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**6 (each 60°)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

**True or false?**

The vertices of an inscribed equilateral triangle are 120° apart around the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

1227

QUESTION ID

**S34-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE 360 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

Joining every other vertex of a regular hexagon gives

**A.** a square

**B.** a rectangle

**C.** an equilateral triangle

**D.** a pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

1228

**QUESTION ID**

## S34-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. an equilateral triangle

**GRADING NOTE** Every other vertex: 120° apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The central angles of an inscribed equilateral triangle are 60°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

1229

QUESTION ID

**S34-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

# BACKWARD 2

Copy a length with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

**ANSWER KEY** Reduced copy of the question page, with its answer

The compass is set to the radius of a circle. What length does each step around the circle mark?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

1231

QUESTION ID

**S34-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**A chord equal to the radius**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

**True or false?**

Keeping the compass opening fixed is required when stepping around the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

1232

QUESTION ID

**S34-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** All chords must be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

Each radius-length chord and the two radii to its ends form

**A.** a square

**B.** an equilateral triangle

**C.** a right triangle

**D.** an isosceles triangle with a 90° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

1233

**QUESTION ID**

**S34-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. an equilateral triangle**

**GRADING NOTE** All three sides equal the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has radius 4 cm. How long is each chord marked by stepping the radius?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

1234

QUESTION ID

**S34-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**4 cm**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

**True or false?**

Stepping a compass set to the diameter marks six points.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

1235

QUESTION ID

**S34-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE The diameter fits only twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSG-CO.D.13

# BACKWARD 3

Circle vocabulary: center, radius, chord, diameter • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

What is a chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

1237

QUESTION ID

**S34-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**A segment with both endpoints on the circle**

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

**True or false?**

A diameter is a chord through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5**

1238

QUESTION ID

**S34-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5**

A circle has radius 6. Its diameter is

**A.** 3

**B.** 36

**C.** 6

**D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

1239

QUESTION ID

**S34-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 12**

**GRADING NOTE** Twice the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What does "inscribed in a circle" mean for a polygon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

1240

QUESTION ID

**S34-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**All its vertices lie on the circle**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

**True or false?**

An inscribed triangle can have a vertex inside the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

1241

QUESTION ID

**S34-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** All vertices are on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSG-CO.D.13

# FORWARD 1

Justify that the inscribed triangle is equilateral • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Points A, B, C, D, E, F are marked around a circle with center O by stepping the radius. Prove that triangle ACE is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

1243

**QUESTION ID**

## S34-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

## AC = CE = EA

**GRADING NOTE** REQUIRED WORK: Show each central angle between consecutive marks is 60°, so AOC, COE, EOA are each 120°, then use congruent triangles (SAS).

Any valid proof. Essential: each small triangle (O, two consecutive marks) is equilateral, so each central angle is 60°; ∠AOC = ∠COE = ∠EOA = 120°; triangles AOC, COE, EOA are congruent by SAS (radii); so AC = CE = EA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

An equilateral triangle is inscribed in a circle of radius 6. Find its side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

1244

QUESTION ID

**S34-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**6√3**

GRADING NOTE Form required: exact value. Side = r√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

**True or false?**

Congruent central angles cut off congruent chords.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

1245

QUESTION ID

**S34-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE SAS with radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

Each angle of the inscribed triangle ACE intercepts an arc (not containing its vertex) of

**A.** 120°

**B.** 240°

**C.** 180°

**D.** 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

1246

**QUESTION ID**

## S34-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 120°**

**GRADING NOTE** Each 60° inscribed angle intercepts an arc twice its measure, 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

Find the area of an equilateral triangle inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

1247

QUESTION ID

**S34-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**3√3**

**GRADING NOTE** Form required: exact. Side 2√3; area (√3/4)(12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

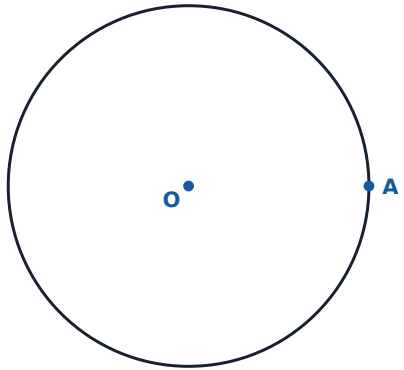
# FORWARD 2

Construct a regular dodecagon by bisecting arcs • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular 12-sided polygon inscribed in the circle.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

1249

QUESTION ID

## S34-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Mark six points by stepping the radius, then bisect each of the six arcs (perpendicular bisector of each chord) to get six more; join the 12 points in order.**

GRADING NOTE Grade the construction marks, not only the final figure: the six stepped points and the bisector constructions for the arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Each side of the inscribed regular hexagon subtends 60° at the center. What does each side of the dodecagon subtend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

1250

**QUESTION ID**
**S34-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**30°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

**True or false?**

Bisecting the arcs of an inscribed regular hexagon gives the vertices of a regular nonagon.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

1251

QUESTION ID

**S34-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Doubling the vertices gives a regular 12-gon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

Starting from an inscribed square, bisecting arcs gives a regular

**A.** pentagon

**B.** hexagon

**C.** decagon

**D.** octagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

1252

**QUESTION ID**

## S34-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. octagon

**GRADING NOTE** 4 → 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

How many sides does the polygon have after bisecting the arcs of an inscribed equilateral triangle twice?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

1253

**QUESTION ID**

**S34-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**12**

**GRADING NOTE** 3 → 6 → 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

# Construct a square inscribed in a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.13** |
| **BACKWARD 1** | Construct a perpendicular bisector | **HSG-CO.D.12** |
| **BACKWARD 2** | Diagonals of a square | **HSG-CO.C.11** |
| **BACKWARD 3** | Diameters and the center of a circle | **7.G.B.4** |
| **FORWARD 1** | Justify that the inscribed quadrilateral is a square | **HSG-CO.C.11** |
| **FORWARD 2** | Construct a regular octagon inscribed in a circle | **HSG-CO.D.13** |

SET 35 • HSG-CO.D.13

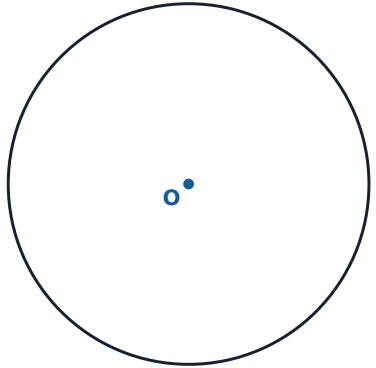
# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

1256

QUESTION ID

## S35-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Draw a diameter AC through O; construct the perpendicular bisector of AC (a second diameter BD); join A, B, C, D.**

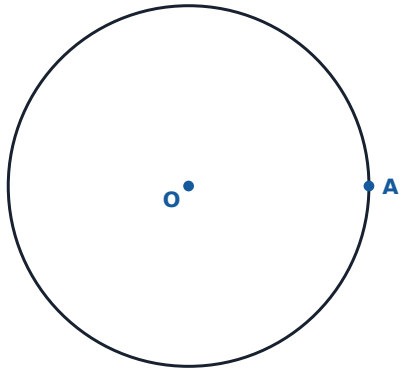
GRADING NOTE Grade the construction marks, not only the final figure: a diameter, the perpendicular bisector construction for it (arcs), and the square joining the four endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one vertex at A.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

1257

QUESTION ID

## S35-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Draw diameter AC through O; construct the perpendicular to AC at O, meeting the circle at B and D; join ABCD.**

**GRADING NOTE** Grade the construction marks, not only the final figure: diameter through A and the perpendicular construction.

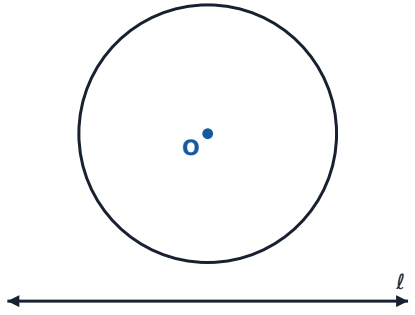
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one side parallel to the given line $\ell$.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

1258

**QUESTION ID**

## S35-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the perpendicular from O to $\ell$ (one diameter); bisect the right angles at O to get diagonals at 45°; their endpoints form a square with sides parallel to $\ell$.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the perpendicular, the angle bisectors, and the square. Accept any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

Write the steps for inscribing a square in a circle with center O.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

1259

**QUESTION ID**

## S35-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1. Draw a diameter AC. 2. Construct the perpendicular bisector of AC, meeting the circle at B and D. 3. Join A, B, C, D in order.**

**GRADING NOTE** Any valid steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

Which property of a square's diagonals does the construction use?

**A.** They are congruent, perpendicular, and bisect each other

**B.** They are parallel

**C.** They are equal to the sides

**D.** They meet outside the square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

1260

**QUESTION ID**
**S35-M-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. They are congruent, perpendicular, and bisect each other**

**GRADING NOTE** Two perpendicular diameters have all three properties.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

SET 35 • HSG-CO.D.13

# BACKWARD 1

Construct a perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

**ANSWER KEY** Reduced copy of the question page, with its answer

Which construction turns one diameter into two perpendicular diameters?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

1262

QUESTION ID

**S35-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**The perpendicular bisector of the diameter**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

**True or false?**

The perpendicular bisector of a diameter passes through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

1263

QUESTION ID

**S35-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The center is the diameter's midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

Two perpendicular diameters cut the circle into

**A.** six 60° arcs

**B.** four 45° arcs

**C.** four 90° arcs

**D.** two 90° arcs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

1264

**QUESTION ID**

**S35-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. four 90° arcs**

**GRADING NOTE** 360 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

A diameter is 10 cm. Where does its perpendicular bisector cross it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

1265

QUESTION ID

**S35-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**At the center, 5 cm from each end**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

**True or false?**

A perpendicular bisector can be drawn by measuring with a ruler only.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

1266

QUESTION ID

**S35-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A construction uses compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

# BACKWARD 2

Diagonals of a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**True or false?**

The diagonals of a square are congruent and perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

1268

QUESTION ID

**S35-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE A square is a rectangle and a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

A quadrilateral's diagonals are congruent, perpendicular, and bisect each other. It is a

**A.** rhombus that is not a square

**B.** square

**C.** trapezoid

**D.** kite

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

1269

**QUESTION ID**

**S35-B2-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. square**

**GRADING NOTE** All three properties.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A square has diagonal 8. Find its side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

1270

QUESTION ID

**S35-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**4√2**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

**True or false?**

A rectangle always has perpendicular diagonals.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

1271

QUESTION ID

**S35-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only a square does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The diagonals of a square meet at O. Name the angle between them.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

1272

**QUESTION ID**

**S35-B2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSG-CO.D.13

# BACKWARD 3

Diameters and the center of a circle • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has radius 3. How long is any diameter?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

1274

QUESTION ID

**S35-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

**True or false?**

Every diameter passes through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

1275

QUESTION ID

**S35-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

Where do two different diameters of a circle meet?

**A.** At the midpoint of a radius

**B.** On the circle

**C.** At the center

**D.** Nowhere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

1276

**QUESTION ID**

## S35-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. At the center

**GRADING NOTE** Both pass through the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the longest chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

1277

QUESTION ID

**S35-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**A diameter**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

**True or false?**

All diameters of a circle have different lengths.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

1278

QUESTION ID

**S35-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE All equal 2r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSG-CO.D.13

# FORWARD 1

Justify that the inscribed quadrilateral is a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

Diameters AC and BD of a circle with center O are perpendicular. Prove that ABCD is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

1280

QUESTION ID

## S35-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

**ABCD is a square.**

**GRADING NOTE** REQUIRED WORK: Use that the diagonals are congruent (both diameters), bisect each other (radii), and are perpendicular; or SAS on the four triangles.

Any valid proof. Essential: OA = OB = OC = OD; four right angles at O; SAS gives four congruent sides; each triangle is isosceles right, so each vertex angle is 45° + 45° = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

A square is inscribed in a circle of radius 5. Find its side length and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

1281

QUESTION ID

**S35-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Side 5√2; area 50**

**GRADING NOTE** Form required: exact side and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

**True or false?**

The side of a square inscribed in a circle of radius r is 2r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

1282

QUESTION ID

**S35-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE The side is r√2; 2r is its diagonal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The area of a square inscribed in a circle of radius r is

**A.** $2r^2$

**B.** $\pi r^2$

**C.** $4r^2$

**D.** $r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

1283

QUESTION ID

**S35-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. $2r^2$**

GRADING NOTE $(r\sqrt{2})^2 = 2r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What fraction of the circle's area does the inscribed square cover? Give the exact value.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

1284

**QUESTION ID**

**S35-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**$2/\pi$**

**GRADING NOTE** Form required: exact value. $2r^2 \div \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

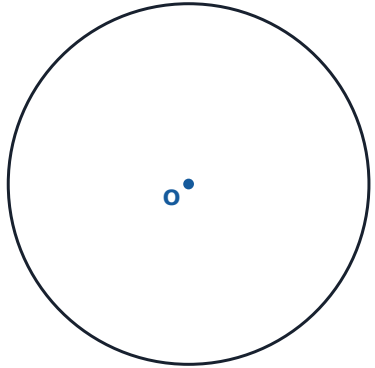
# FORWARD 2

Construct a regular octagon inscribed in a circle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular octagon inscribed in the circle.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

1286

**QUESTION ID**

## S35-F2-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct two perpendicular diameters, then bisect two adjacent right angles at O (or the four arcs); the eight points on the circle are the vertices.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the perpendicular diameters and the bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

What is the central angle for each side of a regular octagon?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

1287

**QUESTION ID**

**S35-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**45°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A regular octagon has interior angles of 120°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

1288

QUESTION ID

**S35-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE (8 − 2) · 180 ÷ 8 = 135°; 120° is the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

Bisecting the arcs of an inscribed regular octagon gives a regular

**A.** hexagon

**B.** 16-gon

**C.** 12-gon

**D.** square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

1289

**QUESTION ID**

**S35-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 16-gon**

**GRADING NOTE** 8 → 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

A regular octagon is constructed in a circle by bisecting the four 90° central angles of an inscribed square. Explain why the eight points are equally spaced.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

1290

**QUESTION ID**

## S35-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Each central angle is half of 90°, which is 45°, and equal central angles give equal chords.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

# Construct a regular hexagon inscribed in a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-CO.D.13** |
| **BACKWARD 1** | Construct an inscribed equilateral triangle | **HSG-CO.D.13** |
| **BACKWARD 2** | An equilateral triangle has equal sides and 60° angles | **4.G.A.2** |
| **BACKWARD 3** | Circle facts: radius and chord | **7.G.B.4** |
| **FORWARD 1** | Justify that the inscribed hexagon is regular | **HSG-C.A.2** |
| **FORWARD 2** | Use inscribed hexagons to estimate π | **HSG-GMD.A.1** |

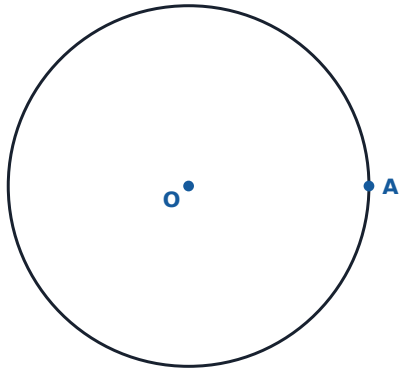
SET 36 • HSG-CO.D.13

# MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle, with one vertex at A.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

1293

**QUESTION ID**

## S36-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Set the compass to the radius OA; from A step around the circle marking 5 more points; join the six points in order.**

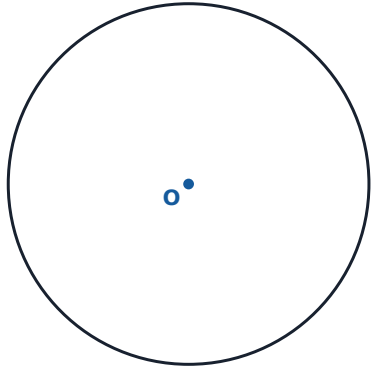
**GRADING NOTE** Grade the construction marks, not only the final figure: radius-length arcs around the circle and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle (choose any starting point).




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

1294

**QUESTION ID**

## S36-M-Q2

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Mark a point, step the radius around the circle six times, and join consecutive points.**

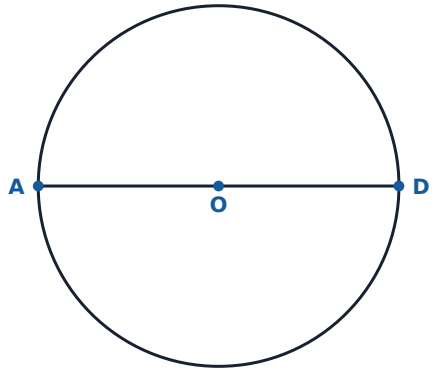
**GRADING NOTE** Grade the construction marks, not only the final figure: all stepping arcs and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle using diameter AD and arcs centered at A and D with radius equal to the circle's radius.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

1295

QUESTION ID

**S36-M-Q3**

TYPE

Drawing / plotting

CORRECT ANSWER

**The arc at A meets the circle at B and F; the arc at D meets it at C and E; join A, B, C, D, E, F.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the two arcs and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

Write the steps for inscribing a regular hexagon in a circle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

1296

QUESTION ID

**S36-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**1. Mark a point A on the circle. 2. Keep the compass set to the radius. 3. Mark B on the circle with an arc centered at A, then C from B, and so on, to F. 4. Join the points in order.**

GRADING NOTE Any valid steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

**True or false?**

In the hexagon construction, the compass is set to the diameter of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

1297

QUESTION ID

**S36-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It is set to the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

SET 36 • HSG-CO.D.13

# BACKWARD 1

Construct an inscribed equilateral triangle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

**ANSWER KEY** Reduced copy of the question page, with its answer

How are the inscribed equilateral triangle and the inscribed regular hexagon related?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

1299

QUESTION ID

**S36-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**The triangle joins every other vertex of the hexagon.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

**True or false?**

Both constructions start by stepping the radius around the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

1300

QUESTION ID

**S36-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Same six points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

How many points does stepping the radius mark?

**A.** 12

**B.** 6

**C.** 4

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

1301

QUESTION ID

**S36-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 6**

GRADING NOTE 60° each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An equilateral triangle is inscribed. How many more points must be added to make a regular hexagon?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

1302

QUESTION ID

**S36-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

**True or false?**

The inscribed equilateral triangle and hexagon have the same side length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

1303

QUESTION ID

**S36-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Triangle side r√3; hexagon side r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

# BACKWARD 2

An equilateral triangle has equal sides and 60° angles • Grade 4 • Geometry • 4.G.A.2

**True or false?**

Each angle of an equilateral triangle measures 60°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

1305

QUESTION ID

**S36-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** 180 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

A triangle has three sides equal to the radius of a circle. What kind of triangle is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

1306

**QUESTION ID**

**S36-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**Equilateral**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Six equilateral triangles meeting at a point fill an angle of

**A.** 300°

**B.** 360°

**C.** 60°

**D.** 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

1307

**QUESTION ID**

**S36-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 360°**

**GRADING NOTE** $6 \times 60$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

1307

**ANSWER KEY** Reduced copy of the question page, with its answer

A regular hexagon has side 5. How long are the segments from its center to its vertices?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

1308

QUESTION ID

**S36-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

**True or false?**

A regular hexagon splits into four equilateral triangles from its center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

1309

**QUESTION ID**

## S36-B2-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Six.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSG-CO.D.13

# BACKWARD 3

Circle facts: radius and chord • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A chord equals the radius. What central angle does it subtend?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

1311

QUESTION ID

**S36-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**60°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

**True or false?**

A chord equal to the radius forms an equilateral triangle with the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

1312

**QUESTION ID**

## S36-B3-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** All sides equal r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

A circle has radius 7. A regular inscribed hexagon has perimeter

**A.** 21

**B.** 14

**C.** 42

**D.** 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

1313

**QUESTION ID**

**S36-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 42**

**GRADING NOTE** $6 \times 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has circumference 31.4 (π ≈ 3.14). Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

1314

QUESTION ID

**S36-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

**True or false?**

Every chord of a circle is as long as the radius.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5**

1315

QUESTION ID

## S36-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Chords range up to the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5**

SET 36 • HSG-CO.D.13

# FORWARD 1

Justify that the inscribed hexagon is regular • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Six points are marked around a circle with center O by stepping the radius. Prove that the hexagon they form is regular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

1317

QUESTION ID

**S36-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**The hexagon is regular.**

GRADING NOTE REQUIRED WORK: Show the six triangles from O are equilateral (all sides r), so all sides equal r and all angles equal 120°.

Any valid proof. Essential: each triangle O-consecutive points has three sides equal to r (equilateral, 60° at O); six 60° angles fill 360°, so the stepping closes; all sides equal r; each hexagon angle is 60° + 60° = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A compass set to the radius of a circle is stepped around the circle from a starting point. Why does the sixth step land exactly back on the starting point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

1318

QUESTION ID

**S36-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Each step makes a 60° central angle, and 6 × 60° = 360°.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

**True or false?**

Each interior angle of a regular hexagon is 60°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

1319

QUESTION ID

**S36-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Each interior angle is 120°; 60° is the central angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

A regular hexagon inscribed in a circle of radius 4 has perimeter

**A.** 12

**B.** 48

**C.** 24

**D.** 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

1320

**QUESTION ID**

**S36-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 24**

**GRADING NOTE** 6 × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

Find the area of a regular hexagon inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

1321

**QUESTION ID**

## S36-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**$6\sqrt{3}$**

**GRADING NOTE** Form required: exact. Six equilateral triangles of side 2: $6 \cdot \sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

# FORWARD 2

Use inscribed hexagons to estimate π • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A regular hexagon is inscribed in a circle of radius r. Compare its perimeter with the circumference. What estimate of π does this give?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

1323

**QUESTION ID**
**S36-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**Perimeter**
**6r < circumference**
**2πr, so π > 3.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

A regular 12-gon inscribed in a circle of radius 1 has perimeter about 6.21. What estimate of π does it give?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

1324

**QUESTION ID**

## S36-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**About 3.11 (π > 3.10)**

**GRADING NOTE** Any value 3.10-3.11 with explanation (half the perimeter).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

**True or false?**

Doubling the number of sides of an inscribed regular polygon makes its perimeter longer than the circumference.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

1325

QUESTION ID

## S36-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The perimeter gets closer to the circumference but stays shorter (each side is a chord).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

Which polygon's perimeter is closest to the circumference of its circle?

**A.** Inscribed regular 96-gon

**B.** Inscribed regular hexagon

**C.** Inscribed square

**D.** Inscribed triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

1326

QUESTION ID

**S36-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. Inscribed regular 96-gon**

**GRADING NOTE** More sides fit the circle more closely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

Explain why an inscribed polygon's perimeter is always less than the circumference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

1327

**QUESTION ID**

## S36-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Each side is a chord, shorter than the arc it cuts off.**

**GRADING NOTE** Any equivalent explanation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-F2-Q5**

**High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5**

# Verify that a dilation maps a line to a parallel line (or to itself through the center)

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.A.1.a** |
| **BACKWARD 1** | Dilate points with coordinates | **8.G.A.3** |
| **BACKWARD 2** | Compare slopes to test whether lines are parallel | **8.EE.B.6** |
| **BACKWARD 3** | Write the equation of a line through two points | **8.F.B.4** |
| **FORWARD 1** | Dilate lines about a center other than the origin | **HSG-SRT.A.1.a** |
| **FORWARD 2** | Prove algebraically that a dilation maps a line to a parallel line | **HSG-SRT.A.1.a** |

SET 37 • HSG-SRT.A.1.a

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

**ANSWER KEY** Reduced copy of the question page, with its answer

Line $\ell$ is $y = 2x + 1$. Dilate it about the origin with scale factor 3: find the images of (0, 1) and (1, 3), and write an equation of the image line. Is it parallel to $\ell$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

1330

QUESTION ID
**S37-M-Q1**

TYPE
Short answer

CORRECT ANSWER
**(0, 3) and (3, 9); y = 2x + 3; yes, parallel (both slopes are 2).**

GRADING NOTE Form required: both images, the equation, and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

Line m is y = −x, which passes through the origin. Dilate it about the origin with scale factor 2: find the images of (1, −1) and (3, −3). What is the image line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

1331

**QUESTION ID**

## S37-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(2, −2) and (6, −6); the image is y = −x, the same line.**

**GRADING NOTE** Form required: the images and the conclusion that the line maps onto itself.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-M-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5**

The horizontal line y = 4 is dilated about the origin with scale factor ½. Find an equation of the image and compare it with the original.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

1332

**QUESTION ID**

## S37-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**y = 2; it is parallel to y = 4 (both horizontal).**

**GRADING NOTE** Form required: the equation and the comparison. (x, 4) → (½x, 2).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-M-Q3**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5**

A dilation with center C maps line $\ell$, which does not pass through C, to line $\ell'$. Which is true?

**A.** $\ell'$ passes through C

**B.** $\ell' \parallel \ell$

**C.** $\ell'$ is the same line as $\ell$

**D.** $\ell' \perp \ell$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

1333

QUESTION ID
**S37-M-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. $\ell' \parallel \ell$**

**GRADING NOTE** Lines not through the center map to parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

**True or false?**

A dilation about point C maps every line through C onto itself.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

1334

QUESTION ID
**S37-M-Q5**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Each point of the line moves along the line, toward or away from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

SET 37 • HSG-SRT.A.1.a

# BACKWARD 1

Dilate points with coordinates • Grade 8 • Geometry • 8.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Dilate (2, −3) about the origin with scale factor 4.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

1336

QUESTION ID

**S37-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(8, −12)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Dilate (6, 10) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

1337

QUESTION ID

**S37-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**(3, 5)**

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

A dilation about the origin sends (3, 1) to (9, 3). What is the scale factor?

**A.** 6

**B.** 9

**C.** 3

**D.** ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

1338

**QUESTION ID**

**S37-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 3**

**GRADING NOTE** Each coordinate is tripled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

**True or false?**

A dilation about the origin with scale factor 1 leaves every point where it is.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

1339

QUESTION ID

**S37-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** (x, y) → (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

**True or false?**

A dilation with scale factor 2 moves the center to a new point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

1340

**QUESTION ID**

## S37-B1-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** The center is fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

# BACKWARD 2

Compare slopes to test whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (3, 9).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

1342

QUESTION ID

**S37-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

**True or false?**

y = 2x + 1 and y = 2x + 3 are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

1343

QUESTION ID

**S37-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Same slope, different intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

Which line is parallel to y = −3x + 2?

**A.** y = ⅓x

**B.** y = −3

**C.** y = −3x − 5

**D.** y = 3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

1344

**QUESTION ID**

**S37-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. y = −3x − 5**

**GRADING NOTE** Slope −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Are y = 5 and y = 2 parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

1345

QUESTION ID

## S37-B2-Q4

TYPE

Short answer

CORRECT ANSWER

**Yes (both horizontal)**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

**True or false?**

Lines with slopes 2 and −2 are parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

1346

QUESTION ID

**S37-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Different slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSG-SRT.A.1.a

# BACKWARD 3

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Write an equation of the line through (0, 3) and (3, 9).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

1348

QUESTION ID

**S37-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**y = 2x + 3**

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write an equation of the line through (2, −2) and (6, −6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

1349

QUESTION ID

**S37-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**y = −x**

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which point is on y = 2x + 3?

**A.** (5, 1)

**B.** (2, 3)

**C.** (0, 2)

**D.** (1, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

1350

QUESTION ID

**S37-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. (1, 5)**

GRADING NOTE 2(1) + 3 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

**True or false?**

The line through (0, 0) and (4, 8) is y = 2x.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

1351

QUESTION ID

**S37-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Slope 2, intercept 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

**True or false?**

The line through (1, 1) and (2, 4) has slope 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

1352

QUESTION ID

**S37-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $(4 - 1) \div (2 - 1) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSG-SRT.A.1.a

# FORWARD 1

Dilate lines about a center other than the origin • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

**ANSWER KEY** Reduced copy of the question page, with its answer

Dilate the line y = x + 2 about the center (2, 0) with scale factor 2. Find the images of (0, 2) and (2, 4) and the image line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

1354

QUESTION ID

**S37-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(−2, 4) and (2, 8); y = x + 6, parallel to y = x + 2.**

GRADING NOTE Form required: images, equation, and conclusion. Image = center + 2(point − center).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Does the line y = x − 2 pass through the center (2, 0)? What happens to it under any dilation about (2, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

1355

**QUESTION ID**

**S37-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes, 0 = 2 − 2; it maps onto itself.**

**GRADING NOTE** Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

**True or false?**

Under a dilation about (1, 1) with scale factor 3, the point (2, 1) goes to (4, 1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

1356

**QUESTION ID**

## S37-F1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** (1, 1) + 3(1, 0) = (4, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

A dilation about (0, 2) with scale factor ½ maps (4, 6) to

**A.** (4, 4)

**B.** (8, 10)

**C.** (2, 3)

**D.** (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

1357

**QUESTION ID**

## S37-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (2, 4)**

**GRADING NOTE** (0, 2) + ½(4, 4) = (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line $\ell$: $y = 3x$. A dilation about (1, 0) with scale factor 2 maps $\ell$ to $\ell'$. Find $\ell'$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

1358

**QUESTION ID**

**S37-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = 3x + 3$**

**GRADING NOTE** Any equivalent equation. (0, 0) → (−1, 0) and (1, 3) → (1, 6); slope 3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-F1-Q5**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5**

SET 37 • HSG-SRT.A.1.a

# FORWARD 2

Prove algebraically that a dilation maps a line to a parallel line • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Prove that the dilation about the origin with scale factor k ($k > 0$, $k \neq 1$) maps the line $y = mx + b$ to the line $y = mx + kb$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

1360

**QUESTION ID**

## S37-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The image of (x, mx + b) is (kx, kmx + kb), which satisfies y = mx + kb.**

**GRADING NOTE** REQUIRED WORK: Take any point (x, mx + b) on the line, find its image, and show the image satisfies y = mx + kb (and every point of that line is an image).

Any valid proof (paragraph, two-column, or flow). Essential: image (X, Y) = (kx, k(mx + b)); Y = m(kx) + kb = mX + kb; conversely (X, mX + kb) is the image of (X/k, mX/k + b). Conclusion: same slope m, so parallel when b ≠ 0, the same line when b = 0. Invalid: checking only two points of one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

From the general result, when does the image of $y = mx + b$ coincide with the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

1361

**QUESTION ID**

## S37-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**When b = 0 (the line passes through the center).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

**True or false?**

The dilation $(x, y) \to (kx, ky)$ maps the vertical line $x = c$ to the line $x = c + k$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

1362

QUESTION ID

**S37-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It maps $x = c$ to $x = kc$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Under the dilation about the origin with scale factor 4, the line $y = -2x + 5$ maps to

**A.** $y = -8x + 5$

**B.** $y = -2x + 20$

**C.** $y = -2x + 5$

**D.** $y = -8x + 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

1363

**QUESTION ID**

**S37-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $y = -2x + 20$**

**GRADING NOTE** $y = mx + kb$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

Explain why measuring a few dilated lines is not a proof of the property.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

1364

QUESTION ID
**S37-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**Examples cannot cover every line; a proof must work for all lines (with variables).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

# Verify that a dilation multiplies lengths by the scale factor

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.A.1.b** |
| **BACKWARD 1** | Find lengths with the distance formula | **8.G.B.8** |
| **BACKWARD 2** | Scale factor in scale drawings | **7.G.A.1** |
| **BACKWARD 3** | Dilate points about the origin | **8.G.A.3** |
| **FORWARD 1** | Find the center or scale factor from a segment and its image | **HSG-SRT.A.1.b** |
| **FORWARD 2** | Perimeter and area of dilated figures | **HSG-SRT.A.2** |

SET 38 • HSG-SRT.A.1.b

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

Segment AB has A(1, 2) and B(4, 6). Dilate it about the origin with scale factor 3. Find A′B′ and AB, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

1367

**QUESTION ID**

## S38-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**A′(3, 6), B′(12, 18); AB = 5 and A′B′ = 15 = 3 · AB.**

**GRADING NOTE** Form required: images, both lengths, and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

Segment PQ has P(−4, 2) and Q(8, −6). Dilate it about the origin with scale factor ½. Find PQ and P′Q′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

1368

QUESTION ID
**S38-M-Q2**

TYPE
Short answer

CORRECT ANSWER
**P′(−2, 1), Q′(4, −3); PQ = √208 = 4√13 ≈ 14.42 and P′Q′ = √52 = 2√13 ≈ 7.21, half as long.**

**GRADING NOTE** Form required: both lengths (exact or to 0.01) and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

Segment CD has C(2, 1) and D(5, 5). Dilate it about the center (2, 1) with scale factor 2. Find C′D′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

1369

**QUESTION ID**
**S38-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**C′ = (2, 1), D′ = (8, 9); C′D′ = 10 = 2 · CD (CD = 5).**

**GRADING NOTE** Form required: images and lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

A segment 8 cm long is dilated with scale factor 0.75. How long is the image?

**A.** 8.75 cm

**B.** 6 cm

**C.** 10.67 cm

**D.** 7.25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

1370

QUESTION ID

**S38-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 6 cm**

**GRADING NOTE** $0.75 \times 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

**True or false?**

A dilation with scale factor 2.5 makes every segment 2.5 units longer.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

1371

QUESTION ID
**S38-M-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** It multiplies lengths by 2.5; it does not add 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

SET 38 • HSG-SRT.A.1.b

# BACKWARD 1

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (3, 6) and (12, 18).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

1373

QUESTION ID

**S38-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (−2, 1) and (4, −3).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

1374

QUESTION ID

**S38-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**√52 = 2√13**

**GRADING NOTE** Any equivalent exact form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

**A.** √17

**B.** 7

**C.** 17

**D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

1375

QUESTION ID

**S38-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 13**

**GRADING NOTE** √169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

**True or false?**

The distance between (1, 2) and (4, 6) is 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5**

1376

QUESTION ID

## S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** √(9 + 16).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5**

**True or false?**

The distance between (0, 0) and (3, 3) is 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

1377

QUESTION ID

**S38-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $3\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

# BACKWARD 2

Scale factor in scale drawings • Grade 7 • Geometry • 7.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A drawing uses scale factor 4. A 3 cm segment in the original is how long in the drawing?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

1379

QUESTION ID

**S38-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**12 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 15 m wall is drawn 5 cm long. What scale factor (drawing ÷ actual, in the same units) was used?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

1380

QUESTION ID

**S38-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5/1500 = 1/300**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

**True or false?**

A scale factor less than 1 makes a figure smaller.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

1381

QUESTION ID

**S38-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Lengths shrink.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

A rectangle 4 by 6 is enlarged by a scale factor of 1.5. Its new dimensions are

**A.** 5.5 by 7.5

**B.** 4 by 9

**C.** 8 by 12

**D.** 6 by 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

1382

**QUESTION ID**

**S38-B2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 6 by 9**

**GRADING NOTE** Multiply each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

**True or false?**

Enlarging by a scale factor of 2 doubles the area.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

1383

QUESTION ID

**S38-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Area is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSG-SRT.A.1.b

# BACKWARD 3

Dilate points about the origin • Grade 8 • Geometry • 8.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Dilate (4, 6) about the origin with scale factor 3.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

1385

QUESTION ID

**S38-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**(12, 18)**

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Dilate (−4, 2) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

1386

**QUESTION ID**

**S38-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(−2, 1)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

Dilation about the origin sends (2, 5) to (5, 12.5). Scale factor?

**A.** 2.5

**B.** 3

**C.** 0.4

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

1387

**QUESTION ID**

**S38-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 2.5**

**GRADING NOTE** 5 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

**True or false?**

Dilating about the origin multiplies both coordinates by the scale factor.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

1388

**QUESTION ID**

## S38-B3-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** (kx, ky).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

**True or false?**

Dilating (3, 0) by 2 about the origin gives (5, 0).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

1389

QUESTION ID

**S38-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE (6, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSG-SRT.A.1.b

# FORWARD 1

Find the center or scale factor from a segment and its image • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

**ANSWER KEY** Reduced copy of the question page, with its answer

A dilation about the origin maps a segment of length 6 to a segment of length 15. What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

1391

**QUESTION ID**

**S38-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**2.5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

A dilation maps A(1, 1) to A′(3, 3) and B(2, 1) to B′(5, 3). Find the scale factor and the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

1392

QUESTION ID

**S38-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Scale factor 2; center (−1, −1).**

**GRADING NOTE** Form required: both. A′B′ = 2 = 2AB; lines AA′ and BB′ meet at (−1, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

**True or false?**

Lines through each point and its image under a dilation all pass through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

1393

QUESTION ID

**S38-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Each image is on the ray from the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A dilation maps P(4, 0) to P′(1, 0) and its center is the origin. Scale factor?

**A.** 4

**B.** −3

**C.** ¼

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

1394

**QUESTION ID**

**S38-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. ¼**

**GRADING NOTE** 1 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

A segment of length 9 is dilated by an unknown factor k, and the image is 3 units shorter than the original. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

1395

**QUESTION ID**

**S38-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

$\frac{2}{3}$

**GRADING NOTE** Any equivalent form. 9k = 6.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-F1-Q5**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5**

SET 38 • HSG-SRT.A.1.b

# FORWARD 2

Perimeter and area of dilated figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A triangle with perimeter 18 and area 12 is dilated with scale factor 3. Find the new perimeter and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

1397

**QUESTION ID**

**S38-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**Perimeter 54; area 108**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

A square of area 50 is dilated so its area becomes 8. Find the scale factor.

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

1398

**QUESTION ID**

**S38-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

$\frac{2}{5}$

**GRADING NOTE** Any equivalent form. $k^2 = 8/50 = 4/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

**True or false?**

If lengths are multiplied by k, areas are multiplied by 2k.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

1399

QUESTION ID
**S38-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Areas are multiplied by $k^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

A figure is dilated with scale factor ½. Its area is multiplied by

**A.** ½

**B.** 4

**C.** 2

**D.** ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

1400

**QUESTION ID**

**S38-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. ¼**

**GRADING NOTE** $(½)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

Explain why a dilation with factor k multiplies the perimeter by k.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

1401

**QUESTION ID**

## S38-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Every side is multiplied by k, and the perimeter is the sum of the sides.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

# Decide whether two figures are similar using similarity transformations

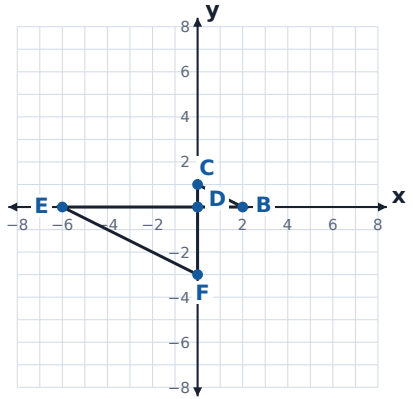
| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.A.2** |
| **BACKWARD 1** | Similar figures as a sequence of transformations including a dilati… | **8.G.A.4** |
| **BACKWARD 2** | Specify a sequence of rigid motions | **HSG-CO.A.5** |
| **BACKWARD 3** | Dilate figures about the origin | **8.G.A.3** |
| **FORWARD 1** | Find the scale factor and correspondence when the image is turne | **HSG-SRT.A.2** |
| **FORWARD 2** | Prove that all circles are similar | **HSG-C.A.1** |

SET 39 • HSG-SRT.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Is triangle ABC similar to triangle DEF? If so, describe a similarity transformation that maps ABC onto DEF; if not, explain.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

1404

**QUESTION ID**

## S39-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes. Dilate by 3 about the origin, then rotate 180° about the origin.**

**GRADING NOTE** Any valid similarity transformation. Check: (2, 0) → (6, 0) → (−6, 0); (0, 1) → (0, 3) → (0, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

Is rectangle R (4 by 6) similar to rectangle S (6 by 8)? Explain using similarity transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

1405

**QUESTION ID**

## S39-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

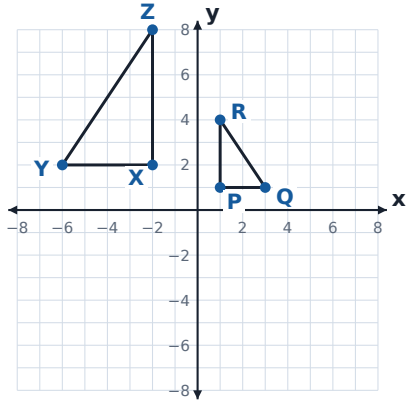
**No. A dilation would need factor 6/4 = 1.5 for one side and 8/6 ≈ 1.33 for the other; no single scale factor works.**

**GRADING NOTE** Any equivalent explanation comparing the ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

Is triangle PQR similar to triangle XYZ? If so, describe a similarity transformation; if not, explain.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

1406

QUESTION ID

**S39-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Yes. Dilate by 2 about the origin, then reflect across the y-axis.**

**GRADING NOTE** Any valid sequence. (1, 1) → (2, 2) → (−2, 2); (3, 1) → (6, 2) → (−6, 2); (1, 4) → (2, 8) → (−2, 8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which pair of figures must be similar?

**A.** Any two isosceles triangles

**B.** Any two rectangles

**C.** Any two rhombuses

**D.** Any two squares

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

1407

**QUESTION ID**

## S39-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. Any two squares

**GRADING NOTE** Two squares differ only by a scale factor (and position); the others can have different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

**True or false?**

If a sequence of rigid motions and one dilation carries figure F onto figure G, then F and G are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

1408

QUESTION ID
**S39-M-Q5**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Definition of similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

SET 39 • HSG-SRT.A.2

# BACKWARD 1

Similar figures as a sequence of transformations including a dilation • Grade 8 • Geometry • 8.G.A.4

**True or false?**

A figure and its image after a dilation and a translation are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

1410

QUESTION ID

**S39-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE A similarity sequence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two figures are similar. What may be different?

**A.** Their shape

**B.** Their size

**C.** Their angle measures

**D.** The ratios of their sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

1411

**QUESTION ID**

## S39-B1-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. Their size

**GRADING NOTE** Size may change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle is dilated by 2 and then rotated. Its sides were 3, 4, 5. What are the image's sides?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

1412

QUESTION ID

**S39-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**6, 8, 10**

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

**True or false?**

Two figures with the same angles but different side ratios are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

1413

QUESTION ID

**S39-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Side ratios must all be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name the transformation that changes size.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

1414

QUESTION ID

**S39-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**A dilation**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

# BACKWARD 2

Specify a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Which rigid motion maps (x, y) to (−x, y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

1416

QUESTION ID

**S39-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Reflection across the y-axis**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

Which rigid motion maps (x, y) to (−x, −y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

1417

QUESTION ID

**S39-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Rotation of 180° about the origin**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

Which rigid motion maps (2, 3) to (−3, 2)?

**A.** Translation 5 left

**B.** Rotation 90° counterclockwise about the origin

**C.** Rotation 180°

**D.** Reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

1418

QUESTION ID

**S39-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. Rotation 90° counterclockwise about the origin**

GRADING NOTE (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

**True or false?**

A translation then a reflection is a valid sequence of rigid motions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

1419

QUESTION ID

**S39-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Any composition of rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

**True or false?**

A dilation is a rigid motion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

1420

QUESTION ID

**S39-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It changes lengths unless k = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSG-SRT.A.2

# BACKWARD 3

Dilate figures about the origin • Grade 8 • Geometry • 8.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Dilate triangle (0, 0), (2, 0), (0, 1) by 3 about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

1422

QUESTION ID

**S39-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**(0, 0), (6, 0), (0, 3)**

GRADING NOTE Form required: the three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

**True or false?**

A dilation about the origin with factor 3 triples each coordinate.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

1423

QUESTION ID

**S39-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** (3x, 3y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

(4, −2) dilated by ½ about the origin is

**A.** (8, −4)

**B.** (2, −1)

**C.** (4.5, −1.5)

**D.** (2, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

1424

QUESTION ID

**S39-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. (2, −1)**

**GRADING NOTE** Halve each coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

A dilation about the origin maps (5, 5) to (2, 2). What is the factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

1425

**QUESTION ID**

## S39-B3-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**0.4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

**True or false?**

A dilation can change the angles of a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

1426

QUESTION ID

**S39-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Angles are preserved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSG-SRT.A.2

# FORWARD 1

Find the scale factor and correspondence when the image is turned or flipped • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides AB = 4, BC = 6, CA = 8. Triangle PQR has PQ = 9, QR = 12, RP = 6. Are they similar? If so, give the correspondence and the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

1428

**QUESTION ID**

**S39-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes: A ↔ R, B ↔ P, C ↔ Q (AB ↔ RP = 6, BC ↔ PQ = 9, CA ↔ QR = 12); scale factor 1.5.**

**GRADING NOTE** Form required: the correspondence and the factor. Match shortest to shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

A 3-4-5 triangle and a triangle with sides 7.5, 10, 12.5: similar? Scale factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

1429

**QUESTION ID**

**S39-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes; 2.5**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-F1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5**

**True or false?**

Matching the shortest side with the shortest side, middle with middle, and longest with longest gives the correspondence for similar triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

1430

QUESTION ID
**S39-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Similarity preserves the order of side lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

Triangles with sides 5, 6, 7 and 10, 12, 15: are they similar?

**A.** Cannot tell

**B.** Yes, factor 2.5

**C.** Yes, factor 2

**D.** No, 15/7 ≠ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

1431

**QUESTION ID**

## S39-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. No, 15/7 ≠ 2**

**GRADING NOTE** Ratios 2, 2, 15/7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

Triangle ABC ~ triangle DEF with a factor of 4 from ABC to DEF. DE = 10. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

1432

**QUESTION ID**

**S39-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**2.5**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-F1-Q5**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5**

SET 39 • HSG-SRT.A.2

# FORWARD 2

Prove that all circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle $C_1$ has center A and radius r; circle $C_2$ has center B and radius s. Prove that $C_1$ and $C_2$ are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

1434

**QUESTION ID**

## S39-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**A translation and a dilation map $C_1$ onto $C_2$.**

**GRADING NOTE** REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r; show the image of $C_1$ is $C_2$.

Any valid proof (paragraph, two-column, or flow). Essential: the translation maps $C_1$ to the circle with center B and radius r; the dilation about B with factor s/r sends each point at distance r from B to a point at distance s, so the image is $C_2$ (and every point of $C_2$ is reached).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

Circle P has radius 4 and circle Q has radius 10. What scale factor maps P onto Q (after a translation)?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

1435

**QUESTION ID**

**S39-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**2.5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

**True or false?**

Any two rectangles are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

1436

QUESTION ID

**S39-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE A 1 × 2 and a 1 × 3 rectangle do not have proportional sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

Which pair of figures is NOT always similar?

**A.** Two equilateral triangles

**B.** Two squares

**C.** Two circles

**D.** Two ellipses

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

1437

**QUESTION ID**

## S39-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. Two ellipses

**GRADING NOTE** Ellipses can have different shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

Explain why all equilateral triangles are similar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

1438

QUESTION ID

**S39-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**All their angles are 60°, so AA applies (or all side ratios are equal).**

**GRADING NOTE** Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

# Explain similarity of triangles as equal angles and proportional sides

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.A.2** |
| **BACKWARD 1** | Read correspondence from a similarity or congruence statement | **HSG-CO.B.7** |
| **BACKWARD 2** | Solve proportions | **7.RP.A.2** |
| **BACKWARD 3** | Scale factor between figures | **7.G.A.1** |
| **FORWARD 1** | Use the AA criterion to recognize similar triangles | **HSG-SRT.A.3** |
| **FORWARD 2** | Use similarity to find unknown lengths in problems | **HSG-SRT.B.5** |

SET 40 • HSG-SRT.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A similarity transformation (a dilation with scale factor k followed by rigid motions) maps △ABC onto △DEF with A → D, B → E, C → F. Explain why ∠A ≅ ∠D, ∠B ≅ ∠E, ∠C ≅ ∠F and DE/AB = EF/BC = FD/CA = k.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

1441

QUESTION ID

**S40-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**Angles are preserved and every length is multiplied by k.**

**GRADING NOTE** REQUIRED WORK: Use what dilations and rigid motions do to angles and to lengths.

Any valid explanation. Essential: dilations and rigid motions preserve angle measure, so corresponding angles are congruent; the dilation multiplies every length by k and rigid motions keep lengths, so each image side is k times its original side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

△ABC ~ △DEF with A ↔ D, B ↔ E, C ↔ F. AB = 6, BC = 9, DE = 10. Find EF and the scale factor from ABC to DEF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

1442

**QUESTION ID**

## S40-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**EF = 15; scale factor 5/3**

**GRADING NOTE** Form required: both. 10/6 = 5/3; EF = 9 · 5/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△PQR ~ △STU with P ↔ S, Q ↔ T, R ↔ U. m∠P = 42° and m∠Q = 71°. Find m∠U.

m∠U = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

1443

**QUESTION ID**

## S40-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**67°**

**GRADING NOTE** Any equivalent form. ∠U ≅ ∠R = 180 − 113.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

Which information shows that △ABC is NOT similar to △XYZ (A ↔ X, B ↔ Y, C ↔ Z)?

**A.** ∠A ≅ ∠X

**B.** AB/XY = BC/YZ = CA/ZX = 2

**C.** AB/XY = 2 but BC/YZ = 3

**D.** All angles of both are 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

1444

**QUESTION ID**

**S40-M-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. AB/XY = 2 but BC/YZ = 3**

**GRADING NOTE** All three side ratios must be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

**True or false?**

If △ABC ~ △DEF, then AB/DE = BC/EF.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

1445

QUESTION ID
**S40-M-Q5**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Corresponding sides are proportional.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

SET 40 • HSG-SRT.A.2

# BACKWARD 1

Read correspondence from a similarity or congruence statement • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

△ABC ~ △XYZ. Which angle corresponds to ∠C?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

1447

QUESTION ID

**S40-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**∠Z**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ABC ~ △XYZ. Which side corresponds to BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

1448

QUESTION ID

**S40-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**YZ**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△KLM ~ △RST. Which ratio equals KL/RS?

**A.** LM/RT

**B.** RS/LM

**C.** LM/ST

**D.** KM/ST

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

1449

QUESTION ID

**S40-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. LM/ST**

GRADING NOTE Second pair of letters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

**True or false?**

△DEF ~ △GHI means ∠E ≅ ∠H.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

1450

QUESTION ID

**S40-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Middle letters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

**True or false?**

△DEF ~ △GHI means DF corresponds to GH.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

1451

QUESTION ID

**S40-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** DF ↔ GI.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

# BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 6/9 = 10/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

1453

QUESTION ID

**S40-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve x/4 = 15/6.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

1454

QUESTION ID

**S40-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which ratio is equal to 3 : 5?

**A.** 9 : 12

**B.** 6 : 8

**C.** 12 : 20

**D.** 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

1455

**QUESTION ID**

## S40-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 12 : 20**

**GRADING NOTE** Multiply both by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

2/3 = 8/12

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

1456

QUESTION ID

**S40-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Both reduce to 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

4/6 = 6/9 = 10/14

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

1457

QUESTION ID

**S40-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 10/14 = 5/7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSG-SRT.A.2

# BACKWARD 3

Scale factor between figures • Grade 7 • Geometry • 7.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A figure is enlarged so that a 4 cm side becomes 10 cm. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

1459

QUESTION ID

**S40-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**2.5**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

**True or false?**

All lengths of a scale drawing are multiplied by the same number.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

**Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5**

1460

QUESTION ID

## S40-B3-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

**Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5**

With scale factor 3, a 7 cm side becomes

**A.** 7 cm

**B.** 10 cm

**C.** 73 cm

**D.** 21 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

1461

QUESTION ID

**S40-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 21 cm**

**GRADING NOTE** Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A scale drawing's 12 cm side represents 60 cm. Scale factor (drawing to actual)?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

1462

QUESTION ID

**S40-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

**True or false?**

A scale drawing changes the angles of a figure.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

1463

QUESTION ID

**S40-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Angles stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSG-SRT.A.2

# FORWARD 1

Use the AA criterion to recognize similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Two triangles have angles of 35° and 80°. Are they similar? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

1465

**QUESTION ID**

## S40-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes, by AA: two pairs of congruent angles (the third angles are both 65°).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

Triangle ABC has m∠A = 50°, m∠B = 60°. Triangle DEF has m∠D = 60°, m∠F = 70°. Are they similar? Give the correspondence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

1466

QUESTION ID

**S40-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Yes: ∠C = 70°, ∠E = 50°; A ↔ E, B ↔ D, C ↔ F.**

**GRADING NOTE** Form required: the decision and the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

**True or false?**

Two right triangles with one pair of congruent acute angles are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

1467

QUESTION ID

**S40-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Two pairs of congruent angles (the right angles and the acute angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

Which pair is NOT necessarily similar?

**A.** Two triangles with angles 30° and 60°

**B.** Two right isosceles triangles

**C.** Two equilateral triangles

**D.** Two isosceles triangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

1468

QUESTION ID

## S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

## D. Two isosceles triangles

**GRADING NOTE** Isosceles triangles can have different vertex angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In a figure, DE ∥ BC with D on AB and E on AC. Explain why △ADE ~ △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

1469

**QUESTION ID**
**S40-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**∠A is shared and ∠ADE ≅ ∠ABC (corresponding angles), so AA.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

# FORWARD 2

Use similarity to find unknown lengths in problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 1.8 m person casts a 2.4 m shadow when a tree casts a 16 m shadow. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

1471

**QUESTION ID**

## S40-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**12 m**

**GRADING NOTE** Any equivalent form. 1.8/2.4 = h/16.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-F2-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with AD = 4, DB = 6, and DE = 5. Find BC.

BC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

1472

**QUESTION ID**

## S40-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**12.5**

**GRADING NOTE** Any equivalent form. AD/AB = DE/BC: 4/10 = 5/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

**True or false?**

In similar triangles with scale factor 3, a 7 cm side corresponds to a 10 cm side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

1473

QUESTION ID

## S40-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** 7 × 3 = 21 cm; a scale factor multiplies, it does not add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 6 ft pole casts a 4 ft shadow. At the same time, a building casts a 30 ft shadow. How tall is the building?

**A.** 36 ft

**B.** 45 ft

**C.** 20 ft

**D.** 24 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

1474

**QUESTION ID**

**S40-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 45 ft**

**GRADING NOTE** 6/4 = h/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two similar triangles have perimeters 24 and 36. A side of the smaller is 10. Find the corresponding side of the larger.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

1475

**QUESTION ID**

## S40-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Any equivalent form. Factor 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

# Establish the AA criterion with similarity transformations

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.A.3** |
| **BACKWARD 1** | Prove triangles congruent with ASA | **HSG-CO.B.8** |
| **BACKWARD 2** | Dilations multiply lengths and keep angles | **HSG-SRT.A.1.b** |
| **BACKWARD 3** | Use the angle sum to find a third angle | **8.G.A.5** |
| **FORWARD 1** | Establish the SAS and SSS similarity criteria | **HSG-SRT.A.3** |
| **FORWARD 2** | Use AA to set up and solve problems | **HSG-SRT.B.5** |

SET 41 • HSG-SRT.A.3

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In △ABC and △DEF, ∠A ≅ ∠D and ∠B ≅ ∠E. Prove that △ABC ~ △DEF using similarity transformations.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

1478

**QUESTION ID**

## S41-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**△ABC ~ △DEF.**

**GRADING NOTE** REQUIRED WORK: Dilate △ABC by k = DE/AB; show the image is congruent to △DEF by ASA; conclude a dilation followed by rigid motions maps ABC onto DEF.

Any valid proof (paragraph, two-column, or flow). Essential: (1) dilate by k = DE/AB to get △A′B′C′ with A′B′ = DE; (2) dilations keep angles, so ∠A′ ≅ ∠D and ∠B′ ≅ ∠E; (3) △A′B′C′ ≅ △DEF by ASA; (4) so rigid motions map A′B′C′ onto DEF, and the dilation followed by those motions maps ABC onto DEF. Invalid: assuming the sides are proportional.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

In the AA proof, why is the scale factor chosen as DE/AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

1479

QUESTION ID
**S41-M-Q2**

TYPE
Short answer

CORRECT ANSWER
**So the image of AB has the same length as DE, making ASA possible.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In the AA proof, after dilating $\triangle ABC$ to $\triangle A'B'C'$, which criterion shows $\triangle A'B'C' \cong \triangle DEF$, and which parts does it use?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

1480

**QUESTION ID**
**S41-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**ASA: $\angle A' \cong \angle D$, $A'B' = DE$, $\angle B' \cong \angle E$.**

**GRADING NOTE** Form required: the criterion and the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

Why is it enough to know TWO pairs of congruent angles?

**A.** The third angle can be different

**B.** Two angles fix the side lengths

**C.** The angle sum makes the third pair congruent too

**D.** Any two triangles are similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

1481

**QUESTION ID**

## S41-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The angle sum makes the third pair congruent too**

**GRADING NOTE** Both third angles equal 180° minus the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

**True or false?**

The AA argument uses the fact that dilations preserve angle measure.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

1482

QUESTION ID

## S41-M-Q5

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The dilated triangle keeps ∠A and ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

SET 41 • HSG-SRT.A.3

# BACKWARD 1

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Two triangles have ∠A ≅ ∠D, AB = DE, ∠B ≅ ∠E. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

1484

**QUESTION ID**

**S41-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**ASA**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

**True or false?**

ASA needs the side between the two angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

1485

QUESTION ID

**S41-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Included side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

Which set of parts gives ASA?

**A.** Three angles

**B.** Three sides

**C.** Two sides and the angle between them

**D.** Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

1486

**QUESTION ID**

## S41-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Two angles and the side between them**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

After proving △PQR ≅ △XYZ, are their corresponding sides equal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

1487

**QUESTION ID**

**S41-B1-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes**

**GRADING NOTE** Corresponding parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

**True or false?**

ASA works only for right triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

1488

QUESTION ID

**S41-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Any triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

# BACKWARD 2

Dilations multiply lengths and keep angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

**ANSWER KEY** Reduced copy of the question page, with its answer

A 5 cm segment is dilated by 1.6. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

1490

**QUESTION ID**

## S41-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**8 cm**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-B2-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5**

**True or false?**

A dilation preserves angle measures.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

1491

QUESTION ID

## S41-B2-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Similar shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A dilation maps AB = 4 to A′B′ = 10. Factor?

**A.** 2.5

**B.** 6

**C.** 0.4

**D.** 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

1492

**QUESTION ID**

**S41-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 2.5**

**GRADING NOTE** 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What scale factor makes a 6 cm segment 6 cm long?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

1493

**QUESTION ID**

## S41-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-B2-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5**

**True or false?**

A dilation by 2 doubles the angles of a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

1494

QUESTION ID

## S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Angles are unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSG-SRT.A.3

# BACKWARD 3

Use the angle sum to find a third angle • Grade 8 • Geometry • 8.G.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has angles 47° and 88°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

1496

QUESTION ID

**S41-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**45°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

**True or false?**

If two angles of one triangle equal two angles of another, the third angles are equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

1497

QUESTION ID

**S41-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Angle sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

Triangles with angles 30°, 70°, x and 70°, 80°, y. Which is true?

**A.** x = 80, y = 30; the triangles are similar

**B.** x = 80, y = 30; not similar

**C.** x = 70, y = 30

**D.** x = 30, y = 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

1498

**QUESTION ID**

## S41-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. x = 80, y = 30; the triangles are similar**

**GRADING NOTE** 180 − 100 = 80; 180 − 150 = 30; same angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

Two angles of a triangle are equal and the third is 40°. Find the equal angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

1499

**QUESTION ID**

## S41-B3-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**70° each**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

**True or false?**

Knowing one angle of each triangle is enough for similarity.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

1500

QUESTION ID

**S41-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Need two pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSG-SRT.A.3

# FORWARD 1

Establish the SAS and SSS similarity criteria • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC and △DEF, DE/AB = DF/AC = k and ∠A ≅ ∠D. Prove that △ABC ~ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

1502

QUESTION ID

**S41-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**△ABC ~ △DEF (SAS similarity).**

GRADING NOTE REQUIRED WORK: Dilate △ABC by k, then use SAS congruence with △DEF.

Any valid proof (paragraph, two-column, or flow). Essential: dilate by k: A′B′ = DE, A′C′ = DF, ∠A′ ≅ ∠A ≅ ∠D; SAS congruence; dilation then rigid motions map ABC onto DEF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

Triangles with sides 4, 6, 8 and 6, 9, 12. Similar? Which criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

1503

**QUESTION ID**

## S41-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes, SSS similarity (ratio 1.5).**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-F1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5**

**True or false?**

Two triangles with two pairs of proportional sides and congruent included angles are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

1504

QUESTION ID

## S41-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** SAS similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is NOT a similarity criterion?

**A.** SAS

**B.** SSA

**C.** SSS

**D.** AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

1505

QUESTION ID

**S41-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. SSA**

**GRADING NOTE** SSA does not determine shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△PQR: PQ = 5, PR = 7, ∠P = 40°. △XYZ: XY = 10, XZ = 14, ∠X = 40°. Similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

1506

QUESTION ID

**S41-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**Yes (SAS similarity); factor 2**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

# FORWARD 2

Use AA to set up and solve problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two vertical poles are 3 m and 7 m tall. A wire from the top of each to the bottom of the other crosses at a point P. How high above the ground is P? (Use similar triangles.)

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

1508

QUESTION ID

**S41-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2.1 m**

GRADING NOTE Any equivalent form. h = (3 · 7)/(3 + 7) = 2.1, from two pairs of AA-similar triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with ∠ADE ≅ ∠ACB. AD = 4, AE = 6, AB = 12. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

1509

**QUESTION ID**

**S41-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**8**

**GRADING NOTE** Any equivalent form. △ADE ~ △ACB (AA): AD/AC = AE/AB.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-F2-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5**

**True or false?**

A mirror on the ground can be used to find a building's height with similar triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

1510

QUESTION ID

## S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Equal angles of incidence and reflection give AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Diagonals of trapezoid ABCD (AB ∥ DC) meet at E. AB = 12, DC = 4, AE = 9. Find EC.

**A.** 6

**B.** 27

**C.** 4

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

1511

QUESTION ID

**S41-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 3**

**GRADING NOTE** △ABE ~ △CDE (AA): EC = 9 · 4/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

A 2 m stick casts a 1.5 m shadow. A flagpole casts a 9 m shadow. How tall is the flagpole?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

1512

**QUESTION ID**

## S41-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**12 m**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

# Prove the triangle proportionality theorem

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.B.4** |
| **BACKWARD 1** | Use AA to show triangles are similar | **HSG-SRT.A.3** |
| **BACKWARD 2** | Corresponding angles of parallel lines | **HSG-CO.C.9** |
| **BACKWARD 3** | Rewrite and solve proportions algebraically | **7.RP.A.2** |
| **FORWARD 1** | Prove that parallel lines cut transversals proportionally | **HSG-SRT.B.4** |
| **FORWARD 2** | Prove the angle bisector theorem | **HSG-SRT.B.5** |

SET 42 • HSG-SRT.B.4

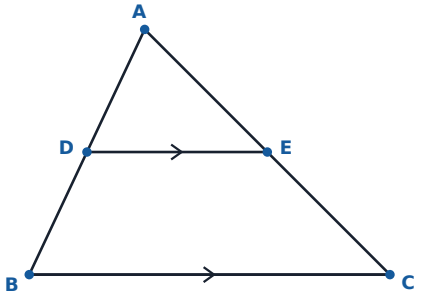
# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

1515

QUESTION ID

## S42-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

**AD/DB = AE/EC**

**GRADING NOTE** REQUIRED WORK: Prove △ADE ~ △ABC by AA, write AD/AB = AE/AC, then rewrite with AB = AD + DB and AC = AE + EC.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A shared; ∠ADE ≅ ∠ABC (corresponding angles, DE ∥ BC); AA; AD/AB = AE/AC; so AB/AD = AC/AE, (AD + DB)/AD = (AE + EC)/AE, 1 + DB/AD = 1 + EC/AE, DB/AD = EC/AE, and AD/DB = AE/EC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

In the proof of the triangle proportionality theorem, the similarity gives AD/AB = AE/AC. Show the algebra that turns this into AD/DB = AE/EC.

Work: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

1516

**QUESTION ID**

## S42-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Take reciprocals: AB/AD = AC/AE; write AB = AD + DB and AC = AE + EC; subtract 1 from both sides: DB/AD = EC/AE; take reciprocals: AD/DB = AE/EC.**

**GRADING NOTE** Any correct algebraic route.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △PQR, a line parallel to QR meets PQ at S and PR at T. Prove that PS/SQ = PT/TR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

1517

**QUESTION ID**

## S42-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**PS/SQ = PT/TR**

**GRADING NOTE** REQUIRED WORK: AA similarity of △PST and △PQR, then the same algebra.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and corresponding angles give △PST ~ △PQR; then the proportion algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

A proof of the triangle proportionality theorem:

1. ∠ADE ≅ ∠ABC (?)
2. ∠A ≅ ∠A (?)
3. △ADE ~ △ABC (?)
4. AD/AB = AE/AC (?)

Give the reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

1518

**QUESTION ID**

## S42-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**1: corresponding angles (DE ∥ BC); 2: reflexive; 3: AA; 4: corresponding sides of similar triangles are proportional.**

**GRADING NOTE** Any equivalent names for all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

In the proof of the triangle proportionality theorem (DE ∥ BC in △ABC, with D on AB and E on AC), which fact does the proof use first?

**A.** AB = AC

**B.** D and E are midpoints

**C.** DE ∥ BC makes corresponding angles congruent

**D.** DE = ½BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

1519

**QUESTION ID**

## S42-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. DE ∥ BC makes corresponding angles congruent**

**GRADING NOTE** Only DE ∥ BC is given.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSG-SRT.B.4

# BACKWARD 1

Use AA to show triangles are similar • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**True or false?**

Two pairs of congruent angles make two triangles similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

1521

QUESTION ID

**S42-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

DE ∥ BC in △ABC. Name two pairs of congruent angles in △ADE and △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

1522

QUESTION ID

**S42-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**∠A ≅ ∠A and ∠ADE ≅ ∠ABC (or ∠AED ≅ ∠ACB)**

**GRADING NOTE** Any two valid pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

△ADE ~ △ABC. Which ratio equals AD/AB?

**A.** DB/AB

**B.** AE/AC

**C.** DE/AB

**D.** AE/EC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

1523

QUESTION ID

**S42-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. AE/AC**

**GRADING NOTE** Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ADE ~ △ABC with AD = 3, AB = 9, AE = 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

1524

QUESTION ID

**S42-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

**True or false?**

△ADE ~ △ABC means AD/DB = DE/BC.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

1525

QUESTION ID
**S42-B1-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** DB is not a side of either triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

# BACKWARD 2

Corresponding angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

When a transversal crosses parallel lines, corresponding angles are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

1527

QUESTION ID

**S42-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

DE ∥ BC and AB is a transversal. Which angle corresponds to ∠ADE?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

1528

QUESTION ID

**S42-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**∠ABC**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

DE ∥ BC, AC a transversal. ∠AED corresponds to

**A.** ∠BAC

**B.** ∠ABC

**C.** ∠DEC

**D.** ∠ACB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

1529

QUESTION ID

**S42-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. ∠ACB**

**GRADING NOTE** Same position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

∠ABC = 65° and DE ∥ BC. Find ∠ADE.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

1530

**QUESTION ID**

**S42-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**65°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

**True or false?**

Corresponding angles are always supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

1531

QUESTION ID

**S42-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Congruent for parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSG-SRT.B.4

# BACKWARD 3

Rewrite and solve proportions algebraically • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

If a/b = c/d, write the equation you get by taking reciprocals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

1533

QUESTION ID

**S42-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**b/a = d/c**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 4/(4 + x) = 6/15.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

1534

QUESTION ID

**S42-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

GRADING NOTE Any equivalent form. 60 = 24 + 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If (a + b)/a = (c + d)/c, then

**A.** b/a = d/c

**B.** b = d

**C.** a = c

**D.** a/b = c/d is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

1535

QUESTION ID

**S42-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. b/a = d/c**

**GRADING NOTE** Subtract 1 from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

3/5 = 6/10 implies 5/3 = 10/6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

1536

QUESTION ID

**S42-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

(2 + 3)/2 = 5/2 implies 3/2 = 5/2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

1537

QUESTION ID

## S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Subtracting 1 gives 3/2 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSG-SRT.B.4

# FORWARD 1

Prove that parallel lines cut transversals proportionally • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Three parallel lines cut transversal t at A, B, C and transversal u at D, E, F (in order). Prove that AB/BC = DE/EF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

1539

QUESTION ID

**S42-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**AB/BC = DE/EF**

GRADING NOTE

REQUIRED WORK: Draw segment AF; apply the triangle proportionality theorem in △ACF and in △FDA.

Any valid proof (paragraph, two-column, or flow). Essential: AF crosses the middle line at G; in △ACF, BG ∥ CF so AB/BC = AG/GF; in △FAD, GE ∥ AD so FG/GA = FE/ED, so AG/GF = DE/EF; combine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Three parallel lines cut one transversal into segments 4 and 6 and another into segments 5 and x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

1540

QUESTION ID

**S42-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**7.5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

**True or false?**

Parallel lines that cut equal segments on one transversal cut equal segments on every transversal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

1541

QUESTION ID

## S42-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Ratio 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC, AD = 5, DB = 3, AE = 7.5. Find EC.

**A.** 12.5

**B.** 4.5

**C.** 6

**D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

1542

**QUESTION ID**

**S42-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 4.5**

**GRADING NOTE** 5/3 = 7.5/EC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Lots between parallel streets have frontages 30 m and 45 m on one avenue. On another avenue the first frontage is 40 m. Find the second.

Frontage: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

1543

QUESTION ID
**S42-F1-Q5**

TYPE
Short answer

CORRECT ANSWER
**60 m**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

# FORWARD 2

Prove the angle bisector theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A meets BC at D. Prove that BD/DC = AB/AC. (Draw the line through C parallel to AD, meeting line AB at E.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

1545

**QUESTION ID**

## S42-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

## BD/DC = AB/AC

**GRADING NOTE** REQUIRED WORK: Show △ACE is isosceles (AE = AC), then apply the triangle proportionality theorem in △BCE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAD ≅ ∠AEC (corresponding) and ∠DAC ≅ ∠ACE (alternate interior), and ∠BAD ≅ ∠DAC, so ∠AEC ≅ ∠ACE and AE = AC; in △BCE, AD ∥ EC gives BD/DC = BA/AE = AB/AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

In △ABC, the bisector of ∠A meets BC at D. AB = 9, AC = 6, BC = 10. Find BD and DC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

1546

**QUESTION ID**

## S42-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**BD = 6, DC = 4**

**GRADING NOTE** Form required: both. BD/DC = 9/6.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-F2-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5**

**True or false?**

The bisector of an angle of a triangle always meets the opposite side at its midpoint.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

1547

QUESTION ID

## S42-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Only when the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The bisector of ∠P in △PQR divides QR into 3 and 5. PQ = 6. Find PR.

**A.** 10

**B.** 15

**C.** 3.6

**D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

1548

**QUESTION ID**

## S42-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 10**

**GRADING NOTE** 3/5 = 6/PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

Explain why the angle bisector theorem is a relationship that similarity or proportionality, not congruence, proves.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

1549

**QUESTION ID**

## S42-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**It states a ratio of lengths, which comes from proportional segments (parallel lines), not from equal lengths.**

**GRADING NOTE** Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

# Prove the converse of the triangle proportionality theorem

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.B.4** |
| **BACKWARD 1** | Use the triangle proportionality theorem | **HSG-SRT.B.4** |
| **BACKWARD 2** | SAS similarity | **HSG-SRT.A.3** |
| **BACKWARD 3** | Converse of the corresponding angles theorem | **HSG-CO.C.9** |
| **FORWARD 1** | Use the converse to decide parallelism in figures | **HSG-SRT.B.4** |
| **FORWARD 2** | Prove the trapezoid midsegment theorem | **HSG-SRT.B.5** |

SET 43 • HSG-SRT.B.4

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In △ABC, D is on AB and E is on AC with AD/DB = AE/EC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

1552

**QUESTION ID**

## S43-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**DE ∥ BC**

**GRADING NOTE** REQUIRED WORK: Turn the given ratio into AD/AB = AE/AC, prove △ADE ~ △ABC by SAS similarity, then use congruent corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AD/DB = AE/EC gives AD/AB = AE/AC (add 1 to the reciprocals); ∠A shared; SAS similarity; ∠ADE ≅ ∠ABC; congruent corresponding angles make DE ∥ BC (converse). Invalid: assuming DE ∥ BC to get the similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △PQR, S is on PQ and T is on PR with PS = 4, SQ = 6, PT = 6, TR = 9. Prove that ST ∥ QR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

1553

QUESTION ID

**S43-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**ST ∥ QR**

GRADING NOTE REQUIRED WORK: Show PS/PQ = PT/PR, then SAS similarity and corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: PS/PQ = 4/10 = 0.4 and PT/PR = 6/15 = 0.4; ∠P shared; SAS similarity; corresponding angles congruent; converse of corresponding angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Show the algebra that turns AD/DB = AE/EC into AD/AB = AE/AC.

Work: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

1554

QUESTION ID

**S43-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Reciprocals: DB/AD = EC/AE; add 1: (AD + DB)/AD = (AE + EC)/AE, so AB/AD = AC/AE; reciprocals: AD/AB = AE/AC.**

GRADING NOTE Any correct route.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

In △ABC, AD = 3, DB = 5, AE = 4.5, EC = 7.5 (D on AB, E on AC). Is DE ∥ BC?

**A.** Yes, because 3/5 = 4.5/7.5

**B.** No, because 3 ≠ 4.5

**C.** Yes, because D and E are midpoints

**D.** No, because 5 ≠ 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

1555

**QUESTION ID**

## S43-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. Yes, because 3/5 = 4.5/7.5**

**GRADING NOTE** Both ratios equal 0.6, so the converse applies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

**True or false?**

In △ABC, D is on AB and E is on AC with AD = 2, DB = 3, AE = 3, EC = 4. Then DE ∥ BC.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

1556

QUESTION ID

## S43-M-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** 2/3 ≠ 3/4, so DE is not parallel to BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSG-SRT.B.4

# BACKWARD 1

Use the triangle proportionality theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC, AD = 6, DB = 4, AE = 9. Find EC.

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

1558

**QUESTION ID**

**S43-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S43-B1-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5**

**True or false?**

If DE ∥ BC in △ABC, then AD/DB = AE/EC.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

1559

QUESTION ID
**S43-B1-Q2**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** The theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

DE ∥ BC, AD = 2, AB = 8, AC = 12. Find AE.

**A.** 9

**B.** 4

**C.** 6

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

1560

**QUESTION ID**

**S43-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 3**

**GRADING NOTE** 2/8 = AE/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

State the converse of "If DE ∥ BC, then AD/DB = AE/EC."

Converse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

1561

QUESTION ID
**S43-B1-Q4**

TYPE
Short answer

CORRECT ANSWER
**If AD/DB = AE/EC, then DE ∥ BC.**

**GRADING NOTE** Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

**True or false?**

A theorem and its converse are always both true.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

1562

QUESTION ID

## S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Some converses are false; this one must be proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

# BACKWARD 2

SAS similarity • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**True or false?**

Two pairs of proportional sides with congruent included angles make triangles similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

1564

QUESTION ID

**S43-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** SAS similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

△ADE and △ABC share ∠A, AD/AB = AE/AC = 0.4. Are they similar? Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

1565

**QUESTION ID**

## S43-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

## Yes, SAS similarity

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

After △ADE ~ △ABC, which angle is congruent to ∠ADE?

**A.** ∠ACB

**B.** ∠BAC

**C.** ∠ABC

**D.** ∠DEA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

1566

**QUESTION ID**

**S43-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. ∠ABC**

**GRADING NOTE** D ↔ B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

Ratios 3/6 and 4/8 with a shared included angle: similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

1567

**QUESTION ID**

## S43-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes; factor 2 (or ½)**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S43-B2-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5**

**True or false?**

SAS similarity needs the angle to be between the two pairs of sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

1568

QUESTION ID

## S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSG-SRT.B.4

# BACKWARD 3

Converse of the corresponding angles theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

Congruent corresponding angles make two lines parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

1570

QUESTION ID

**S43-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

∠ADE ≅ ∠ABC with transversal AB. What can you conclude about DE and BC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

1571

**QUESTION ID**

**S43-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**DE ∥ BC**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

Which angles, if congruent, prove DE ∥ BC?

**A.** ∠A and ∠B

**B.** ∠B and ∠C

**C.** ∠AED and ∠ACB

**D.** ∠ADE and ∠AED

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

1572

QUESTION ID

**S43-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. ∠AED and ∠ACB**

GRADING NOTE Corresponding angles along AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

Corresponding angles measure 58° and 62°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

1573

QUESTION ID

**S43-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**No**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

**True or false?**

Vertical angles prove lines parallel.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

1574

QUESTION ID

**S43-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE They are always congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSG-SRT.B.4

# FORWARD 1

Use the converse to decide parallelism in figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, D is on AB with AD = 5, DB = 10, and E is on AC with AE = 4. How long must EC be for DE ∥ BC?

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

1576

**QUESTION ID**

**S43-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △PQR, S on PQ, T on PR: PS = x, SQ = 6, PT = 4, TR = 8. For which x is ST ∥ QR?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

1577

**QUESTION ID**

**S43-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

**True or false?**

The midsegment theorem's parallel part is a special case of the converse (ratio 1 : 1).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

1578

QUESTION ID

## S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** AD/DB = AE/EC = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

Points divide two sides in ratios 2 : 3 and 4 : 6. The segment joining them is

**A.** perpendicular to the third side

**B.** half the third side

**C.** parallel to the third side

**D.** not parallel to the third side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

1579

**QUESTION ID**

## S43-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. parallel to the third side**

**GRADING NOTE** 2 : 3 = 4 : 6.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S43-F1-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, D is on AB with AD : DB = 1 : 2 and DE ∥ BC with E on AC. Find DE : BC.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

1580

QUESTION ID

**S43-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**1 : 3**

**GRADING NOTE** Any equivalent form. AD : AB = 1 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

# FORWARD 2

Prove the trapezoid midsegment theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC. M and N are the midpoints of legs AD and BC. Prove that MN ∥ AB and MN = ½(AB + DC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1582

QUESTION ID

**S43-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**MN ∥ AB and MN = ½(AB + DC)**

**GRADING NOTE** REQUIRED WORK: Draw diagonal AC; let P be the midpoint of AC; use the midsegment theorem (or the converse) in △ACD and △ABC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ADC, MP ∥ DC and MP = ½DC; in △ABC, PN ∥ AB and PN = ½AB; AB ∥ DC so MP and PN are both parallel to AB through P, so M, P, N are collinear; MN = ½(AB + DC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A trapezoid has bases 10 and 16. Find the length of the segment joining the midpoints of the legs.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1583

QUESTION ID
**S43-F2-Q2**

TYPE
Short answer

CORRECT ANSWER
**13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

**True or false?**

The midsegment of a trapezoid is half the length of the longer base.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1584

QUESTION ID
**S43-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** It is half the sum of the bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A trapezoid's midsegment is 12 and one base is 9. Find the other base.

**A.** 3

**B.** 15

**C.** 6

**D.** 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1585

**QUESTION ID**

**S43-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 15**

**GRADING NOTE** $(9 + b)/2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why M, P, N are on one line in the trapezoid midsegment proof.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1586

**QUESTION ID**

## S43-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**MP and PN are both parallel to the bases through P, and only one line through P is parallel to them.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

# Prove the Pythagorean Theorem using similar triangles

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.B.4** |
| **BACKWARD 1** | Show triangles similar with AA | **HSG-SRT.A.3** |
| **BACKWARD 2** | Cross-multiply proportions | **7.RP.A.2** |
| **BACKWARD 3** | Explain a proof of the Pythagorean Theorem informally | **8.G.B.6** |
| **FORWARD 1** | Use the geometric mean relationships in a right triangle | **HSG-SRT.B.5** |
| **FORWARD 2** | The Law of Cosines as a generalization of the Pythagorean Theor | **HSG-SRT.D.10** |

SET 44 • HSG-SRT.B.4

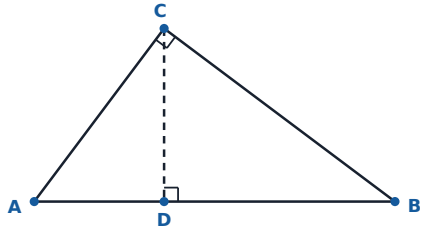
# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangle ABC has the right angle at C, legs a = BC, b = AC, and hypotenuse c = AB. The altitude from C meets AB at D. Prove that $a^2 + b^2 = c^2$.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

1589

QUESTION ID

## S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

**$a^2 + b^2 = c^2$**

**GRADING NOTE** REQUIRED WORK: Show △ACD ~ △ABC and △CBD ~ △ABC (AA), write two proportions, and add.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (shared ∠A, right angles), so AD/AC = AC/AB and $b^2 = c \cdot AD$; △CDB ~ △ACB (shared ∠B), so BD/BC = BC/AB and $a^2 = c \cdot BD$; adding, $a^2 + b^2 = c(AD + BD) = c \cdot c = c^2$. Invalid: assuming $a^2 + b^2 = c^2$ anywhere; using the distance formula (it depends on the theorem).

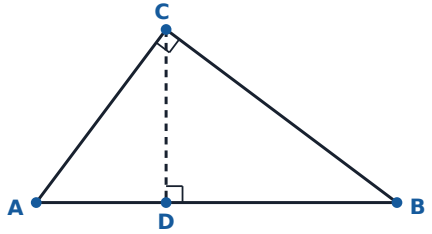
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In the similarity proof of the Pythagorean Theorem, why is △ADC ~ △ACB?

Answer: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

1590

QUESTION ID

**S44-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**They share ∠A and both have a right angle (at D and at C), so AA applies.**

GRADING NOTE Any equivalent answer naming both pairs of angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC with the right angle at C, legs a and b, and hypotenuse c, the altitude from C meets AB at D. Similar triangles give $b^2 = c \cdot AD$ and $a^2 = c \cdot BD$. Show how these give $a^2 + b^2 = c^2$.

Work: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

1591

QUESTION ID

**S44-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**$a^2 + b^2 = c \cdot BD + c \cdot AD = c(BD + AD) = c \cdot c = c^2$, since $AD + DB = AB = c$.**

**GRADING NOTE** Any correct algebra, including why AD + BD = c (D is between A and B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has legs 6 and 8, and the altitude to the hypotenuse divides it into parts x and y. Use similar triangles (not the Pythagorean Theorem) to find x and y, and show that the hypotenuse is 10.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

1592

QUESTION ID

**S44-M-Q4**

TYPE

Answer with required work

CORRECT ANSWER

**c = 10, x = 3.6, y = 6.4**

GRADING NOTE

REQUIRED WORK: Use $6^2 = c \cdot x$ and $8^2 = c \cdot y$ with $x + y = c$.

Any valid proof (paragraph, two-column, or flow). Essential: from similarity $36 = cx$ and $64 = cy$; adding, $100 = c(x + y) = c^2$, so $c = 10$, $x = 3.6$, $y = 6.4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

Right triangle ABC has its right angle at C, and the altitude from C meets hypotenuse AB at D. Which pair of triangles is NOT similar?

**A.** △ACD and △ABC

**B.** △BCD and △ABC

**C.** △ACD and △CBD

**D.** None; all three triangles are similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

1593

**QUESTION ID**
**S44-M-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. None; all three triangles are similar**

**GRADING NOTE** All three triangles are similar (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

SET 44 • HSG-SRT.B.4

# BACKWARD 1

Show triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**True or false?**

Two right triangles that share an acute angle are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

1595

QUESTION ID
**S44-B1-Q1**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ADC and △ACB: name two pairs of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

1596

QUESTION ID
**S44-B1-Q2**

TYPE
Short answer

CORRECT ANSWER
**∠A ≅ ∠A; ∠ADC ≅ ∠ACB (right angles)**

**GRADING NOTE** Any two valid pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ADC ~ △ACB. Which ratio equals AD/AC?

**A.** DC/AB

**B.** AD/AB

**C.** AC/AB

**D.** CB/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

1597

QUESTION ID

**S44-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. AC/AB**

**GRADING NOTE** AD ↔ AC, AC ↔ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If △ADC ~ △ACB with AD = 4 and AB = 9, find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

1598

QUESTION ID

**S44-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form. $AC^2 = AD \cdot AB = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

**True or false?**

Similar triangles must be the same size.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

1599

QUESTION ID

## S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

GRADING NOTE Only the shape matches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

# BACKWARD 2

Cross-multiply proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

From AD/AC = AC/AB, write an equation without fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

1601

QUESTION ID

**S44-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$AC^2 = AD \cdot AB$**

**GRADING NOTE** Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $4/x = x/9$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

1602

QUESTION ID

**S44-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If $p/q = q/r$, then

**A.** $q^2 = pr$

**B.** $p = r$

**C.** $r^2 = pq$

**D.** $p^2 = qr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

1603

QUESTION ID

**S44-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. $q^2 = pr$**

**GRADING NOTE** Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

From $a/b = b/c$ we get $b^2 = ac$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

1604

QUESTION ID

**S44-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

From x/5 = 5/x we get x = 5 only.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

1605

QUESTION ID

**S44-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $x^2 = 25$, so $x = \pm 5$ ($x = 5$ for a length).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSG-SRT.B.4

# BACKWARD 3

Explain a proof of the Pythagorean Theorem informally • Grade 8 • Geometry • 8.G.B.6

**True or false?**

The Pythagorean Theorem says $a^2 + b^2 = c^2$ for every right triangle with legs a, b and hypotenuse c.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

1607

**QUESTION ID**

## S44-B3-Q1

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name a way to show the theorem by rearranging area pieces.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

1608

QUESTION ID

**S44-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**Arrange four copies of the triangle inside a square of side a + b (the leftover is $c^2$, or $a^2 + b^2$).**

GRADING NOTE Any valid area argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

Which triangle satisfies the theorem?

**A.** 5, 12, 13

**B.** 6, 7, 9

**C.** 5, 6, 7

**D.** 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

1609

**QUESTION ID**

**S44-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 5, 12, 13**

**GRADING NOTE** 25 + 144 = 169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Legs 9 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

1610

QUESTION ID

**S44-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

**True or false?**

Checking the theorem with three triangles proves it for all right triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

1611

**QUESTION ID**

## S44-B3-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Examples are not proofs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSG-SRT.B.4

# FORWARD 1

Use the geometric mean relationships in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC with altitude CD to hypotenuse AB, AD = 4 and DB = 9. Find CD.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

1613

QUESTION ID

**S44-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form. $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C and altitude CD, prove that $CD^2 = AD \cdot DB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

1614

**QUESTION ID**

## S44-F1-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$CD^2 = AD \cdot DB$**

**GRADING NOTE** REQUIRED WORK: Show △ACD ~ △CBD and write a proportion.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △CDB (AA: right angles at D, and ∠A ≅ ∠BCD since both are complementary to ∠B); AD/CD = CD/DB.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5**

**True or false?**

The altitude to the hypotenuse is the geometric mean of the two parts of the hypotenuse.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

1615

QUESTION ID

## S44-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $CD^2 = AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

Right triangle ABC has its right angle at C, and the altitude from C meets hypotenuse AB at D. AD = 2 and AB = 8. Find AC.

**A.** 4

**B.** $2\sqrt{2}$

**C.** 6

**D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

1616

QUESTION ID
**S44-F1-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**A. 4**

**GRADING NOTE** $AC^2 = AD \cdot AB = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle's altitude to the hypotenuse is 12 and one part of the hypotenuse is 9. Find the other part.

Part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

1617

**QUESTION ID**

**S44-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**16**

**GRADING NOTE** Any equivalent form. 144 = 9x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

# FORWARD 2

The Law of Cosines as a generalization of the Pythagorean Theorem • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

For a triangle with sides a, b, c and angle C opposite c, the Law of Cosines says $c^2 = a^2 + b^2 - 2ab \cos C$. What does it become when $C = 90°$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

1619

**QUESTION ID**

## S44-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$c^2 = a^2 + b^2$ (the Pythagorean Theorem), since $\cos 90° = 0$.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

1620

QUESTION ID

**S44-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**7**

GRADING NOTE Any equivalent form. 25 + 64 − 40 = 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

**True or false?**

If $a^2 + b^2 > c^2$, the angle opposite c is obtuse.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

1621

QUESTION ID

**S44-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is acute; the angle is obtuse when $a^2 + b^2 < c^2$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F2-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5**

A triangle has sides 4, 5, 7. The largest angle is

**A.** right

**B.** impossible to tell

**C.** obtuse

**D.** acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

1622

**QUESTION ID**

**S44-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. obtuse**

**GRADING NOTE** $16 + 25 = 41 < 49$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5**

Use the converse idea: is a triangle with sides 7, 24, 25 right, acute, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

1623

**QUESTION ID**

## S44-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Right**

**(49 + 576 = 625)**

**GRADING NOTE** Form required: the classification with the check.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F2-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5**

# Use similarity criteria to solve problems in geometric figures

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.B.5** |
| **BACKWARD 1** | Similarity criteria: AA, SAS, SSS | **HSG-SRT.A.3** |
| **BACKWARD 2** | Solve proportions | **7.RP.A.2** |
| **BACKWARD 3** | Identify corresponding sides in overlapping triangles | **HSG-SRT.A.2** |
| **FORWARD 1** | Solve multistep problems with two or more pairs of similar triangl… | **HSG-SRT.B.5** |
| **FORWARD 2** | Solve similarity problems with algebra (unknowns on both sides) | **HSG-SRT.B.5** |

SET 45 • HSG-SRT.B.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 9, DE = 8. Find BC. Name the criterion you used for similar triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

1626

**QUESTION ID**

**S45-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**BC = 20; △ADE ~ △ABC by AA.**

**GRADING NOTE** Form required: the length and the criterion. 6/15 = 8/BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A lamppost lights a student who is 1.6 m tall and stands 3 m from the post. The student's shadow is 2 m long, measured away from the post. How tall is the lamppost? Name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

1627

QUESTION ID

**S45-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**4 m; AA (the shared angle at the tip of the shadow, and two right angles).**

GRADING NOTE Form required: the height and the criterion. The tip of the shadow is 5 m from the post: 1.6/2 = h/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

Two triangles: △PQR with PQ = 8, QR = 12, ∠Q = 50°; △STU with ST = 6, TU = 9, ∠T = 50°. Are they similar? If so, find the ratio of their perimeters.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

1628

QUESTION ID

**S45-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Yes, SAS similarity (8/6 = 12/9 = 4/3); perimeter ratio 4 : 3.**

GRADING NOTE Form required: the criterion and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

A surveyor finds a river's width w with similar right triangles: legs 12 m and 18 m on the small triangle, 12 m corresponding to 40 m on the large one. What is w (the side matching 18 m)?

**A.** 26.7 m

**B.** 46 m

**C.** 54 m

**D.** 60 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

1629

**QUESTION ID**

**S45-M-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 60 m**

**GRADING NOTE** $18 \cdot 40/12 = 60$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

**True or false?**

If two triangles have sides 3, 5, 7 and 6, 10, 15, they are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

1630

QUESTION ID

**S45-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $15/7 \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

SET 45 • HSG-SRT.B.5

# BACKWARD 1

Similarity criteria: AA, SAS, SSS • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Which criterion uses two pairs of congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

1632

QUESTION ID

**S45-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**AA**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-B1-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5**

Sides 2, 3, 4 and 4, 6, 8. Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

1633

QUESTION ID

**S45-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**SSS similarity**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

Which information is enough for similarity?

**A.** Two pairs of congruent angles

**B.** One pair of congruent angles

**C.** One pair of proportional sides

**D.** Two sides of each triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

1634

QUESTION ID

**S45-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. Two pairs of congruent angles**

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

**True or false?**

SAS similarity requires the congruent angle to be between the proportional sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

1635

QUESTION ID

## S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

**True or false?**

Any two right triangles are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

1636

QUESTION ID

## S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** They need another congruent angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

# BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 6/15 = 8/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

1638

QUESTION ID

**S45-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**20**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 1.6/2 = h/5.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

1639

QUESTION ID

**S45-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 12/40 = 18/x.

**A.** 26.7

**B.** 54

**C.** 46

**D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

1640

QUESTION ID

**S45-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 60**

**GRADING NOTE** x = 18 · 40/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

3/4 = 9/12

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

1641

QUESTION ID

**S45-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

5/8 = 10/14

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

1642

QUESTION ID

**S45-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 10/16 would be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5

# BACKWARD 3

Identify corresponding sides in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

△ADE ~ △ABC. Which side of △ABC corresponds to DE?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

1644

QUESTION ID

**S45-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**BC**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

**True or false?**

In △ADE ~ △ABC, side AD corresponds to side AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

1645

QUESTION ID

## S45-B3-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC. In △ADE ~ △ABC, the side of the large triangle that corresponds to AD is

**A.** DB

**B.** AC

**C.** BC

**D.** AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

1646

**QUESTION ID**

## S45-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. AB

**GRADING NOTE** Whole side, not the piece DB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

AD = 6, DB = 9. What is AB?

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

1647

**QUESTION ID**

**S45-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-B3-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

In △ADE ~ △ABC, DB corresponds to DE.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

1648

QUESTION ID

**S45-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE DB is not a side of △ADE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSG-SRT.B.5

# FORWARD 1

Solve multistep problems with two or more pairs of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Trapezoid ABCD has AB ∥ DC, AB = 15, DC = 10. The diagonals meet at E, and AC = 20. Find AE and EC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

1650

**QUESTION ID**

## S45-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**AE = 12, EC = 8**

**GRADING NOTE** Form required: both. AE/EC = 15/10.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-F1-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A pinhole camera: an object 2 m tall is 8 m from the pinhole; the image is 5 cm from the pinhole on the screen. How tall is the image?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

1651

**QUESTION ID**

## S45-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**1.25 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

**True or false?**

In a figure with two pairs of similar triangles, one ratio found in the first pair can be reused in the second.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

1652

QUESTION ID

## S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Shared sides link them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with AD = 2, DB = 6; and FG ∥ BC farther down with AF = 5. If BC = 16, find FG.

**A.** 10

**B.** 12

**C.** 4

**D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

1653

QUESTION ID

**S45-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 10**

**GRADING NOTE** FG/BC = AF/AB = 5/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Using a mirror on the ground, Ana (eyes 1.5 m high) stands 2 m from the mirror and sees the top of a tree 12 m beyond the mirror. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

1654

**QUESTION ID**
**S45-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**9 m**

**GRADING NOTE** Any equivalent form. 1.5/2 = h/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

# FORWARD 2

Solve similarity problems with algebra (unknowns on both sides) • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with AD = x, DB = 6, AE = 4, EC = x + 2. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

1656

QUESTION ID

**S45-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form. x(x + 2) = 24, x > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Similar triangles have corresponding sides x + 1 and 2x − 3 with scale factor 1.5 (larger to smaller is 2x − 3 to x + 1). Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

1657

**QUESTION ID**

**S45-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**9**

**GRADING NOTE** Any equivalent form. 2x − 3 = 1.5(x + 1).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-F2-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5**

**True or false?**

A proportion with the unknown in two places can lead to a quadratic equation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

1658

QUESTION ID

## S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Cross-multiplying gives products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

In right triangle ABC with the right angle at C, the altitude CD meets hypotenuse AB at D. AD = x, DB = x + 5, and CD = 6. Find x.

**A.** 4

**B.** 3

**C.** 6

**D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

1659

**QUESTION ID**

## S45-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 4**

**GRADING NOTE** x(x + 5) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two similar rectangles: the first is 4 by x, the second is x by 9. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

1660

QUESTION ID

**S45-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form. 4/x = x/9.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-F2-Q5**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5**

# Use congruence criteria to prove relationships in figures

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.B.5** |
| **BACKWARD 1** | Choose and apply SSS, SAS, ASA, AAS | **HSG-CO.B.8** |
| **BACKWARD 2** | Find hidden parts: shared sides, vertical angles, right angles | **HSG-CO.C.9** |
| **BACKWARD 3** | Use corresponding parts to finish a proof | **HSG-CO.B.7** |
| **FORWARD 1** | Prove relationships that need two congruences in a row | **HSG-SRT.B.5** |
| **FORWARD 2** | Decide which given data are sufficient for a proof, and supply a m… | **HSG-SRT.B.5** |

SET 46 • HSG-SRT.B.5

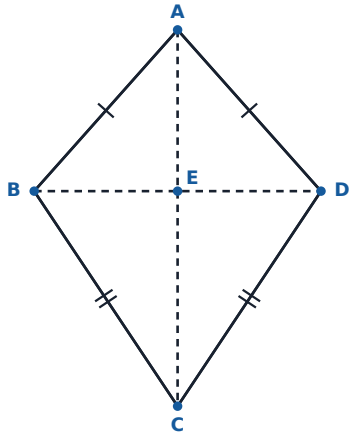
# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Kite ABCD has AB = AD and CB = CD. Its diagonals meet at E. Prove that AC ⊥ BD.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

1663

QUESTION ID

**S46-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**AC ⊥ BD**

**GRADING NOTE** REQUIRED WORK: Prove △ABC ≅ △ADC (SSS) to get ∠BAE ≅ ∠DAE; then △ABE ≅ △ADE (SAS); then the angles at E form a congruent linear pair.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = AD, CB = CD, AC shared); ∠BAC ≅ ∠DAC; SAS on △ABE and △ADE (AE shared); ∠AEB ≅ ∠AED, a linear pair, so 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

Segments AB and CD bisect each other at M. Prove that AC ∥ DB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

1664

QUESTION ID

**S46-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**AC ∥ DB**

**GRADING NOTE** REQUIRED WORK: Prove △AMC ≅ △BMD (SAS with vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM, CM = DM (bisect), ∠AMC ≅ ∠BMD (vertical); SAS; ∠CAM ≅ ∠DBM; these are alternate interior angles for AC and DB with transversal AB; converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Point P is on the bisector of ∠ABC. PD ⊥ BA at D and PE ⊥ BC at E. Prove that PD = PE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

1665

**QUESTION ID**

## S46-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**PD = PE**

**GRADING NOTE** REQUIRED WORK: Prove △PBD ≅ △PBE by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PBD ≅ ∠PBE (bisector), ∠PDB ≅ ∠PEB (right angles), PB shared; AAS; PD = PE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, AB = AC. M is the midpoint of BC. Prove that AM bisects ∠BAC and AM ⊥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

1666

QUESTION ID

**S46-M-Q4**

TYPE

Answer with required work

CORRECT ANSWER

**∠BAM ≅ ∠CAM and AM ⊥ BC**

GRADING NOTE REQUIRED WORK: Prove △ABM ≅ △ACM (SSS), then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: SSS; ∠BAM ≅ ∠CAM (bisects); ∠AMB ≅ ∠AMC form a linear pair, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

To prove that two segments in a figure are congruent, the most common first step is to

**A.** prove two triangles containing them are congruent

**B.** measure them carefully

**C.** show the figure is symmetric by looking

**D.** assume they are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

1667

**QUESTION ID**
**S46-M-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. prove two triangles containing them are congruent**

**GRADING NOTE** Then use corresponding parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

SET 46 • HSG-SRT.B.5

# BACKWARD 1

Choose and apply SSS, SAS, ASA, AAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Two triangles share a side and have two more pairs of congruent sides. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

1669

QUESTION ID

**S46-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**SSS**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two angles and a non-included side match. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

1670

QUESTION ID

**S46-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**AAS**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

Which pair of facts plus a shared side gives SAS?

**A.** A pair of congruent sides and the congruent included angles

**B.** Two pairs of congruent sides not including the angle

**C.** Nothing more is needed

**D.** Two pairs of congruent angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

1671

QUESTION ID

**S46-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. A pair of congruent sides and the congruent included angles**

**GRADING NOTE** Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

**True or false?**

A shared side is a valid pair for any criterion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

1672

QUESTION ID

**S46-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Reflexive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

AAA proves congruence.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

1673

QUESTION ID

**S46-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

# BACKWARD 2

Find hidden parts: shared sides, vertical angles, right angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Segments AB and CD cross at M. Which angles are vertical to ∠AMC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

1675

QUESTION ID

**S46-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**∠BMD**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

**True or false?**

Perpendicular lines give congruent right angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

1676

QUESTION ID

**S46-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE All 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

Which is a hidden fact in figures where two triangles share a side?

**A.** All angles are equal

**B.** The shared side is congruent to itself

**C.** The triangles are right

**D.** Nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

1677

**GRADING NOTE** Reflexive property.

**QUESTION ID**

## S46-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. The shared side is congruent to itself**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

A bisector divides ∠ABC into which two congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

1678

**QUESTION ID**

## S46-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**∠ABP and ∠PBC (for bisector ray BP)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Vertical angles are supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

1679

QUESTION ID

**S46-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSG-SRT.B.5

# BACKWARD 3

Use corresponding parts to finish a proof • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△ABC ≅ △ADC. Which angle corresponds to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

1681

QUESTION ID

**S46-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**∠DAC**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

**True or false?**

Once triangles are congruent, all corresponding parts are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

1682

QUESTION ID

**S46-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

△PBD ≅ △PBE. Which conclusion follows?

**A.** PD = PE

**B.** PB = PD

**C.** BD = PE

**D.** ∠D ≅ ∠B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

1683

QUESTION ID

**S46-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. PD = PE**

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two congruent angles form a linear pair. Each measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

1684

QUESTION ID

**S46-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

**True or false?**

Corresponding parts can be used before the triangles are proved congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

1685

QUESTION ID

**S46-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only after.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSG-SRT.B.5

# FORWARD 1

Prove relationships that need two congruences in a row • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB = CD and AD = CB. Diagonals AC and BD meet at E. Prove that E is the midpoint of AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

1687

**QUESTION ID**

## S46-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

## AE = CE

**GRADING NOTE** REQUIRED WORK: Prove △ABC ≅ △CDA (SSS) to get parallel sides, then △ABE ≅ △CDE (ASA).

Any valid proof (paragraph, two-column, or flow). Essential: SSS gives alternate interior angles congruent (AB ∥ DC); then ASA (or AAS) for △ABE and △CDE gives AE = CE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, AB = AC. D is on AB and E is on AC with BD = CE. Prove that BE = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

1688

QUESTION ID

**S46-F1-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**BE = CD**

GRADING NOTE REQUIRED WORK: Show AD = AE, then prove △ABE ≅ △ACD by SAS (shared ∠A).

Any valid proof (paragraph, two-column, or flow). Essential: AD = AB − BD = AC − CE = AE; ∠A shared; AB = AC; SAS; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

**True or false?**

Some proofs need one congruence to find a part used in a second congruence.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

1689

QUESTION ID

## S46-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Chained congruences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

In a two-step proof, the first congruence gives $\angle 1 \cong \angle 2$. This is used to

**A.** measure

**B.** end the proof

**C.** supply a pair of parts for the second congruence

**D.** restart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

1690

QUESTION ID

**S46-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. supply a pair of parts for the second congruence**

**GRADING NOTE** Chaining.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

Explain why you cannot use corresponding parts of the second pair of triangles before proving them congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

1691

**QUESTION ID**
## S46-F1-Q5

**TYPE**
Short answer

**CORRECT ANSWER**

**That would assume what must be proved (circular reasoning).**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

# FORWARD 2

Decide which given data are sufficient for a proof, and supply a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △PQR and △PSR (sharing PR), PQ = PS. What one more fact would prove ∠QPR ≅ ∠SPR? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

1693

**QUESTION ID**
**S46-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**QR = SR (then SSS gives the congruence and the angles correspond).**

**GRADING NOTE** Any sufficient condition with explanation (∠PRQ ≅ ∠PRS would give SSA, which is not enough).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A student claims a kite's diagonals bisect each other. Prove or disprove.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

1694

**QUESTION ID**

## S46-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**False in general: only the diagonal BD is bisected (by AC); AC is bisected only if the kite is a rhombus. Example: A(0, 3), B(−1, 0), C(0, −1), D(1, 0).**

**GRADING NOTE** Any valid counterexample or explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

**True or false?**

Given AB = DE, BC = EF, and ∠C ≅ ∠F, the triangles must be congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

1695

QUESTION ID

**S46-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** SSA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

Given ∠A ≅ ∠D and AB = DE in △ABC and △DEF, which extra fact gives ASA?

**A.** ∠C ≅ ∠F

**B.** BC = EF

**C.** AC = DF

**D.** ∠B ≅ ∠E

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

1696

**QUESTION ID**

## S46-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. ∠B ≅ ∠E**

**GRADING NOTE** The angle at the other end of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

Given AB = DE and AC = DF, name an angle condition that guarantees congruence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

1697

**QUESTION ID**

## S46-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**∠A ≅ ∠D (the included angle)**

**GRADING NOTE** Form required: the included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

# Use similarity criteria to prove relationships in figures

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.B.5** |
| **BACKWARD 1** | Prove triangles similar with AA | **HSG-SRT.A.3** |
| **BACKWARD 2** | Turn a proportion into a product | **7.RP.A.2** |
| **BACKWARD 3** | Match corresponding sides of similar triangles | **HSG-SRT.A.2** |
| **FORWARD 1** | Use proved product relationships to compute | **HSG-SRT.B.5** |
| **FORWARD 2** | Prove relationships in circles with similar triangles | **HSG-C.A.2** |

SET 47 • HSG-SRT.B.5

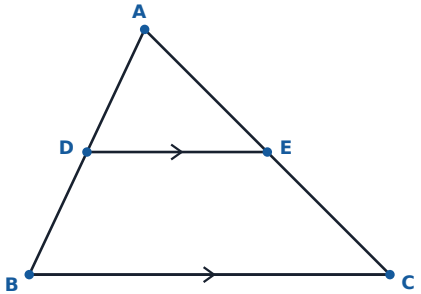
# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, DE ∥ BC with D on AB and E on AC. Prove that AD · AC = AE · AB.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

1700

QUESTION ID

## S47-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

**AD · AC = AE · AB**

**GRADING NOTE** REQUIRED WORK: Prove △ADE ~ △ABC (AA), write the proportion, and cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: AA (shared ∠A; corresponding angles); AD/AB = AE/AC; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Chords AB and CD of a circle meet at P, and ∠BAD ≅ ∠BCD (given). Prove that PA · PB = PC · PD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

1701

QUESTION ID

**S47-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**PA · PB = PC · PD**

GRADING NOTE REQUIRED WORK: Prove △PAD ~ △PCB (AA: the given angles and vertical angles), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PAD ≅ ∠PCB (given), ∠APD ≅ ∠CPB (vertical); AA; PA/PC = PD/PB; cross-multiply.

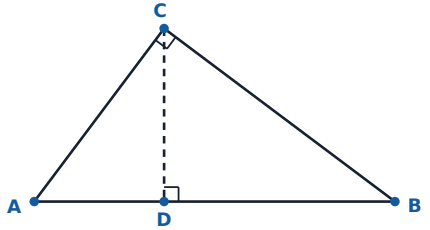
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), CD is the altitude to AB. Prove that AC · BC = AB · CD.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

1702

QUESTION ID

**S47-M-Q3**

TYPE

Answer with required work

CORRECT ANSWER

**AC · BC = AB · CD**

**GRADING NOTE** REQUIRED WORK: Prove △ACB ~ △ADC (or △CDB) and use the ratio of corresponding sides.

Any valid proof (paragraph, two-column, or flow). Essential: △ADC ~ △ACB (AA); DC/CB = AC/AB; cross-multiply. Alternative: both products equal twice the area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In trapezoid ABCD with AB ∥ DC, the diagonals AC and BD meet at E. Prove that AE/EC = AB/DC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

1703

QUESTION ID

**S47-M-Q4**

TYPE

Answer with required work

CORRECT ANSWER

**AE/EC = AB/DC**

GRADING NOTE REQUIRED WORK: Prove △ABE ~ △CDE by AA (alternate interior angles or vertical angles).

Any valid proof (paragraph, two-column, or flow). Essential: ∠EAB ≅ ∠ECD and ∠EBA ≅ ∠EDC (alternate interior, AB ∥ DC); AA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

To prove a product relationship such as PA · PB = PC · PD, which is the usual strategy?

**A.** Use the Pythagorean Theorem only

**B.** Measure the four lengths

**C.** Find two similar triangles whose sides appear in the products

**D.** Assume the product is true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

1704

**QUESTION ID**

## S47-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. Find two similar triangles whose sides appear in the products**

**GRADING NOTE** A proportion from similar triangles becomes a product by cross-multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

SET 47 • HSG-SRT.B.5

# BACKWARD 1

Prove triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**True or false?**

Vertical angles can be one pair of congruent angles in an AA proof.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

1706

QUESTION ID

**S47-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two triangles share ∠A and have another pair of congruent angles. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

1707

QUESTION ID
**S47-B1-Q2**

TYPE
Short answer

CORRECT ANSWER
**AA**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S47-B1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5**

Parallel lines in a figure usually supply

**A.** equal side lengths

**B.** a scale factor

**C.** right angles

**D.** congruent corresponding or alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

1708

**QUESTION ID**

**S47-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. congruent corresponding or alternate interior angles**

**GRADING NOTE** Angle pairs.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S47-B1-Q3**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

△PAD ~ △PCB. Which side of △PCB corresponds to PA?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

1709

**QUESTION ID**

## S47-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**PC**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S47-B1-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5**

**True or false?**

Similarity requires all three pairs of sides to be equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

1710

QUESTION ID

**S47-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

# BACKWARD 2

Turn a proportion into a product • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

From a/b = c/d, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

1712

QUESTION ID

**S47-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**ad = bc**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From PA/PC = PD/PB, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

1713

**QUESTION ID**

**S47-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**PA · PB = PC · PD**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If x/3 = 4/6, then

**A.** 6x = 12

**B.** x = 8

**C.** 3x = 24

**D.** 4x = 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

1714

QUESTION ID

**S47-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 6x = 12**

**GRADING NOTE** Cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

a/b = c/d implies ab = cd.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

1715

QUESTION ID

**S47-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It implies ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

2/5 = 6/15 implies 2 · 15 = 5 · 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

1716

QUESTION ID

**S47-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** 30 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSG-SRT.B.5

# BACKWARD 3

Match corresponding sides of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

△ADE ~ △ABC. Write the proportion with AD, AB, AE, AC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

1718

QUESTION ID

**S47-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**AD/AB = AE/AC**

**GRADING NOTE** Any equivalent proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

**True or false?**

In △ABE ~ △CDE, AB corresponds to CD.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

1719

QUESTION ID
**S47-B3-Q2**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ACB ~ △ADC. Which side corresponds to CB?

**A.** DC

**B.** AD

**C.** AB

**D.** AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

1720

**QUESTION ID**

## S47-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. DC

**GRADING NOTE** A ↔ A, C ↔ D, B ↔ C, so CB ↔ DC.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S47-B3-Q3**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

△XYZ ~ △UVW. Write all three equal ratios.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

1721

QUESTION ID

**S47-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**XY/UV = YZ/VW = XZ/UW**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

**True or false?**

Any two sides of similar triangles are proportional.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

1722

QUESTION ID
**S47-B3-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Only corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSG-SRT.B.5

# FORWARD 1

Use proved product relationships to compute • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Chords AB and CD meet at P with PA = 4, PB = 9, PC = 6. Find PD.

PD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

1724

**QUESTION ID**

**S47-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form. 4 · 9 = 6 · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), AC = 6, BC = 8, AB = 10. Find the altitude CD to AB.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

1725

**QUESTION ID**

**S47-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**4.8**

**GRADING NOTE** Any equivalent form. 6 · 8 = 10 · CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

**True or false?**

If two chords meet inside a circle, the products of their segments are equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

1726

QUESTION ID
**S47-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Proved by similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Trapezoid diagonals: AB = 18, DC = 12, AE = 9. Find EC.

**A.** 8

**B.** 4

**C.** 13.5

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

1727

**QUESTION ID**

**S47-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 6**

**GRADING NOTE** AE/EC = 18/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

DE ∥ BC in △ABC with AD = 3, AB = 9, AC = 12. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

1728

**QUESTION ID**

## S47-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**4**

**GRADING NOTE** Any equivalent form. AD · AC = AE · AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

# FORWARD 2

Prove relationships in circles with similar triangles • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

From point P outside a circle, a tangent touches the circle at T and a secant meets it at A and B (A between P and B). Given ∠PTA ≅ ∠PBT, prove that $PT^2 = PA \cdot PB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

1730

QUESTION ID

**S47-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**$PT^2 = PA \cdot PB$**

GRADING NOTE REQUIRED WORK: Prove △PTA ~ △PBT (AA), then cross-multiply.

Any valid proof (paragraph, two-column, or flow). Essential: shared ∠P and the given angles; AA; PT/PB = PA/PT; cross-multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

A tangent segment from P is 8 and the secant from P has outer part 4. Find the whole secant length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

1731

**QUESTION ID**

**S47-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**16**

**GRADING NOTE** Any equivalent form. 64 = 4 · PB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

**True or false?**

Two secants from an outside point P satisfy PA + PB = PC + PD.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

1732

QUESTION ID

**S47-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The products are equal: PA · PB = PC · PD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From P, secants meet a circle at A, B and at C, D with PA = 3, PB = 12, PC = 4. Find PD.

**A.** 7

**B.** 1

**C.** 16

**D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

1733

QUESTION ID

**S47-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 9**

**GRADING NOTE** $36 = 4 \cdot PD$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

Explain why the angle facts used in these circle proofs depend on inscribed angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

1734

**QUESTION ID**
**S47-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**Angles that intercept the same arc are congruent (inscribed angle theorem), which supplies the AA pairs.**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

# Understand trigonometric ratios as properties of an angle, by similarity

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.C.6** |
| **BACKWARD 1** | Similar triangles have equal side ratios within each triangle | **HSG-SRT.A.2** |
| **BACKWARD 2** | Name the opposite side, adjacent side, and hypotenuse for an ang | **HSG-SRT.A.3** |
| **BACKWARD 3** | Write and simplify ratios | **6.RP.A.1** |
| **FORWARD 1** | Find trig ratios from one ratio, using a similar right triangle | **HSG-SRT.C.6** |
| **FORWARD 2** | The unit circle extends the ratios beyond acute angles | **HSF-TF.A.2** |

SET 48 • HSG-SRT.C.6

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Right triangles ABC and DEF have right angles at C and F and ∠A ≅ ∠D. Explain why BC/AB = EF/DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

1737

QUESTION ID

**S48-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**BC/AB = EF/DE**

GRADING NOTE

REQUIRED WORK: Use AA similarity, then the proportion of corresponding sides, rearranged.

Any valid proof (paragraph, two-column, or flow). Essential: right angles and ∠A ≅ ∠D give AA similarity; BC/EF = AB/DE; rearranging gives BC/AB = EF/DE. So the ratio depends only on the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two right triangles each have a 30° angle. In one, the side opposite 30° is 4 and the hypotenuse is 8. In the other, the hypotenuse is 15. Find the side opposite 30°, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

1738

**QUESTION ID**

## S48-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**7.5; the triangles are similar (AA), so opposite/hypotenuse = 4/8 = ½ in both.**

**GRADING NOTE** Form required: the length and the similarity reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle PQR, the right angle is at R, PR = 5, QR = 12, PQ = 13. Write sin P, cos P, and tan P.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

1739

QUESTION ID

**S48-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**sin P = 12/13, cos P = 5/13, tan P = 12/5**

**GRADING NOTE** Form required: all three ratios (any equivalent fractions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

Why is sin 40° the same in every right triangle with a 40° angle?

**A.** All such triangles are similar, so the opposite-to-hypotenuse ratio is the same

**B.** Every hypotenuse is 1

**C.** The sides are always 3, 4, 5

**D.** Sine depends on the triangle's size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

1740

**QUESTION ID**

## S48-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. All such triangles are similar, so the opposite-to-hypotenuse ratio is the same**

**GRADING NOTE** Similarity fixes the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

**True or false?**

Doubling every side of a right triangle doubles the tangent of its acute angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

1741

QUESTION ID

## S48-M-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The ratio stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

SET 48 • HSG-SRT.C.6

# BACKWARD 1

Similar triangles have equal side ratios within each triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangles with sides 3, 4, 5 and 6, 8, 10 are similar.
Compare 3/5 and 6/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

1743

**QUESTION ID**
**S48-B1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**Both equal 0.6.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

**True or false?**

If △ABC ~ △DEF, then AB/AC = DE/DF.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

1744

QUESTION ID

## S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Ratios within each triangle match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

△ABC ~ △DEF with BC/AB = 0.8. Find EF/DE.

**A.** Cannot tell

**B.** 0.8

**C.** 1.25

**D.** 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

1745

**QUESTION ID**

**S48-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 0.8**

**GRADING NOTE** Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has legs 2 and 3. A similar one has shorter leg 8. Find the longer leg.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

1746

QUESTION ID
**S48-B1-Q4**

TYPE
Short answer

CORRECT ANSWER
**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

**True or false?**

Similar triangles have equal side lengths.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

1747

QUESTION ID

## S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Proportional, not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

# BACKWARD 2

Name the opposite side, adjacent side, and hypotenuse for an angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is opposite ∠A?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

1749

QUESTION ID

**S48-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**BC**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC with right angle at C, which side is adjacent to ∠A (not the hypotenuse)?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

1750

QUESTION ID
**S48-B2-Q2**

TYPE
Short answer

CORRECT ANSWER
**AC**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

The hypotenuse is always

**A.** opposite the right angle

**B.** next to the right angle

**C.** the shortest side

**D.** opposite the smallest angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

1751

QUESTION ID

**S48-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. opposite the right angle**

**GRADING NOTE** Longest side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

**True or false?**

The side opposite ∠A is adjacent to ∠B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

1752

QUESTION ID

**S48-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE BC is next to ∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

**True or false?**

The opposite side of an angle depends on the triangle's size.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

1753

QUESTION ID

**S48-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It depends on position.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSG-SRT.C.6

# BACKWARD 3

Write and simplify ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Write 12 : 13 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

1755

QUESTION ID

**S48-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**12/13**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify 8/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

1756

QUESTION ID

**S48-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**4/5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

Which equals 0.75?

**A.** 7/5

**B.** 3/7

**C.** 3/4

**D.** 4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

1757

**QUESTION ID**

## S48-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 3/4**

**GRADING NOTE** 3 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

5/13 ≈ 0.385

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

1758

QUESTION ID

**S48-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

**True or false?**

A ratio changes when both quantities are tripled.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

1759

QUESTION ID

## S48-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSG-SRT.C.6

# FORWARD 1

Find trig ratios from one ratio, using a similar right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

For an acute angle θ, sin θ = 3/5. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

1761

QUESTION ID

**S48-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**cos θ = 4/5, tan θ = 3/4**

**GRADING NOTE** Form required: both. A 3-4-5 triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

tan θ = 5/12 for an acute angle θ. Find sin θ.

sin θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

1762

QUESTION ID

**S48-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**5/13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

**True or false?**

For any acute angle, $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

1763

QUESTION ID

**S48-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

cos θ = 0.6. Which is sin θ (θ acute)?

**A.** 0.8

**B.** 1.6

**C.** 0.4

**D.** 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

1764

**QUESTION ID**

## S48-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 0.8**

**GRADING NOTE** √(1 − 0.36).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

Explain why $\sin\theta < 1$ for every acute angle $\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

1765

**QUESTION ID**

## S48-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The opposite leg is shorter than the hypotenuse.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

# FORWARD 2

The unit circle extends the ratios beyond acute angles • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

On a unit circle, the point for angle θ is (cos θ, sin θ).
Find the point for 90°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

1767

QUESTION ID
**S48-F2-Q1**

TYPE
Short answer

CORRECT ANSWER
**(0, 1)**

**GRADING NOTE** Form required: an ordered pair.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S48-F2-Q1**

**High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the coordinates of the point on the unit circle at 30°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

1768

QUESTION ID
**S48-F2-Q2**

TYPE
Short answer

CORRECT ANSWER
**(√3/2, ½)**

**GRADING NOTE** Form required: exact values.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S48-F2-Q2**

**High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5**

**True or false?**

For acute angles, the unit-circle definition of sine gives a different value from the right-triangle definition.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

1769

QUESTION ID
**S48-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** They agree: the hypotenuse is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin 150° on the unit circle equals

**A.** −½

**B.** −√3/2

**C.** √3/2

**D.** ½

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

1770

QUESTION ID
**S48-F2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**D. ½**

**GRADING NOTE** Point (−√3/2, ½).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S48-F2-Q4**

**High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the unit circle lets us define sine for 120°, when no right triangle has a 120° acute angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

1771

QUESTION ID

## S48-F2-Q5

TYPE

Short answer

CORRECT ANSWER

**The point on the unit circle at 120° has a y-coordinate, which defines sin 120°.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

# Explain and use the relationship between the sine and cosine of complementary angles

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.C.7** |
| **BACKWARD 1** | Complementary angles add to 90° | **7.G.B.5** |
| **BACKWARD 2** | Opposite and adjacent sides depend on the chosen angle | **HSG-SRT.C.6** |
| **BACKWARD 3** | Angle sum in a right triangle | **8.G.A.5** |
| **FORWARD 1** | Relationships between tangents of complementary angles | **HSG-SRT.C.7** |
| **FORWARD 2** | The Pythagorean identity | **HSF-TF.C.8** |

SET 49 • HSG-SRT.C.7

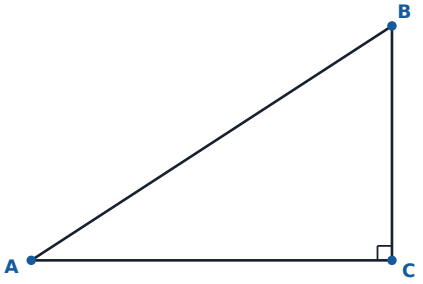
# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C. Explain why sin A = cos B.

Work space




Conclusion: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

1774

**QUESTION ID**

## S49-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**sin A = BC/AB = cos B, and A and B are complementary.**

**GRADING NOTE** REQUIRED WORK: Name the side opposite ∠A and the side adjacent to ∠B, and use the angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = BC/AB (opposite over hypotenuse); cos B = BC/AB (BC is adjacent to ∠B); so they are equal; ∠A + ∠B = 90°, so sin A = cos(90° − A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin 25° = cos x°, where 0 < x < 90. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

1775

QUESTION ID

**S49-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**65**

**GRADING NOTE** Any equivalent form. Complementary angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

θ is acute and cos θ = 0.6. Find sin(90° − θ).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

1776

QUESTION ID

**S49-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**0.6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Both angles are acute and sin(2x + 10)° = cos(3x)°.
Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

1777

**QUESTION ID**
**S49-M-Q4**

**TYPE**
Short answer

**CORRECT ANSWER**
**16**

**GRADING NOTE** Any equivalent form. 2x + 10 + 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which expression equals cos 72°?

**A.** cos 18°

**B.** sin 72°

**C.** sin 18°

**D.** tan 18°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

1778

QUESTION ID
**S49-M-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**C. sin 18°**

GRADING NOTE 72 + 18 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

SET 49 • HSG-SRT.C.7

# BACKWARD 1

Complementary angles add to 90° • Grade 7 • Geometry • 7.G.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the complement of 37°.

Complement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

1780

QUESTION ID

**S49-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**53°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

**True or false?**

The two acute angles of a right triangle are complementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

1781

**QUESTION ID**

## S49-B1-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** They add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Angles x and 2x are complementary. x =

**A.** 90°

**B.** 45°

**C.** 60°

**D.** 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

1782

**QUESTION ID**

**S49-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 30°**

**GRADING NOTE** 3x = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The complement of θ is 90° − θ. If θ = 64°, find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

1783

QUESTION ID

**S49-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**26°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Complementary angles must be adjacent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

1784

QUESTION ID

**S49-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They need only add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

# BACKWARD 2

Opposite and adjacent sides depend on the chosen angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

In right triangle ABC (right angle at C), name the side adjacent to ∠B.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

1786

**QUESTION ID**

## S49-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**BC**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-B2-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5**

**True or false?**

The side opposite ∠A is the side adjacent to ∠B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

1787

**QUESTION ID**

## S49-B2-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Both are BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

cos B equals

**A.** AC/AB

**B.** BC/AB

**C.** AC/BC

**D.** BC/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

1788

**QUESTION ID**

## S49-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. BC/AB

**GRADING NOTE** Adjacent over hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

Write sin A using the sides of right triangle ABC (right angle at C).

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

1789

**QUESTION ID**

**S49-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**BC/AB**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-B2-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

sin A and cos A are always equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

1790

QUESTION ID

**S49-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only when A = 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSG-SRT.C.7

# BACKWARD 3

Angle sum in a right triangle • Grade 8 • Geometry • 8.G.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has an acute angle of 28°. Find the other acute angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

1792

QUESTION ID

**S49-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**62°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

**True or false?**

A right triangle can have two right angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

**Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5**

1793

QUESTION ID

## S49-B3-Q2

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

**Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5**

In a right triangle, the acute angles are x and x + 20. x =

**A.** 70°

**B.** 45°

**C.** 55°

**D.** 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

1794

QUESTION ID

**S49-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 35°**

**GRADING NOTE** 2x + 20 = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Why do the acute angles of a right triangle add to 90°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

1795

QUESTION ID

**S49-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**The angles add to 180° and the right angle uses 90°.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

**True or false?**

The two acute angles of an isosceles right triangle are 45° each.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

1796

QUESTION ID

**S49-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Equal and complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSG-SRT.C.7

# FORWARD 1

Relationships between tangents of complementary angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

Explain why $\tan\theta \cdot \tan(90° - \theta) = 1$ for an acute angle $\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

1798

QUESTION ID

**S49-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**tan θ = opposite/adjacent and tan(90° − θ) = adjacent/opposite, which are reciprocals.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

tan 20° ≈ 0.364. Estimate tan 70°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

1799

QUESTION ID

**S49-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**About 2.75**

**(1 ÷ 0.364)**

**GRADING NOTE** Any value 2.74-2.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

sin θ = cos θ exactly when θ = 45° (for acute θ).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

1800

QUESTION ID

**S49-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Then 90° − θ = θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which equals 1 for every acute θ?

**A.** $\tan\theta + \tan(90° - \theta)$

**B.** $\tan\theta \cdot \tan(90° - \theta)$

**C.** $\sin\theta \cdot \cos\theta$

**D.** $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

1801

**QUESTION ID**

**S49-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $\tan\theta \cdot \tan(90° - \theta)$**

**GRADING NOTE** Reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Show that sin θ / cos(90° − θ) = 1 for acute θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

1802

**QUESTION ID**

## S49-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**cos(90° − θ) = sin θ, so the quotient is sin θ / sin θ = 1.**

**GRADING NOTE** Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

# FORWARD 2

The Pythagorean identity • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ for every acute angle $\theta$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

1804

**QUESTION ID**

## S49-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$(\sin \theta)^2 + (\cos \theta)^2 = 1$**

**GRADING NOTE** REQUIRED WORK: Use a right triangle with legs a, b and hypotenuse c and the Pythagorean Theorem.

Any valid proof (paragraph, two-column, or flow). Essential: $\sin \theta = a/c$, $\cos \theta = b/c$; $a^2 + b^2 = c^2$; divide by $c^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin θ = 0.28 for acute θ. Find cos θ.

cos θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

1805

QUESTION ID

**S49-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.96**

**GRADING NOTE** Any equivalent form. √(1 − 0.0784).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-F2-Q2**

**High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sin 35° + \cos 35° = 1$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

1806

QUESTION ID

## S49-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The squares add to 1; $\sin 35° + \cos 35° \approx 1.39$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

cos θ = 12/13 (θ acute). sin θ =

**A.** 1/13

**B.** 5/13

**C.** 12/5

**D.** 13/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

1807

**QUESTION ID**

## S49-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 5/13**

**GRADING NOTE** √(1 − 144/169).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-F2-Q4**

**High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5**

Use complementary angles to show $(\sin 20°)^2 + (\sin 70°)^2 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

1808

**QUESTION ID**

## S49-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**sin 70° = cos 20°, so the sum is (sin 20°)² + (cos 20°)² = 1.**

**GRADING NOTE** Any equivalent argument.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-F2-Q5**

**High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5**

# Find unknown sides of right triangles with trigonometric ratios in applied problems

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.C.8** |
| **BACKWARD 1** | Choose the trigonometric ratio that links the known and unknown. | **HSG-SRT.C.6** |
| **BACKWARD 2** | Solve one-variable equations with a multiplication or division | **HSA-REI.B.3** |
| **BACKWARD 3** | Round to a stated precision and keep track of units | **HSN-Q.A.3** |
| **FORWARD 1** | Find a height from two angle-of-elevation observations | **HSG-SRT.C.8** |
| **FORWARD 2** | Find a feasible location under a clearance constraint | **HSG-MG.A.3** |

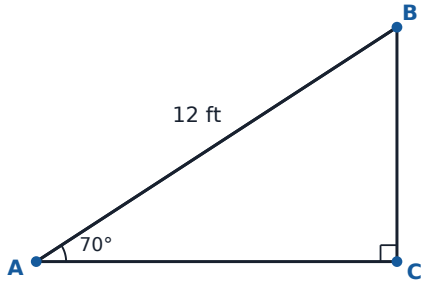
SET 50 • HSG-SRT.C.8

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

A 12 ft ladder leans against a wall, making a 70° angle with the level ground. How high up the wall does it reach, to the nearest tenth of a foot?




Not drawn to scale.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

1811

QUESTION ID
**S50-M-Q1**

TYPE
Short answer

CORRECT ANSWER
**11.3 ft**

**GRADING NOTE** Rounding to 0.1 required. 12 sin 70° = 11.2763.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A wheelchair ramp rises 2.5 ft and makes an 8° angle with the ground. How long is the ramp surface, to the nearest tenth of a foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

1812

**QUESTION ID**

## S50-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**18.0 ft**

**GRADING NOTE** Rounding to 0.1 required. 2.5 ÷ sin 8° = 17.9632.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From a point 25 m from the base of a tree, on level ground, the angle of elevation to the top is 38°. How tall is the tree, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

1813

**QUESTION ID**

## S50-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**19.5 m**

**GRADING NOTE** Rounding to 0.1 required. 25 tan 38° = 19.5321.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

A kite string 60 m long makes a 50° angle with the level ground. About how high is the kite above the point where the string is held (nearest meter)?

**A.** 78 m

**B.** 46 m

**C.** 39 m

**D.** 72 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

1814

**QUESTION ID**
**S50-M-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. 46 m**

**GRADING NOTE** Opposite = hypotenuse × sin 50° = 45.96.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has hypotenuse 20 and a 32° angle. Find the leg adjacent to the 32° angle, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

1815

QUESTION ID

**S50-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**17.0**

**GRADING NOTE** Rounding to 0.1 required. 20 cos 32° = 16.9610.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

SET 50 • HSG-SRT.C.8

# BACKWARD 1

Choose the trigonometric ratio that links the known and unknown sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

You know the hypotenuse and an angle and want the opposite side. Which ratio?

**A.** none

**B.** sine

**C.** cosine

**D.** tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

1817

QUESTION ID

**S50-B1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**B. sine**

**GRADING NOTE** Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

You know the adjacent side and an angle and want the opposite side. Which ratio?

**A.** none

**B.** tangent

**C.** cosine

**D.** sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

1818

**QUESTION ID**

## S50-B1-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. tangent

**GRADING NOTE** Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which ratio uses the adjacent side and the hypotenuse?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

1819

QUESTION ID

**S50-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**Cosine**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

**True or false?**

To find the hypotenuse from the opposite side and the angle, use sine.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

1820

QUESTION ID

## S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** sin A = opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

**True or false?**

Tangent uses the hypotenuse.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

1821

QUESTION ID

## S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Tangent uses the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

# BACKWARD 2

Solve one-variable equations with a multiplication or division • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve x/12 = 0.94.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

1823

QUESTION ID

**S50-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**11.28**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S50-B2-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 0.14 = 2.5/x.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

1824

QUESTION ID

**S50-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**17.86**

**GRADING NOTE** Any value 17.8-17.9.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S50-B2-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

If sin 30° = h/10, then h =

**A.** 10 sin 30°

**B.** 10 ÷ sin 30°

**C.** sin 30° ÷ 10

**D.** 30 ÷ 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

1825

QUESTION ID

**S50-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 10 sin 30°**

**GRADING NOTE** Multiply both sides by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

From 0.5 = 4/x we get x = 8.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

1826

QUESTION ID

## S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** x = 4 ÷ 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

**True or false?**

From tan 40° = 20/x we get x = 20 tan 40°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

1827

QUESTION ID

**S50-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** x = 20 ÷ tan 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8

# BACKWARD 3

Round to a stated precision and keep track of units • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Round 11.2763 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

1829

QUESTION ID

**S50-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**11.3**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S50-B3-Q1**

**High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A calculation gives 19.532 m. Report it to the nearest tenth of a meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

1830

**QUESTION ID**

## S50-B3-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**19.5 m**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

**True or false?**

Rounding in the middle of a calculation can change the final answer.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

1831

QUESTION ID
**S50-B3-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Round at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

A height is measured in feet and the angle in degrees. The answer's unit is

**A.** degrees

**B.** feet per degree

**C.** feet

**D.** no unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

1832

QUESTION ID
**S50-B3-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**C. feet**

**GRADING NOTE** Lengths keep their units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

17.96 rounded to the nearest tenth is 17.9.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

1833

QUESTION ID

## S50-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It is 18.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSG-SRT.C.8

# FORWARD 1

Find a height from two angle-of-elevation observations • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. Find the tower's height to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

1835

**QUESTION ID**

## S50-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**42.0 m**

**GRADING NOTE** Rounding to 0.1 required. h = 40 ÷ (1/tan 25° − 1/tan 40°) = 41.984. Two equations: h = x tan 40° and h = (x + 40) tan 25°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S50-F1-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. How far is point B from the base of the tower, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

1836

**QUESTION ID**

## S50-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**50.0 m**

**GRADING NOTE** Rounding to 0.1 required. x = h ÷ tan 40° = 50.034.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

**True or false?**

With only one angle of elevation and no distance, the height cannot be found.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

1837

QUESTION ID

**S50-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** One equation has two unknowns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

Which equations describe two observations 40 m apart (angles 25° and 40°, distance x from the closer point)?

**A.** h = 40 tan 25°

**B.** h = x tan 25° and h = (x + 40) tan 40°

**C.** h = x tan 40° and h = (x + 40) tan 25°

**D.** h = x sin 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

1838

**QUESTION ID**

## S50-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. h = x tan 40° and h = (x + 40) tan 25°**

**GRADING NOTE** The closer point has the larger angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

Explain why the closer observation point has the larger angle of elevation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

1839

QUESTION ID

**S50-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**The same height over a shorter horizontal distance gives a larger tangent, so a larger angle.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S50-F1-Q5**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5**

SET 50 • HSG-SRT.C.8

# FORWARD 2

Find a feasible location under a clearance constraint • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A ramp must rise 3 ft, and its angle with the ground may be at most 4.8°. What is the shortest allowed horizontal run, to the nearest foot?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

1841

QUESTION ID

**S50-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**36 ft**

GRADING NOTE Form required: a whole number of feet that meets the constraint (round up). 3 ÷ tan 4.8° = 35.73.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A camera on a 6 m pole must see a point on the ground at an angle of depression of at least 20°. How far from the pole can that point be, at most, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

1842

QUESTION ID

**S50-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**16.5 m**

GRADING NOTE Rounding to 0.1 required. 6 ÷ tan 20° = 16.485; farther points need smaller angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

**True or false?**

Rounding UP the run of a ramp with a maximum angle can make the ramp too steep.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

1843

QUESTION ID

## S50-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** A longer run makes the ramp less steep; rounding the run down is the risk.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A ladder must reach 10 ft up and make an angle of at most 75° with the ground. The shortest ladder that works is about

**A.** 10.0 ft

**B.** 9.7 ft

**C.** 2.7 ft

**D.** 10.4 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

1844

QUESTION ID

**S50-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 10.4 ft**

**GRADING NOTE** At exactly 75°, length = 10 ÷ sin 75° = 10.35; a smaller angle needs a longer ladder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A building code allows stairs with an angle of at most 37°. A staircase must rise 9 ft. What is the shortest horizontal run, to the nearest tenth of a foot?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

1845

**QUESTION ID**

**S50-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**11.9 ft**

**GRADING NOTE** Rounding to 0.1 required. 9 ÷ tan 37° = 11.943.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

# Find unknown angles with inverse trigonometric ratios in applied problems

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.C.8** |
| **BACKWARD 1** | Inverse trig undoes a trig ratio | **HSG-SRT.C.6** |
| **BACKWARD 2** | Choose the ratio from the two known sides | **HSG-SRT.C.6** |
| **BACKWARD 3** | Use degree mode and round angles | **HSN-Q.A.3** |
| **FORWARD 1** | Solve a right triangle completely from two sides | **HSG-SRT.C.8** |
| **FORWARD 2** | Find directions (bearings) with inverse trigonometry | **HSG-SRT.C.8** |

SET 51 • HSG-SRT.C.8

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

A ramp rises 1 m over a horizontal distance of 12 m. What angle does it make with the ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

1848

**QUESTION ID**

## S51-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**4.8°**

**GRADING NOTE** Rounding to 0.1° required. $\tan^{-1}(1/12) = 4.764°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 6 m ladder reaches 5.5 m up a vertical wall. What angle does it make with the level ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

1849

**QUESTION ID**
**S51-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**66.4°**

**GRADING NOTE** Rounding to 0.1° required. $\sin^{-1}(5.5/6) = 66.444°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 10 m flagpole casts a 14 m shadow on level ground. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

1850

**QUESTION ID**
**S51-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**35.5°**

**GRADING NOTE** Rounding to 0.1° required. $\tan^{-1}(10/14) = 35.538°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

A right triangle has legs 7 and 24. What is its smaller acute angle, to the nearest tenth of a degree?

**A.** 73.7°

**B.** 16.3°

**C.** 32.5°

**D.** 26.3°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

1851

QUESTION ID

**S51-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 16.3°**

**GRADING NOTE** $\tan^{-1}(7/24) = 16.260°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A road rises 6 m for every 100 m of horizontal distance (a 6% grade). What angle does it make with the horizontal, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

1852

**QUESTION ID**

## S51-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**3.4°**

**GRADING NOTE** Rounding to 0.1° required. $\tan^{-1}(0.06) = 3.434°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

SET 51 • HSG-SRT.C.8

# BACKWARD 1

Inverse trig undoes a trig ratio • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

sin θ = 0.5 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

1854

**QUESTION ID**

## S51-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**30°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

tan θ = 1 for an acute angle θ. Find θ.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

1855

QUESTION ID

**S51-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**45°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

Which gives an angle from a ratio?

**A.** cos

**B.** tan

**C.** sin

**D.** $\sin^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

1856

**QUESTION ID**

## S51-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $\sin^{-1}$**

**GRADING NOTE** Inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

**True or false?**

$\cos^{-1}(0.5) = 60°$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

1857

QUESTION ID

## S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** cos 60° = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sin^{-1}(x)$ means $1 \div \sin(x)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

1858

QUESTION ID

## S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It is the angle whose sine is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

# BACKWARD 2

Choose the ratio from the two known sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

You know the opposite side and the hypotenuse. Which inverse function finds the angle?

**A.** none

**B.** $\cos^{-1}$

**C.** $\sin^{-1}$

**D.** $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

1860

QUESTION ID
**S51-B2-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**C. $\sin^{-1}$**

**GRADING NOTE** Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

You know both legs. Which inverse function finds an acute angle?

**A.** $\sin^{-1}$

**B.** $\tan^{-1}$

**C.** $\cos^{-1}$

**D.** none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

1861

QUESTION ID
**S51-B2-Q2**

TYPE
Multiple choice

CORRECT ANSWER
**B. $\tan^{-1}$**

**GRADING NOTE** Opposite/adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rise 5 and run 12. Write the ratio for the angle with the ground.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

1862

QUESTION ID

**S51-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**tan θ = 5/12**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S51-B2-Q3**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5**

**True or false?**

Adjacent side and hypotenuse give the angle by $\cos^{-1}$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

1863

QUESTION ID

## S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

**True or false?**

Opposite side and hypotenuse give the angle by $\tan^{-1}$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

1864

QUESTION ID
**S51-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE $\sin^{-1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8

# BACKWARD 3

Use degree mode and round angles • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Round 4.7636° to the nearest tenth of a degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

1866

QUESTION ID

**S51-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4.8°**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S51-B3-Q1**

**High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5**

**True or false?**

A calculator in radian mode gives $\sin^{-1}(0.5) \approx 0.524$, not 30.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

1867

QUESTION ID

**S51-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An acute angle of a right triangle must be

**A.** more than 90°

**B.** any angle

**C.** exactly 45°

**D.** between 0° and 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

1868

**QUESTION ID**

**S51-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. between 0° and 90°**

**GRADING NOTE** Acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A computed angle is 112°. Can it be an acute angle of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

1869

**QUESTION ID**

**S51-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**No**

**GRADING NOTE** Any equivalent answer; check the calculation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

35.537° to the nearest tenth is 35.6°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

1870

QUESTION ID

**S51-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 35.5°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSG-SRT.C.8

# FORWARD 1

Solve a right triangle completely from two sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has legs 9 and 14. Find the hypotenuse and both acute angles (nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

1872

**QUESTION ID**
**S51-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**Hypotenuse 16.6; angles 32.7° and 57.3°**

**GRADING NOTE** Rounding to 0.1 required for all three; the angles add to 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

A right triangle has hypotenuse 17 and one leg 8. Find the other leg and both acute angles (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

1873

**QUESTION ID**

## S51-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Other leg 15; angles 28.1° and 61.9°**

**GRADING NOTE** Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

**True or false?**

Once one acute angle is known, the other is 90° minus it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

1874

QUESTION ID

**S51-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Complementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

Legs 5 and 5. Angles?

**A.** 45° and 45°

**B.** 50° and 40°

**C.** 30° and 60°

**D.** 5° and 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

1875

**QUESTION ID**

## S51-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 45° and 45°**

**GRADING NOTE** Isosceles right triangle.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S51-F1-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5**

Explain how to check a solved right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

1876

**QUESTION ID**

## S51-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The acute angles add to 90°, $a^2 + b^2 = c^2$, and each ratio matches its angle.**

**GRADING NOTE** Any reasonable checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

# FORWARD 2

Find directions (bearings) with inverse trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A boat travels 8 km east and then 5 km north. Find the angle north of east of its straight-line direction from the start, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

1878

**QUESTION ID**

## S51-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**32.0°**

**GRADING NOTE** Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A hiker walks 3 km north and 4 km west. What is the bearing angle west of north from the start, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

1879

**QUESTION ID**

**S51-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**53.1°**

**GRADING NOTE** Rounding to 0.1° required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

**True or false?**

Bearing problems become right-triangle problems when the motions are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

1880

QUESTION ID

## S51-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** East-west and north-south legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

A drone flies 120 m horizontally and rises 50 m. Its angle of climb is about

**A.** 22.6°

**B.** 24.6°

**C.** 67.4°

**D.** 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

1881

**QUESTION ID**

**S51-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 22.6°**

**GRADING NOTE** $\tan^{-1}(50/120)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A ship is 10 km east and 10 km south of port. Find the straight-line distance and the angle south of east.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

1882

**QUESTION ID**

## S51-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**14.1 km; 45°**

**GRADING NOTE** Form required: both. $10\sqrt{2} \approx 14.1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

# Solve right triangles in applied problems with trigonometry and the Pythagorean Theorem

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.C.8** |
| **BACKWARD 1** | Use the Pythagorean Theorem to find a side | **8.G.B.7** |
| **BACKWARD 2** | Interpret angles of elevation and depression | **HSG-SRT.C.6** |
| **BACKWARD 3** | Choose sine, cosine, or tangent | **HSG-SRT.C.6** |
| **FORWARD 1** | Solve problems that need two right triangles | **HSG-SRT.C.8** |
| **FORWARD 2** | Solve non-right triangles with the Law of Sines | **HSG-SRT.D.11** |

SET 52 • HSG-SRT.C.8

# MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

From the top of an 80 m cliff, the angle of depression to a boat is 25°. How far is the boat from the base of the cliff, to the nearest tenth of a meter? How far is it from the observer?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

1885

**QUESTION ID**

## S52-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**171.6 m from the base; 189.3 m from the observer**

**GRADING NOTE** Rounding to 0.1 required for both. Horizontal = 80 ÷ tan 25°; line of sight = 80 ÷ sin 25°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane flying at an altitude of 2,000 m is seen at an angle of elevation of 14°. Find the horizontal ground distance and the line-of-sight distance to the plane, to the nearest meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

1886

**QUESTION ID**

## S52-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**8022 m horizontal; 8267 m line of sight**

**GRADING NOTE** Rounding to the nearest meter required for both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A ramp is 15 ft long and rises 2 ft. Find its horizontal run (nearest tenth of a foot) and its angle with the ground (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

1887

QUESTION ID

**S52-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Run 14.9 ft; angle 7.7°**

**GRADING NOTE** Rounding required as stated. Run = $\sqrt{(15^2 - 2^2)}$; angle = $\sin^{-1}(2/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

A 25 ft guy wire from the top of a pole meets the level ground 9 ft from the pole. How tall is the pole and what angle does the wire make with the ground (nearest tenth)?

**A.** 26.6 ft; 68.9°

**B.** 16.0 ft; 21.1°

**C.** 23.3 ft; 21.1°

**D.** 23.3 ft; 68.9°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

1888

**QUESTION ID**

**S52-M-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 23.3 ft; 68.9°**

**GRADING NOTE** Height $\sqrt{(25^2 - 9^2)}$; angle $\cos^{-1}(9/25)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has hypotenuse 25 and a 40° angle. Find both legs and the other acute angle (legs to the nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

1889

**QUESTION ID**
**S52-M-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**Legs 16.1 and 19.2; other angle 50°**

**GRADING NOTE** Form required: both legs and the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

SET 52 • HSG-SRT.C.8

# BACKWARD 1

Use the Pythagorean Theorem to find a side • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 25 and a leg 9. Find the other leg, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

1891

**QUESTION ID**

**S52-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**23.3**

**GRADING NOTE** Any equivalent form. √(625 − 81).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Legs 8 and 15. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

1892

QUESTION ID

**S52-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**17**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

**True or false?**

In a right triangle, the hypotenuse is the longest side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

1893

QUESTION ID

**S52-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE It is opposite the largest angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

Hypotenuse 13, leg 5. Other leg?

**A.** √194

**B.** 12

**C.** 8

**D.** 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

1894

QUESTION ID

**S52-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 12**

**GRADING NOTE** √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$a^2 + b^2 = c^2$ works for every triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

1895

QUESTION ID

**S52-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

# BACKWARD 2

Interpret angles of elevation and depression • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**True or false?**

An angle of depression is measured down from the horizontal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

1897

QUESTION ID

**S52-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

The angle of depression from a cliff top to a boat is 25°. What is the angle of elevation from the boat to the cliff top?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

1898

**QUESTION ID**
**S52-B2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**25°**

**GRADING NOTE** Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

Which side is horizontal in an angle-of-elevation triangle?

**A.** The hypotenuse

**B.** The leg along the ground

**C.** The vertical leg

**D.** None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

1899

**QUESTION ID**

## S52-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. The leg along the ground

**GRADING NOTE** Level ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An observer looks up at 30° to a bird. Is that an angle of elevation or depression?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

1900

**QUESTION ID**

## S52-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

## Elevation

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The angle of depression is measured from the vertical line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

1901

QUESTION ID

**S52-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** From the horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSG-SRT.C.8

# BACKWARD 3

Choose sine, cosine, or tangent • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

Known: opposite side and angle. Wanted: hypotenuse. Ratio?

**A.** none

**B.** tangent

**C.** sine

**D.** cosine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

1903

**QUESTION ID**

## S52-B3-Q1

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. sine

**GRADING NOTE** sin = opp/hyp.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

Known: adjacent side and angle. Wanted: opposite side. Ratio?

**A.** cosine

**B.** tangent

**C.** none

**D.** sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

1904

**QUESTION ID**

**S52-B3-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. tangent**

**GRADING NOTE** tan = opp/adj.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Known: hypotenuse and angle. Wanted: adjacent side. Ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

1905

QUESTION ID

**S52-B3-Q3**

TYPE

Short answer

CORRECT ANSWER

**Cosine**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

**True or false?**

Two different ratios can lead to the same correct side if used correctly.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

1906

QUESTION ID

**S52-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE For example, finding the third side by trig or Pythagoras.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

**True or false?**

Tangent relates the hypotenuse and a leg.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

1907

QUESTION ID

## S52-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Tangent relates the two legs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSG-SRT.C.8

# FORWARD 1

Solve problems that need two right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

From a window 20 m up, the angle of elevation to the top of a building across the street is 30° and the angle of depression to its base is 40°. How tall is the building, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

1909

**QUESTION ID**

## S52-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**33.8 m**

**GRADING NOTE** Rounding to 0.1 required. Street width = 20 ÷ tan 40° = 23.835; height = 20 + width · tan 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower see its top at 35° and 50°. Find the tower's height, to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

1910

**QUESTION ID**

## S52-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**44.1 m**

**GRADING NOTE** Rounding to 0.1 required. h/tan 35° + h/tan 50° = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

**True or false?**

A shared side links two right triangles in such problems.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

1911

QUESTION ID

## S52-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** For example, a common height or distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two observers 100 m apart on opposite sides of a tower of height h see its top at angles of elevation 35° and 50°. Which equation is correct?

**A.** h = 100 tan 85°

**B.** h/sin 35° + h/sin 50° = 100

**C.** h tan 35° + h tan 50° = 100

**D.** h/tan 35° + h/tan 50° = 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

1912

**QUESTION ID**

**S52-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. h/tan 35° + h/tan 50° = 100**

**GRADING NOTE** Horizontal distances add to 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

Two observers on opposite sides of a tower each measure an angle of elevation to its top, and the distance between them is known. Explain why the tower's height cannot be found from one of the two right triangles alone.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

1913

**GRADING NOTE** Any equivalent answer.

**QUESTION ID**

## S52-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Each triangle alone has two unknowns (h and its distance); the shared height and total distance give the needed second equation.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

# FORWARD 2

Solve non-right triangles with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, A = 40°, B = 65°, and a = 12. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

1915

**QUESTION ID**

## S52-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**16.9**

**GRADING NOTE** Rounding to 0.1 required. b = a sin B ÷ sin A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, A = 52°, C = 70°, and c = 20. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

1916

**QUESTION ID**

## S52-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**16.8**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-F2-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5**

**True or false?**

SOH-CAH-TOA applies directly to a triangle with no right angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

1917

**QUESTION ID**

## S52-F2-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Those ratios are defined in right triangles; use the Law of Sines or Cosines, or draw an altitude.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-F2-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5**

Which data allow the Law of Sines directly?

**A.** Two sides and the included angle

**B.** Two angles and any side

**C.** Three sides

**D.** Three angles only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

1918

QUESTION ID

**S52-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. Two angles and any side**

GRADING NOTE AAS or ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

Explain how dropping an altitude turns a non-right triangle into two right triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

1919

**QUESTION ID**

## S52-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The altitude creates two right triangles that share it, so h = b sin A = a sin B.**

**GRADING NOTE** Any equivalent explanation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-F2-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5**

SET 53 • HSG-C.A.1 • CIRCLES

# Prove that all circles are similar

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.1** |
| **BACKWARD 1** | A dilation multiplies distances by the scale factor | **HSG-SRT.A.1.b** |
| **BACKWARD 2** | Translate a point to another point | **8.G.A.3** |
| **BACKWARD 3** | A circle is all points at one distance from its center | **HSG-CO.A.1** |
| **FORWARD 1** | Find a single dilation that maps one circle onto another | **HSG-C.A.1** |
| **FORWARD 2** | Arcs of similar circles scale with the radius | **HSG-C.B.5** |

SET 53 • HSG-C.A.1

# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Circle C has center A and radius r, and circle D has center B and radius s. Prove that the two circles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

1922

**QUESTION ID**

## S53-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**A translation followed by a dilation maps C onto D, so they are similar.**

**GRADING NOTE** REQUIRED WORK: Translate A to B, then dilate about B with scale factor s/r. Show the image of C is exactly D.

Any valid proof (paragraph, two-column, or flow). Essential: (1) the translation maps C onto the circle with center B and radius r (distances kept); (2) the dilation about B with factor s/r maps each point at distance r from B to a point at distance s on the same ray, so onto D; (3) every point of D is reached (the inverse dilation); a similarity transformation exists. Invalid: "circles all look alike"; only checking two specific circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Circle $C_1$ has center (1, 2) and radius 3. Circle $C_2$ has center (−4, 0) and radius 6. Describe a similarity transformation that maps $C_1$ onto $C_2$ and explain why it works.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

1923

QUESTION ID

**S53-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**Translate 5 units left and 2 units down, then dilate about (−4, 0) with scale factor 2.**

GRADING NOTE REQUIRED WORK: Give the translation that moves the centers together and the dilation (center and factor).

Any valid proof (paragraph, two-column, or flow). Essential: the translation (x, y) → (x − 5, y − 2) moves the center to (−4, 0) keeping the radius 3; the dilation about (−4, 0) by 2 doubles every distance from the center, so radius 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

In the proof that all circles are similar, why must the scale factor be the ratio of the radii (s/r)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

1924

QUESTION ID

**S53-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**A dilation about the center multiplies every distance from the center by the factor, so the radius r must become s: k · r = s.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why two circles with the same center but different radii, r and s, are similar using only ONE transformation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

1925

QUESTION ID

**S53-M-Q4**

TYPE

Answer with required work

CORRECT ANSWER

**The dilation about the common center with factor s/r maps the first circle onto the second.**

GRADING NOTE REQUIRED WORK: Name the transformation and its center and factor.

Any valid proof (paragraph, two-column, or flow). Essential: dilation about the shared center; factor s/r; each point at distance r goes to distance s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

**True or false?**

A circle can be similar to an ellipse that is not a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

1926

QUESTION ID

**S53-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** A similarity keeps the ratio of all distances from the center, so the image of a circle is a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

SET 53 • HSG-C.A.1

# BACKWARD 1

A dilation multiplies distances by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

**ANSWER KEY** Reduced copy of the question page, with its answer

A dilation with factor 2.5 is applied to a segment of length 4. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

1928

QUESTION ID

**S53-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**10**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

**True or false?**

A dilation about point O keeps O fixed and multiplies each distance from O by the factor.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

1929

QUESTION ID

## S53-B1-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Definition.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S53-B1-Q2**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Which factor maps a circle of radius 4 about its center to one of radius 10?

**A.** 2.5

**B.** 0.4

**C.** 6

**D.** 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

1930

**QUESTION ID**

**S53-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 2.5**

**GRADING NOTE** 10 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle of radius 9 is dilated about its center by ⅓. New radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

1931

QUESTION ID
**S53-B1-Q4**

TYPE
Short answer

CORRECT ANSWER
**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A dilation can turn a circle into a square.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

1932

QUESTION ID

## S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It keeps shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSG-C.A.1

# BACKWARD 2

Translate a point to another point • Grade 8 • Geometry • 8.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Which translation moves (1, 2) onto (−4, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

1934

QUESTION ID

**S53-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**5 units left and 2 units down**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

**True or false?**

A translation keeps the radius of a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

**Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5**

1935

QUESTION ID

**S53-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** It is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

**Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle centered at (3, 3) is translated by (x, y) → (x − 3, y + 1). New center?

**A.** (6, 2)

**B.** (0, 4)

**C.** (0, 3)

**D.** (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

1936

QUESTION ID

**S53-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. (0, 4)**

GRADING NOTE Apply the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A translation moves (2, −1) to (7, 4). Describe it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

1937

**GRADING NOTE** Any equivalent form.

**QUESTION ID**

## S53-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**5 units right and 5 units up**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

**True or false?**

A translation changes the size of a figure.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

**Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5**

1938

QUESTION ID

## S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

**Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5**

SET 53 • HSG-C.A.1

# BACKWARD 3

A circle is all points at one distance from its center • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**True or false?**

Every point of a circle is the same distance from its center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

1940

QUESTION ID

**S53-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

A circle has center (0, 0) and passes through (3, 4). Its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

1941

**QUESTION ID**

**S53-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

A circle is completely determined by

**A.** its color

**B.** one point on it

**C.** its center only

**D.** its center and radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

1942

**QUESTION ID**

**S53-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. its center and radius**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two circles have the same center and the same radius. What can you say?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

1943

QUESTION ID

**S53-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**They are the same circle.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

**True or false?**

A circle's radius can be different in different directions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

1944

QUESTION ID

**S53-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The definition requires one distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSG-C.A.1

# FORWARD 1

Find a single dilation that maps one circle onto another • High School — Geometry • Geometry: Circles • HSG-C.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Circle P: center (0, 0), radius 2. Circle Q: center (6, 0), radius 4. Find the center of the dilation with a positive factor that maps P onto Q.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

1946

QUESTION ID

**S53-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(−6, 0)**

GRADING NOTE Form required: an ordered pair. The center (c, 0) satisfies 6 = c + 2(0 − c), so c = −6; check: (0, 0) → −6 + 2(0 + 6) = 6, and the radius 2 → 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

Circle P has center (0, 0) and radius 2. Circle Q has center (6, 0) and radius 4. What is the scale factor of a dilation that maps P onto Q?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

1947

**QUESTION ID**

**S53-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**2**

**GRADING NOTE** Any equivalent form. 4 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

**True or false?**

Two circles with different radii can always be mapped onto each other by one dilation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

**High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5**

1948

QUESTION ID

## S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Choose the center on the line of centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

**High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5**

Circles with equal radii and different centers are related by

**A.** nothing

**B.** a translation (no dilation needed)

**C.** a dilation with factor 2

**D.** a reflection only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

1949

**QUESTION ID**

**S53-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. a translation (no dilation needed)**

**GRADING NOTE** Same size: congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

Circle A has radius 3 and circle B has radius 12, both centered at the origin. Which dilation maps A onto B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

1950

**QUESTION ID**

**S53-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**Center (0, 0), factor 4**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSG-C.A.1

# FORWARD 2

Arcs of similar circles scale with the radius • High School — Geometry • Geometry: Circles • HSG-C.B.5

Two circles have radii 3 and 7. A 60° central angle cuts off an arc in each. What is the ratio of the arc lengths?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

1952

**QUESTION ID**

**S53-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**3 : 7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 90° arc of a circle with radius 4 has length 2π. How long is a 90° arc of a circle with radius 10?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

1953

**QUESTION ID**

**S53-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**5π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

**True or false?**

For a fixed central angle, the arc length is the same in every circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

1954

QUESTION ID

**S53-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Arc length is proportional to the radius; s/r is what stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

An arc of length 6 in a circle of radius 2 corresponds, by similarity, to what arc length in a circle of radius 5 (same angle)?

**A.** 2.4

**B.** 15

**C.** 30

**D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

1955

**QUESTION ID**

## S53-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 15**

**GRADING NOTE** Multiply by 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

Explain how the similarity of circles leads to radian measure.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

1956

**QUESTION ID**

## S53-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Since arc length is proportional to radius for a fixed angle, arc/radius depends only on the angle; that number is the angle in radians.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

SET 54 • HSG-C.A.2 • CIRCLES

# Relate inscribed angles and central angles

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.2** |
| **BACKWARD 1** | Base angles of an isosceles triangle are congruent | **HSG-CO.C.10** |
| **BACKWARD 2** | The exterior angle theorem | **HSG-CO.C.10** |
| **BACKWARD 3** | A central angle and its arc have the same measure | **4.MD.C.5.a** |
| **FORWARD 1** | Prove the inscribed angle theorem when the center is inside or out… | **HSG-C.A.2** |
| **FORWARD 2** | Inscribed angles on the same arc, and arc applications | **HSG-C.A.2** |

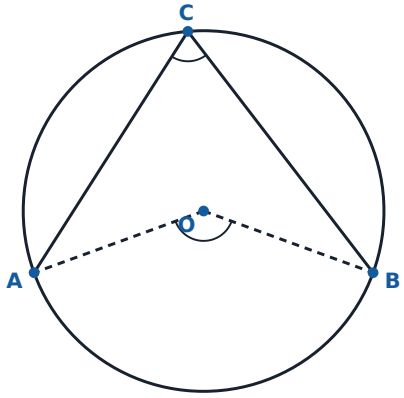
SET 54 • HSG-C.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

In circle O, central angle AOB measures 110°. Inscribed angle ACB intercepts the same arc AB. Find m∠ACB.




m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

1959

QUESTION ID

**S54-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**55°**

**GRADING NOTE** Any equivalent form. An inscribed angle is half the central angle on the same arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In circle O, inscribed angle PQR measures 38°. Find the measure of arc PR (the arc it intercepts) and of central angle POR.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

1960

QUESTION ID

**S54-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**Arc PR = 76°;**
**∠POR = 76°**

GRADING NOTE Form required: both. The central angle equals its arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In circle O, AC is a diameter and B is another point on the circle. Prove that m∠BAC = ½ m∠BOC (the inscribed angle theorem when one side of the angle is a diameter).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

1961

**QUESTION ID**

## S54-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠BAC = ½ m∠BOC**

**GRADING NOTE** REQUIRED WORK: Use isosceles triangle AOB (OA = OB) and the exterior angle theorem at O.

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii), so ∠OAB ≅ ∠OBA; ∠BOC is an exterior angle of △AOB, so m∠BOC = m∠OAB + m∠OBA = 2m∠BAC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Inscribed angle XYZ and central angle XOZ intercept the same arc XZ, and m∠XYZ = 47°. What is m∠XOZ?

**A.** 23.5°

**B.** 94°

**C.** 47°

**D.** 133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

1962

**QUESTION ID**

**S54-M-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 94°**

**GRADING NOTE** Double the inscribed angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

**True or false?**

An inscribed angle and a central angle that intercept the same arc have equal measures.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

1963

QUESTION ID

**S54-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The inscribed angle is half the central angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

SET 54 • HSG-C.A.2

# BACKWARD 1

Base angles of an isosceles triangle are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In △AOB, OA = OB and m∠OAB = 35°. Find m∠OBA and m∠AOB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

1965

**QUESTION ID**
**S54-B1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**35° and 110°**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

**True or false?**

Two radii of a circle and the chord joining their ends form an isosceles triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

1966

QUESTION ID

**S54-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The radii are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Isosceles triangle with vertex angle 80°. Each base angle?

**A.** 100°

**B.** 50°

**C.** 80°

**D.** 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

1967

**QUESTION ID**

**S54-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 50°**

**GRADING NOTE** (180 − 80)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △OPQ with OP = OQ, m∠P = 2x and m∠Q = 50°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

1968

QUESTION ID

**S54-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**25**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

**True or false?**

In an isosceles triangle, the vertex angle always equals a base angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

1969

QUESTION ID

**S54-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only when equilateral.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSG-C.A.2

# BACKWARD 2

The exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

A triangle has remote interior angles 30° and 30°. Find the exterior angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

1971

**QUESTION ID**

**S54-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**60°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

**True or false?**

An exterior angle equals the sum of the two remote interior angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

1972

QUESTION ID

**S54-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Exterior angle 100°, one remote interior angle 45°. The other?

**A.** 45°

**B.** 80°

**C.** 55°

**D.** 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

1973

**QUESTION ID**

**S54-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 55°**

**GRADING NOTE** 100 − 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If both remote interior angles equal x, the exterior angle is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

1974

**QUESTION ID**

**S54-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**2x**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

**True or false?**

An exterior angle equals the adjacent interior angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

1975

QUESTION ID
**S54-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSG-C.A.2

# BACKWARD 3

A central angle and its arc have the same measure • Grade 4 • Measurement & Data • 4.MD.C.5.a

**ANSWER KEY** Reduced copy of the question page, with its answer

A central angle cuts off ¼ of a circle. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

1977

**QUESTION ID**

**S54-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

**True or false?**

The measure of a minor arc equals its central angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

1978

QUESTION ID

**S54-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition of arc measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An arc measures 70°. Its central angle measures

**A.** 70°

**B.** 140°

**C.** 290°

**D.** 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

1979

**QUESTION ID**

## S54-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 70°**

**GRADING NOTE** Equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A central angle of 150°. What fraction of the circle is its arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

1980

**QUESTION ID**

**S54-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**5/12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

**True or false?**

All arcs of 60° have the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

1981

QUESTION ID

**S54-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Length depends on the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSG-C.A.2

# FORWARD 1

Prove the inscribed angle theorem when the center is inside or outside the angle • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Inscribed angle ABC has the center O inside it. Prove that m∠ABC = ½ m∠AOC. (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

1983

**QUESTION ID**

## S54-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠ABC = ½ m∠AOC**

**GRADING NOTE** REQUIRED WORK: Split ∠ABC and ∠AOC along BD and apply the diameter case twice, then add.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ABD = ½m∠AOD and m∠DBC = ½m∠DOC (diameter case); add: m∠ABC = ½(m∠AOD + m∠DOC) = ½m∠AOC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Inscribed angle ABC has the center O outside it. Prove that m∠ABC = ½ m∠AOC. (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

1984

QUESTION ID

**S54-F1-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**m∠ABC = ½ m∠AOC**

GRADING NOTE REQUIRED WORK: Use the diameter case twice and SUBTRACT.

Any valid proof (paragraph, two-column, or flow). Essential: with D such that A and C are on the same side of BD: m∠DBC = ½m∠DOC and m∠DBA = ½m∠DOA; subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

**True or false?**

The inscribed angle theorem needs three cases for a complete proof.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

1985

QUESTION ID

**S54-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Center on a side, inside, or outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In the proof of the inscribed angle theorem for the case where the center lies inside the inscribed angle, a diameter splits the angle in two, and the two diameter-case results are

**A.** averaged

**B.** multiplied

**C.** subtracted

**D.** added

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

1986

QUESTION ID

**S54-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. added**

GRADING NOTE The angle is the sum of two pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

Why can't the proof of the inscribed angle theorem stop after the case in which one side of the angle is a diameter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

1987

**QUESTION ID**

**S54-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**Most inscribed angles do not have a diameter as a side; the other positions must be covered too.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSG-C.A.2

# FORWARD 2

Inscribed angles on the same arc, and arc applications • High School — Geometry • Geometry: Circles • HSG-C.A.2

Inscribed angles ∠ADB and ∠ACB intercept the same arc AB, and m∠ADB = 41°. Find m∠ACB.

m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

1989

**QUESTION ID**

## S54-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**41°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove that two inscribed angles that intercept the same arc are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

1990

QUESTION ID

**S54-F2-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**They are congruent.**

GRADING NOTE

REQUIRED WORK: Each is half the same central angle.

Any valid proof (paragraph, two-column, or flow). Essential: each equals ½ of the central angle on that arc (inscribed angle theorem).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

**True or false?**

An inscribed angle that intercepts a 200° arc measures 200°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

1991

**QUESTION ID**

## S54-F2-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** It is half the arc: 100°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

Points A, B, C, D are on a circle with arc AB = 80° and arc CD = 50° (arcs not containing the other points). Chords AC and BD meet at E inside. m∠AEB?

**A.** 30°

**B.** 65°

**C.** 15°

**D.** 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

1992

**QUESTION ID**

**S54-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 65°**

**GRADING NOTE** Angle formed by two chords = half the sum of the intercepted arcs: (80 + 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An inscribed angle measures 3x + 10 and its intercepted arc measures 110°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

1993

QUESTION ID

**S54-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**15**

GRADING NOTE Any equivalent form. 3x + 10 = 55.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

SET 55 • HSG-C.A.2 • CIRCLES

# Inscribed angles on a diameter are right angles

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.2** |
| **BACKWARD 1** | Use the inscribed angle theorem | **HSG-C.A.2** |
| **BACKWARD 2** | Isosceles triangles from radii | **HSG-CO.C.10** |
| **BACKWARD 3** | Triangle angle sum | **8.G.A.5** |
| **FORWARD 1** | Use the converse to find a circle through a right triangle | **HSG-C.A.2** |
| **FORWARD 2** | Use the diameter property to construct tangents from an external… | **HSG-C.A.4** |

SET 55 • HSG-C.A.2

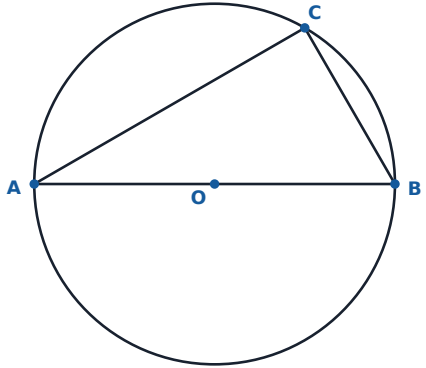
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is any other point on the circle. Prove that ∠ACB is a right angle.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

**High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5**

1996

**QUESTION ID**

## S55-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠ACB = 90°**

**GRADING NOTE** REQUIRED WORK: Use the inscribed angle theorem (the intercepted arc is 180°), or the two isosceles triangles AOC and BOC.

Any valid proof (paragraph, two-column, or flow). Essential (theorem): ∠ACB intercepts a semicircle, 180°, so it measures 90°. Alternative: OA = OC = OB, so △AOC and △BOC are isosceles; with base angles x and y, the angle sum of △ABC gives 2x + 2y = 180°, so x + y = 90° = m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

**High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

AB is a diameter of a circle and C is on the circle. AC = 6 and BC = 8. Find AB and the radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

1997

**QUESTION ID**

## S55-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**AB = 10; radius 5**

**GRADING NOTE** Form required: both. ∠C = 90°, so AB = √(36 + 64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

AB is a diameter of circle O and C is on the circle with m∠CAB = 34°. Find m∠CBA.

m∠CBA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

1998

**QUESTION ID**
**S55-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**56°**

**GRADING NOTE** Any equivalent form. ∠C = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle PQR is inscribed in a circle with PR a diameter. Which statement must be true?

**A.** PQ = QR

**B.** PR = PQ

**C.** ∠P = 90°

**D.** ∠Q = 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

1999

**QUESTION ID**

**S55-M-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. ∠Q = 90°**

**GRADING NOTE** The angle opposite the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Every triangle inscribed in a circle is a right triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

2000

QUESTION ID

**S55-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only when one side is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

SET 55 • HSG-C.A.2

# BACKWARD 1

Use the inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

An inscribed angle intercepts an arc of 140°. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

2002

**QUESTION ID**

**S55-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**70°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

**True or false?**

An inscribed angle is half its intercepted arc.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

2003

QUESTION ID

**S55-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A semicircle measures

**A.** 360°

**B.** 180°

**C.** 90°

**D.** 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

2004

QUESTION ID

**S55-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 180°**

GRADING NOTE Half the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An inscribed angle measures 25°. Its arc?

Arc: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

2005

QUESTION ID

**S55-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**50°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

**True or false?**

An inscribed angle equals its intercepted arc.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

2006

QUESTION ID

**S55-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSG-C.A.2

# BACKWARD 2

Isosceles triangles from radii • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

OA = OC (radii) and m∠OAC = 40°. Find m∠OCA.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

2008

**QUESTION ID**

## S55-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**40°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Any two radii and the chord between them form an isosceles triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

2009

QUESTION ID

**S55-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In △OBC with OB = OC, m∠BOC = 100°. m∠OBC =

**A.** 80°

**B.** 40°

**C.** 100°

**D.** 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

2010

**QUESTION ID**

**S55-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 40°**

**GRADING NOTE** (180 − 100)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

2010

**ANSWER KEY** Reduced copy of the question page, with its answer

Why are ∠OAC and ∠OCA congruent when OA = OC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

2011

**QUESTION ID**

## S55-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Base angles of an isosceles triangle are congruent.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

2011

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A radius is always longer than a chord.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

2012

QUESTION ID

**S55-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Chords can be longer (up to the diameter).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSG-C.A.2

# BACKWARD 3

Triangle angle sum • Grade 8 • Geometry • 8.G.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two angles of a triangle are 90° and 34°. The third?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

2014

QUESTION ID

**S55-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**56°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

**True or false?**

The acute angles of a right triangle add to 90°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

2015

**QUESTION ID**

## S55-B3-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** 180 − 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

2015

**ANSWER KEY** Reduced copy of the question page, with its answer

2x + 2y = 180. Then x + y =

**A.** 45

**B.** 360

**C.** 90

**D.** 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2016

QUESTION ID

**S55-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 90**

**GRADING NOTE** Divide by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

2016

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has angles x, x, and 2x. Find 2x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

2017

**QUESTION ID**

**S55-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A triangle can have angles 90°, 60°, and 40°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

2018

QUESTION ID

**S55-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE They add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSG-C.A.2

# FORWARD 1

Use the converse to find a circle through a right triangle • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, with A(0, 0) and B(8, 6). Where is the center of the circle through A, B, and C, and what is its radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

2020

QUESTION ID

**S55-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Center (4, 3), the midpoint of AB; radius 5.**

GRADING NOTE Form required: both. The hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

2020

**ANSWER KEY** Reduced copy of the question page, with its answer

In △ABC, m∠C = 90°. Prove that C lies on the circle with diameter AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

2021

QUESTION ID

**S55-F1-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**MC = MA = MB, so C is on the circle with center M and diameter AB.**

GRADING NOTE REQUIRED WORK: Let M be the midpoint of AB and show MC = MA (for example, by completing a rectangle).

Any valid proof (paragraph, two-column, or flow). Essential: reflect C through M to D; ACBD is a parallelogram with a right angle, so a rectangle; its diagonals are congruent and bisect each other, so MC = MA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The hypotenuse of a right triangle is a diameter of its circumscribed circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

2022

QUESTION ID

**S55-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

2022

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has legs 5 and 12. Radius of its circumscribed circle?

**A.** 6.5

**B.** 8.5

**C.** 6

**D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

2023

QUESTION ID

**S55-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 6.5**

**GRADING NOTE** Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A carpenter's square (a right angle) is placed with its corner touching a circular board and its edges crossing the edge at P and Q. What is segment PQ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

2024

**QUESTION ID**
**S55-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**A diameter of the board.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

2024

SET 55 • HSG-C.A.2

# FORWARD 2

Use the diameter property to construct tangents from an external point • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the two tangent lines from point P to circle O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

2026

**QUESTION ID**

## S55-F2-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Construct the midpoint M of OP; draw the circle with center M through O and P; it meets circle O at $T_1$ and $T_2$; lines $PT_1$ and $PT_2$ are tangents.**

**GRADING NOTE** Grade the construction marks, not only the final figure: the perpendicular bisector of OP, the circle on diameter OP, and both tangent lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

To construct tangents from P to circle O, a circle with diameter OP is drawn; it meets circle O at $T_1$ and $T_2$. Why are lines $PT_1$ and $PT_2$ tangent to circle O?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

2027

QUESTION ID

**S55-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$T_1$ is on the circle with diameter OP, so $\angle OT_1P = 90°$; a line perpendicular to a radius at its endpoint is tangent.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

**True or false?**

From a point outside a circle there is exactly one tangent line to the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

2028

QUESTION ID

## S55-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** There are two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, a circle with diameter OP is drawn. That circle is used because

**A.** it passes through the center only

**B.** every angle inscribed in it on OP is a right angle

**C.** it has the same radius as circle O

**D.** it is tangent to circle O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

2029

QUESTION ID

**S55-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. every angle inscribed in it on OP is a right angle**

**GRADING NOTE** Right angle at the tangent point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

O is the center, OP = 10, and the radius is 6. How long is each tangent segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

2030

QUESTION ID

**S55-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**8**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 56 • HSG-C.A.2 • CIRCLES

# A radius is perpendicular to the tangent at the point of tangency

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.2** |
| **BACKWARD 1** | Use the Pythagorean Theorem and its converse | **8.G.B.7** |
| **BACKWARD 2** | The perpendicular is the shortest segment from a point to a line | **7.G.A.2** |
| **BACKWARD 3** | A tangent touches a circle at exactly one point | **HSG-CO.A.1** |
| **FORWARD 1** | Tangent segments from an external point are congruent | **HSG-C.A.2** |
| **FORWARD 2** | Write the equation of a tangent line in the coordinate plane | **HSG-GPE.B.5** |

SET 56 • HSG-C.A.2

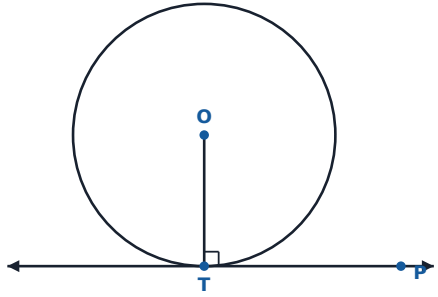
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Line $\ell$ is tangent to circle O at T. Prove that OT $\perp \ell$. (Use an indirect argument.)

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

**High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5**

2033

**QUESTION ID**

## S56-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**OT $\perp \ell$**

**GRADING NOTE** REQUIRED WORK: Suppose OT is not perpendicular; drop the perpendicular from O to $\ell$ at Q ≠ T; show a second point of $\ell$ is on the circle or inside it.

Any valid proof (paragraph, two-column, or flow). Essential: if OQ $\perp \ell$ with Q ≠ T, then OQ < OT (the perpendicular is the shortest segment, as the hypotenuse OT is longest in right △OQT), so Q is inside the circle, and then $\ell$ would cross the circle twice, contradicting tangency. Invalid: "it looks perpendicular".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

**High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5**

2033

**ANSWER KEY** Reduced copy of the question page, with its answer

Line PT is tangent to circle O at T. OT = 5 and PT = 12. Find OP.

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

2034

**QUESTION ID**
**S56-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**13**

**GRADING NOTE** Any equivalent form. ∠OTP = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A tangent from P touches circle O at T. OP = 17 and the radius is 8. Find PT.

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

2035

QUESTION ID
**S56-M-Q3**

TYPE
Short answer

CORRECT ANSWER
**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Segment PT touches circle O at T. OT = 7, PT = 24, OP = 26. Is PT tangent to the circle?

**A.** Yes, because 26 > 24

**B.** Yes, because it touches at T

**C.** No, because 7 < 24

**D.** No, because $7^2 + 24^2 \neq 26^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

2036

**QUESTION ID**

## S56-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. No, because $7^2 + 24^2 \neq 26^2$**

**GRADING NOTE** Tangency requires ∠OTP = 90°, but 49 + 576 = 625 ≠ 676.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

**True or false?**

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

2037

QUESTION ID

**S56-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSG-C.A.2

# BACKWARD 1

Use the Pythagorean Theorem and its converse • Grade 8 • Geometry • 8.G.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Legs 5 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

2039

QUESTION ID

**S56-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Hypotenuse 17, leg 8. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

2040

QUESTION ID

**S56-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

**True or false?**

A triangle with sides 7, 24, 25 is a right triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

2041

QUESTION ID

**S56-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE 49 + 576 = 625.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

Sides 6, 8, 11. Right triangle?

**A.** Yes

**B.** No

**C.** Only if 11 is a leg

**D.** Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

2042

QUESTION ID

**S56-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. No**

GRADING NOTE 36 + 64 ≠ 121.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The Pythagorean Theorem applies to any triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

2043

QUESTION ID

**S56-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSG-C.A.2

# BACKWARD 2

The perpendicular is the shortest segment from a point to a line • Grade 7 • Geometry • 7.G.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The shortest segment from a point to a line is perpendicular to the line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

2045

QUESTION ID

## S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Any other segment is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In a right triangle, which side is longest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

2046

QUESTION ID

**S56-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**The hypotenuse**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

O is 4 cm from line $\ell$ (perpendicular distance). Another segment from O to $\ell$ could be

**A.** 4 cm only

**B.** 5 cm

**C.** 2 cm

**D.** 3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

2047

**QUESTION ID**

## S56-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 5 cm**

**GRADING NOTE** At least 4 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

Why is the perpendicular segment shorter than any slanted one?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

2048

QUESTION ID

**S56-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**Each slanted segment is the hypotenuse of a right triangle with the perpendicular as a leg.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

**True or false?**

The distance from a point to a line is measured along any segment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2049

**QUESTION ID**

## S56-B2-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Along the perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSG-C.A.2

# BACKWARD 3

A tangent touches a circle at exactly one point • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

**True or false?**

A tangent line meets the circle at exactly one point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

2051

QUESTION ID

**S56-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

A line meets a circle at two points. It is a

**A.** secant

**B.** radius

**C.** diameter

**D.** tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

2052

QUESTION ID

**S56-B3-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**A. secant**

**GRADING NOTE** Two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

Point Q on a line is inside the circle. Can the line be tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

2053

**QUESTION ID**

**S56-B3-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**No; it would cross the circle twice.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the point where a tangent touches the circle called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

2054

QUESTION ID

**S56-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**The point of tangency**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

**True or false?**

A tangent can pass through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

2055

QUESTION ID

**S56-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Then it would meet the circle twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSG-C.A.2

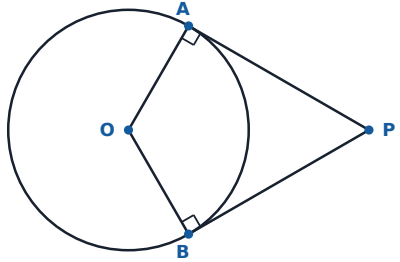
# FORWARD 1

Tangent segments from an external point are congruent • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

From point P outside circle O, tangent segments PA and PB touch the circle at A and B. Prove that PA = PB.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

2057

QUESTION ID

**S56-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**PA = PB**

**GRADING NOTE** REQUIRED WORK: Use the right angles at A and B, OA = OB, and the shared hypotenuse OP (HL, or the Pythagorean Theorem).

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90°; OA = OB; OP shared; HL (or PA = $\sqrt{(OP^2 - r^2)}$ = PB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Tangent segments from P to a circle are 3x + 4 and 25. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

2058

QUESTION ID

**S56-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**7**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

**True or false?**

The segment from P to the center bisects the angle between the two tangents.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

2059

QUESTION ID

**S56-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

A circle is inscribed in a triangle with sides tangent to it. From one vertex, the two tangent segments are

**A.** equal

**B.** parallel

**C.** unequal

**D.** perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

2060

QUESTION ID

**S56-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. equal**

GRADING NOTE Same external point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

A circle is tangent to all sides of a quadrilateral ABCD. AB = 9, BC = 11, CD = 13. Find DA.

DA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

2061

**QUESTION ID**

**S56-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**11**

**GRADING NOTE** Any equivalent form. AB + CD = BC + DA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSG-C.A.2

# FORWARD 2

Write the equation of a tangent line in the coordinate plane • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find an equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

2063

**QUESTION ID**

## S56-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -\frac{3}{4}x + 25/4$**

**GRADING NOTE** Any equivalent form, such as 3x + 4y = 25. The radius has slope 4/3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S56-F2-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5**

Find an equation of the tangent to $x^2 + y^2 = 13$ at $(2, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

2064

**QUESTION ID**

**S56-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$2x - 3y = 13$**

**GRADING NOTE** Any equivalent form. Radius slope $-3/2$; tangent slope $2/3$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S56-F2-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5**

**True or false?**

A tangent line has the same slope as the radius to its point of tangency.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

2065

**QUESTION ID**
**S56-F2-Q3**

**TYPE**
True or false

**CORRECT ANSWER**
**B. False**

**GRADING NOTE** Its slope is the negative reciprocal (when neither line is vertical or horizontal).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The tangent to $x^2 + y^2 = 16$ at (0, 4) is

**A.** $x = 0$

**B.** $y = x + 4$

**C.** $x = 4$

**D.** $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

2066

**QUESTION ID**

**S56-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. y = 4**

**GRADING NOTE** The radius is vertical, so the tangent is horizontal.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S56-F2-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Is y = 7 tangent to the circle with center (2, 3) and radius 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

2067

**GRADING NOTE** Any equivalent answer.

**QUESTION ID**

## S56-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes: the distance from (2, 3) to the line y = 7 is 4, the radius.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

SET 57 • HSG-C.A.2 • CIRCLES

# Relate circumscribed angles and central angles

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.2** |
| **BACKWARD 1** | A tangent is perpendicular to the radius at the point of tangency | **HSG-C.A.2** |
| **BACKWARD 2** | The angles of a quadrilateral add to 360° | **8.G.A.5** |
| **BACKWARD 3** | Supplementary angles add to 180° | **7.G.B.5** |
| **FORWARD 1** | Use the circumscribed angle with arcs, including major arcs | **HSG-C.A.2** |
| **FORWARD 2** | Angles formed by secants and tangents outside a circle | **HSG-C.A.2** |

SET 57 • HSG-C.A.2

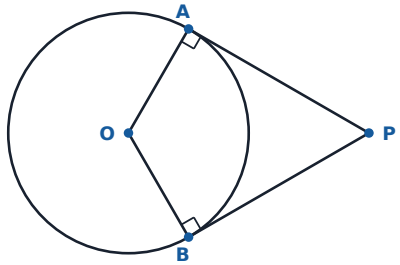
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Tangents from P touch circle O at A and B. Prove that m∠APB + m∠AOB = 180°.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

2070

QUESTION ID

**S57-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**m∠APB + m∠AOB = 180°**

GRADING NOTE REQUIRED WORK: Use the right angles at A and B and the angle sum of quadrilateral OAPB.

Any valid proof (paragraph, two-column, or flow). Essential: ∠OAP = ∠OBP = 90° (radius ⊥ tangent); the angles of quadrilateral OAPB add to 360°; so m∠APB + m∠AOB = 360° − 180° = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The circumscribed angle APB measures 52°. Find the central angle AOB and the minor arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

2071

**QUESTION ID**

## S57-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**∠AOB = 128°; arc AB = 128°**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Minor arc AB measures 100°. Find the circumscribed angle formed by the tangents at A and B.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

2072

**QUESTION ID**
**S57-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**80°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

A circumscribed angle measures 90°. The quadrilateral formed by the center, the two tangent points, and the vertex is a

**A.** kite that is not a square

**B.** trapezoid

**C.** rhombus that is not a square

**D.** square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

2073

**QUESTION ID**
**S57-M-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. square**

**GRADING NOTE** All four angles are 90°, so OAPB is a rectangle; its opposite sides give OA = PB, and with PA = PB and OA = OB all four sides are equal, so it is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

2073

**True or false?**

A circumscribed angle and its central angle are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

2074

QUESTION ID

**S57-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

SET 57 • HSG-C.A.2

# BACKWARD 1

A tangent is perpendicular to the radius at the point of tangency • High School — Geometry • Geometry: Circles • HSG-C.A.2

**True or false?**

The radius to the point of tangency is perpendicular to the tangent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

2076

QUESTION ID

**S57-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is m∠OAP when PA is tangent at A?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

2077

**QUESTION ID**

**S57-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In quadrilateral OAPB with tangents PA and PB, how many right angles are certain?

**A.** 4

**B.** 0

**C.** 2

**D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

2078

**QUESTION ID**

**S57-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 2**

**GRADING NOTE** At A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Tangent PA, radius OA = 9, PA = 12. OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

2079

QUESTION ID

**S57-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The angle between a tangent and a radius can be 60°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

2080

QUESTION ID

**S57-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Always 90° at the point of tangency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSG-C.A.2

# BACKWARD 2

The angles of a quadrilateral add to 360° • Grade 8 • Geometry • 8.G.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A quadrilateral has angles 90°, 90°, and 70°. The fourth?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

2082

QUESTION ID

**S57-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**110°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

**True or false?**

A diagonal splits a quadrilateral into two triangles, so its angles add to 360°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

2083

QUESTION ID

**S57-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE 2 × 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two angles of a quadrilateral are right angles. The other two must

**A.** each be 90°

**B.** add to 360°

**C.** add to 180°

**D.** add to 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

2084

**QUESTION ID**

**S57-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. add to 180°**

**GRADING NOTE** 360 − 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Angles x, 90°, 90°, 2x in a quadrilateral. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

2085

QUESTION ID

**S57-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**60**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

**True or false?**

The angles of a quadrilateral add to 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

2086

QUESTION ID

**S57-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSG-C.A.2

# BACKWARD 3

Supplementary angles add to 180° • Grade 7 • Geometry • 7.G.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the supplement of 52°.

Supplement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

2088

QUESTION ID

**S57-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**128°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

2088

**True or false?**

Supplementary angles add to 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

2089

QUESTION ID

**S57-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Angles x and 3x are supplementary. x =

**A.** 60°

**B.** 90°

**C.** 45°

**D.** 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

2090

**QUESTION ID**

**S57-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 45°**

**GRADING NOTE** 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

2090

**ANSWER KEY** Reduced copy of the question page, with its answer

If two angles are supplementary and one is obtuse, the other is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

2091

**QUESTION ID**

**S57-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**acute**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

**True or false?**

Supplementary angles must form a linear pair.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

2092

QUESTION ID

**S57-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSG-C.A.2

# FORWARD 1

Use the circumscribed angle with arcs, including major arcs • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

The circumscribed angle at P is 40°. Find the minor arc AB and the major arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

2094

**QUESTION ID**
**S57-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**Minor arc 140°; major arc 220°**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

Show that a circumscribed angle equals ½(major arc − minor arc).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

2095

QUESTION ID

**S57-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**If the minor arc is m, the major arc is 360 − m and the angle is 180 − m = ½((360 − m) − m).**

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

**True or false?**

A circumscribed angle can measure more than 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

2096

QUESTION ID

**S57-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is 180° minus the minor arc, so it is less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

2096

**ANSWER KEY** Reduced copy of the question page, with its answer

Tangents from P make a 70° angle. An inscribed angle intercepting the minor arc AB measures

**A.** 110°

**B.** 70°

**C.** 55°

**D.** 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

2097

QUESTION ID

**S57-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 55°**

**GRADING NOTE** Minor arc = 110°, inscribed angle = 55°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The minor arc between two tangent points is x°, and the circumscribed angle is (x − 20)°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

2098

**QUESTION ID**
**S57-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**100**

**GRADING NOTE** Any equivalent form. x + x − 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSG-C.A.2

# FORWARD 2

Angles formed by secants and tangents outside a circle • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Two secants from P meet a circle at A, B and at C, D (A between P and B, C between P and D). Prove that m∠P = ½(arc BD − arc AC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

2100

**QUESTION ID**

**S57-F2-Q1**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**m∠P = ½(arc BD − arc AC)**

**GRADING NOTE** REQUIRED WORK: Draw chord BC; use the exterior angle theorem in triangle PBC and the inscribed angle theorem.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BCD is exterior to △PBC, so m∠BCD = m∠P + m∠PBC; m∠BCD = ½ arc BD and m∠PBC = ½ arc AC; subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two secants from P intercept arcs of 110° and 30°. Find m∠P.

m∠P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

2101

QUESTION ID

**S57-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**40°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

**True or false?**

The angle outside a circle formed by a tangent and a secant is half the sum of the intercepted arcs.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

2102

QUESTION ID

**S57-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is half the difference; half the sum is for two chords meeting inside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An angle outside a circle intercepts arcs of 160° and 50°. Its measure?

**A.** 110°

**B.** 55°

**C.** 210°

**D.** 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

2103

**QUESTION ID**

**S57-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 55°**

**GRADING NOTE** (160 − 50)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the formula for two tangents is a special case of the secant formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

2104

**QUESTION ID**
**S57-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**The tangent case is the limit where each secant's two intersection points merge; the arcs are the major and minor arcs.**

**GRADING NOTE** Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

SET 58 • HSG-C.A.2 • CIRCLES

# Relationships among radii and chords

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.2** |
| **BACKWARD 1** | Right triangles: Hypotenuse-Leg and the Pythagorean Theorem | **HSG-SRT.B.5** |
| **BACKWARD 2** | The perpendicular bisector theorem | **HSG-CO.C.9** |
| **BACKWARD 3** | Radii of a circle are equal | **HSG-CO.A.1** |
| **FORWARD 1** | Locate the center of a circle from chords | **HSG-C.A.2** |
| **FORWARD 2** | Find chord lengths and central angles with trigonometry | **HSG-SRT.C.8** |

SET 58 • HSG-C.A.2

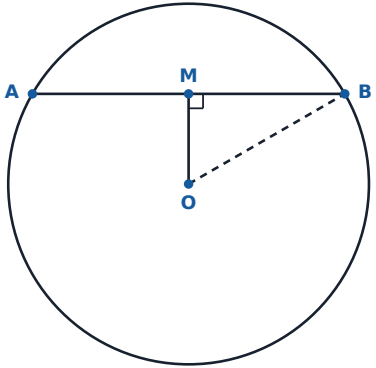
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

In circle O, OM ⊥ chord AB at M. Prove that M is the midpoint of AB.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

2107

QUESTION ID

**S58-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**AM = MB**

GRADING NOTE REQUIRED WORK: Prove △OMA ≅ △OMB (HL, using OA = OB).

Any valid proof (paragraph, two-column, or flow). Essential: OA = OB (radii); OM shared; right angles at M; HL (or the Pythagorean Theorem: AM = √(r² − OM²) = MB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

A circle has radius 13. A chord is 5 units from the center. How long is the chord?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

2108

QUESTION ID

**S58-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**24**

**GRADING NOTE** Any equivalent form. Half-chord $\sqrt{(169 - 25)} = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

2109

**QUESTION ID**
**S58-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

Two chords of a circle are each 7 units from the center. Which is true?

**A.** The chords are perpendicular

**B.** The chords are congruent

**C.** The chords are parallel

**D.** The chords meet at the center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

2110

QUESTION ID

**S58-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. The chords are congruent**

**GRADING NOTE** Equidistant chords are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

2110

**True or false?**

The perpendicular bisector of any chord passes through the center of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

2111

QUESTION ID

**S58-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE The center is equidistant from the chord's endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

SET 58 • HSG-C.A.2

# BACKWARD 1

Right triangles: Hypotenuse-Leg and the Pythagorean Theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangles with equal hypotenuses and one pair of equal legs are congruent by

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

2113

QUESTION ID
**S58-B1-Q1**

TYPE
Short answer

CORRECT ANSWER
**HL**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Hypotenuse 10, leg 6. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

2114

QUESTION ID

**S58-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**8**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

**True or false?**

In circle O, OM is perpendicular to chord AB at M. The radii OA and OB are congruent hypotenuses of right triangles OMA and OMB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

2115

QUESTION ID
**S58-B1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** OA = OB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

OM ⊥ AB in circle O. △OMA and △OMB share

**A.** nothing

**B.** OM

**C.** AB

**D.** OA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

2116

**QUESTION ID**
**S58-B1-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. OM**

**GRADING NOTE** Common leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

**True or false?**

HL works for any triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

2117

QUESTION ID

## S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSG-C.A.2

# BACKWARD 2

The perpendicular bisector theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

A point equidistant from A and B lies on the perpendicular bisector of AB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

2119

QUESTION ID

**S58-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

The center O of a circle is equidistant from the endpoints of chord AB. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

2120

**QUESTION ID**

**S58-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**OA and OB are radii.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

The center of a circle is equidistant from the endpoints of each of its chords. So the center lies on

**A.** the perpendicular bisector of every chord

**B.** a tangent

**C.** the circle

**D.** every chord

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

2121

QUESTION ID

**S58-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. the perpendicular bisector of every chord**

**GRADING NOTE** Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two chords' perpendicular bisectors meet at P. What is P?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

2122

QUESTION ID

**S58-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**The center of the circle**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

**True or false?**

The perpendicular bisector of a chord can miss the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

2123

QUESTION ID

**S58-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It always passes through it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSG-C.A.2

# BACKWARD 3

Radii of a circle are equal • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

A circle has center O. OA = 7. Find OB if B is on the circle.

OB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

2125

**QUESTION ID**

## S58-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

**True or false?**

All radii of one circle have the same length.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

2126

QUESTION ID

**S58-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

AB is a chord of circle O, and OA = 5. OB =

**A.** unknown

**B.** 10

**C.** 2.5

**D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

2127

QUESTION ID

**S58-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 5**

**GRADING NOTE** Radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

What is the longest chord in a circle of radius 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

2128

QUESTION ID

**S58-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**A diameter, 12**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

**True or false?**

A chord always passes through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

2129

QUESTION ID

**S58-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

SET 58 • HSG-C.A.2

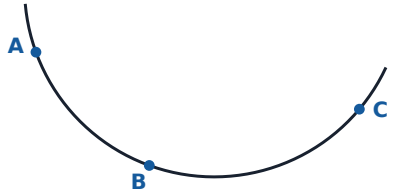
# FORWARD 1

Locate the center of a circle from chords • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

The arc below is part of a circle. Construct its center.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

2131

QUESTION ID

## S58-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Draw chords AB and BC; construct their perpendicular bisectors; they meet at the center.**

**GRADING NOTE** Grade the construction marks, not only the final figure: two chords and both perpendicular bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

A circle passes through (0, 0), (8, 0), and (0, 6). Find its center.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

2132

QUESTION ID

**S58-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**(4, 3)**

**GRADING NOTE** Form required: an ordered pair. Perpendicular bisectors x = 4 and y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

**True or false?**

Three points not on one line determine exactly one circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

2133

QUESTION ID

**S58-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE The perpendicular bisectors meet in one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

Why can't three points on one line lie on a circle?

**A.** Three points on a line always lie on a circle

**B.** Their perpendicular bisectors are parallel, so no center exists

**C.** Only two points can lie on a circle

**D.** They are too close

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

2134

QUESTION ID

**S58-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. Their perpendicular bisectors are parallel, so no center exists**

GRADING NOTE Parallel bisectors never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

A circular plate is broken. Describe how to find its original radius from the piece.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

2135

**QUESTION ID**

## S58-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Mark three points on the edge, construct the perpendicular bisectors of two chords to find the center, then measure to the edge.**

**GRADING NOTE** Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSG-C.A.2

# FORWARD 2

Find chord lengths and central angles with trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

A chord subtends a central angle of 80° in a circle of radius 10. Find its length to the nearest tenth.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

2137

**QUESTION ID**
**S58-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**12.9**

**GRADING NOTE** Rounding to 0.1 required. Half the chord = 10 sin 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

A chord of length 12 is in a circle of radius 10. Find the central angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

2138

**QUESTION ID**

## S58-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**73.7°**

**GRADING NOTE** Rounding to 0.1° required. Half-angle $\sin^{-1}(6/10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

**True or false?**

A chord's length is 2r sin θ for central angle θ.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

2139

QUESTION ID

**S58-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is 2r sin(θ/2), from the right triangle with half the chord.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Central angle 60°, radius 9. Chord length?

**A.** 4.5

**B.** 18

**C.** 9

**D.** 9√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

2140

QUESTION ID

**S58-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 9**

GRADING NOTE $2 \cdot 9 \cdot \sin 30° = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A chord equals the radius. Find its central angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

2141

QUESTION ID

**S58-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**60°**

**GRADING NOTE** Any equivalent form. Equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSG-C.A.3 • CIRCLES

# Construct the inscribed circle of a triangle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.3** |
| **BACKWARD 1** | Construct an angle bisector | **HSG-CO.D.12** |
| **BACKWARD 2** | Construct a perpendicular from a point to a line | **HSG-CO.D.12** |
| **BACKWARD 3** | Points on an angle bisector are equidistant from the sides | **HSG-SRT.B.5** |
| **FORWARD 1** | Prove that the incenter is equidistant from the sides | **HSG-C.A.3** |
| **FORWARD 2** | Locate a point equidistant from three lines in a design | **HSG-MG.A.3** |

SET 59 • HSG-C.A.3

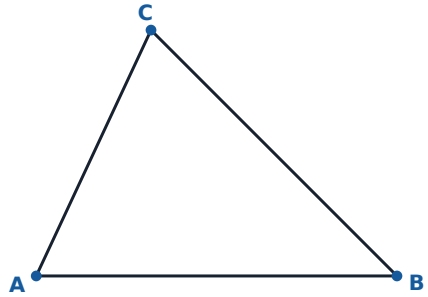
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

2144

QUESTION ID

## S59-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Bisect two angles (for example ∠A and ∠B); they meet at the incenter I; construct the perpendicular from I to one side, meeting it at D; draw the circle with center I and radius ID.**

GRADING NOTE Grade the construction marks, not only the final figure: two angle bisector constructions, the perpendicular from I to a side, and the circle touching all three sides.

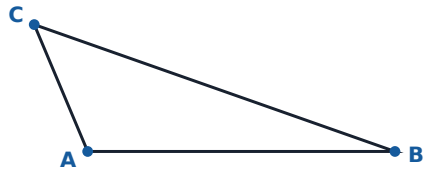
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of obtuse triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

2145

QUESTION ID

**S59-M-Q2**

TYPE

Drawing / plotting

CORRECT ANSWER

**Same steps: two angle bisectors meet at I (inside the triangle), perpendicular from I to a side gives the radius, then the circle.**

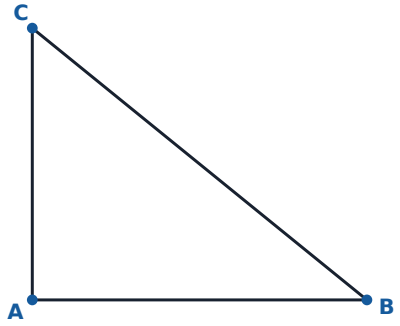
**GRADING NOTE** Grade the construction marks, not only the final figure: all constructions and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of right triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

2146

QUESTION ID

## S59-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Two angle bisectors, the perpendicular from their intersection to a side, and the circle.**

**GRADING NOTE** Grade the construction marks, not only the final figure: all constructions and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

Write the steps for constructing the inscribed circle of a triangle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

2147

QUESTION ID

**S59-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**1. Construct the bisectors of two angles; label their intersection I. 2. Construct the perpendicular from I to one side; label the foot D. 3. Draw the circle with center I through D.**

**GRADING NOTE** Any equivalent steps. The radius must come from a PERPENDICULAR to a side, not from I to a vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

**High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5**

Why is the radius of the inscribed circle measured along a perpendicular from the incenter to a side?

**A.** The radius goes to a vertex

**B.** Any segment from I works

**C.** The circle must be tangent to the side, and a tangent is perpendicular to the radius

**D.** The perpendicular is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

2148

**QUESTION ID**

## S59-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The circle must be tangent to the side, and a tangent is perpendicular to the radius**

**GRADING NOTE** Tangent ⊥ radius at the point of tangency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

SET 59 • HSG-C.A.3

# BACKWARD 1

Construct an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

How many arcs are needed to bisect an angle with compass and straightedge?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

2150

QUESTION ID

**S59-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Three (one at the vertex, then two equal arcs from the crossing points)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

**True or false?**

An angle bisector construction ends with a ray from the vertex.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

2151

QUESTION ID

**S59-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Through the crossing of the equal arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The incenter is where the

**A.** medians meet

**B.** perpendicular bisectors meet

**C.** angle bisectors meet

**D.** altitudes meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

2152

QUESTION ID

**S59-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. angle bisectors meet**

GRADING NOTE Equidistant from sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

Why construct only two angle bisectors to find the incenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

2153

QUESTION ID

**S59-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**The third bisector passes through the same point.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

**True or false?**

Bisecting the sides locates the incenter.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

2154

QUESTION ID

**S59-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** That locates the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSG-C.A.3

# BACKWARD 2

Construct a perpendicular from a point to a line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

**ANSWER KEY** Reduced copy of the question page, with its answer

Which construction finds the point where the incircle touches a side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

2156

QUESTION ID

**S59-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**The perpendicular from the incenter to the side**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

**True or false?**

The perpendicular from a point to a line is the shortest segment to the line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

2157

QUESTION ID

**S59-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

From point I, an arc meets side AB at two points. Next step?

**A.** Measure AB

**B.** Draw a circle at A

**C.** Bisect ∠I

**D.** Arcs of equal radius from those two points, crossing on the other side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

2158

**QUESTION ID**
**S59-B2-Q3**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. Arcs of equal radius from those two points, crossing on the other side**

**GRADING NOTE** Then draw the line through I and that crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

Why not measure from I to a vertex for the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

2159

QUESTION ID

**S59-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**Vertices are farther than the sides; the circle must touch the sides, not pass through vertices.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

**True or false?**

The incircle passes through the vertices.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

2160

QUESTION ID

**S59-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It touches the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSG-C.A.3

# BACKWARD 3

Points on an angle bisector are equidistant from the sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

**True or false?**

A point on the bisector of ∠A is equidistant from the two sides of ∠A.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

2162

QUESTION ID

**S59-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The incenter is 3 cm from side AB. How far is it from side BC?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

2163

QUESTION ID
**S59-B3-Q2**

TYPE
Short answer

CORRECT ANSWER
**3 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

Distance from a point to a side is measured

**A.** to the nearest vertex

**B.** along the perpendicular

**C.** along the bisector

**D.** along any segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

2164

QUESTION ID

**S59-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. along the perpendicular**

GRADING NOTE Shortest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

Why is the incenter the same distance from all three sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

2165

**QUESTION ID**
**S59-B3-Q4**

**TYPE**
Short answer

**CORRECT ANSWER**
**It is on all three angle bisectors, and each bisector point is equidistant from that angle's sides.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

**True or false?**

The incenter is equidistant from the three vertices.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

2166

QUESTION ID
**S59-B3-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** That is the circumcenter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSG-C.A.3

# FORWARD 1

Prove that the incenter is equidistant from the sides • High School — Geometry • Geometry: Circles • HSG-C.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

The bisectors of ∠A and ∠B of △ABC meet at I. Prove that I is equidistant from all three sides and lies on the bisector of ∠C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

2168

QUESTION ID

**S59-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**I is equidistant from AB, BC, CA and lies on the bisector of ∠C.**

GRADING NOTE REQUIRED WORK: Use the angle bisector distance property twice, then its converse at C.

Any valid proof (paragraph, two-column, or flow). Essential: on the bisector of ∠A: d(I, AB) = d(I, AC); on the bisector of ∠B: d(I, AB) = d(I, BC); so d(I, AC) = d(I, BC); the converse puts I on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

A triangle has area 30 and perimeter 30. Find the radius of its inscribed circle.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

2169

QUESTION ID

**S59-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form. Area = r · (perimeter ÷ 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

**High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5**

**True or false?**

The incenter of an obtuse triangle lies outside the triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

2170

QUESTION ID

**S59-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The incenter is always inside; the circumcenter is the one that can be outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 3-4-5 right triangle has inradius

**A.** 2.5

**B.** 1

**C.** 1.5

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

2171

**QUESTION ID**

## S59-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 1**

**GRADING NOTE** Area 6 = r · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why area = r · s, where s is half the perimeter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

2172

**QUESTION ID**

## S59-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Joining I to the vertices splits the triangle into three triangles with heights r on bases a, b, c: area = ½r(a + b + c) = r · s.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSG-C.A.3

# FORWARD 2

Locate a point equidistant from three lines in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A circular fountain must touch three straight paths that form a triangle with sides 13, 14, and 15 m (area 84 m²). What is the largest possible radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

2174

**QUESTION ID**

**S59-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**4 m**

**GRADING NOTE** Any equivalent form. s = 21, r = 84/21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

Explain why the incircle gives the largest circle that fits inside a triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

2175

**QUESTION ID**

## S59-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**A larger circle would cross some side; the incircle touches all three sides.**

**GRADING NOTE** Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

**True or false?**

A sprinkler equidistant from three straight fences forming a triangle should be at the circumcenter.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

2176

QUESTION ID

**S59-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Use the incenter (equidistant from lines); the circumcenter is equidistant from the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

To be equidistant from three towns (points), use the

**A.** orthocenter

**B.** centroid

**C.** incenter

**D.** circumcenter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

2177

**QUESTION ID**

## S59-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. circumcenter

**GRADING NOTE** Points, not lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An equilateral triangle has side 6. Find its inradius exactly.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

2178

**QUESTION ID**
**S59-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**√3**

**GRADING NOTE** Form required: exact. Area 9√3, s = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSG-C.A.3 • CIRCLES

# Construct the circumscribed circle of a triangle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.3** |
| **BACKWARD 1** | Construct the perpendicular bisector of a segment | **HSG-CO.D.12** |
| **BACKWARD 2** | Points on the perpendicular bisector are equidistant from the end… | **HSG-CO.C.9** |
| **BACKWARD 3** | Triangle types: acute, right, obtuse | **4.G.A.2** |
| **FORWARD 1** | Find the circumcenter and radius with coordinates | **HSG-GPE.B.5** |
| **FORWARD 2** | Find the circumradius with trigonometry (extended Law of Sines) | **HSG-SRT.D.10** |

SET 60 • HSG-C.A.3

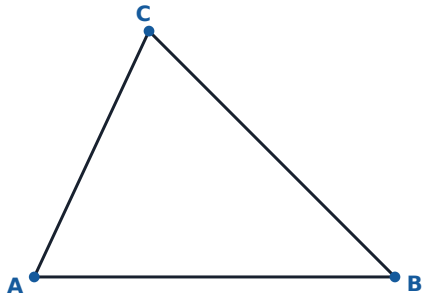
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

2181

QUESTION ID

## S60-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Construct the perpendicular bisectors of two sides; they meet at O; draw the circle with center O through A (it also passes through B and C).**

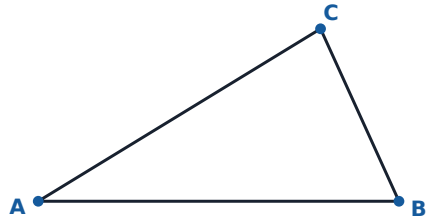
GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular bisector constructions and the circle through all three vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of obtuse triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

2182

QUESTION ID

## S60-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Same steps; the circumcenter is outside the triangle.**

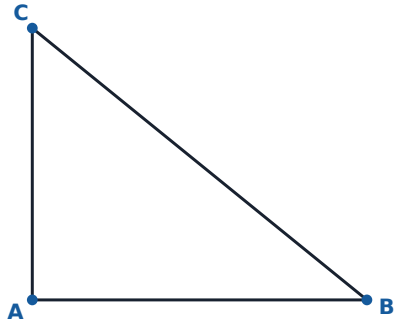
**GRADING NOTE** Grade the construction marks, not only the final figure: all constructions and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of right triangle ABC.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

2183

**QUESTION ID**

## S60-M-Q3

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**The perpendicular bisectors meet at the midpoint of hypotenuse BC; circle centered there through A.**

**GRADING NOTE** Grade the construction marks, not only the final figure: constructions and the circle; the center is on BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the steps for constructing the circumscribed circle of a triangle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

2184

QUESTION ID

**S60-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**1. Construct the perpendicular bisectors of two sides. 2. Label their intersection O. 3. Draw the circle with center O through one vertex.**

GRADING NOTE Any equivalent steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

**True or false?**

The circumcenter of an obtuse triangle lies inside the triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

2185

QUESTION ID

**S60-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It lies outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

SET 60 • HSG-C.A.3

# BACKWARD 1

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

**ANSWER KEY** Reduced copy of the question page, with its answer

Which construction gives all points equidistant from A and B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

2187

QUESTION ID

**S60-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**The perpendicular bisector of AB**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

**True or false?**

The perpendicular bisector construction uses equal arcs from both endpoints.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

2188

QUESTION ID

**S60-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The circumcenter is where the

**A.** angle bisectors meet

**B.** sides meet

**C.** perpendicular bisectors of the sides meet

**D.** medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

2189

QUESTION ID

**S60-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. perpendicular bisectors of the sides meet**

GRADING NOTE Equidistant from vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

After constructing two perpendicular bisectors, why is the third unnecessary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

2190

**QUESTION ID**

## S60-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**It passes through the same point.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

**True or false?**

The arcs in a perpendicular bisector construction can have any two different radii.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

2191

QUESTION ID

**S60-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSG-C.A.3

# BACKWARD 2

Points on the perpendicular bisector are equidistant from the endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

Every point on the perpendicular bisector of AB is equidistant from A and B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

2193

QUESTION ID

**S60-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

O is on the perpendicular bisectors of AB and BC.
Compare OA, OB, OC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

2194

QUESTION ID

**S60-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**OA = OB = OC**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The circumcircle's radius is

**A.** the distance from O to any vertex

**B.** the longest side

**C.** the distance from O to a side

**D.** half a side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

2195

**QUESTION ID**

## S60-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. the distance from O to any vertex**

**GRADING NOTE** Through the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle passes through A, B, C. Where must its center lie relative to AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

2196

QUESTION ID

**S60-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**On the perpendicular bisector of AB**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

**True or false?**

The circumcenter is equidistant from the sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

2197

QUESTION ID

**S60-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** From the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSG-C.A.3

# BACKWARD 3

Triangle types: acute, right, obtuse • Grade 4 • Geometry • 4.G.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has angles 30°, 45°, 105°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

2199

QUESTION ID

**S60-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**Obtuse**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A right triangle has one 90° angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

2200

QUESTION ID

**S60-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Where is the circumcenter of a right triangle?

**A.** Inside

**B.** Outside

**C.** On the hypotenuse (its midpoint)

**D.** At the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

2201

QUESTION ID

**S60-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. On the hypotenuse (its midpoint)**

**GRADING NOTE** Hypotenuse is a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has angles 50°, 60°, 70°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

2202

QUESTION ID

**S60-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**Acute**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

An acute triangle can have an obtuse angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

2203

QUESTION ID

**S60-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE All angles are acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSG-C.A.3

# FORWARD 1

Find the circumcenter and radius with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle with A(0, 0), B(6, 0), C(2, 4). Find the circumcenter and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

2205

**QUESTION ID**

**S60-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**(3, 1); radius √10**

**GRADING NOTE** Form required: both. Bisector of AB: x = 3; of AC: y − 2 = −½(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle with A(−2, 0), B(2, 0), C(0, 2). Find the circumcenter.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

2206

**QUESTION ID**
**S60-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**(0, 0)**

**GRADING NOTE** Form required: an ordered pair. All three are 2 from the origin.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5**

**True or false?**

The circumcenter is the intersection of two perpendicular bisector equations.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

2207

QUESTION ID

## S60-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Solve the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q3/5

A right triangle has vertices (0, 0), (8, 0), (0, 6). Circumradius?

**A.** 5

**B.** 10

**C.** 4

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

2208

QUESTION ID

**S60-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 5**

**GRADING NOTE** Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

Explain how to check a computed circumcenter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

2209

QUESTION ID

## S60-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**Verify it is the same distance from all three vertices.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5**

SET 60 • HSG-C.A.3

# FORWARD 2

Find the circumradius with trigonometry (extended Law of Sines) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In a triangle, a = 10 and A = 30°. Find the circumradius R using a/sin A = 2R.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

2211

**QUESTION ID**
**S60-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**10**

**GRADING NOTE** Any equivalent form. 10 / 0.5 = 20 = 2R.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Equilateral triangle with side 6. Find the circumradius exactly.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

2212

QUESTION ID
**S60-F2-Q2**

TYPE
Short answer

CORRECT ANSWER
**$2\sqrt{3}$**

**GRADING NOTE** Form required: exact. $6/\sin 60° = 4\sqrt{3} = 2R$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F2-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5**

**True or false?**

In any triangle, a/sin A equals the radius of the circumcircle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

2213

QUESTION ID

## S60-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It equals the diameter, 2R.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F2-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle with a 90° angle opposite side 14 has circumradius

**A.** 3.5

**B.** 14

**C.** 28

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

2214

QUESTION ID

**S60-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 7**

**GRADING NOTE** 14/sin 90° = 14 = 2R.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5**

Explain why a/sin A = 2R using an inscribed angle on a diameter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

2215

**QUESTION ID**

## S60-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Draw the diameter from B; the inscribed angle at its far end equals A (same arc) and is in a right triangle with hypotenuse 2R, so sin A = a/(2R).**

**GRADING NOTE** Any equivalent explanation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S60-F2-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5**

SET 61 • HSG-C.A.3 • CIRCLES

# Prove that opposite angles of an inscribed quadrilateral are supplementary

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.3** |
| **BACKWARD 1** | The inscribed angle theorem | **HSG-C.A.2** |
| **BACKWARD 2** | Supplementary angles | **7.G.B.5** |
| **BACKWARD 3** | Arcs of a circle add to 360° | **4.MD.C.7** |
| **FORWARD 1** | Use the converse: decide whether a quadrilateral can be inscribed | **HSG-C.A.3** |
| **FORWARD 2** | Exterior angle of a cyclic quadrilateral | **HSG-C.A.3** |

SET 61 • HSG-C.A.3

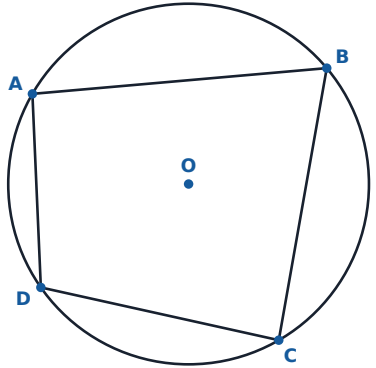
# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in circle O. Prove that m∠A + m∠C = 180°.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

2218

QUESTION ID

**S61-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**m∠A + m∠C = 180°**

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠A and ∠C intercept the two arcs BCD and BAD, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: m∠A = ½ arc BCD and m∠C = ½ arc BAD (inscribed angle theorem); the two arcs add to 360°; so the angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Quadrilateral PQRS is inscribed in a circle. Prove that ∠Q and ∠S are supplementary.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

2219

QUESTION ID

**S61-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**m∠Q + m∠S = 180°**

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠Q and ∠S intercept arcs PSR and PQR, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: each angle is half its intercepted arc, and the two intercepted arcs make up the whole circle (360°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

**High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in a circle with m∠A = 72° and m∠B = 105°. Find m∠C and m∠D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

2220

**QUESTION ID**

**S61-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**m∠C = 108°, m∠D = 75°**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

2220

**ANSWER KEY** Reduced copy of the question page, with its answer

Which quadrilateral can always be inscribed in a circle?

**A.** Any rhombus

**B.** Any trapezoid

**C.** Any parallelogram

**D.** Any rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

2221

**QUESTION ID**

## S61-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. Any rectangle

**GRADING NOTE** Opposite angles 90° + 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A parallelogram that is not a rectangle can be inscribed in a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

2222

QUESTION ID

**S61-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Its opposite angles are equal but not 90°, so they are not supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSG-C.A.3

# BACKWARD 1

The inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

An inscribed angle intercepts a 220° arc. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

2224

QUESTION ID

**S61-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**110°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

**True or false?**

An inscribed angle can intercept an arc greater than 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

2225

QUESTION ID

**S61-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Then the angle is obtuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An inscribed angle of 70° intercepts

**A.** 70°

**B.** 290°

**C.** 35°

**D.** 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

2226

**QUESTION ID**

**S61-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 140°**

**GRADING NOTE** Double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two arcs together make a whole circle. Their sum?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

2227

**QUESTION ID**

**S61-B1-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**360°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

**True or false?**

An inscribed angle equals half the central angle only when it is acute.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

2228

QUESTION ID

**S61-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Always.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSG-C.A.3

# BACKWARD 2

Supplementary angles • Grade 7 • Geometry • 7.G.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Supplement of 72°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

2230

QUESTION ID

**S61-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**108°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

**True or false?**

Two angles whose measures add to 180° are supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

2231

QUESTION ID

**S61-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Angles 3x and 2x are supplementary. x =

**A.** 30°

**B.** 45°

**C.** 36°

**D.** 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

2232

**QUESTION ID**

**S61-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 36°**

**GRADING NOTE** 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If ∠A = ∠C and they are supplementary, find each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

2233

**QUESTION ID**

**S61-B2-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Supplementary angles are equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

2234

QUESTION ID

**S61-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only if both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSG-C.A.3

# BACKWARD 3

Arcs of a circle add to 360° • Grade 4 • Measurement & Data • 4.MD.C.7

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle is split into arcs of 100°, 80°, and x°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

2236

QUESTION ID

**S61-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**180**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

**True or false?**

The arcs between four points on a circle add to 360°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

2237

QUESTION ID

**S61-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two arcs make a whole circle; one is 150°. The other is

**A.** 210°

**B.** 150°

**C.** 30°

**D.** 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

2238

**QUESTION ID**

## S61-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 210°**

**GRADING NOTE** 360 − 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Half of 360° is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

2239

**QUESTION ID**

**S61-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**180°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The arcs of a circle add to 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

2240

QUESTION ID

**S61-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

2240

SET 61 • HSG-C.A.3

# FORWARD 1

Use the converse: decide whether a quadrilateral can be inscribed • High School — Geometry • Geometry: Circles • HSG-C.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Quadrilateral ABCD has angles A = 85°, B = 70°, C = 95°, D = 110°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

2242

QUESTION ID

**S61-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Yes: A + C = 180° and B + D = 180°.**

GRADING NOTE Form required: the conclusion with the check. (The converse is true: such a quadrilateral is cyclic.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

An isosceles trapezoid has base angles of 64°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

2243

**QUESTION ID**

## S61-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes: its other angles are 116°, and 64° + 116° = 180° for each pair of opposite angles.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Every kite can be inscribed in a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

2244

QUESTION ID

**S61-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only kites with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Inscribed quadrilateral with angles (2x)°, (3x + 5)°, (2x + 20)°, and y°. If 2x and 2x + 20 are opposite, x =

**A.** 40

**B.** 50

**C.** 45

**D.** 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

2245

QUESTION ID

**S61-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 40**

GRADING NOTE 4x + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Inscribed quadrilateral: ∠A = (5y − 10)°,
∠C = (3y + 30)°, A and C opposite. Find y and ∠A.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

2246

QUESTION ID
**S61-F1-Q5**

TYPE
Short answer

CORRECT ANSWER
**y = 20; ∠A = 90°**

GRADING NOTE Form required: both. 8y + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSG-C.A.3

# FORWARD 2

Exterior angle of a cyclic quadrilateral • High School — Geometry • Geometry: Circles • HSG-C.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

ABCD is inscribed in a circle and side DC is extended past C to E. Prove that m∠BCE = m∠A.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

2248

QUESTION ID

**S61-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**m∠BCE = m∠A**

GRADING NOTE REQUIRED WORK: Use the linear pair at C and the supplementary opposite angles.

Any valid proof (paragraph, two-column, or flow). Essential: m∠BCE = 180° − m∠BCD (linear pair); m∠A = 180° − m∠BCD (opposite angles); so equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In inscribed ABCD, ∠A = 77°. Side DC is extended past C to E. Find ∠BCE.

∠BCE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

2249

QUESTION ID

**S61-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**77°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

**True or false?**

An exterior angle of a cyclic quadrilateral equals the interior angle next to it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

2250

QUESTION ID

**S61-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is supplementary to the adjacent interior angle and equals the opposite one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Inscribed ABCD with ∠B = 100°. The exterior angle at D (extending CD past D) is

**A.** 50°

**B.** 180°

**C.** 100°

**D.** 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

2251

QUESTION ID

**S61-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 100°**

GRADING NOTE Equals the opposite interior angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

ABCD is inscribed in a circle, and side DC is extended past C. Why is the exterior angle at C equal to ∠A, not ∠B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

2252

QUESTION ID

**S61-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**It is supplementary to ∠C, and the angle supplementary to ∠C is its opposite angle, ∠A.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSG-C.B.5 • CIRCLES

# Derive arc length from similarity and define radian measure

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.B.5** |
| **BACKWARD 1** | Similar figures: lengths scale by the scale factor | **HSG-SRT.A.2** |
| **BACKWARD 2** | Circumference of a circle | **7.G.B.4** |
| **BACKWARD 3** | Find a fraction of a whole | **7.RP.A.3** |
| **FORWARD 1** | Convert between degrees and radians | **HSG-C.B.5** |
| **FORWARD 2** | Radian measure for angles of rotation and the unit circle | **HSF-TF.A.1** |

SET 62 • HSG-C.B.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A central angle θ cuts off arc length $s_1$ on a circle of radius $r_1$ and arc length $s_2$ on a circle of radius $r_2$ (same center). Explain, using similarity, why $s_1/r_1 = s_2/r_2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

**High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5**

2255

**QUESTION ID**

## S62-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$s_2 = (r_2/r_1)s_1$, so $s_1/r_1 = s_2/r_2$.**

**GRADING NOTE** REQUIRED WORK: Use the dilation about the center with factor $r_2/r_1$: it maps the first circle to the second and the first arc to the second arc.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation about the center with factor $k = r_2/r_1$ maps the first circle onto the second and the arc of angle θ onto the arc of angle θ; it multiplies all lengths, including arc length, by k; so $s_2 = k\, s_1$ and s/r is the same for both. This constant is θ in radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

**High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5**

2255

**ANSWER KEY** Reduced copy of the question page, with its answer

A central angle cuts off an arc of length 12 cm on a circle of radius 4 cm. What is the angle in radians?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

2256

**QUESTION ID**
**S62-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**3 radians**

**GRADING NOTE** Any equivalent form. θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the length of the arc cut off by a central angle of 2.5 radians in a circle of radius 6 m.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

2257

**QUESTION ID**
**S62-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**15 m**

**GRADING NOTE** Any equivalent form. s = rθ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the exact length of the arc cut off by a 120° central angle in a circle of radius 9.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

2258

QUESTION ID

**S62-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**6π**

**GRADING NOTE** Form required: exact value. 120° = 2π/3 radians; s = 9 · 2π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Why is a full circle 2π radians?

**A.** There are 360 degrees

**B.** Every circle has radius 1

**C.** The circumference is 2πr, so s/r = 2π for the whole circle

**D.** π ≈ 3.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

2259

QUESTION ID

**S62-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. The circumference is 2πr, so s/r = 2π for the whole circle**

**GRADING NOTE** Arc length ÷ radius for the whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSG-C.B.5

# BACKWARD 1

Similar figures: lengths scale by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

**True or false?**

Under a dilation with factor k, every length (straight or curved) is multiplied by k.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

2261

QUESTION ID

## S62-B1-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two similar circles have radii 3 and 12. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

2262

QUESTION ID

**S62-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An arc of length 5 on the small circle (radius 3) matches which arc length on the large one (radius 12)?

**A.** 20

**B.** 15

**C.** 8

**D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

2263

**QUESTION ID**

**S62-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 20**

**GRADING NOTE** × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Are all circles similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

2264

QUESTION ID

**S62-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**Yes**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

**True or false?**

Arc lengths do not change under dilation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

2265

QUESTION ID

## S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** They scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSG-C.B.5

# BACKWARD 2

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the circumference of a circle with radius 5 (exact).

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

2267

QUESTION ID

**S62-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**10π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

**True or false?**

$C = 2\pi r$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5**

2268

QUESTION ID

## S62-B2-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5**

A circle has diameter 8. Its circumference is

**A.** 64π

**B.** 4π

**C.** 16π

**D.** 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

2269

**QUESTION ID**

**S62-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 8π**

**GRADING NOTE** πd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has circumference 18π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

2270

QUESTION ID

**S62-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

**True or false?**

$C = \pi r^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5**

2271

QUESTION ID

**S62-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** That is area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5**

SET 62 • HSG-C.B.5

# BACKWARD 3

Find a fraction of a whole • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

What fraction of a circle is a 90° arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

2273

QUESTION ID

**S62-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**¼**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find ⅓ of 18π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

2274

**QUESTION ID**

**S62-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**6π**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An arc of 45° is what fraction of the circumference?

**A.** 45/100

**B.** ⅛

**C.** 1/45

**D.** ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

2275

QUESTION ID

**S62-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. ⅛**

**GRADING NOTE** 45/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

**True or false?**

A 120° arc is ⅓ of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

2276

QUESTION ID

**S62-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE 120/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

**True or false?**

A 60° arc is 1/60 of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

2277

QUESTION ID

## S62-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSG-C.B.5

# FORWARD 1

Convert between degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Convert 150° to radians (exact).

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

2279

QUESTION ID

**S62-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5π/6**

GRADING NOTE Form required: exact value. Multiply by π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Convert 3π/4 radians to degrees.

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

2280

**QUESTION ID**

## S62-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**135°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

1 radian is about 3.14°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

2281

QUESTION ID

**S62-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 1 radian ≈ 57.3°; π radians = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

Which equals π/3 radians?

**A.** 60°

**B.** 120°

**C.** 30°

**D.** 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

2282

QUESTION ID

**S62-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 60°**

GRADING NOTE 180/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

An arc in a circle of radius 10 has length 5π. Find the central angle in degrees.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

2283

**QUESTION ID**

## S62-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form. θ = π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSG-C.B.5

# FORWARD 2

Radian measure for angles of rotation and the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

On the unit circle, an angle of 2 radians cuts off an arc of what length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

2285

QUESTION ID
**S62-F2-Q1**

TYPE
Short answer

CORRECT ANSWER
**2**

**GRADING NOTE** Any equivalent form. r = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A wheel of radius 0.3 m turns through 10 radians. How far does a point on its rim travel?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

2286

**QUESTION ID**
**S62-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**3 m**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

An angle of 6 radians is more than one full turn.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

2287

QUESTION ID
**S62-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** One full turn is $2\pi \approx 6.28$ radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

A point on the unit circle travels an arc of length π. The angle is

**A.** 1 radian

**B.** 2π radians

**C.** π°

**D.** π radians (180°)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

2288

QUESTION ID

**S62-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. π radians (180°)**

GRADING NOTE θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the radian measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

2289

QUESTION ID
**S62-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**Arc length is proportional to radius, so s/r is the same for every circle.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSG-C.B.5 • CIRCLES

# Derive and use the formula for the area of a sector

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.B.5** |
| **BACKWARD 1** | Area of a circle | **7.G.B.4** |
| **BACKWARD 2** | The radius is half the diameter | **7.G.B.4** |
| **BACKWARD 3** | Fractions of a circle in degrees and radians | **HSG-C.B.5** |
| **FORWARD 1** | Solve inverse sector problems | **HSG-C.B.5** |
| **FORWARD 2** | Areas of segments and composite circular regions | **HSG-MG.A.1** |

SET 63 • HSG-C.B.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Derive a formula for the area of a sector with central angle θ (in degrees) in a circle of radius r. Then rewrite it for θ in radians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

2292

QUESTION ID

**S63-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**A = (θ/360)πr² for degrees; A = ½r²θ for radians.**

GRADING NOTE REQUIRED WORK: Use the fraction of the whole circle that the sector is, times the circle's area.

Any valid proof (paragraph, two-column, or flow). Essential: the sector is θ/360 of the circle (central angles split the area in proportion); area = (θ/360)πr²; with θ radians, the fraction is θ/(2π), so A = (θ/(2π))πr² = ½r²θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the exact area of a sector with a 60° central angle in a circle of radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

2293

QUESTION ID

**S63-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**6π**

GRADING NOTE Form required: exact value. (60/360)π(36).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the area of a sector with central angle 1.2 radians in a circle of radius 5.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

2294

**QUESTION ID**
**S63-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**15**

**GRADING NOTE** Any equivalent form. ½(25)(1.2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pizza with diameter 14 in. is cut into 8 equal slices. Find the area of one slice to the nearest tenth of a square inch.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

2295

**QUESTION ID**

## S63-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**19.2 in²**

**GRADING NOTE** Rounding to 0.1 required. Radius 7 (not 14): (1/8)π(49).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sector has area 12π in a circle of radius 6. What is its central angle?

**A.** 240°

**B.** 12°

**C.** 120°

**D.** 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

2296

**QUESTION ID**

**S63-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 120°**

**GRADING NOTE** 12π/36π = ⅓ of 360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

2296

SET 63 • HSG-C.B.5

# BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the exact area of a circle with radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

2298

QUESTION ID

**S63-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**36π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has diameter 10. Exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

2299

QUESTION ID

**S63-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**25π**

**GRADING NOTE** Form required: exact value. Use the radius 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$A = \pi r^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

2300

QUESTION ID

**S63-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

A circle has area 49π. Radius?

**A.** 14

**B.** 7

**C.** 24.5

**D.** 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

2301

**QUESTION ID**

**S63-B1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 7**

**GRADING NOTE** $r^2 = 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

**True or false?**

Doubling the radius doubles the area.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5**

2302

QUESTION ID

**S63-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It multiplies the area by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5**

SET 63 • HSG-C.B.5

# BACKWARD 2

The radius is half the diameter • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has diameter 14 in. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

2304

QUESTION ID

**S63-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**7 in.**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

**True or false?**

Using the diameter in $A = \pi r^2$ gives an area four times too large.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

2305

QUESTION ID

**S63-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE $(2r)^2 = 4r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sector of a circle with diameter 20 uses r =

**A.** 10

**B.** 20

**C.** 5

**D.** 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

2306

**QUESTION ID**

**S63-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 10**

**GRADING NOTE** Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle's radius is 2.5 cm. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

2307

**QUESTION ID**

## S63-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**5 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

**True or false?**

The radius and diameter can be used interchangeably in area formulas.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

2308

QUESTION ID

**S63-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** They differ by a factor of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSG-C.B.5

# BACKWARD 3

Fractions of a circle in degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle 45°?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

2310

**QUESTION ID**

## S63-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

$\frac{1}{8}$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle π/2 radians?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

2311

**QUESTION ID**

**S63-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**¼**

**GRADING NOTE** Any equivalent form. (π/2)/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A full circle measures

**A.** 360° or 2π radians

**B.** 180° or π radians

**C.** 360 radians

**D.** 2π°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

2312

QUESTION ID

**S63-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 360° or 2π radians**

**GRADING NOTE** Both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

**True or false?**

A central angle of 1 radian is 1/(2π) of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

2313

QUESTION ID

**S63-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Fraction of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

**True or false?**

A 90° sector is ⅓ of the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

2314

**QUESTION ID**

**S63-B3-Q5**

**TYPE**

True or false

**CORRECT ANSWER**

**B. False**

**GRADING NOTE** ¼.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSG-C.B.5

# FORWARD 1

Solve inverse sector problems • High School — Geometry • Geometry: Circles • HSG-C.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sector with a 40° angle has area 4π. Find the radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

2316

QUESTION ID

**S63-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form. $(40/360)\pi r^2 = 4\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

A sector of a circle with radius 10 has area 30. Find the central angle in radians.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

2317

**QUESTION ID**

**S63-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**0.6 radians**

**GRADING NOTE** Any equivalent form. ½(100)θ = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

**True or false?**

If the radius is tripled and the angle stays the same, the sector area is tripled.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

2318

QUESTION ID

**S63-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Area depends on $r^2$, so it is multiplied by 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sector has area 50 and arc length 10. Radius?

**A.** 20

**B.** 5

**C.** 2.5

**D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

2319

QUESTION ID

**S63-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 10**

GRADING NOTE A = ½rs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

Show that the area of a sector is A = ½rs, where s is the arc length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

2320

QUESTION ID

**S63-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**A = ½r²θ and s = rθ, so A = ½r(rθ) = ½rs.**

GRADING NOTE Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSG-C.B.5

# FORWARD 2

Areas of segments and composite circular regions • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the exact area of the region between a 90° arc and its chord in a circle of radius 4 (a circular segment).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

2322

**QUESTION ID**
**S63-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**4π − 8**

**GRADING NOTE** Form required: exact value. Sector 4π minus right triangle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sprinkler waters a 120° sector of radius 9 m. Find the watered area to the nearest square meter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

2323

QUESTION ID
**S63-F2-Q2**

TYPE
Short answer

CORRECT ANSWER
**85 m²**

**GRADING NOTE** Rounding to the nearest whole number required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

**True or false?**

The area of a circular segment (smaller than a semicircle) equals the sector area plus the area of the triangle formed by the two radii and the chord.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

2324

QUESTION ID

## S63-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Subtract the triangle: segment = sector − triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An annulus (ring) has inner radius 3 and outer radius 5. Its area is

**A.** 8π

**B.** 4π

**C.** 16π

**D.** 25π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

2325

**QUESTION ID**

**S63-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 16π**

**GRADING NOTE** 25π − 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

A windshield wiper blade covers the part of its arm from 20 cm to 60 cm from the pivot and sweeps through 120°. Find the area it wipes to the nearest square centimeter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

2326

**QUESTION ID**
**S63-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**3351 cm²**

**GRADING NOTE** Rounding to the nearest whole number required. $\frac{1}{3}\pi(60^2 - 20^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

SET 64 • HSG-GPE.A.1 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

# Derive the equation of a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.A.1** |
| **BACKWARD 1** | Distance between two points with the Pythagorean Theorem | **8.G.B.8** |
| **BACKWARD 2** | Use the Pythagorean Theorem with expressions | **8.G.B.7** |
| **BACKWARD 3** | Shifting a graph: reading h and k | **HSF-BF.B.3** |
| **FORWARD 1** | Write circle equations from other given information | **HSG-GPE.A.1** |
| **FORWARD 2** | Prove or disprove that points lie on a circle | **HSG-GPE.B.4** |

SET 64 • HSG-GPE.A.1

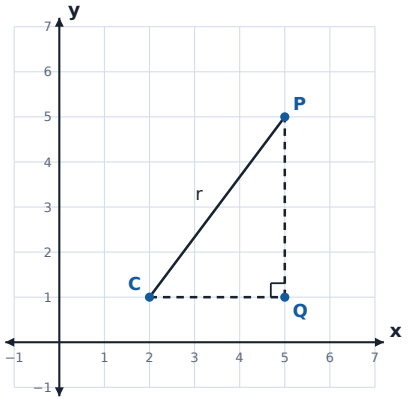
# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has center (h, k) and radius r. Use the Pythagorean Theorem to derive the equation satisfied by every point (x, y) on the circle. The diagram shows one example, with center C(2, 1).

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

2329

QUESTION ID

**S64-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**$(x - h)^2 + (y - k)^2 = r^2$**

**GRADING NOTE** REQUIRED WORK: Draw the right triangle with legs $|x - h|$ and $|y - k|$ and hypotenuse r.

Any valid proof (paragraph, two-column, or flow). Essential: a right triangle with horizontal leg $|x - h|$, vertical leg $|y - k|$, hypotenuse r (the distance CP); the Pythagorean Theorem gives $(x - h)^2 + (y - k)^2 = r^2$; squaring removes the absolute values; the reasoning holds for every point on the circle (degenerate legs of length 0 included). Invalid: only checking the one example point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the circle with center (3, −2) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

2330

QUESTION ID

**S64-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**$(x - 3)^2 + (y + 2)^2 = 25$**

**GRADING NOTE** Form required: an equation; equivalent expanded forms accepted. Sign check: $y - (-2) = y + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

Write the equation of the circle with center at the origin that passes through (6, −8).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

2331

**QUESTION ID**

## S64-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2 + y^2 = 100$**

**GRADING NOTE** Form required: an equation. $r^2 = 36 + 64 = 100$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S64-M-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5**

Write the equation of the circle with center (−1, 4) that passes through (2, 8).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

2332

**QUESTION ID**

**S64-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**$(x + 1)^2 + (y - 4)^2 = 25$**

**GRADING NOTE** Form required: an equation. $r^2 = 3^2 + 4^2 = 25$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S64-M-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5**

Which is the equation of the circle with center (−2, 5) and radius 3?

**A.** $(x - 2)^2 + (y + 5)^2 = 9$

**B.** $(x + 2)^2 + (y - 5)^2 = 9$

**C.** $(x + 2)^2 + (y - 5)^2 = 3$

**D.** $(x - 2)^2 + (y - 5)^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

2333

QUESTION ID

**S64-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. $(x + 2)^2 + (y - 5)^2 = 9$**

GRADING NOTE Signs are opposite to the center coordinates; the right side is $r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

SET 64 • HSG-GPE.A.1

# BACKWARD 1

Distance between two points with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

2335

QUESTION ID

**S64-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form. Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

2336

QUESTION ID

**S64-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

The legs of the right triangle joining (2, 1) and (7, 13) are

**A.** 2 and 1

**B.** 7 and 13

**C.** 9 and 14

**D.** 5 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

2337

**QUESTION ID**

## S64-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. 5 and 12

**GRADING NOTE** Differences of coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

**True or false?**

The distance from $(-3, 0)$ to $(3, 0)$ is 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5**

2338

QUESTION ID

**S64-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Horizontal distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5**

**True or false?**

The distance from (0, 0) to (3, 4) is 7.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

2339

QUESTION ID

## S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It is 5; the legs are not added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

# BACKWARD 2

Use the Pythagorean Theorem with expressions • Grade 8 • Geometry • 8.G.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has legs 6 and 8. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

2341

QUESTION ID

**S64-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

A right triangle has legs a and b and hypotenuse c.
Write the Pythagorean relationship.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

2342

**QUESTION ID**

## S64-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$a^2 + b^2 = c^2$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

A right triangle has legs $(x - 2)$ and $(y - 1)$ and hypotenuse 4. Which equation is true?

**A.** $(x - 2)^2 + (y - 1)^2 = 16$

**B.** $x^2 + y^2 = 16$

**C.** $(x - 2)^2 + (y - 1)^2 = 4$

**D.** $(x - 2) + (y - 1) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

2343

**QUESTION ID**

**S64-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. $(x - 2)^2 + (y - 1)^2 = 16$**

**GRADING NOTE** Square the legs and the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

**True or false?**

In a right triangle, the hypotenuse is opposite the right angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

2344

QUESTION ID

**S64-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$a^2 + b^2 = c^2$ holds for every triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

2345

QUESTION ID

**S64-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSG-GPE.A.1

# BACKWARD 3

Shifting a graph: reading h and k • High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3

The graph of $y = (x - 3)^2$ is the graph of $y = x^2$ shifted in which direction and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

2347

**QUESTION ID**

## S64-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

## Right 3

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write x − (−4) without the double sign.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

2348

QUESTION ID

**S64-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**x + 4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In $(x + 2)^2$, the shift is

**A.** 2 down

**B.** 2 right

**C.** 2 left

**D.** 2 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

2349

QUESTION ID
**S64-B3-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**C. 2 left**

**GRADING NOTE** $x - (-2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

**True or false?**

Replacing x with x − 5 shifts a graph 5 units right.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

2350

QUESTION ID

## S64-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Horizontal translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

**True or false?**

Replacing y with y + 1 shifts a graph 1 unit up.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

2351

QUESTION ID

## S64-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It shifts it 1 unit down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSG-GPE.A.1

# FORWARD 1

Write circle equations from other given information • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A diameter of a circle has endpoints (−2, 3) and (6, −3). Write the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

2353

**QUESTION ID**
**S64-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**$(x - 2)^2 + y^2 = 25$**

**GRADING NOTE** Form required: an equation. Center (2, 0) is the midpoint; radius 5 (half of 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has center (4, −3) and is tangent to the x-axis. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

2354

QUESTION ID
**S64-F1-Q2**

TYPE
Short answer

CORRECT ANSWER
**$(x - 4)^2 + (y + 3)^2 = 9$**

**GRADING NOTE** Form required: an equation. Radius = distance to the x-axis = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

**True or false?**

The circle $(x - 1)^2 + (y - 1)^2 = 2$ passes through the origin.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

2355

QUESTION ID
**S64-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE $1 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

A circle with center (0, 5) passes through (0, −1). Its radius is

**A.** 4

**B.** 36

**C.** 6

**D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

2356

QUESTION ID

**S64-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 6**

GRADING NOTE Vertical distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle is centered at (3, 3) and is tangent to both axes. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

2357

**QUESTION ID**
**S64-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**$(x - 3)^2 + (y - 3)^2 = 9$**

**GRADING NOTE** Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

# FORWARD 2

Prove or disprove that points lie on a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Does (−4, 6) lie on the circle with center (2, −2) passing through (8, 6)? Show the check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

2359

**QUESTION ID**

## S64-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes: both points are 10 from the center (36 + 64 = 100).**

**GRADING NOTE** Form required: the conclusion with the distance check.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S64-F2-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Is (7, 1) inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

2360

**QUESTION ID**

## S64-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**On (16 + 9 = 25)**

**GRADING NOTE** Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

**True or false?**

A point is inside a circle when its distance from the center is greater than the radius.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

2361

QUESTION ID

## S64-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Greater than the radius means outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The point (1, 1) is ___ the circle $x^2 + y^2 = 4$.

**A.** on

**B.** outside

**C.** inside

**D.** the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

2362

**QUESTION ID**

## S64-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. inside

**GRADING NOTE** $1 + 1 = 2 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has the diameter with endpoints (−2, 3) and (6, −3). Does (5, 4) lie on it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

2363

**QUESTION ID**

## S64-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes:**

**$(5 - 2)^2 + 4^2 = 25$.**

**GRADING NOTE** Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

# Complete the square to find the center and radius of a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.A.1** |
| **BACKWARD 1** | Complete the square in one variable | **HSA-REI.B.4.a** |
| **BACKWARD 2** | Square a binomial | **HSA-APR.A.1** |
| **BACKWARD 3** | Keep an equation balanced | **HSA-REI.A.1** |
| **FORWARD 1** | Leading coefficients and equations that are not circles | **HSG-GPE.A.1** |
| **FORWARD 2** | Complete the square for an ellipse | **HSG-GPE.A.3** |

SET 65 • HSG-GPE.A.1

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle
$x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

2366

**QUESTION ID**
**S65-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**Center (3, −2); radius 5**

**GRADING NOTE** Form required: both. $(x - 3)^2 + (y + 2)^2 = 25$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-M-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5**

Find the center and radius of the circle
$x^2 + y^2 + 8x - 10y + 5 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

2367

**QUESTION ID**
**S65-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**Center (−4, 5); radius 6**

**GRADING NOTE** Form required: both. $(x + 4)^2 + (y - 5)^2 = -5 + 16 + 25 = 36$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Complete the square to write $x^2 + y^2 - 2x - 14y + 25 = 0$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

2368

**QUESTION ID**

## S65-M-Q3

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$(x - 1)^2 + (y - 7)^2 = 25$; center (1, 7); radius 5**

**GRADING NOTE** REQUIRED WORK: Group x and y terms, add $(b/2)^2$ for each to both sides, and factor.

Form required: the standard form, center, and radius. Added 1 and 49 to both sides: $-25 + 1 + 49 = 25$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-M-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5**

Find the center and radius of the circle
$x^2 + y^2 + 10y = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

2369

QUESTION ID
**S65-M-Q4**

TYPE
Short answer

CORRECT ANSWER
**Center (0, −5); radius 5**

**GRADING NOTE** Form required: both. $x^2 + (y + 5)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the radius of the circle $x^2 + y^2 - 4x + 6y = 3$?

**A.** 2

**B.** $\sqrt{3}$

**C.** 4

**D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

2370

**QUESTION ID**

**S65-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 4**

**GRADING NOTE** $(x - 2)^2 + (y + 3)^2 = 3 + 4 + 9 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

SET 65 • HSG-GPE.A.1

# BACKWARD 1

Complete the square in one variable • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

**ANSWER KEY** Reduced copy of the question page, with its answer

What number completes the square: $x^2 - 6x + ___?$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

2372

QUESTION ID

**S65-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**9**

**GRADING NOTE** Any equivalent form. $(6/2)^2$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B1-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Write $x^2 + 10x + 25$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

2373

QUESTION ID
**S65-B1-Q2**

TYPE
Short answer

CORRECT ANSWER
**$(x + 5)^2$**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B1-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$x^2 - 8x + 16$ equals

**A.** $(x + 4)^2$

**B.** $(x - 8)^2$

**C.** $(x - 4)^2$

**D.** $(x - 16)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

2374

QUESTION ID

**S65-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. $(x - 4)^2$**

**GRADING NOTE** Half of 8 is 4.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B1-Q3**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

To complete $x^2 + bx$, add $(b/2)^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

2375

QUESTION ID
**S65-B1-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

**True or false?**

$x^2 + 6x + 6$ is a perfect square.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

2376

QUESTION ID

**S65-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It needs + 9.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B1-Q5**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5**

SET 65 • HSG-GPE.A.1

# BACKWARD 2

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

2378

QUESTION ID
**S65-B2-Q1**

TYPE
Short answer

CORRECT ANSWER
**$x^2 - 6x + 9$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(y + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

2379

QUESTION ID
**S65-B2-Q2**

TYPE
Short answer

CORRECT ANSWER
**$y^2 + 4y + 4$**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B2-Q2**
**High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions**
**HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$(x + 5)^2 =$

**A.** $x^2 + 10x + 25$

**B.** $x^2 + 25$

**C.** $x^2 + 10x + 10$

**D.** $x^2 + 5x + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

2380

QUESTION ID
**S65-B2-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**A. $x^2 + 10x + 25$**

**GRADING NOTE** Middle term 2 · 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(a - b)^2 = a^2 - 2ab + b^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

2381

QUESTION ID
**S65-B2-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(x - 3)^2 = x^2 - 9$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

2382

QUESTION ID
**S65-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE Missing the middle term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSG-GPE.A.1

# BACKWARD 3

Keep an equation balanced • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If 9 is added to the left side of an equation, what must be done to the right side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

2384

QUESTION ID
**S65-B3-Q1**

TYPE
Short answer

CORRECT ANSWER
**Add 9**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B3-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$x^2 - 6x = 7$. Add 9 to both sides and factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

2385

QUESTION ID

**S65-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**$(x - 3)^2 = 16$**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B3-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$x^2 + y^2 - 4x = 5$ becomes $(x - 2)^2 + y^2 =$

**A.** 4

**B.** 5

**C.** 1

**D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

2386

**QUESTION ID**

## S65-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 9**

**GRADING NOTE** 5 + 4.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B3-Q3**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5**

**True or false?**

Adding the same number to both sides keeps the solutions the same.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

2387

QUESTION ID

**S65-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Equality property.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-B3-Q4**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5**

**True or false?**

A number added to one side need not be added to the other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

2388

QUESTION ID
**S65-B3-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** That changes the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSG-GPE.A.1

# FORWARD 1

Leading coefficients and equations that are not circles • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the center and radius of
$2x^2 + 2y^2 - 8x + 12y - 6 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

2390

QUESTION ID
**S65-F1-Q1**

TYPE
Short answer

CORRECT ANSWER
**Center (2, −3); radius 4**

GRADING NOTE Form required: both. Divide by 2 first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

Does $x^2 + y^2 + 2x + 4y + 10 = 0$ represent a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

2391

**QUESTION ID**

## S65-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**No: it becomes $(x + 1)^2 + (y + 2)^2 = -5$, and a sum of squares cannot be negative.**

**GRADING NOTE** Form required: the conclusion with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

**True or false?**

$x^2 + y^2 - 6x + 9 = 0$ represents a single point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

2392

QUESTION ID

**S65-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** $(x - 3)^2 + y^2 = 0$: only (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$3x^2 + 3y^2 - 12y = 0$ has radius

**A.** $\sqrt{12}$

**B.** 2

**C.** 12

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

2393

**QUESTION ID**

**S65-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 2**

**GRADING NOTE** $x^2 + (y - 2)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For what values of c is $x^2 + y^2 - 4x + c = 0$ a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

2394

QUESTION ID

**S65-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**c < 4**

**GRADING NOTE** Form required: an inequality. $r^2 = 4 - c > 0$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5**

2394

SET 65 • HSG-GPE.A.1

# FORWARD 2

Complete the square for an ellipse • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Write $4x^2 + 9y^2 - 16x + 18y - 11 = 0$ in standard form and give the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

2396

QUESTION ID

**S65-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$(x - 2)^2/9 + (y + 1)^2/4 = 1$; center $(2, -1)$**

**GRADING NOTE** Form required: both. $4(x - 2)^2 + 9(y + 1)^2 = 11 + 16 + 9 = 36$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the center of $x^2 + 4y^2 + 6x - 8y + 9 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

2397

**QUESTION ID**

**S65-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(−3, 1)**

**GRADING NOTE** Form required: an ordered pair. $(x + 3)^2 + 4(y - 1)^2 = 4$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-F2-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5**

**True or false?**

After factoring 4 out of $4x^2 - 16x$, add 16 inside the parentheses to complete the square.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

2398

QUESTION ID

**S65-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Add 4 inside: $4(x^2 - 4x + 4)$, which adds 16 to the expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is NOT a circle?

**A.** $x^2 + 4y^2 = 4$

**B.** $2x^2 + 2y^2 = 8$

**C.** $x^2 + y^2 = 4$

**D.** $(x - 1)^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

2399

QUESTION ID

**S65-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. $x^2 + 4y^2 = 4$**

**GRADING NOTE** Unequal coefficients give an ellipse.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S65-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why $x^2 + 4y^2 = 4$ is not a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

2400

QUESTION ID

**S65-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**The $x^2$ and $y^2$ coefficients differ, so the "radius" differs by direction (2 across, 1 up).**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

# Derive the equation of a parabola from a focus and directrix

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.A.2** |
| **BACKWARD 1** | Distance between two points | **8.G.B.8** |
| **BACKWARD 2** | Distance from a point to a horizontal line | **6.NS.C.8** |
| **BACKWARD 3** | Points equidistant from two objects | **HSG-CO.C.9** |
| **BACKWARD 4** | Expand and simplify squares | **HSA-APR.A.1** |
| **FORWARD 1** | Find the focus and directrix from an equation | **HSG-GPE.A.2** |
| **FORWARD 2** | Design with the reflective property of parabolas | **HSG-MG.A.3** |

SET 66 • HSG-GPE.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Derive the equation of the parabola with focus (0, 2) and directrix y = −2.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

2403

**QUESTION ID**

## S66-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$x^2 = 8y$ ($y = x^2/8$)**

**GRADING NOTE** REQUIRED WORK: Set the distance from (x, y) to the focus equal to its distance to the directrix, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$; square: $x^2 + y^2 - 4y + 4 = y^2 + 4y + 4$; simplify to $x^2 = 8y$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S66-M-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the parabola with focus (0, −3) and directrix y = 3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

2404

**QUESTION ID**

## S66-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2 = -12y$**

**GRADING NOTE** Form required: an equation; equivalent forms ($y = -x^2/12$) accepted.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S66-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the parabola with focus (2, 5) and directrix y = 1.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

2405

QUESTION ID

**S66-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**$(x - 2)^2 = 8(y - 3)$**

**GRADING NOTE** Form required: an equation; equivalent forms accepted. Vertex (2, 3) midway; p = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the parabola with focus (3, 0) and directrix x = −3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

2406

QUESTION ID

**S66-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**$y^2 = 12x$**

**GRADING NOTE** Form required: an equation; equivalent forms accepted.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S66-M-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5**

A parabola has focus (1, 4) and directrix y = 0. Its vertex is

**A.** (1, 4)

**B.** (0, 2)

**C.** (1, 0)

**D.** (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

2407

**QUESTION ID**

## S66-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (1, 2)**

**GRADING NOTE** Midway between focus and directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

SET 66 • HSG-GPE.A.2

# BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance from (3, 6) to (0, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

2409

QUESTION ID

**S66-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the distance from (x, y) to (0, 2) as an expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

2410

QUESTION ID

**S66-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$\sqrt{(x^2 + (y - 2)^2)}$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The distance from (4, 1) to (1, 5) is

**A.** 7

**B.** 5

**C.** √7

**D.** 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

2411

**QUESTION ID**

## S66-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 5**

**GRADING NOTE** Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

**True or false?**

The distance formula comes from the Pythagorean Theorem.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5**

2412

QUESTION ID

## S66-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5**

**True or false?**

$\sqrt{(9 + 16)} = 3 + 4$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

2413

QUESTION ID

**S66-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $\sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

# BACKWARD 2

Distance from a point to a horizontal line • Grade 6 • The Number System • 6.NS.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

How far is (5, 7) from the line y = 2?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

2415

QUESTION ID

**S66-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How far is (1, −3) from the line y = 4?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

2416

QUESTION ID

**S66-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The distance from (x, y) to y = −2 is

**A.** |x + 2|

**B.** y

**C.** |y − 2|

**D.** |y + 2|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

2417

**QUESTION ID**

**S66-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. |y + 2|**

**GRADING NOTE** Vertical difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

**True or false?**

The distance from a point to a horizontal line is measured straight up or down.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

2418

QUESTION ID

**S66-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

**True or false?**

The distance from (6, 1) to y = 4 is 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

2419

QUESTION ID

**S66-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSG-GPE.A.2

# BACKWARD 3

Points equidistant from two objects • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

**True or false?**

The set of points equidistant from two points is the perpendicular bisector of the segment joining them.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

2421

QUESTION ID

**S66-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

A parabola is the set of points equidistant from what two objects?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

2422

QUESTION ID

**S66-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**A point (the focus) and a line (the directrix)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

The vertex of a parabola is

**A.** at the focus

**B.** twice as far from the focus

**C.** halfway between the focus and the directrix

**D.** on the directrix

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

2423

QUESTION ID

**S66-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. halfway between the focus and the directrix**

GRADING NOTE Equidistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Focus (0, 6), directrix y = 0. Find the vertex.

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

2424

QUESTION ID

**S66-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**(0, 3)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

**True or false?**

Every point of a parabola is closer to the focus than to the directrix.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

2425

QUESTION ID

**S66-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSG-GPE.A.2

# BACKWARD 4

Expand and simplify squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

2427

QUESTION ID
**S66-B4-Q1**

TYPE
Short answer

CORRECT ANSWER
**$y^2 - 4y + 4$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify $(y + 2)^2 - (y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

2428

QUESTION ID
**S66-B4-Q2**

TYPE
Short answer

CORRECT ANSWER
**8y**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$y^2 + 6y + 9 - (y^2 - 6y + 9) =$

**A.** 18

**B.** $2y^2$

**C.** 0

**D.** 12y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

2429

QUESTION ID
**S66-B4-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**D. 12y**

**GRADING NOTE** The squares cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(y + 3)^2 = y^2 + 6y + 9$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

2430

QUESTION ID
**S66-B4-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(y - 2)^2 - y^2 = 4$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

2431

QUESTION ID
**S66-B4-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE $-4y + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

SET 66 • HSG-GPE.A.2

# FORWARD 1

Find the focus and directrix from an equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = x^2/12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

2433

QUESTION ID

**S66-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Focus (0, 3); directrix y = −3**

**GRADING NOTE** Form required: both. $x^2 = 12y = 4py$, $p = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

Find the vertex, focus, and directrix of $(x - 1)^2 = -8(y + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

2434

QUESTION ID

**S66-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Vertex (1, −2); focus (1, −4); directrix y = 0**

GRADING NOTE Form required: all three. p = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

**True or false?**

In $x^2 = 4py$, the focus is p units from the vertex.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

2435

QUESTION ID
**S66-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The parabola $y^2 = -20x$ opens

**A.** left

**B.** up

**C.** down

**D.** right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

2436

QUESTION ID

**S66-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. left**

**GRADING NOTE** Negative coefficient with $y^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the focus of $y = 2x^2$.

Focus: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

2437

QUESTION ID

**S66-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**(0, ⅛)**

**GRADING NOTE** Form required: an ordered pair. $x^2 = \frac{1}{2}y$, $4p = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

# FORWARD 2

Design with the reflective property of parabolas • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A satellite dish is 4 ft across and 1 ft deep. How far from the vertex should the receiver (at the focus) be placed?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

2439

QUESTION ID

**S66-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**1 ft**

GRADING NOTE Any equivalent form. $x^2 = 4py$ through (2, 1): $4 = 4p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A headlight reflector has equation $x^2 = 12y$ (inches). Where should the bulb go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

2440

**QUESTION ID**
**S66-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**3 in. above the vertex, at (0, 3)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

**True or false?**

Rays parallel to the axis of a parabolic mirror reflect through the vertex.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

2441

QUESTION ID
**S66-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** They reflect through the focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

A dish has focus 2 m above its vertex. Its equation (vertex at the origin) is

**A.** $x^2 = 4y$

**B.** $x^2 = 8y$

**C.** $y = 8x^2$

**D.** $x^2 = 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

2442

QUESTION ID

**S66-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. $x^2 = 8y$**

GRADING NOTE $4p = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

A dish has equation $x^2 = 16y$ (meters) and is 8 m across. How deep is it?

Depth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

2443

**QUESTION ID**
**S66-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**1 m**

**GRADING NOTE** Any equivalent form. 16 = 16y at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

# Prove or disprove the type of a quadrilateral with coordinates

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.4** |
| **BACKWARD 1** | Find the slope between two points | **8.F.B.4** |
| **BACKWARD 2** | Find the distance between two points | **8.G.B.8** |
| **BACKWARD 3** | Properties that define special quadrilaterals | **HSG-CO.C.11** |
| **FORWARD 1** | Coordinate proofs with variables | **HSG-GPE.B.4** |
| **FORWARD 2** | Use the classification to find area | **HSG-GPE.B.7** |

SET 67 • HSG-GPE.B.4

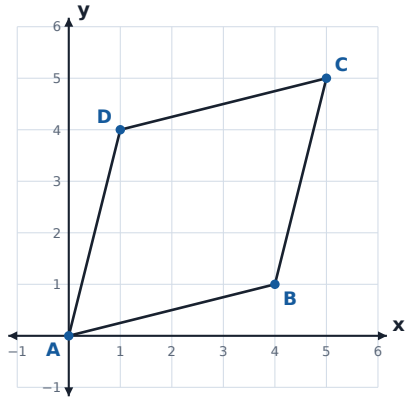
# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

2446

**QUESTION ID**

## S67-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**AB = BC = CD = DA = √17, so ABCD is a rhombus.**

**GRADING NOTE** REQUIRED WORK: Compute all four side lengths with the distance formula.

Any valid coordinate proof. Essential: all four distances computed correctly (√17 each) and the definition of a rhombus cited. Alternative: show it is a parallelogram (slopes) with perpendicular diagonals (slopes 1 and −1). Invalid: "it looks like a rhombus".

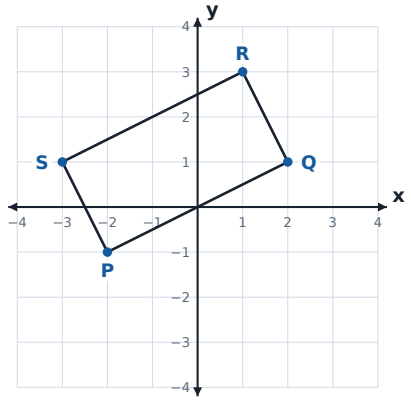
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove that PQRS with P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1) is a rectangle. Is it a square?

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

2447

QUESTION ID

## S67-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

**Slopes PQ = RS = ½ and QR = SP = −2; adjacent products are −1, so all angles are right: a rectangle. Not a square: PQ = √20 but QR = √5.**

GRADING NOTE REQUIRED WORK: Compute the slopes of all four sides; compare adjacent slopes.

Any valid coordinate proof. Essential: four slopes; opposite sides parallel; adjacent slopes multiply to −1 (right angles); side lengths √20 ≠ √5 for "not a square".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

Prove or disprove: KLMN with K(0, 0), L(5, 0), M(6, 3), N(0, 3) is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

2448

QUESTION ID

**S67-M-Q3**

TYPE

Answer with required work

CORRECT ANSWER

**Disproved: KL ∥ MN (slope 0), but LM has slope 3 while NK is vertical, so LM ∦ NK. KLMN is a trapezoid, not a parallelogram.**

**GRADING NOTE** REQUIRED WORK: Compare the slopes of both pairs of opposite sides.

Any valid coordinate proof. Essential: one pair of opposite sides shown NOT parallel (a single counterexample disproves). Alternative: KL = 5 but MN = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Classify the triangle with vertices A(1, 1), B(4, 5), C(8, 2) by its sides and angles. Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

2449

QUESTION ID

**S67-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**Isosceles right triangle: AB = BC = 5 and AC = √50;**
**$5^2 + 5^2 = 50$, so the angle at B is right.**

**GRADING NOTE** Form required: the classification with distances. Alternative: slopes AB = 4/3 and BC = −3/4 multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

A quadrilateral is shown to be a parallelogram. Which additional calculation proves it is a rectangle?

**A.** Two adjacent sides have slopes whose product is −1

**B.** Two opposite sides have equal slopes

**C.** The diagonals have equal slopes

**D.** Two adjacent sides have equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

2450

QUESTION ID

**S67-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. Two adjacent sides have slopes whose product is −1**

**GRADING NOTE** A parallelogram with one right angle is a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSG-GPE.B.4

# BACKWARD 1

Find the slope between two points • Grade 8 • Functions • 8.F.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (5, 4).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

2452

QUESTION ID

**S67-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (2, −1).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

2453

QUESTION ID

**S67-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**−2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The slope through (2, 5) and (6, 5) is

**A.** undefined

**B.** 0

**C.** 5

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

2454

QUESTION ID

**S67-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0**

**GRADING NOTE** Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

**True or false?**

A vertical segment has an undefined slope.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5**

2455

QUESTION ID

**S67-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Run 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Slope is run divided by rise.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5**

2456

QUESTION ID

## S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Rise ÷ run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5**

SET 67 • HSG-GPE.B.4

# BACKWARD 2

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (4, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

2458

QUESTION ID

**S67-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**√17**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance between (1, 1) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

2459

QUESTION ID

**S67-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

Which pair of points is 10 apart?

**A.** (2, 3) and (8, 3)

**B.** (0, 0) and (6, 8)

**C.** (0, 0) and (10, 10)

**D.** (1, 1) and (5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

2460

**QUESTION ID**

## S67-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. (0, 0) and (6, 8)**

**GRADING NOTE** 36 + 64 = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

**True or false?**

Two segments with the same length are congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5**

2461

QUESTION ID

## S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5**

**True or false?**

$\sqrt{10} + \sqrt{10} = \sqrt{20}$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

2462

QUESTION ID

**S67-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $\sqrt{10} + \sqrt{10} = 2\sqrt{10} = \sqrt{40}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4

# BACKWARD 3

Properties that define special quadrilaterals • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Which quadrilateral has four congruent sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

2464

QUESTION ID

**S67-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**A rhombus (a square is a special rhombus)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A parallelogram with one right angle is a ___.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

2465

QUESTION ID

**S67-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**Rectangle**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

Which shows a quadrilateral is a parallelogram?

**A.** One pair of sides is parallel

**B.** Two adjacent sides are equal

**C.** One right angle

**D.** Both pairs of opposite sides are parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

2466

**QUESTION ID**

**S67-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Both pairs of opposite sides are parallel**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Every square is a rhombus.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

2467

QUESTION ID

**S67-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Four congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

**True or false?**

Every rhombus is a square.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

2468

QUESTION ID

**S67-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Its angles need not be right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSG-GPE.B.4

# FORWARD 1

Coordinate proofs with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangle has vertices O(0, 0), A(a, 0), B(a, b), C(0, b). Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

2470

QUESTION ID

**S67-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**OB = AC = √(a² + b²)**

GRADING NOTE

REQUIRED WORK: Compute OB and AC with the distance formula.

Any valid coordinate proof. Essential: OB = √(a² + b²), AC = √((0 − a)² + (b − 0)²) = √(a² + b²); equal for all a, b. Invalid: using numbers for a and b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle with vertices O(0, 0), A(2a, 0), B(2b, 2c). M and N are the midpoints of OB and AB. Prove that MN ∥ OA and MN = ½OA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

2471

**QUESTION ID**

## S67-F1-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**M(b, c), N(a + b, c); MN is horizontal like OA, and MN = a = ½(2a).**

**GRADING NOTE** REQUIRED WORK: Find M and N, then compare slope and length.

Any valid coordinate proof. Essential: correct midpoints; slope 0 for both; length a vs 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

**True or false?**

Placing a figure with a vertex at the origin and a side on the x-axis is allowed in a coordinate proof.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

2472

QUESTION ID

**S67-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Any figure can be moved there by a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

Why do coordinates like (2a, 0) and (2b, 2c) make the midpoint proof easier?

**A.** The midpoints have no fractions

**B.** They make the triangle isosceles

**C.** They make a right angle

**D.** They make the slopes 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

2473

QUESTION ID

## S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

**A. The midpoints have no fractions**

**GRADING NOTE** Halves of even coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

Why is a proof using A(6, 0) and B(2, 4) not enough to prove the midsegment theorem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5



**QUESTION ID**

## S67-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**It shows only one triangle; the theorem is about all triangles.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S67-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5**

SET 67 • HSG-GPE.B.4

# FORWARD 2

Use the classification to find area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus. Find its area using its diagonals.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

2476

**QUESTION ID**

## S67-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Any equivalent form. AC = 5√2, BD = 3√2; ½ · 5√2 · 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rectangle PQRS has P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

2477

**QUESTION ID**
**S67-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**10**

**GRADING NOTE** Any equivalent form. $\sqrt{20} \cdot \sqrt{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

**True or false?**

Once a quadrilateral is proved a rhombus, its area is the product of two adjacent side lengths.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

2478

QUESTION ID

## S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Only for a square; use $\frac{1}{2}d_1d_2$ or base × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its area is

**A.** 16

**B.** √10

**C.** 10

**D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

2479

**QUESTION ID**

**S67-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 10**

**GRADING NOTE** Side √10.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S67-F2-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5**

Explain why proving the shape first matters before using a special area formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

2480

**QUESTION ID**
**S67-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**The formula (such as ½$d_1d_2$ or length × width) is valid only for that type of figure.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

# Prove or disprove that a point lies on a circle, and other coordinate proofs

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.4** |
| **BACKWARD 1** | Distance between two points | **8.G.B.8** |
| **BACKWARD 2** | The equation of a circle | **HSG-GPE.A.1** |
| **BACKWARD 3** | Substitute values into an expression | **6.EE.A.2** |
| **FORWARD 1** | Coordinate proofs about circles with variables | **HSG-GPE.B.4** |
| **FORWARD 2** | Find the tangent line to a circle at a point | **HSG-GPE.B.5** |

SET 68 • HSG-GPE.B.4

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and contains the point (0, 2). Prove or disprove that the point (1, √3) lies on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

2483

QUESTION ID

## S68-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

**Proved: the radius is 2 and the distance to (1, √3) is √(1 + 3) = 2.**

GRADING NOTE REQUIRED WORK: Find the radius, then compare the distance from the origin to (1, √3).

Any valid coordinate proof. Essential: radius 2; distance $\sqrt{(1^2 + (\sqrt{3})^2)} = 2$; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove or disprove that (5, 2) lies on the circle with center (1, −1) and radius 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

2484

QUESTION ID

**S68-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**Proved:**

$$(5 - 1)^2 + (2 + 1)^2 = 16 + 9 = 25 = 5^2.$$

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

Prove or disprove that $(-3, 6)$ lies on the circle $(x - 1)^2 + (y - 2)^2 = 30$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

2485

**QUESTION ID**

**S68-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**Disproved: 16 + 16 = 32 ≠ 30 (the point is outside).**

**GRADING NOTE** Form required: the conclusion with the check.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S68-M-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangle ABC has A(0, 0), B(8, 0), C(0, 6). Prove that the midpoint M of hypotenuse BC is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

2486

QUESTION ID

**S68-M-Q4**

TYPE

Answer with required work

CORRECT ANSWER

**M(4, 3);**

**MA = MB = MC = 5**

GRADING NOTE

REQUIRED WORK: Find M, then compute MA, MB, MC.

Any valid coordinate proof. Essential: midpoint (4, 3); three distances of 5. Consequence: A, B, C lie on a circle centered at M.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The point (4, −3) is ___ the circle $x^2 + y^2 = 20$.

**A.** on

**B.** inside

**C.** outside

**D.** the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

2487

**QUESTION ID**

**S68-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. outside**

**GRADING NOTE** 16 + 9 = 25 > 20.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S68-M-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5**

SET 68 • HSG-GPE.B.4

# BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (1, −1) to (5, 2)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

2489

QUESTION ID

**S68-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (0, 0) to (1, √3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

2490

QUESTION ID

**S68-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (0, 0) to (−3, 4) is

**A.** 5

**B.** 1

**C.** 7

**D.** 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

2491

QUESTION ID

**S68-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 5**

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

**True or false?**

$(-3)^2 = 9$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

2492

QUESTION ID

## S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Square of a negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

**True or false?**

The distance from (2, 2) to (2, 7) is 9.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

2493

QUESTION ID

**S68-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

# BACKWARD 2

The equation of a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the circle with center (1, −1) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

2495

**QUESTION ID**

## S68-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$(x - 1)^2 + (y + 1)^2 = 25$**

**GRADING NOTE** Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the radius of $(x - 1)^2 + (y - 2)^2 = 30$?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

2496

**QUESTION ID**

**S68-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$\sqrt{30}$**

**GRADING NOTE** Form required: exact value.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S68-B2-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Center of $(x + 4)^2 + y^2 = 9$?

**A.** (4, 0)

**B.** (−4, 0)

**C.** (−4, 9)

**D.** (0, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

2497

QUESTION ID

**S68-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. (−4, 0)**

**GRADING NOTE** Opposite sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

**True or false?**

A point satisfies a circle's equation exactly when it is on the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

2498

QUESTION ID

## S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Meaning of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

**True or false?**

The radius of $x^2 + y^2 = 20$ is 20.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

2499

QUESTION ID
**S68-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** $\sqrt{20}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSG-GPE.B.4

# BACKWARD 3

Substitute values into an expression • Grade 6 • Expressions & Equations • 6.EE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Evaluate $(x - 1)^2 + (y - 2)^2$ when $x = -3$ and $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

2501

QUESTION ID

**S68-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**32**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Evaluate $x^2 + y^2$ when $x = 4$, $y = -3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

2502

QUESTION ID

**S68-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**25**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$(x + 1)^2$ when $x = -4$ is

**A.** $-9$

**B.** 9

**C.** 25

**D.** $-3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

2503

QUESTION ID

**S68-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 9**

**GRADING NOTE** $(-3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

**True or false?**

$(-5 - 1)^2 = 36$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

2504

QUESTION ID

## S68-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $(-6)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(2 + 1)^2 = 2^2 + 1^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

2505

QUESTION ID

**S68-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $9 \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSG-GPE.B.4

# FORWARD 1

Coordinate proofs about circles with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A(−r, 0) and B(r, 0) are endpoints of a diameter of the circle $x^2 + y^2 = r^2$, and P(x, y) is any other point on the circle. Prove that ∠APB is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

2507

QUESTION ID

**S68-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**slope PA · slope PB = $y^2/(x^2 - r^2)$ = $y^2/(-y^2) = -1$, so PA ⊥ PB.**

GRADING NOTE REQUIRED WORK: Compute the slopes of PA and PB and use $x^2 + y^2 = r^2$.

Any valid coordinate proof. Essential: slopes $y/(x + r)$ and $y/(x - r)$; product $y^2/(x^2 - r^2)$; substitute $x^2 - r^2 = -y^2$; product −1. (P is not A or B, so $y \neq 0$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Show that if (a, b) is on $x^2 + y^2 = r^2$, then (b, a) and (−a, b) are too.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

2508

**QUESTION ID**

**S68-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$b^2 + a^2 = r^2$ and $(-a)^2 + b^2 = r^2$, the same sum.**

**GRADING NOTE** Any equivalent argument.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S68-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5**

**True or false?**

A coordinate proof with variables covers every case at once.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

2509

QUESTION ID
**S68-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** General.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

To prove four points lie on one circle, show they are

**A.** all the same distance from one point

**B.** the same distance from the origin only

**C.** vertices of a rectangle only

**D.** all on one line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

2510

QUESTION ID

## S68-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

**A. all the same distance from one point**

**GRADING NOTE** Common center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove or disprove: A(1, 0), B(0, 1), C(−1, 0), D(0, −2) lie on one circle centered at the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

2511

QUESTION ID

## S68-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**Disproved: D is 2 from the origin, the others 1.**

**GRADING NOTE** Form required: the conclusion with distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

# FORWARD 2

Find the tangent line to a circle at a point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

2513

**QUESTION ID**

## S68-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -\frac{3}{4}x + 25/4$**

**GRADING NOTE** Form required: an equation; equivalent forms (3x + 4y = 25) accepted. Radius slope 4/3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S68-F2-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the tangent to $(x - 1)^2 + (y - 1)^2 = 25$ at (4, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

2514

**QUESTION ID**

## S68-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$y - 5 = -\frac{3}{4}(x - 4)$**

**GRADING NOTE** Form required: an equation; equivalent forms accepted.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S68-F2-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5**

**True or false?**

The tangent at P is parallel to the radius drawn to P.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

2515

QUESTION ID

**S68-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is perpendicular to that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

The radius to (0, 5) on $x^2 + y^2 = 25$ is vertical, so the tangent there is

**A.** $x = 0$

**B.** $x = 5$

**C.** $y = x + 5$

**D.** $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

2516

QUESTION ID

**S68-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. y = 5**

**GRADING NOTE** Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

Explain how the slope of the tangent at (a, b) on $x^2 + y^2 = r^2$ is found.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

2517

**QUESTION ID**

## S68-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The radius has slope b/a, so the tangent has slope −a/b (negative reciprocal), for a, b ≠ 0.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

# Prove the slope criteria for parallel and perpendicular lines

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.5** |
| **BACKWARD 1** | Similar slope triangles give the same slope | **8.EE.B.6** |
| **BACKWARD 2** | Rotate a point 90° about the origin | **HSG-CO.A.5** |
| **BACKWARD 3** | Negative reciprocals | **7.NS.A.2** |
| **FORWARD 1** | Use the criteria to classify pairs of lines | **HSG-GPE.B.5** |
| **FORWARD 2** | Use the criteria inside a general coordinate proof | **HSG-GPE.B.4** |

SET 69 • HSG-GPE.B.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Prove that two distinct nonvertical lines with the same slope m are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

2520

QUESTION ID

**S69-M-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**$mx + b_1 = mx + b_2$ gives $b_1 = b_2$, a contradiction; so the lines never meet and are parallel.**

**GRADING NOTE** REQUIRED WORK: Write the lines as $y = mx + b_1$ and $y = mx + b_2$ with $b_1 \neq b_2$ and show there is no intersection.

Any valid proof (paragraph, two-column, or flow). Essential: equations with equal slopes and different intercepts; solving for an intersection leads to $b_1 = b_2$ (false); conclusion. Alternative: a translation maps one line to the other; or congruent slope triangles with corresponding angles equal.

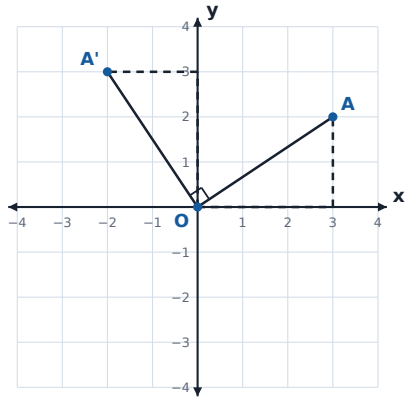
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line OA passes through O(0, 0) and A(a, b) with a, b ≠ 0. Rotating A 90° counterclockwise about O gives A'(−b, a). Use this to prove that perpendicular lines have slopes whose product is −1. The diagram shows the case A(3, 2).

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

2521

QUESTION ID

**S69-M-Q2**

TYPE

Answer with required work

CORRECT ANSWER

**slope OA = b/a, slope OA' = a/(−b) = −a/b, product −1.**

**GRADING NOTE** REQUIRED WORK: Find the slopes of OA and OA', then their product.

Any valid proof (paragraph, two-column, or flow). Essential: a 90° rotation maps OA onto a perpendicular line; the image of (a, b) is (−b, a); slopes b/a and −a/b; product −1. Any pair of perpendicular nonvertical lines can be translated to pass through O, keeping slopes. Invalid: checking only the one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

Explain why a 90° rotation of the slope triangle with run 3 and rise 2 gives a slope triangle with run −2 and rise 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

2522

**QUESTION ID**
## S69-M-Q3

**TYPE**
Short answer

**CORRECT ANSWER**

**Rotating 90° counterclockwise sends (x, y) to (−y, x): the run becomes the old rise (negated) and the rise becomes the old run.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

Why do two nonvertical lines with equal slopes and different y-intercepts never meet?

**A.** Their slopes are negative reciprocals

**B.** Their intercepts are equal

**C.** Setting the equations equal leads to a false statement

**D.** They meet at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

2523

QUESTION ID

**S69-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. Setting the equations equal leads to a false statement**

**GRADING NOTE** $mx + b_1 = mx + b_2$ has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

**True or false?**

Lines with slopes 2 and −2 are perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

2524

QUESTION ID

**S69-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Product −4, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSG-GPE.B.5

# BACKWARD 1

Similar slope triangles give the same slope • Grade 8 • Expressions & Equations • 8.EE.B.6

**True or false?**

Any two slope triangles on the same line are similar.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

**Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5**

2526

QUESTION ID

## S69-B1-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

**Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A line passes through (0, 0) and (2, 3). Find the rise for a run of 4.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

2527

QUESTION ID

**S69-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

Slope triangles with run 2, rise 3 and run 6, rise 9 give slopes

**A.** 6 and 18

**B.** 3/2 and 9/6 that differ

**C.** both 3/2

**D.** 2/3 and 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

2528

QUESTION ID

**S69-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. both 3/2**

**GRADING NOTE** 9/6 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the slope of a line is the same between any two of its points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

2529

QUESTION ID

**S69-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**The slope triangles are similar, so rise/run is the same ratio.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

**True or false?**

The slope between two points on a line depends on which points are chosen.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

2530

QUESTION ID

**S69-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

# BACKWARD 2

Rotate a point 90° about the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rotate (3, 2) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

2532

QUESTION ID

**S69-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**(−2, 3)**

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rotate (1, 4) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

2533

QUESTION ID

**S69-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**(−4, 1)**

**GRADING NOTE** Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

The rule for a 90° counterclockwise rotation about the origin is

**A.** (x, y) → (−y, x)

**B.** (x, y) → (−x, −y)

**C.** (x, y) → (y, x)

**D.** (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

2534

**QUESTION ID**

## S69-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. (x, y) → (−y, x)**

**GRADING NOTE** Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

**True or false?**

A 90° rotation maps a line to a perpendicular line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

2535

QUESTION ID

**S69-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Angle of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

**True or false?**

A 90° rotation maps (2, 0) to (0, −2).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

2536

QUESTION ID

**S69-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Counterclockwise gives (0, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5

# BACKWARD 3

Negative reciprocals • Grade 7 • The Number System • 7.NS.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Negative reciprocal of 2/3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

2538

QUESTION ID

**S69-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**−3/2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Multiply (b/a)(−a/b).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

2539

QUESTION ID

**S69-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**−1**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Negative reciprocal of −4?

**A.** ¼

**B.** −¼

**C.** 4

**D.** −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

2540

**QUESTION ID**

## S69-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. ¼**

**GRADING NOTE** −1 ÷ (−4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A number times its negative reciprocal is −1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

2541

QUESTION ID

**S69-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The negative reciprocal of 1 is 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

2542

QUESTION ID

**S69-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSG-GPE.B.5

# FORWARD 1

Use the criteria to classify pairs of lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are y = 3x − 2 and 6x − 2y = 7 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

2544

QUESTION ID

**S69-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Parallel (both slopes 3)**

**GRADING NOTE** Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Are 2x + 5y = 1 and 5x − 2y = 4 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

2545

QUESTION ID

## S69-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**Perpendicular (slopes −2/5 and 5/2)**

**GRADING NOTE** Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

**True or false?**

A horizontal line and a vertical line are perpendicular even though the slope product rule cannot be used.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

2546

QUESTION ID

## S69-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Vertical slope is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$y = \frac{1}{2}x + 1$ and $y = 2x - 1$ are

**A.** neither

**B.** the same line

**C.** perpendicular

**D.** parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

2547

**QUESTION ID**

## S69-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. neither

**GRADING NOTE** Product 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find k so that y = kx + 1 is perpendicular to y = −⅓x.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

2548

**QUESTION ID**

## S69-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**3**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S69-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5**

SET 69 • HSG-GPE.B.5

# FORWARD 2

Use the criteria inside a general coordinate proof • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Rhombus OABC has O(0, 0), A(a, 0), B(a + b, c), C(b, c), with $b^2 + c^2 = a^2$. Prove that its diagonals are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

2550

**QUESTION ID**

## S69-F2-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**slope OB · slope AC = c/(a + b) · c/(b − a) = c²/(b² − a²) = c²/(−c²) = −1**

**GRADING NOTE** REQUIRED WORK: Find the slopes of OB and AC and multiply.

Any valid coordinate proof. Essential: both slopes; product simplified with $b^2 + c^2 = a^2$; conclusion. (Assume $b \neq \pm a$, so neither diagonal is vertical.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A coordinate proof places rhombus OABC at O(0, 0), A(a, 0), B(a + b, c), C(b, c). Why must the proof assume $b^2 + c^2 = a^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

2551

**QUESTION ID**
**S69-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**It says OC = a = OA, the equal sides that make OABC a rhombus.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

**True or false?**

The slope criterion turns "perpendicular" into an algebra statement.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

2552

QUESTION ID
**S69-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

In a coordinate proof that lines are parallel, you show

**A.** equal slopes

**B.** equal intercepts

**C.** slope product −1

**D.** equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

2553

QUESTION ID

**S69-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. equal slopes**

**GRADING NOTE** Criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

Prove or disprove with slopes: A(0, 0), B(4, 2), C(3, 4) form a right angle at B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

2554

QUESTION ID

**S69-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**Proved: slope AB = ½ and slope BC = −2; the product is −1, so AB ⊥ BC.**

**GRADING NOTE** Form required: the conclusion with slopes. Slope BC = (4 − 2)/(3 − 4) = −2; product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

# Write equations of parallel and perpendicular lines

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.5** |
| **BACKWARD 1** | Read the slope of a line | **8.F.B.4** |
| **BACKWARD 2** | Rewrite an equation in slope-intercept form | **HSA-CED.A.4** |
| **BACKWARD 3** | Write a line from a point and a slope | **HSF-LE.A.2** |
| **BACKWARD 4** | Opposite reciprocals | **7.NS.A.2** |
| **FORWARD 1** | Write perpendicular bisectors and altitudes | **HSG-GPE.B.5** |
| **FORWARD 2** | Find the distance from a point to a line | **HSG-GPE.B.4** |

SET 70 • HSG-GPE.B.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the line parallel to y = 3x − 5 that passes through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

2557

QUESTION ID

**S70-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**y = 3x − 2**

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

Write the equation of the line perpendicular to $y = \frac{2}{3}x + 1$ that passes through $(4, -1)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

2558

**QUESTION ID**

## S70-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -3/2\ x + 5$**

**GRADING NOTE** Form required: an equation; equivalent forms accepted.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the line parallel to 2x + 5y = 10 that passes through (5, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

2559

QUESTION ID
**S70-M-Q3**

TYPE
Short answer

CORRECT ANSWER
**y = −2/5 x + 5**

**GRADING NOTE** Form required: an equation; equivalent forms (2x + 5y = 25) accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to x = 4 that passes through (1, −3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

2560

QUESTION ID
**S70-M-Q4**

TYPE
Short answer

CORRECT ANSWER
**y = −3**

GRADING NOTE Form required: an equation. x = 4 is vertical, so the line is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

Which line is perpendicular to y = −4x + 1 and passes through the origin?

**A.** y = −4x

**B.** y = −¼x

**C.** y = ¼x

**D.** y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

2561

QUESTION ID

**S70-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. y = ¼x**

**GRADING NOTE** Negative reciprocal of −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

SET 70 • HSG-GPE.B.5

# BACKWARD 1

Read the slope of a line • Grade 8 • Functions • 8.F.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope of $y = -2x + 7$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

2563

QUESTION ID

**S70-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**−2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope of y = 5?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

2564

QUESTION ID

**S70-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**0**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope of $y = \frac{2}{3}x - 4$?

**A.** 3/2

**B.** ⅔

**C.** −4

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

2565

QUESTION ID

**S70-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. ⅔**

GRADING NOTE Coefficient of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

**True or false?**

In y = mx + b, m is the slope.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5**

2566

QUESTION ID

**S70-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Slope-intercept form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5**

**True or false?**

The slope of x = 3 is 3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

2567

QUESTION ID

## S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

# BACKWARD 2

Rewrite an equation in slope-intercept form • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 2x + 5y = 10 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

2569

QUESTION ID

**S70-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**y = −2/5 x + 2**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-B2-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 3x − y = 4 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

2570

QUESTION ID

**S70-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**y = 3x − 4**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-B2-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

6x + 3y = 9 in slope-intercept form is

**A.** y = 2x + 3

**B.** y = −6x + 9

**C.** y = −2x + 9

**D.** y = −2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

2571

**QUESTION ID**

**S70-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. y = −2x + 3**

**GRADING NOTE** Divide by 3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-B2-Q3**

**High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5**

2571

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$x - 2y = 6$ has slope ½.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

2572

QUESTION ID

## S70-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $y = ½x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

4x + 2y = 8 has slope 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

2573

QUESTION ID
**S70-B2-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5

# BACKWARD 3

Write a line from a point and a slope • High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the line with slope 3 through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

2575

QUESTION ID

**S70-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**y = 3x − 2**

**GRADING NOTE** Form required: an equation; equivalent forms accepted.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-B3-Q1**

**High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the line with slope −1 through (0, 6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

2576

QUESTION ID

**S70-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**y = −x + 6**

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope 2 through (1, 5): b =

**A.** 5

**B.** 7

**C.** −3

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

2577

**QUESTION ID**

**S70-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 3**

**GRADING NOTE** 5 = 2 + b.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-B3-Q3**

**High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5**

**True or false?**

$y - 4 = 3(x - 2)$ is the line with slope 3 through (2, 4).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

2578

QUESTION ID

**S70-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Point-slope form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A line through (0, 6) has y-intercept 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

2579

QUESTION ID

**S70-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSG-GPE.B.5

# BACKWARD 4

Opposite reciprocals • Grade 7 • The Number System • 7.NS.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Negative reciprocal of ⅔?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

2581

QUESTION ID

**S70-B4-Q1**

TYPE

Short answer

CORRECT ANSWER

**−3/2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Negative reciprocal of −5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

2582

QUESTION ID

**S70-B4-Q2**

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Negative reciprocal of −2/5?

**A.** 5/2

**B.** 2/5

**C.** −2/5

**D.** −5/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

2583

**QUESTION ID**

## S70-B4-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 5/2**

**GRADING NOTE** Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

**True or false?**

The negative reciprocal of ¼ is −4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

2584

QUESTION ID

**S70-B4-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

**True or false?**

The negative reciprocal of 3 is −3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

2585

**QUESTION ID**

## S70-B4-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** $-\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

SET 70 • HSG-GPE.B.5

# FORWARD 1

Write perpendicular bisectors and altitudes • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the perpendicular bisector of the segment from (2, 1) to (6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

2587

**QUESTION ID**

## S70-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = -x + 7$**

**GRADING NOTE** Form required: an equation. Midpoint (4, 3), slope −1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5**

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Write the equation of the altitude from A to BC.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

2588

**QUESTION ID**

## S70-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**y = x**

**GRADING NOTE** Form required: an equation. BC slope −1, so the altitude has slope 1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5**

**True or false?**

A perpendicular bisector passes through the midpoint of the segment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

2589

**QUESTION ID**

## S70-F1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The perpendicular bisector of the segment from (0, 0) to (0, 8) is

**A.** x = 0

**B.** y = 4

**C.** x = 4

**D.** y = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

2590

**QUESTION ID**

## S70-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. y = 4**

**GRADING NOTE** Horizontal through (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the perpendicular bisector of the segment from (−3, 2) to (5, 2).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

2591

**QUESTION ID**

## S70-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**x = 1**

**GRADING NOTE** Form required: an equation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5**

SET 70 • HSG-GPE.B.5

# FORWARD 2

Find the distance from a point to a line • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance from P(5, 5) to the line y = −x + 2.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

2593

QUESTION ID

**S70-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**4√2 (about 5.66); the foot is (1, 1)**

GRADING NOTE REQUIRED WORK: Write the perpendicular through P, find where it meets the line, then use the distance formula. Form required: exact value or rounding to 0.01. Perpendicular y = x meets y = −x + 2 at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance from (0, 0) to the line y = 4.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

2594

QUESTION ID

**S70-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-F2-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5**

**True or false?**

The distance from a point to a line can be measured along any segment from the point to the line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

2595

QUESTION ID
**S70-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** It is measured along the perpendicular, the shortest segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The foot of the perpendicular from (3, 0) to y = x is

**A.** (0, 3)

**B.** (0, 0)

**C.** (3, 3)

**D.** (1.5, 1.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

2596

**QUESTION ID**

## S70-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (1.5, 1.5)**

**GRADING NOTE** y = −x + 3 meets y = x.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S70-F2-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the distance from (6, 1) to the line y = 2x − 1.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

2597

QUESTION ID

**S70-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**2√5 (about 4.47)**

**GRADING NOTE** Form required: exact value or rounding to 0.01. Foot (2, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

# Solve geometric problems with the slope criteria

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.5** |
| **BACKWARD 1** | Slope from two points | **8.F.B.4** |
| **BACKWARD 2** | Solve a linear equation for an unknown | **HSA-REI.B.3** |
| **BACKWARD 3** | Equations of parallel and perpendicular lines | **HSG-GPE.B.5** |
| **BACKWARD 4** | Opposite sides of a parallelogram are parallel | **HSG-CO.C.11** |
| **FORWARD 1** | Find the orthocenter of a triangle | **HSG-GPE.B.5** |
| **FORWARD 2** | Shortest path to a straight line in a design | **HSG-MG.A.3** |

SET 71 • HSG-GPE.B.5

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

The line through (1, 2) and (4, k) is parallel to $y = 2x + 7$. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

2600

**QUESTION ID**

**S71-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8**

**GRADING NOTE** Any equivalent form. $(k - 2)/3 = 2$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-M-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5**

ABCD is a parallelogram with A(0, 0), B(5, 1), C(7, 4). Find D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

2601

**QUESTION ID**
**S71-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**(2, 3)**

**GRADING NOTE** Form required: an ordered pair. AD ∥ BC and DC ∥ AB.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5**

The segment from (2, 1) to (5, k) is perpendicular to the line y = 3x. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

2602

**QUESTION ID**
**S71-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**0**

**GRADING NOTE** Any equivalent form. (k − 1)/3 = −⅓.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-M-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(8, 4), C(4, 6). Find the foot of the altitude from C to AB.

Foot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

2603

**QUESTION ID**

**S71-M-Q4**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**(5.6, 2.8)**

**GRADING NOTE** REQUIRED WORK: Write line AB and the perpendicular through C, then solve the system.

Form required: an ordered pair. AB: y = ½x; altitude: y = −2x + 14.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-M-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has A(1, 1), B(4, 2), C(3, 5). At which vertex is the right angle?

**A.** C

**B.** B

**C.** A

**D.** There is no right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

2604

**QUESTION ID**

**S71-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. B**

**GRADING NOTE** Slope AB = ⅓, slope BC = −3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-M-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5**

SET 71 • HSG-GPE.B.5

# BACKWARD 1

Slope from two points • Grade 8 • Functions • 8.F.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope through (1, 2) and (4, 8)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

2606

QUESTION ID

**S71-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope through (4, 2) and (3, 5)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

2607

QUESTION ID

**S71-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**−3**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Slope through (0, 0) and (8, 4)?

**A.** ½

**B.** 8

**C.** 4

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

2608

QUESTION ID

**S71-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. ½**

**GRADING NOTE** 4/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

**True or false?**

The slope from A to B equals the slope from B to A.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

2609

QUESTION ID
**S71-B1-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The slope through (1, 1) and (1, 5) is 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

2610

QUESTION ID

**S71-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

# BACKWARD 2

Solve a linear equation for an unknown • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $(k - 2)/3 = 2$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

2612

QUESTION ID

**S71-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**8**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-B2-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve ½x = −2x + 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

2613

QUESTION ID

**S71-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5.6**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-B2-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $(k - 1)/3 = -\frac{1}{3}$.

**A.** $-2$

**B.** 1

**C.** 2

**D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

2614

**QUESTION ID**

**S71-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 0**

**GRADING NOTE** $k - 1 = -1$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-B2-Q3**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5**

**True or false?**

Multiplying both sides by 3 is a valid first step in $(k - 2)/3 = 2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

2615

QUESTION ID

**S71-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Inverse operation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

**True or false?**

The solution of $2x + 1 = 7$ is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

2616

QUESTION ID

## S71-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5

# BACKWARD 3

Equations of parallel and perpendicular lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line perpendicular to y = ½x through (4, 6)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

2618

QUESTION ID

**S71-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**y = −2x + 14**

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Line parallel to y = −x through (0, 3)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

2619

**QUESTION ID**

**S71-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**y = −x + 3**

**GRADING NOTE** Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Perpendicular slope to ⅓?

**A.** 3

**B.** −3

**C.** −⅓

**D.** ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

2620

QUESTION ID

**S71-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. −3**

GRADING NOTE Negative reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

An altitude is perpendicular to the side it meets.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

2621

QUESTION ID

**S71-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

**True or false?**

An altitude passes through the midpoint of the opposite side.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

2622

**QUESTION ID**

## S71-B3-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** That is a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSG-GPE.B.5

# BACKWARD 4

Opposite sides of a parallelogram are parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**True or false?**

In a parallelogram, both pairs of opposite sides are parallel and congruent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

2624

QUESTION ID

**S71-B4-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A parallelogram has A(0, 0), B(3, 0), C(4, 2). Find D (in order ABCD).

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

2625

**QUESTION ID**

**S71-B4-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(1, 2)**

**GRADING NOTE** Form required: an ordered pair.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-B4-Q2**

**High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In parallelogram ABCD, side AB is parallel to

**A.** AC

**B.** BC

**C.** AD

**D.** DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

2626

**QUESTION ID**

**S71-B4-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. DC**

**GRADING NOTE** Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Going from B to C is "right 2, up 3". Going from A to D is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

2627

**QUESTION ID**

**S71-B4-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**Right 2, up 3**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A parallelogram's adjacent sides are always perpendicular.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

2628

QUESTION ID

**S71-B4-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Only rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

SET 71 • HSG-GPE.B.5

# FORWARD 1

Find the orthocenter of a triangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Find the orthocenter (the intersection of the altitudes).

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

2630

QUESTION ID

**S71-F1-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**(2, 2)**

**GRADING NOTE** REQUIRED WORK: Write two altitude equations and solve the system.

Form required: an ordered pair. Altitude from C: x = 2; from A: y = x.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5**

Triangle with A(0, 0), B(4, 0), C(0, 3). Where is the orthocenter?

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

2631

**QUESTION ID**

## S71-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**(0, 0)**

**GRADING NOTE** Form required: an ordered pair. Right angle at A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5**

**True or false?**

The orthocenter of a right triangle is at the right-angle vertex.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

2632

QUESTION ID

## S71-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Two legs are altitudes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

To find the orthocenter, you need at least

**A.** three medians

**B.** the circumcenter

**C.** one altitude

**D.** two altitudes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

2633

**QUESTION ID**

## S71-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. two altitudes

**GRADING NOTE** Two lines meet in a point.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S71-F1-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5**

Check that the third altitude of the triangle A(0, 0), B(6, 0), C(2, 4) also passes through (2, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

2634

QUESTION ID

**S71-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**AC has slope 2, so the altitude from B has slope −½: y = −½(x − 6); at x = 2, y = 2. Yes.**

**GRADING NOTE** Any equivalent check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

# FORWARD 2

Shortest path to a straight line in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A well is at W(2, 9). A straight canal runs along $y = 2x - 1$ (units: 100 m). Where should a pipe from the well meet the canal so the pipe is as short as possible, and how long is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

2636

QUESTION ID

**S71-F2-Q1**

TYPE

Answer with required work

CORRECT ANSWER

**At (4.4, 7.8); about 2.68 units (268 m)**

GRADING NOTE REQUIRED WORK: Use the perpendicular from W to the canal.

Form required: the point and the length (rounding to 0.01 unit or 1 m). Perpendicular $y = -\frac{1}{2}x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pipe will connect a well to a straight canal. Why is the shortest possible pipe perpendicular to the canal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

2637

**QUESTION ID**
**S71-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**The perpendicular segment is the shortest distance from a point to a line.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

**True or false?**

A pipe meeting the canal at a 60° angle is shorter than one meeting it at 90°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

2638

QUESTION ID

**S71-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The perpendicular is shortest; the slanted pipe is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

2638

**ANSWER KEY** Reduced copy of the question page, with its answer

A path from (0, 5) to the road y = 0 is shortest at

**A.** (0, 0)

**B.** (5, 0)

**C.** (−5, 0)

**D.** (0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

2639

**QUESTION ID**

**S71-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. (0, 0)**

**GRADING NOTE** Straight down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sensor at (3, 4) must connect to the line y = x. Where should the wire meet the line?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2640

QUESTION ID
**S71-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**(3.5, 3.5)**

**GRADING NOTE** Form required: an ordered pair. y = −x + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

# Partition a directed segment in a given ratio

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.6** |
| **BACKWARD 1** | A ratio as a fraction of the whole | **6.RP.A.1** |
| **BACKWARD 2** | Horizontal and vertical change | **6.NS.C.8** |
| **BACKWARD 3** | Find a fraction of a number | **6.NS.A.1** |
| **FORWARD 1** | Work backward: find an endpoint or the ratio | **HSG-GPE.B.6** |
| **FORWARD 2** | The centroid as a partition point | **HSG-GPE.B.4** |

SET 72 • HSG-GPE.B.6

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point P on the directed segment from A(1, 2) to B(13, 8) that partitions it in the ratio 1 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

2643

**QUESTION ID**
**S72-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**(5, 4)**

**GRADING NOTE** Form required: an ordered pair. ⅓ of the way from A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-M-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5**

Find the point P on the directed segment from A(−4, 6) to B(6, 1) that partitions it in the ratio 3 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

2644

**QUESTION ID**
**S72-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**(2, 3)**

**GRADING NOTE** Form required: an ordered pair. 3/5 of the way from A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5**

Find the point P on the directed segment from A(2, −3) to B(−6, 9) that partitions it in the ratio 1 : 3.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

2645

**QUESTION ID**
**S72-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**(0, 0)**

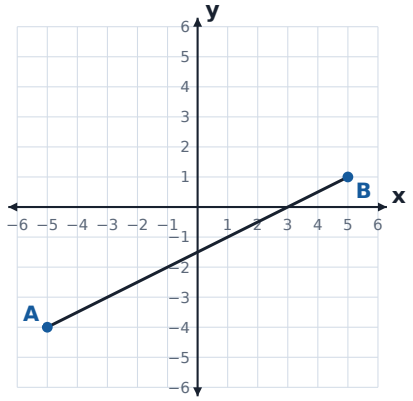
**GRADING NOTE** Form required: an ordered pair. ¼ of the way from A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-M-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the point P on the directed segment from A to B that partitions it in the ratio 2 : 3.




P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

2646

**QUESTION ID**

## S72-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**(−1, −2)**

**GRADING NOTE** Form required: an ordered pair. A(−5, −4), B(5, 1); 2/5 of (10, 5) is (4, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Point P partitions the directed segment from B(10, 5) to A(0, 0) in the ratio 1 : 4. P is

**A.** (8, 4)

**B.** (2, 1)

**C.** (5, 2.5)

**D.** (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

2647

QUESTION ID
**S72-M-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**A. (8, 4)**

**GRADING NOTE** Start at B: ⅕ of the way toward A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

SET 72 • HSG-GPE.B.6

# BACKWARD 1

A ratio as a fraction of the whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A ratio of 2 : 3 splits a whole into how many equal parts?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

2649

**QUESTION ID**

**S72-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In a 1 : 2 partition, the first part is what fraction of the whole?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

2650

QUESTION ID

**S72-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**1/3**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In a 3 : 2 partition, the first part is

**A.** 3/2 of the whole

**B.** 2/3 of the whole

**C.** 3/5 of the whole

**D.** 2/5 of the whole

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2651

**QUESTION ID**

## S72-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. 3/5 of the whole

**GRADING NOTE** 3 of 5 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2651

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A ratio of 1 : 3 means the point is ¼ of the way along.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

2652

QUESTION ID

**S72-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE 1 of 4 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A ratio of 1 : 2 means the point is halfway.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

2653

QUESTION ID

**S72-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

# BACKWARD 2

Horizontal and vertical change • Grade 6 • The Number System • 6.NS.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

From (1, 2) to (13, 8), what is the change in x?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

2655

QUESTION ID

**S72-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From (−4, 6) to (6, 1), what is the change in y?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

2656

QUESTION ID

**S72-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**−5**

GRADING NOTE Any equivalent form (a decrease of 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From (2, −3) to (−6, 9), the change in x is

**A.** −8

**B.** −4

**C.** 4

**D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

2657

**QUESTION ID**

**S72-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. −8**

**GRADING NOTE** −6 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

**True or false?**

Change in x is the ending x minus the starting x.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

2658

QUESTION ID

**S72-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Direction matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

**True or false?**

The change in y from (0, 5) to (0, 1) is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

2659

QUESTION ID

**S72-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSG-GPE.B.6

# BACKWARD 3

Find a fraction of a number • Grade 6 • The Number System • 6.NS.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Find ⅓ of 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

2661

**QUESTION ID**

**S72-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find 3/5 of −5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

2662

QUESTION ID

**S72-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**−3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

2/5 of 10 is

**A.** 5

**B.** 25

**C.** 4

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

2663

**QUESTION ID**

## S72-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 4**

**GRADING NOTE** $10 \div 5 \times 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

**True or false?**

¼ of −8 is −2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

2664

QUESTION ID

**S72-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

⅔ of 9 is 3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

2665

QUESTION ID

**S72-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSG-GPE.B.6

# FORWARD 1

Work backward: find an endpoint or the ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

**ANSWER KEY** Reduced copy of the question page, with its answer

P(3, 4) partitions the directed segment from A(1, 2) to B in the ratio 1 : 2. Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

2667

**QUESTION ID**

## S72-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(7, 8)**

**GRADING NOTE** Form required: an ordered pair. AP is ⅓ of AB.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

P(4, 3) is on the segment from A(0, 1) to B(10, 6). In what ratio does P partition AB?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

2668

**QUESTION ID**
**S72-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**2 : 3**

**GRADING NOTE** Form required: a ratio. 4/10 = 2/5 of the way.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5**

**True or false?**

The midpoint partitions a segment in the ratio $1 : 1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

2669

**QUESTION ID**

## S72-F1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Halfway.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

P partitions AB in 3 : 1. Then P is closer to

**A.** B

**B.** neither

**C.** A

**D.** the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

2670

**QUESTION ID**

## S72-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. B

**GRADING NOTE** ¾ of the way from A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F1-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

M(2, 5) is the midpoint of AB with A(−1, 1). Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

2671

**QUESTION ID**

## S72-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**(5, 9)**

**GRADING NOTE** Form required: an ordered pair.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5**

SET 72 • HSG-GPE.B.6

# FORWARD 2

The centroid as a partition point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle with A(0, 0), B(6, 0), C(3, 9). Find the point ⅔ of the way from A to the midpoint of BC.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

2673

**QUESTION ID**

## S72-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(3, 3)**

**GRADING NOTE** Form required: an ordered pair. Midpoint (4.5, 4.5).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F2-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5**

Show that the centroid (the point 2 : 1 along a median from the vertex) of the triangle with vertices $(x_1, y_1)$, $(x_2, y_2)$, $(x_3, y_3)$ is the average of the vertices.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

2674

**QUESTION ID**

**S72-F2-Q2**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$**

**GRADING NOTE** REQUIRED WORK: Partition the median from $(x_1, y_1)$ to the midpoint of the opposite side in the ratio 2 : 1.

Any valid coordinate proof. Essential: midpoint $((x_2 + x_3)/2, (y_2 + y_3)/2)$; $x_1 + \frac{2}{3}((x_2 + x_3)/2 - x_1) = (x_1 + x_2 + x_3)/3$; the same for y.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F2-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5**

**True or false?**

The centroid of the triangle (0, 0), (6, 0), (3, 9) is (3, 4.5).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

2675

QUESTION ID
**S72-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** It is (3, 3), the average of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

The centroid of (1, 2), (4, 8), (7, −1) is

**A.** (4, 3)

**B.** (4, 9)

**C.** (3, 3)

**D.** (12, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

2676

**QUESTION ID**

## S72-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. (4, 3)**

**GRADING NOTE** (12/3, 9/3).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F2-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the centroid of the triangle with vertices (−2, 5), (4, 1), (7, 0).

Centroid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

2677

**QUESTION ID**
**S72-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**(3, 2)**

**GRADING NOTE** Form required: an ordered pair.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S72-F2-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5**

# Compute perimeters of polygons with coordinates

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.7** |
| **BACKWARD 1** | Distance between two points | **8.G.B.8** |
| **BACKWARD 2** | Lengths of horizontal and vertical sides | **6.G.A.3** |
| **BACKWARD 3** | Simplify square roots | **HSN-RN.A.2** |
| **FORWARD 1** | Find a missing coordinate from a perimeter | **HSG-GPE.B.7** |
| **FORWARD 2** | Fencing and cost from a coordinate plan | **HSG-MG.A.3** |

SET 73 • HSG-GPE.B.7

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the perimeter of the triangle with vertices (0, 0), (3, 4), and (3, 0).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

2680

**QUESTION ID**
**S73-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**12 units**

**GRADING NOTE** Any equivalent form. 5 + 4 + 3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-M-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the perimeter of the quadrilateral with vertices A(1, 1), B(4, 5), C(10, −3), D(7, −7).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

2681

**QUESTION ID**
**S73-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**30 units**

**GRADING NOTE** Any equivalent form. 5 + 10 + 5 + 10.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the perimeter of the triangle with vertices (0, 0), (4, 0), (2, 4). Give the exact value and a decimal to the nearest tenth.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

2682

**QUESTION ID**
**S73-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**$4 + 4\sqrt{5} \approx 12.9$ units**

**GRADING NOTE** Form required: both. Two sides $\sqrt{20} = 2\sqrt{5}$.

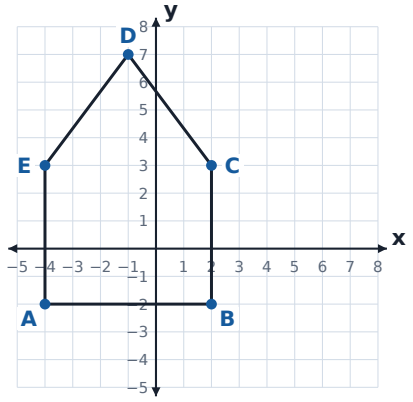
**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-M-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the perimeter of pentagon ABCDE.

Perimeter: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

2683

**QUESTION ID**

**S73-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**26 units**

**GRADING NOTE** Any equivalent form. 6 + 5 + 5 + 5 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its perimeter is

**A.** 4√10

**B.** 10

**C.** 40

**D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

2684

**QUESTION ID**
**S73-M-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. 4√10**

**GRADING NOTE** Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

SET 73 • HSG-GPE.B.7

# BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (2, 3) to (−1, 7)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

2686

QUESTION ID

**S73-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (4, 5) to (10, −3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

2687

QUESTION ID

**S73-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (0, 0) to (2, 4)?

**A.** 20

**B.** 2√5

**C.** 6

**D.** √6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

2688

QUESTION ID

**S73-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 2√5**

**GRADING NOTE** √20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

**True or false?**

The distance from (0, 0) to (3, 1) is √10.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5**

2689

**QUESTION ID**

## S73-B1-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** 9 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

**Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Distances can be negative.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

2690

QUESTION ID

**S73-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Never.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

# BACKWARD 2

Lengths of horizontal and vertical sides • Grade 6 • Geometry • 6.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Length of the side from (−4, −2) to (2, −2)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

2692

QUESTION ID

**S73-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Length of the side from (2, −2) to (2, 3)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

2693

QUESTION ID

**S73-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

Length from $(-3, 4)$ to $(-3, -1)$?

**A.** 7

**B.** 3

**C.** 5

**D.** $-5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

2694

**QUESTION ID**

**S73-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 5**

**GRADING NOTE** $|4 - (-1)|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

**True or false?**

A horizontal side's length is the difference of the x-coordinates.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

2695

QUESTION ID

**S73-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Absolute difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

**True or false?**

The side from (1, 1) to (1, 9) has length 10.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

2696

QUESTION ID

**S73-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7

# BACKWARD 3

Simplify square roots • High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify √20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

2698

QUESTION ID

**S73-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**2√5**

**GRADING NOTE** Form required: simplest radical form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-B3-Q1**

**High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

2699

**QUESTION ID**

## S73-B3-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**5√2**

**GRADING NOTE** Form required: simplest radical form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-B3-Q2**

**High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$2\sqrt{5} + 2\sqrt{5} =$

**A.** $2\sqrt{10}$

**B.** $8\sqrt{5}$

**C.** $4\sqrt{10}$

**D.** $4\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S73-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

2700

**QUESTION ID**

**S73-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $4\sqrt{5}$**

**GRADING NOTE** Like terms.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-B3-Q3**

**High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sqrt{10} \approx 3.16$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

2701

QUESTION ID
**S73-B3-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Calculator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sqrt{9} + \sqrt{16} = \sqrt{25}$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

2702

QUESTION ID
**S73-B3-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE $3 + 4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSG-GPE.B.7

# FORWARD 1

Find a missing coordinate from a perimeter • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (k, 0), (k, 3), (0, 3) with k > 0 and perimeter 22. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

2704

**QUESTION ID**

**S73-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8**

**GRADING NOTE** Any equivalent form. 2k + 6 = 22.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (3, h), h > 0, has perimeter 16. Find h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

2705

**QUESTION ID**
**S73-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**4**

**GRADING NOTE** Any equivalent form. Each slanted side is 5.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5**

**True or false?**

Doubling every coordinate of a polygon multiplies its perimeter by 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

2706

QUESTION ID

## S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The perimeter doubles; the area is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A square with one side from (0, 0) to (4, 3) has perimeter

**A.** 28

**B.** 25

**C.** 14

**D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

2707

QUESTION ID

**S73-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 20**

**GRADING NOTE** Side 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

Is the triangle (0, 0), (6, 0), (3, 4) isosceles? Justify with lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

2708

**QUESTION ID**

## S73-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes: two sides are 5.**

**GRADING NOTE** Form required: the conclusion with lengths.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S73-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5**

SET 73 • HSG-GPE.B.7

# FORWARD 2

Fencing and cost from a coordinate plan • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A garden plan has corners (0, 0), (8, 0), (8, 6), (0, 6), in meters. Fencing costs $12 per meter. Find the cost to fence it.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

2710

**QUESTION ID**
**S73-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**$336**

**GRADING NOTE** Any equivalent form. 28 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangular lot has corners (0, 0), (30, 0), (30, 40) in feet. Edging costs $2.50 per foot. Find the total cost.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

2711

**QUESTION ID**

**S73-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$300**

**GRADING NOTE** Any equivalent form. 120 ft.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

On a map with 1 unit = 5 m, an area of 12 square units is 60 m².

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

2712

QUESTION ID

**S73-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Areas scale by 5²: 12 × 25 = 300 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A path has corners (0, 0), (3, 4), (9, 12) (units: 10 m). Its length is

**A.** 150 m

**B.** 210 m

**C.** 15 m

**D.** 21 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

2713

**QUESTION ID**
**S73-F2-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. 150 m**

**GRADING NOTE** 5 + 10 = 15 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A run around a lot with corners (0, 0), (5, 12), (5, 0) in units of 100 m: how many meters?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

2714

**QUESTION ID**
**S73-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**3000 m**

**GRADING NOTE** Any equivalent form. 13 + 12 + 5 = 30 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

# Compute areas of triangles and rectangles with coordinates

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.B.7** |
| **BACKWARD 1** | Area of a triangle from base and height | **6.G.A.1** |
| **BACKWARD 2** | Lengths on the coordinate plane | **6.G.A.3** |
| **BACKWARD 3** | Distance between two points | **8.G.B.8** |
| **FORWARD 1** | Find a missing vertex from an area | **HSG-GPE.B.7** |
| **FORWARD 2** | Use area from coordinates with density | **HSG-MG.A.2** |

SET 74 • HSG-GPE.B.7

# MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the area of the triangle with vertices (0, 0), (8, 0), and (3, 5).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

2717

**QUESTION ID**
**S74-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**20 square units**

**GRADING NOTE** Any equivalent form. Base 8, height 5.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S74-M-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Rectangle ABCD has A(1, 1), B(5, 3), C(4, 5), D(0, 3). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

2718

**QUESTION ID**
**S74-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**10 square units**

**GRADING NOTE** Any equivalent form. AB = √20, BC = √5.

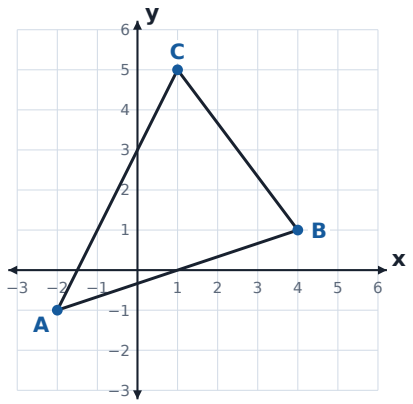
**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S74-M-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the area of triangle ABC.

Area: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

2719

QUESTION ID
**S74-M-Q3**

TYPE
Short answer

CORRECT ANSWER
**15 square units**

**GRADING NOTE** Any equivalent form. Bounding rectangle 36 minus 6, 6, and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the area of the triangle with vertices A(0, 0), B(6, 2), C(2, 6).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

2720

**QUESTION ID**

## S74-M-Q4

**TYPE**

Answer with required work

**CORRECT ANSWER**

**16 square units**

**GRADING NOTE** REQUIRED WORK: Enclose the triangle in a rectangle and subtract the three right triangles.

Any equivalent form. 36 − 6 − 8 − 6. Alternative methods (shoelace) accepted with work.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S74-M-Q4**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5**

A triangle has vertices (0, 0), (10, 0), and (x, 4). What is its area?

**A.** 20, for any x

**B.** It depends on x

**C.** 40

**D.** 20 only if x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

2721

QUESTION ID

**S74-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. 20, for any x**

**GRADING NOTE** Base 10, height 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

SET 74 • HSG-GPE.B.7

# BACKWARD 1

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Area of a triangle with base 8 and height 5?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

2723

QUESTION ID

**S74-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**20**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Area of a right triangle with legs 6 and 2?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

2724

QUESTION ID

**S74-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A =

**A.** ½bh

**B.** b + h

**C.** 2bh

**D.** bh

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

2725

**QUESTION ID**

## S74-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. ½bh**

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

**True or false?**

The height of a triangle is perpendicular to the base.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

2726

QUESTION ID

**S74-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The height must be one of the sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

2727

QUESTION ID

## S74-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Only in right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

# BACKWARD 2

Lengths on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Length from (−2, −1) to (4, −1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

2729

QUESTION ID

**S74-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Height of the point (3, 5) above the x-axis?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

2730

QUESTION ID

**S74-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

Width of the rectangle enclosing x from −2 to 4?

**A.** 2

**B.** 8

**C.** 4

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

2731

**QUESTION ID**

**S74-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 6**

**GRADING NOTE** 4 − (−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

**True or false?**

A point with y-coordinate −3 is 3 units from the x-axis.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

2732

QUESTION ID

**S74-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Absolute value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

**True or false?**

The length from (0, 2) to (0, 9) is 11.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

2733

QUESTION ID

**S74-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7

# BACKWARD 3

Distance between two points • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (1, 1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

2735

**QUESTION ID**

## S74-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**√20**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (5, 3) to (4, 5)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

2736

QUESTION ID

**S74-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**√5**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$\sqrt{20} \cdot \sqrt{5} =$

**A.** 10

**B.** 25

**C.** $\sqrt{25}$

**D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

2737

QUESTION ID

**S74-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 10**

GRADING NOTE $\sqrt{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

**True or false?**

Slanted rectangle sides need the distance formula.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

2738

QUESTION ID

**S74-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Not grid-aligned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sqrt{20} \cdot \sqrt{5} = 5$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

2739

QUESTION ID

**S74-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSG-GPE.B.7

# FORWARD 1

Find a missing vertex from an area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (2, k) has area 18.
Find all possible k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

2741

**QUESTION ID**
**S74-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**6 or −6**

**GRADING NOTE** Form required: both values. ½ · 6 · |k| = 18.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S74-F1-Q1**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (a, 0), (a, 5), (0, 5), $a > 0$, and area 35. Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

2742

**QUESTION ID**

**S74-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**7**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S74-F1-Q2**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5**

**True or false?**

Moving the top vertex of a triangle parallel to its base does not change its area.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

2743

QUESTION ID

## S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Same height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

Triangle (0, 0), (4, 0), (x, 6). For which x is the area 12?

**A.** Any x

**B.** No x

**C.** x = 4 only

**D.** x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

2744

QUESTION ID

**S74-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. Any x**

GRADING NOTE ½ · 4 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle (1, 1), (7, 1), (4, k) has area 21 and k > 1. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

2745

QUESTION ID

**S74-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**8**

**GRADING NOTE** Any equivalent form. Base 6, height 7.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S74-F1-Q5**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5**

SET 74 • HSG-GPE.B.7

# FORWARD 2

Use area from coordinates with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A park on a map has corners (0, 0), (4, 0), (4, 3), (0, 3), with 1 unit = 1 km. 1,800 people live there. Find the population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

2747

**QUESTION ID**

**S74-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**150 people per km²**

**GRADING NOTE** Any equivalent form. Area 12 km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

2747

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangular field has vertices (0, 0), (200, 0), (0, 150) in meters. Seeds are planted at 4 per m². How many seeds?

Seeds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

2748

**QUESTION ID**

**S74-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**60,000**

**GRADING NOTE** Any equivalent form. Area 15,000 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

**True or false?**

Density based on area is an area divided by a quantity.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

2749

QUESTION ID

**S74-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It is a quantity divided by an area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A lot with corners (0, 0), (6, 0), (6, 5), (0, 5) (meters) holds 60 plants. Density?

**A.** 2 plants per m²

**B.** 30 plants per m²

**C.** 11 plants per m²

**D.** 0.5 plants per m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

2750

**QUESTION ID**

## S74-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 2 plants per m²**

**GRADING NOTE** 60/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangle with corners (0, 0), (3, 0), (3, 2), (0, 2) (miles) has 4,200 residents. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

2751

QUESTION ID
**S74-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**700 people per square mile**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

# Informal arguments for the circumference and area of a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.1** |
| **BACKWARD 1** | Circumference and area formulas | **7.G.B.4** |
| **BACKWARD 2** | Area of a parallelogram | **6.G.A.1** |
| **BACKWARD 3** | All circles are similar | **HSG-C.A.1** |
| **FORWARD 1** | Area of a circle as a limit of regular polygons | **HSG-GMD.A.1** |
| **FORWARD 2** | From area to volume: stacking disks | **HSG-GMD.A.1** |

SET 75 • HSG-GMD.A.1

# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle of radius r is cut into many equal sectors, which are rearranged as shown. Explain why this shows that the area of the circle is $\pi r^2$.

Work space

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

2754

**QUESTION ID**

## S75-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**Area = (half the circumference) × r = $\pi r \cdot r = \pi r^2$.**

**GRADING NOTE** REQUIRED WORK: Describe the base and height of the rearranged shape and what happens as the number of sectors grows.

Any valid explanation. Essential: the shape approaches a parallelogram (rectangle); its height approaches r; its base is half the circumference, $\pi r$; area = $\pi r \cdot r = \pi r^2$. Invalid: "the formula says so" or measuring one drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the ratio of circumference to diameter is the same number (π) for every circle, so that $C = \pi d = 2\pi r$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

2755

**QUESTION ID**
**S75-M-Q2**

**TYPE**
Answer with required work

**CORRECT ANSWER**
**Every circle is a dilation of every other, so C and d scale by the same factor and C/d is constant; that constant is called π.**

**GRADING NOTE** REQUIRED WORK: Use the fact that all circles are similar.

Any valid explanation. Essential: circles are similar (dilation); lengths scale by the same factor k; $C_2/d_2 = kC_1/kd_1 = C_1/d_1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

A circle of radius r is cut into many equal sectors, which are rearranged, alternating point up and point down, into a near-parallelogram. Its base gets closer and closer to

**A.** $\pi r^2$

**B.** $2\pi r$ (the whole circumference)

**C.** $r$

**D.** $\pi r$ (half the circumference)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

2756

**QUESTION ID**

## S75-M-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $\pi r$ (half the circumference)**

**GRADING NOTE** Half the arcs are on top, half on the bottom.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A regular hexagon is inscribed in a circle of radius 5. Find the hexagon's perimeter and compare it with the circle's circumference. What happens as the number of sides increases?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

2757

**QUESTION ID**

## S75-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Perimeter 30 < 10π ≈ 31.4; with more sides the perimeter gets closer to the circumference.**

**GRADING NOTE** Form required: both numbers and the trend. The hexagon is six equilateral triangles of side 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

**True or false?**

As the number of sectors increases, the rearranged shape gets closer to a rectangle with height r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

2758

QUESTION ID

## S75-M-Q5

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The scalloped edges flatten.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

SET 75 • HSG-GMD.A.1

# BACKWARD 1

Circumference and area formulas • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the exact circumference of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

2760

**QUESTION ID**

**S75-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the exact area of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

2761

**QUESTION ID**

**S75-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**16π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

2761

**ANSWER KEY** Reduced copy of the question page, with its answer

Half the circumference of a circle of radius r is

**A.** $\pi r^2$

**B.** $\pi r$

**C.** $2\pi r$

**D.** r/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

2762

**QUESTION ID**

**S75-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $\pi r$**

**GRADING NOTE** $\frac{1}{2} \cdot 2\pi r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

**True or false?**

$C = 2\pi r$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

2763

QUESTION ID

**S75-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

**True or false?**

$A = 2\pi r$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5**

2764

QUESTION ID

**S75-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $A = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5**

SET 75 • HSG-GMD.A.1

# BACKWARD 2

Area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Area of a parallelogram with base 9 and height 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q1/5

2766

QUESTION ID

**S75-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**36**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q1/5

**True or false?**

A parallelogram has the same area as a rectangle with the same base and height.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q2/5

2767

QUESTION ID

**S75-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Cut and slide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q2/5

A parallelogram's area is

**A.** base × slanted side

**B.** base × height

**C.** base + height

**D.** ½ base × height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q3/5

2768

QUESTION ID

**S75-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. base × height**

**GRADING NOTE** Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A parallelogram has base πr and height r. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q4/5

2769

QUESTION ID

**S75-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**$\pi r^2$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The slanted side is the height of a parallelogram.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q5/5

2770

QUESTION ID

## S75-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1

# BACKWARD 3

All circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Any circle can be mapped to any other circle by a translation and a dilation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

2772

QUESTION ID

**S75-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle of radius 2 is dilated to radius 6. By what factor is the circumference multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

2773

QUESTION ID

**S75-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

Circles of radii 3 and 12 have circumferences in the ratio

**A.** 1 : 16

**B.** 3 : 4

**C.** 1 : 4

**D.** 1 : 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

2774

**QUESTION ID**

**S75-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 1 : 4**

**GRADING NOTE** Scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

Why does C/d not depend on the size of the circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

2775

QUESTION ID

**S75-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**Both C and d are multiplied by the same scale factor.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

**True or false?**

A larger circle has a larger value of π.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

2776

QUESTION ID

**S75-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE π is constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSG-GMD.A.1

# FORWARD 1

Area of a circle as a limit of regular polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A regular n-gon is inscribed in a circle of radius r. Its area is ½ · perimeter · apothem. Explain why this approaches πr² as n increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

2778

**QUESTION ID**

## S75-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**The perimeter approaches 2πr and the apothem approaches r, so the area approaches ½ · 2πr · r = πr².**

**GRADING NOTE** REQUIRED WORK: Describe what the perimeter and apothem approach.

Any valid explanation. Essential: perimeter → circumference; apothem → radius; substitution.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S75-F1-Q1**

**High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the area of a regular hexagon with side 2 and apothem √3. Compare with the area of its circumscribed circle (radius 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

2779

**QUESTION ID**
**S75-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**$6\sqrt{3} \approx 10.4 < 4\pi \approx 12.6$**

**GRADING NOTE** Form required: both numbers. ½ · 12 · √3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

**True or false?**

An inscribed polygon always has less area than its circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

2780

QUESTION ID

**S75-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** It lies inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

As n grows, the apothem of an inscribed regular n-gon approaches

**A.** πr

**B.** r

**C.** 0

**D.** 2r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

2781

QUESTION ID
**S75-F1-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. r**

**GRADING NOTE** The sides approach the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Why do mathematicians call this an "informal" argument?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

2782

**GRADING NOTE** Any reasonable answer.

**QUESTION ID**

## S75-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**It relies on what the shapes approach (a limit) without a formal proof of the limit.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

# FORWARD 2

From area to volume: stacking disks • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A stack of 40 identical coins, each a disk of radius 1 cm and thickness 0.2 cm, forms a cylinder. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

2784

**QUESTION ID**
**S75-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**$8\pi$ cm³**

**GRADING NOTE** Form required: exact value. $\pi \cdot 1^2 \cdot 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

2784

**True or false?**

A cylinder can be thought of as a stack of congruent disks.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

2785

QUESTION ID
**S75-F2-Q2**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

The volume of a stack of disks of area $\pi r^2$ and total height h is

**A.** $\frac{1}{3}\pi r^2 h$

**B.** $2\pi rh$

**C.** $\pi r^2 h$

**D.** $\pi r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

2786

**QUESTION ID**

**S75-F2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $\pi r^2 h$**

**GRADING NOTE** Base area × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain how the area of a circle is used to find the volume of a cylinder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

2787

QUESTION ID

**S75-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**The volume is the base area $\pi r^2$ times the height h.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Each disk in a stack has area 12 cm² and the stack is 5 cm high. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

2788

**QUESTION ID**
**S75-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**60 cm³**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

2788

SET 76 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

# Informal argument for the volume of a cylinder

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.1** |
| **BACKWARD 1** | Volume of a rectangular prism as layers | **5.MD.C.5** |
| **BACKWARD 2** | Area of a circle | **7.G.B.4** |
| **BACKWARD 3** | Cross-sections parallel to the base | **7.G.A.3** |
| **FORWARD 1** | Cavalieri's principle with different shapes | **HSG-GMD.A.1** |
| **FORWARD 2** | Use the cylinder volume formula to solve problems | **HSG-GMD.A.3** |

SET 76 • HSG-GMD.A.1

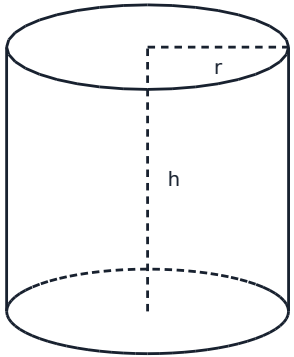
# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the volume of the cylinder shown is $V = \pi r^2 h$.

Work space




Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

2791

**QUESTION ID**

## S76-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**Each layer is a disk of area $\pi r^2$; stacking layers to height h gives volume (base area) × (height) = $\pi r^2 h$.**

**GRADING NOTE** REQUIRED WORK: Think of the cylinder as a stack of thin congruent disks.

Any valid explanation. Essential: congruent circular cross-sections of area $\pi r^2$; volume = base area × height, as for a prism (layers of unit thickness); result $\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right cylinder and an oblique cylinder have the same radius r and the same height h. Explain why they have the same volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

2792

**QUESTION ID**

## S76-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**At every height both cross-sections are disks of area $\pi r^2$, so by Cavalieri's principle their volumes are equal: $\pi r^2 h$.**

**GRADING NOTE**

REQUIRED WORK: Use Cavalieri's principle.

Any valid explanation. Essential: equal heights; equal cross-sectional areas at every level; Cavalieri's principle cited. Invalid: "they look the same size".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

A stack of 50 coins, each of radius 1 cm and thickness 0.2 cm, has volume

**A.** $10\pi$ cm³

**B.** $0.2\pi$ cm³

**C.** $50\pi$ cm³

**D.** $100\pi$ cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

2793

**QUESTION ID**
**S76-M-Q3**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. $10\pi$ cm³**

**GRADING NOTE** $\pi \cdot 1 \cdot 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why V = Bh works for both prisms and cylinders.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

2794

QUESTION ID

**S76-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**Both have congruent cross-sections of area B at every height, so the volume is B times the height.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

**True or false?**

Pushing a straight stack of coins into a slanted stack changes its volume.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

2795

QUESTION ID
**S76-M-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Same coins, same volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSG-GMD.A.1

# BACKWARD 1

Volume of a rectangular prism as layers • Grade 5 • Measurement & Data • 5.MD.C.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A prism has a 4 × 3 base and height 5. How many unit cubes are in one layer, and what is the volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

2797

**QUESTION ID**

**S76-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**12 per layer; 60**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

2797

**True or false?**

V = Bh, where B is the area of the base.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

2798

QUESTION ID

**S76-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A prism has base area 20 and height 7. Volume?

**A.** 27

**B.** 140

**C.** 200

**D.** 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

2799

**QUESTION ID**

**S76-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 140**

**GRADING NOTE** B × h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A prism has 6 layers of 10 cubes each. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

2800

QUESTION ID

**S76-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**60 cubic units**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

2800

**True or false?**

Volume is measured in square units.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

**Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5**

2801

QUESTION ID

## S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Cubic units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

**Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5**

SET 76 • HSG-GMD.A.1

# BACKWARD 2

Area of a circle • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of a circle of radius 3?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

2803

QUESTION ID

**S76-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**9π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of a circle of diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

2804

QUESTION ID

**S76-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**25π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A =

**A.** $2\pi r$

**B.** $\pi r$

**C.** $\pi r^2$

**D.** $\pi d$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

2805

QUESTION ID

**S76-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. $\pi r^2$**

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

**True or false?**

The base of a cylinder is a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

2806

QUESTION ID

**S76-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

**True or false?**

The area of a circle of radius 2 is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

2807

QUESTION ID

**S76-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1

# BACKWARD 3

Cross-sections parallel to the base • Grade 7 • Geometry • 7.G.A.3

**True or false?**

A slice of a right prism parallel to its base is congruent to the base.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

**Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5**

2809

QUESTION ID

## S76-B3-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Same shape and size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

**Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder is sliced parallel to its base. Describe the cross-section.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

2810

**QUESTION ID**
**S76-B3-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**A circle congruent to the base**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

A rectangular prism sliced parallel to its 3 × 4 base gives a

**A.** smaller rectangle

**B.** 3 × 4 rectangle

**C.** triangle

**D.** circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

2811

**QUESTION ID**

## S76-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 3 × 4 rectangle**

**GRADING NOTE** Congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

2811

**ANSWER KEY** Reduced copy of the question page, with its answer

Why do all horizontal slices of a cylinder have the same area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

2812

QUESTION ID

**S76-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**They are all congruent to the base.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

**True or false?**

Horizontal slices of a cylinder get smaller toward the top.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

2813

**QUESTION ID**

## S76-B3-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** That is a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSG-GMD.A.1

# FORWARD 1

Cavalieri's principle with different shapes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder with base area 50 cm² and a square prism with base area 50 cm² both have height 8 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

2815

**QUESTION ID**
**S76-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**Equal: 400 cm³ each**

**GRADING NOTE** Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An oblique cylinder has radius 3 and vertical height 7 (its slanted side is 9). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

2816

QUESTION ID

**S76-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**63π**

GRADING NOTE Form required: exact value. Use the vertical height, not the slant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

**True or false?**

Cavalieri's principle requires equal cross-sectional areas at every height and equal heights.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

2817

QUESTION ID

## S76-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

Two stacks of 30 identical cards, one straight and one slanted, have

**A.** different volumes

**B.** equal volumes

**C.** equal surface areas

**D.** no volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

2818

**QUESTION ID**

## S76-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. equal volumes

**GRADING NOTE** Same cross-sections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

State Cavalieri's principle in your own words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

2819

**GRADING NOTE** Any equivalent statement.

**QUESTION ID**

## S76-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Two solids with the same height and equal cross-sectional areas at every level have the same volume.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

# FORWARD 2

Use the cylinder volume formula to solve problems • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A can has radius 4 cm and height 12 cm. Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

2821

**QUESTION ID**
**S76-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**603 cm³**

**GRADING NOTE** Rounding to the nearest whole number required. 192π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pipe 2 m long has inner radius 0.1 m. Find the volume of water it holds, to the nearest thousandth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

2822

**QUESTION ID**
**S76-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**0.063 m³**

**GRADING NOTE** Rounding to 0.001 required. 0.02π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

**True or false?**

Doubling the height of a cylinder multiplies its volume by 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

2823

QUESTION ID

## S76-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Volume is linear in h, so it doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Doubling the radius of a cylinder multiplies its volume by

**A.** 4

**B.** 8

**C.** 6

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

2824

**QUESTION ID**

## S76-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 4**

**GRADING NOTE** $r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

2824

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylindrical tank has diameter 6 ft and height 10 ft. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

2825

**QUESTION ID**
**S76-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**$90\pi$ ft$^3$**

**GRADING NOTE** Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

# Informal arguments for the volumes of a pyramid and a cone

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.1** |
| **BACKWARD 1** | Volume of a prism | **7.G.B.6** |
| **BACKWARD 2** | Base area of a pyramid | **6.G.A.1** |
| **BACKWARD 3** | The cylinder volume argument | **HSG-GMD.A.1** |
| **FORWARD 1** | Cavalieri's principle for pyramids and cones | **HSG-GMD.A.1** |
| **FORWARD 2** | Use pyramid and cone volume formulas | **HSG-GMD.A.3** |

SET 77 • HSG-GMD.A.1

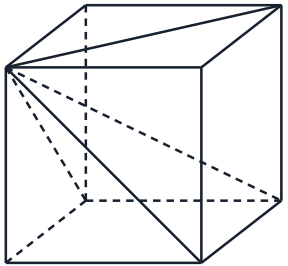
# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A cube is split into three congruent pyramids that share one vertex of the cube as their apex, as shown. Use this to explain why a pyramid's volume is ⅓Bh.

Work space



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

2828

QUESTION ID

## S77-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

**Each pyramid has the cube's base $B = s^2$ and height $h = s$, and the three fill the cube, so each has volume $\frac{1}{3}s^3 = \frac{1}{3}Bh$.**

**GRADING NOTE** REQUIRED WORK: Compare each pyramid's base and height with the cube's.

Any valid explanation. Essential: three congruent pyramids fill the cube; each has base = a face ($B = s^2$) and height = s; so $V = \frac{1}{3}s^3 = \frac{1}{3}Bh$. Extension to other pyramids by Cavalieri (cross-sections scale the same way) earns full credit but is not required.

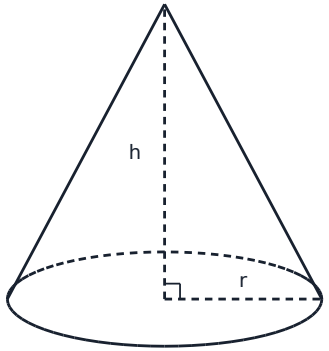
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why the volume of a cone is ⅓πr²h, using pyramids.

Work space




Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

2829

**QUESTION ID**

## S77-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**Pyramids with regular n-gon bases inscribed in the circle have V = ⅓Bh; as n grows, B approaches πr² and the pyramids approach the cone, so V = ⅓πr²h.**

**GRADING NOTE** REQUIRED WORK: Think of pyramids whose bases are regular polygons with more and more sides.

Any valid explanation. Essential: pyramid volume ⅓Bh; base polygons approach the circle (B → πr²); the pyramids approach the cone. Alternative: Cavalieri comparison with a pyramid of equal base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone and a cylinder have the same base and the same height. How many cones full of water fill the cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

2830

**QUESTION ID**
**S77-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

A cube with side 6 is split into three congruent pyramids that share one vertex of the cube as their apex. The volume of each pyramid is

**A.** 36

**B.** 216

**C.** 108

**D.** 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

2831

**QUESTION ID**

**S77-M-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 72**

**GRADING NOTE** 216 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

**True or false?**

A pyramid has half the volume of a prism with the same base and height.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

2832

QUESTION ID

**S77-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** One third.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSG-GMD.A.1

# BACKWARD 1

Volume of a prism • Grade 7 • Geometry • 7.G.B.6

**ANSWER KEY** Reduced copy of the question page, with its answer

A prism has base area 36 and height 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

2834

QUESTION ID

**S77-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**216**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cube has side 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

2835

**QUESTION ID**

**S77-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**216**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

V =

**A.** Bh

**B.** ⅓Bh

**C.** 2Bh

**D.** B + h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

2836

QUESTION ID

**S77-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. Bh**

**GRADING NOTE** Prism.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

**True or false?**

A cube is a prism.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

2837

QUESTION ID

## S77-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Square base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

**True or false?**

A prism's volume depends on the number of its faces.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

2838

**QUESTION ID**

## S77-B1-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

# BACKWARD 2

Base area of a pyramid • Grade 6 • Geometry • 6.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A square base has side 10. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

2840

QUESTION ID

**S77-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**100**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangular base has base 6 and height 4. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

2841

QUESTION ID

**S77-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**12**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The base of a cone is a

**A.** triangle

**B.** circle

**C.** sector

**D.** square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

2842

**QUESTION ID**

## S77-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. circle

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

B in V = ⅓Bh means the area of the base.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

2843

QUESTION ID

**S77-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

**True or false?**

B is the length of a base edge.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

2844

QUESTION ID

**S77-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1

# BACKWARD 3

The cylinder volume argument • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**True or false?**

A cylinder's volume is $\pi r^2 h$ because it is a stack of disks of area $\pi r^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q1/5

2846

QUESTION ID

**S77-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Set 76.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Cylinder radius 3, height 5: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q2/5

2847

**QUESTION ID**

**S77-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**45π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q2/5

A cone fits inside a cylinder with the same base and height. The cone's volume is

**A.** equal to the cylinder's

**B.** ⅔ of the cylinder's

**C.** ½ of the cylinder's

**D.** ⅓ of the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q3/5

2848

**QUESTION ID**
**S77-B3-Q3**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. ⅓ of the cylinder's**

**GRADING NOTE** Cone vs cylinder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder holds 60 cm³. A cone with the same base and height holds

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q4/5

2849

QUESTION ID
**S77-B3-Q4**

TYPE
Short answer

CORRECT ANSWER
**20 cm³**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S77-B3-Q4**

**High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q4/5**

**True or false?**

A cone is a cylinder with the top cut off flat.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q5/5

2850

QUESTION ID

## S77-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It comes to a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSG-GMD.A.1

# FORWARD 1

Cavalieri's principle for pyramids and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A square pyramid and a cone both have base area 50 cm² and height 9 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

2852

QUESTION ID

**S77-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Equal: 150 cm³ each**

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An oblique cone has radius 4 and vertical height 6. Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

2853

QUESTION ID

**S77-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**32π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

**True or false?**

Two pyramids with equal base areas and equal heights have equal volumes even if their apexes are in different positions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

2854

QUESTION ID

## S77-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Why do a pyramid and a cone with equal base areas and heights have equal cross-section areas at every level?

**A.** Their bases have the same shape

**B.** Both cross-sections are scaled copies of the base by the same factor

**C.** They have the same slant height

**D.** They have the same apex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

2855

**QUESTION ID**
**S77-F1-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. Both cross-sections are scaled copies of the base by the same factor**

**GRADING NOTE** Each cross-section at height y is the base scaled by (h − y)/h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cross-section of a pyramid halfway up has what fraction of the base area?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

2856

QUESTION ID
**S77-F1-Q5**

TYPE
Short answer

CORRECT ANSWER
**¼**

**GRADING NOTE** Any equivalent form. Scale factor ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

# FORWARD 2

Use pyramid and cone volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A square pyramid has base side 9 m and height 10 m. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

2858

**QUESTION ID**
**S77-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**270 m³**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has radius 3 in. and height 8 in. Find its volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

2859

**QUESTION ID**
**S77-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**75.4 in³**

**GRADING NOTE** Rounding to 0.1 required. 24π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

**True or false?**

A cone with the same base and height as a cylinder of volume 90 has volume 45.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

2860

QUESTION ID

**S77-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It has ⅓ of the volume: 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pyramid with base area 48 and height 5 has volume

**A.** 120

**B.** 53

**C.** 240

**D.** 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

2861

**QUESTION ID**

**S77-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 80**

**GRADING NOTE** $\frac{1}{3} \cdot 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone-shaped cup has diameter 6 cm and height 9 cm. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

2862

**QUESTION ID**
**S77-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**$27\pi$ cm$^3$**

**GRADING NOTE** Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

# Use volume formulas for cylinders, pyramids, and cones

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.3** |
| **BACKWARD 1** | Area of a circle | **7.G.B.4** |
| **BACKWARD 2** | Find a missing side with the Pythagorean Theorem | **8.G.B.7** |
| **BACKWARD 3** | Evaluate a formula and round | **7.EE.B.3** |
| **BACKWARD 4** | Which solid uses ⅓? | **HSG-GMD.A.1** |
| **FORWARD 1** | Composite and hollow solids | **HSG-GMD.A.3** |
| **FORWARD 2** | Mass from volume and density | **HSG-MG.A.2** |

SET 78 • HSG-GMD.A.3

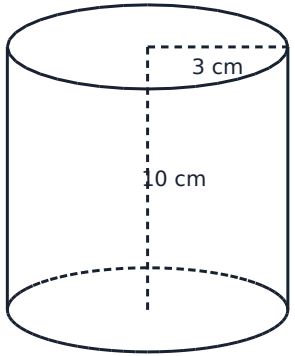
# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder has radius 3 cm and height 10 cm. Find its volume, exact and to the nearest tenth.

Volume: _____




Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

2865

QUESTION ID

**S78-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**$90\pi$ cm³ ≈ 282.7 cm³**

**GRADING NOTE** Form required: both the exact value and rounding to 0.1, with units.

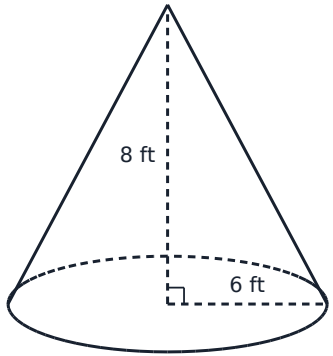
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has radius 6 ft and height 8 ft. Find its volume, exact and to the nearest tenth.

Volume: _____




Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

2866

**QUESTION ID**

## S78-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$96\pi$ ft³ ≈ 301.6 ft³**

**GRADING NOTE** Form required: both, with units. ⅓π · 36 · 8.

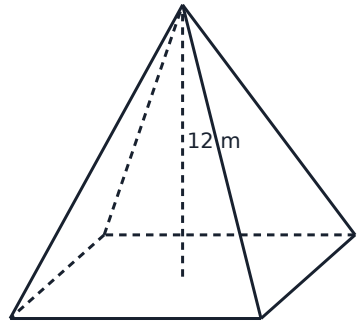
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A square pyramid has base side 10 m and height 12 m. Find its volume.

Volume: _____




Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

2867

**QUESTION ID**

## S78-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**400 m³**

**GRADING NOTE** Any equivalent form, with units. ⅓ · 100 · 12.

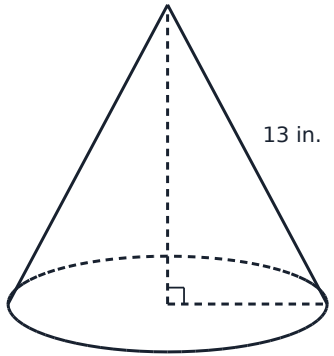
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has diameter 10 in. and slant height 13 in. Find its volume, exact and to the nearest tenth.

Volume: _____




Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

2868

**QUESTION ID**

## S78-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**$100\pi$ in$^3$ $\approx$ 314.2 in$^3$**

**GRADING NOTE** Form required: both, with units. r = 5; height $\sqrt{(169 - 25)} = 12$; $\frac{1}{3}\pi \cdot 25 \cdot 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder and a cone have the same radius and height. The cylinder holds 600 mL. The cone holds

**A.** 300 mL

**B.** 200 mL

**C.** 600 mL

**D.** 1,800 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

2869

**QUESTION ID**

## S78-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 200 mL**

**GRADING NOTE** ⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

SET 78 • HSG-GMD.A.3

# BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of a circle with radius 6?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

2871

QUESTION ID

**S78-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**36π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of a circle with diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

2872

QUESTION ID

**S78-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**25π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

A base of radius 3 has area

**A.** 9

**B.** 9π

**C.** 3π

**D.** 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

2873

**QUESTION ID**

## S78-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 9π**

**GRADING NOTE** $\pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

**True or false?**

Squaring the radius comes before multiplying by π.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

2874

QUESTION ID

**S78-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Order of operations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\pi r^2$ with $r = 3$ is $(3\pi)^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

2875

**QUESTION ID**

## S78-B1-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** $9\pi$, not $9\pi^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

# BACKWARD 2

Find a missing side with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has hypotenuse 13 and one leg 5. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

2877

QUESTION ID

**S78-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has radius 6 and slant height 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

2878

QUESTION ID

**S78-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**8**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Radius, height, and slant height of a cone form a

**A.** square

**B.** right triangle

**C.** equilateral triangle

**D.** circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

2879

QUESTION ID

**S78-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. right triangle**

**GRADING NOTE** Height ⊥ base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The slant height is longer than the height.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

2880

QUESTION ID

**S78-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

**True or false?**

The slant height can be used as the height in $V = \frac{1}{3}\pi r^2 h$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

2881

QUESTION ID

**S78-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Use the perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3

# BACKWARD 3

Evaluate a formula and round • Grade 7 • Expressions & Equations • 7.EE.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Evaluate ⅓ · 36 · 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

2883

QUESTION ID

**S78-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**96**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Round 90π to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

2884

**QUESTION ID**

**S78-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**282.7**

**GRADING NOTE** Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$96\pi \approx$

**A.** 301.6

**B.** 301.5

**C.** 3,016

**D.** 96.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

2885

QUESTION ID

**S78-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 301.6**

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

**True or false?**

Using 3.14 for π may change the last digit of a rounded answer.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

2886

**QUESTION ID**

## S78-B3-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\frac{1}{3} \cdot 100 \cdot 12 = 1{,}200$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

2887

QUESTION ID

**S78-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSG-GMD.A.3

# BACKWARD 4

Which solid uses ⅓? • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Pyramids and cones use $V = \frac{1}{3}Bh$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

2889

QUESTION ID

**S78-B4-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Pointed solids.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which has the formula V = Bh: a cone or a cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

2890

QUESTION ID

**S78-B4-Q2**

TYPE

Short answer

CORRECT ANSWER

**Cylinder**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$V = \frac{1}{3}\pi r^2 h$ is the volume of a

**A.** cylinder

**B.** cone

**C.** prism

**D.** sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

2891

**QUESTION ID**

**S78-B4-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. cone**

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Why does a cone have ⅓ of the matching cylinder's volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

2892

**QUESTION ID**

## S78-B4-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**Three cones fill the cylinder (the informal argument).**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A pyramid uses V = Bh.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

2893

QUESTION ID

## S78-B4-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

SET 78 • HSG-GMD.A.3

# FORWARD 1

Composite and hollow solids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A silo is a cylinder (radius 4 m, height 10 m) topped by a cone (radius 4 m, height 3 m). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

2895

**QUESTION ID**
**S78-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**$176\pi$ m$^3$**

**GRADING NOTE** Form required: exact value. $160\pi + 16\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pipe is a cylinder of length 20 cm with outer radius 5 cm and inner radius 4 cm. Find the volume of the material, exact.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

2896

**QUESTION ID**
**S78-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**$180\pi$ cm$^3$**

**GRADING NOTE** Form required: exact value. $\pi(25 - 16)(20)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

**True or false?**

The volume of a composite solid is the sum of the volumes of its non-overlapping parts.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

2897

QUESTION ID

## S78-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Additivity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder (V = 300) has a cone with the same base and height carved out. Remaining volume?

**A.** 300

**B.** 200

**C.** 0

**D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

2898

QUESTION ID

**S78-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 200**

GRADING NOTE 300 − 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pencil tip is a cone of radius 0.4 cm and height 1.5 cm on a cylinder of radius 0.4 cm and length 15 cm. Find the total volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

2899

**QUESTION ID**
**S78-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**7.8 cm³**

**GRADING NOTE** Rounding to 0.1 required. 2.4π + 0.08π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

# FORWARD 2

Mass from volume and density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A steel cylinder has radius 2 cm and height 5 cm. Steel has density 7.85 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

2901

**QUESTION ID**
**S78-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**493 g**

**GRADING NOTE** Rounding to the nearest gram required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone of ice cream (radius 3 cm, height 10 cm) has density 0.55 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

2902

**QUESTION ID**
**S78-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**52 g**

**GRADING NOTE** Rounding to the nearest gram required. 30π cm³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Mass = volume ÷ density.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

2903

QUESTION ID

**S78-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

A pyramid of volume 400 cm³ is made of material with density 2 g/cm³. Mass?

**A.** 200 g

**B.** 402 g

**C.** 1,600 g

**D.** 800 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

2904

**QUESTION ID**

## S78-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 800 g**

**GRADING NOTE** 2 × 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A gold cube has volume 8 cm³ and mass 154.4 g. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

2905

**QUESTION ID**
**S78-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**19.3 g/cm³**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

# Use the volume formula for spheres, including composite solids

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.3** |
| **BACKWARD 1** | Cubes of numbers | **6.EE.A.1** |
| **BACKWARD 2** | Radius from diameter | **7.G.B.4** |
| **BACKWARD 3** | Multiply fractions by numbers | **7.NS.A.2** |
| **BACKWARD 4** | Cylinder and cone volumes | **HSG-GMD.A.3** |
| **FORWARD 1** | Scaling spheres | **HSG-GMD.A.3** |
| **FORWARD 2** | Model real objects as spheres | **HSG-MG.A.1** |

SET 79 • HSG-GMD.A.3

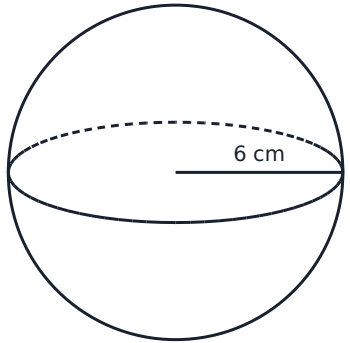
# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the volume of a sphere with radius 6 cm, exact and to the nearest tenth.

Volume: _____




Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

2908

**QUESTION ID**
**S79-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**$288\pi$ cm³ ≈ 904.8 cm³**

**GRADING NOTE** Form required: both, with units. 4/3 π · 216.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the volume of a ball with diameter 10 in., exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

2909

**QUESTION ID**

## S79-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$500\pi/3$ in$^3$ ≈ 523.6 in$^3$**

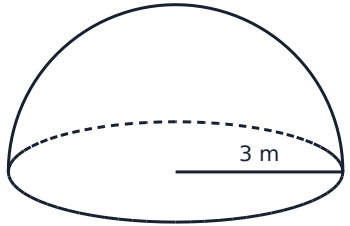
**GRADING NOTE** Form required: both, with units. r = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the volume of a hemisphere with radius 3 m, exact and to the nearest tenth.




Volume: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

2910

**QUESTION ID**
**S79-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**$18\pi$ m³ ≈ 56.5 m³**

**GRADING NOTE** Form required: both, with units. ½ · 4/3 π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A capsule is a cylinder of radius 3 cm and length 10 cm with a hemisphere on each end. Find its volume, exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

2911

**QUESTION ID**

## S79-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**$126\pi$ cm³ ≈ 395.8 cm³**

**GRADING NOTE** Form required: both, with units. 90π + 36π (two hemispheres make one sphere).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An ice cream cone (radius 2, height 9) is topped by a hemisphere of radius 2. The total exact volume is

**A.** 52π/3

**B.** 28π

**C.** 12π

**D.** 44π/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

2912

**QUESTION ID**

**S79-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 52π/3**

**GRADING NOTE** 12π + 16π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

SET 79 • HSG-GMD.A.3

# BACKWARD 1

Cubes of numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

$6^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

2914

QUESTION ID

**S79-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**216**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$5^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

2915

**QUESTION ID**

**S79-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**125**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$3^3 =$

**A.** 9

**B.** 81

**C.** 6

**D.** 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

2916

QUESTION ID

**S79-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 27**

**GRADING NOTE** $3 \cdot 3 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$2^3 = 8$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

2917

QUESTION ID

**S79-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Cube.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

**True or false?**

$4^3 = 12$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

2918

QUESTION ID

**S79-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

# BACKWARD 2

Radius from diameter • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A ball has diameter 10. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

2920

QUESTION ID

**S79-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

**True or false?**

Using the diameter in $4/3\ \pi r^3$ makes the answer 8 times too large.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

2921

QUESTION ID

**S79-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE $2^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere with diameter 12 has $r^3$ =

**A.** 216

**B.** 144

**C.** 36

**D.** 1,728

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

2922

**QUESTION ID**

**S79-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 216**

**GRADING NOTE** $6^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

2922

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere has radius 2.5. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

2923

QUESTION ID

**S79-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The radius of a sphere is twice its diameter.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

2924

QUESTION ID

**S79-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3

# BACKWARD 3

Multiply fractions by numbers • Grade 7 • The Number System • 7.NS.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

$4/3 \cdot 216 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

2926

QUESTION ID

**S79-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**288**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

½ · 4/3 · 27 =

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

2927

**QUESTION ID**

**S79-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**18**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$4/3 \cdot 125 =$

**A.** 375/4

**B.** 500/3

**C.** 125/3

**D.** 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

2928

QUESTION ID

**S79-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 500/3**

GRADING NOTE Numerators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$4/3 \cdot 27 = 36$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

2929

**GRADING NOTE** $27 \div 3 \times 4$.

QUESTION ID

## S79-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

**True or false?**

$4/3 \cdot 6 = 24$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

**Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5**

2930

QUESTION ID

**S79-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

**Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5**

SET 79 • HSG-GMD.A.3

# BACKWARD 4

Cylinder and cone volumes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Cylinder radius 3, length 10: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

2932

**QUESTION ID**

**S79-B4-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**90π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Cone radius 2, height 9: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

2933

**QUESTION ID**

**S79-B4-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**12π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder and a cone with radius 3 and height 4 have volumes

**A.** 36π and 12π

**B.** 12π and 36π

**C.** 12π and 4π

**D.** 36π and 36π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

2934

**QUESTION ID**
**S79-B4-Q3**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. 36π and 12π**

**GRADING NOTE** ⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

**True or false?**

Two hemispheres of radius r make one sphere of radius r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

2935

**QUESTION ID**

## S79-B4-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

**True or false?**

A composite volume is the product of the parts.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

2936

QUESTION ID

**S79-B4-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

SET 79 • HSG-GMD.A.3

# FORWARD 1

Scaling spheres • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

The radius of a sphere is doubled. By what factor is the volume multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

2938

QUESTION ID

**S79-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**8**

**GRADING NOTE** Any equivalent form. $2^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere fits exactly inside a cube with side 6. Find the exact volume of the space in the cube outside the sphere.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

2939

**QUESTION ID**
**S79-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**216 − 36π**

**GRADING NOTE** Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

**True or false?**

A sphere of radius 3 has 9 times the volume of a sphere of radius 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

2940

QUESTION ID

## S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Volume scales by $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

A balloon's radius grows from 10 cm to 15 cm. Its volume is multiplied by

**A.** 1.5

**B.** 2.25

**C.** 5

**D.** 3.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

2941

**QUESTION ID**

## S79-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 3.375**

**GRADING NOTE** $1.5^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r. What fraction of the cylinder's volume is the sphere's?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

2942

QUESTION ID

**S79-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

**GRADING NOTE** Any equivalent form. $(4/3\ \pi r^3)/(2\pi r^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

# FORWARD 2

Model real objects as spheres • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Model Earth as a sphere with radius 6,371 km. Estimate its volume in cubic kilometers, in scientific notation to 3 significant figures.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

2944

**QUESTION ID**

## S79-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$1.08 \times 10^{12}$ km$^3$**

**GRADING NOTE** Rounding to 3 significant figures required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

A basketball has circumference 75 cm. Model it as a sphere and find its volume to the nearest 100 $cm^3$.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

2945

**QUESTION ID**

**S79-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**7100 $cm^3$**

**GRADING NOTE** Rounding to the nearest 100 required. r = 75/(2π) ≈ 11.94 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

**True or false?**

Modeling an orange as a sphere ignores its small bumps.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

2946

QUESTION ID

**S79-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Modeling assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A scoop of ice cream is modeled as a sphere of radius 3 cm. Its volume is about

**A.** 28 cm³

**B.** 113 cm³

**C.** 339 cm³

**D.** 36 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

2947

**QUESTION ID**
**S79-F2-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. 113 cm³**

**GRADING NOTE** 36π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Give one reason a sphere is a good model for a ball bearing and one limitation of the model for a pumpkin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

2948

**QUESTION ID**
**S79-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**A ball bearing is nearly perfectly round; a pumpkin has ridges, a flattened shape, and a stem, so the model is only approximate.**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

# Solve for an unknown dimension from a volume

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.3** |
| **BACKWARD 1** | Solve a one-step equation with a coefficient | **HSA-REI.B.3** |
| **BACKWARD 2** | Square roots and cube roots | **8.EE.A.2** |
| **BACKWARD 3** | Rearrange a formula | **HSA-CED.A.4** |
| **FORWARD 1** | Find dimensions to meet a target volume | **HSG-GMD.A.3** |
| **FORWARD 2** | Design a container that meets constraints | **HSG-MG.A.3** |

SET 80 • HSG-GMD.A.3

# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder has volume $360\pi$ cm³ and radius 6 cm. Find its height.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

2951

**QUESTION ID**
**S80-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**10 cm**

**GRADING NOTE** Any equivalent form, with units. $36\pi h = 360\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has volume $96\pi$ in$^3$ and height 8 in. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

2952

**QUESTION ID**

**S80-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**6 in.**

**GRADING NOTE** Any equivalent form, with units. $\frac{1}{3}\pi r^2 \cdot 8 = 96\pi \rightarrow r^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere has volume $36\pi$ m³. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

2953

**QUESTION ID**
**S80-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**3 m**

**GRADING NOTE** Any equivalent form, with units. $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

2953

**ANSWER KEY** Reduced copy of the question page, with its answer

A square pyramid has volume 256 ft³ and height 12 ft. Find the side of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

2954

QUESTION ID
**S80-M-Q4**

TYPE
Short answer

CORRECT ANSWER
**8 ft**

GRADING NOTE Any equivalent form, with units. $\frac{1}{3}s^2 \cdot 12 = 256 \rightarrow s^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder holds 500 cm³ and has height 10 cm. Its radius is closest to

**A.** 7.1 cm

**B.** 4.0 cm

**C.** 2.0 cm

**D.** 15.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

2955

**QUESTION ID**

## S80-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 4.0 cm**

**GRADING NOTE** $r = \sqrt{(50/\pi)} \approx 3.99$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

SET 80 • HSG-GMD.A.3

# BACKWARD 1

Solve a one-step equation with a coefficient • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $36\pi h = 360\pi$.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

2957

QUESTION ID

**S80-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**10**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S80-B1-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve ⅓x = 12.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

2958

QUESTION ID

**S80-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**36**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S80-B1-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 4x = 108.

**A.** 25

**B.** 104

**C.** 432

**D.** 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

2959

QUESTION ID

**S80-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 27**

**GRADING NOTE** Divide by 4.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S80-B1-Q3**

**High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5**

**True or false?**

To solve $36\pi h = 360\pi$, dividing both sides by $\pi$ is allowed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

2960

**QUESTION ID**

## S80-B1-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** $\pi \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

To solve ⅓x = 12, divide by 3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

2961

QUESTION ID
**S80-B1-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Multiply by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

# BACKWARD 2

Square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $r^2 = 36$ for $r > 0$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

2963

QUESTION ID

**S80-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $r^3 = 27$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

2964

QUESTION ID

**S80-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$\sqrt[3]{125} =$

**A.** 25

**B.** 5

**C.** 41.7

**D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

2965

QUESTION ID

**S80-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 5**

GRADING NOTE $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

**True or false?**

$\sqrt{64} = 8$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5**

2966

QUESTION ID

## S80-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

If $r^3 = 8$, then $r = 4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

2967

QUESTION ID

**S80-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3

# BACKWARD 3

Rearrange a formula • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $V = \pi r^2 h$ for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

2969

QUESTION ID

**S80-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**$V/(\pi r^2)$**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S80-B3-Q1**

**High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve V = ⅓Bh for B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

2970

QUESTION ID

**S80-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**3V/h**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S80-B3-Q2**

**High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

V = 4/3 πr³ solved for r³ is

**A.** 4V/(3π)

**B.** V/(4π)

**C.** 3V/(4π)

**D.** 3πV/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

2971

**QUESTION ID**

**S80-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 3V/(4π)**

**GRADING NOTE** Multiply by 3/(4π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Solving $V = \pi r^2 h$ for r needs a square root.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

2972

QUESTION ID

**S80-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE $r = \sqrt{(V/(\pi h))}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

**True or false?**

Solving V = Bh for h gives h = VB.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

2973

QUESTION ID
**S80-B3-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** V/B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSG-GMD.A.3

# FORWARD 1

Find dimensions to meet a target volume • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A drink can must hold 355 cm³ and have radius 3.3 cm. Find the height to the nearest tenth of a centimeter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

2975

**QUESTION ID**
**S80-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**10.4 cm**

**GRADING NOTE** Rounding to 0.1 required. $h = 355/(\pi \cdot 3.3^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A spherical tank must hold 1,000 m³. Find its radius to the nearest tenth of a meter.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

2976

**QUESTION ID**
**S80-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**6.2 m**

**GRADING NOTE** Rounding to 0.1 required. $r = \sqrt[3]{(750/\pi)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

**True or false?**

To double the volume of a cylinder while keeping its radius, double its height.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

2977

QUESTION ID

## S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Linear in h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has the same volume as a cylinder of equal radius. The cone's height is

**A.** 9 times the cylinder's

**B.** ⅓ of the cylinder's

**C.** 3 times the cylinder's

**D.** equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

2978

**QUESTION ID**

**S80-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 3 times the cylinder's**

**GRADING NOTE** $\frac{1}{3}\pi r^2 h_1 = \pi r^2 h_2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cube has the same volume as a sphere of radius 3. Find the cube's edge to the nearest tenth.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

2979

**QUESTION ID**

## S80-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**4.8**

**GRADING NOTE** Rounding to 0.1 required. $\sqrt[3]{(36\pi)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

# FORWARD 2

Design a container that meets constraints • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylindrical vase must hold 2,000 cm³ and be 20 cm tall. Its radius must be at least 5 cm to fit a bouquet. Does the design work? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

2981

QUESTION ID

**S80-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Yes: r = √(2000/(20π)) ≈ 5.6 cm, which is at least 5 cm.**

**GRADING NOTE** Form required: the conclusion with the computed radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A conical paper cup must hold 150 cm³ and have radius 3.5 cm. Find the height to the nearest tenth.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

2982

**QUESTION ID**
**S80-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**11.7 cm**

**GRADING NOTE** Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

**True or false?**

A container's volume requirement fixes each of its dimensions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

2983

QUESTION ID

**S80-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It fixes only a relationship: many (r, h) pairs give the same volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

A cylinder must hold $100\pi$ cm$^3$. Which (r, h) works?

**A.** (5, 4)

**B.** (5, 20)

**C.** (4, 5)

**D.** (10, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

2984

QUESTION ID

**S80-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. (5, 4)**

GRADING NOTE $25 \cdot 4 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Give two different cylinders (radius and height) that each hold $64\pi$ cm$^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

2985

**QUESTION ID**
**S80-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**For example r = 4, h = 4 and r = 2, h = 16.**

**GRADING NOTE** Any two (r, h) pairs with $r^2h = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

# Identify cross-sections of three-dimensional objects

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.B.4** |
| **BACKWARD 1** | Cross-sections of right rectangular prisms and pyramids | **7.G.A.3** |
| **BACKWARD 2** | Properties of polygons | **5.G.B.4** |
| **BACKWARD 3** | Parts of solids | **6.G.A.4** |
| **FORWARD 1** | Areas of cross-sections | **HSG-GMD.B.4** |
| **FORWARD 2** | Use cross-section areas to compare volumes (Cavalieri) | **HSG-GMD.A.2** |

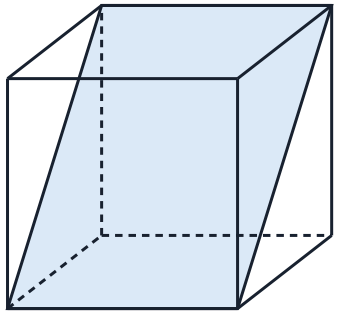
SET 81 • HSG-GMD.B.4

# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane cuts the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

2988

QUESTION ID

## S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

**A rectangle (not a square)**

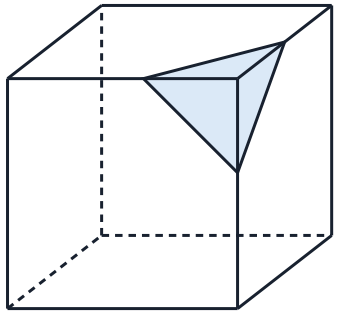
GRADING NOTE Any equivalent answer. Two sides are cube edges (s) and two are face diagonals (s√2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane cuts off one corner of the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

2989

QUESTION ID

**S81-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**A triangle**

**GRADING NOTE** Any equivalent answer. (Equilateral if the three cut points are equally far from the corner.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

A right circular cone is cut by a vertical plane through its apex. The cross-section is

**A.** a circle

**B.** a rectangle

**C.** a triangle (isosceles)

**D.** an ellipse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

2990

**QUESTION ID**

## S81-M-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. a triangle (isosceles)**

**GRADING NOTE** Two slant edges and a diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name the cross-section of a cylinder cut by a plane (a) parallel to its base, (b) perpendicular to its base through its axis, and (c) slanted, cutting the curved side only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

2991

**QUESTION ID**

## S81-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**(a) circle (b) rectangle (c) ellipse**

**GRADING NOTE** Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

**True or false?**

Every plane section of a sphere that cuts through it is a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

2992

QUESTION ID

**S81-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Points of the sphere in the plane are equidistant from the foot of the perpendicular from the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

SET 81 • HSG-GMD.B.4

# BACKWARD 1

Cross-sections of right rectangular prisms and pyramids • Grade 7 • Geometry • 7.G.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangular prism is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

2994

QUESTION ID

**S81-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**A rectangle congruent to the base**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A square pyramid is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

2995

QUESTION ID

**S81-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**A smaller square**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

A square pyramid cut vertically through its apex and two opposite base edges' midpoints gives a

**A.** trapezoid

**B.** square

**C.** circle

**D.** triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

2996

**QUESTION ID**

## S81-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. triangle

**GRADING NOTE** Through the apex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

**True or false?**

A vertical cut of a rectangular prism parallel to a side face gives a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

2997

QUESTION ID

## S81-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Congruent to the face.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A cut of a cube parallel to a face can be a triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

2998

QUESTION ID

**S81-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

# BACKWARD 2

Properties of polygons • Grade 5 • Geometry • 5.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A quadrilateral with four right angles and unequal adjacent sides is a

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

3000

**QUESTION ID**

## S81-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**rectangle**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

3000

**True or false?**

Every square is a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

3001

QUESTION ID

**S81-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Hierarchy.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A polygon with six sides is a

**A.** hexagon

**B.** pentagon

**C.** octagon

**D.** heptagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

3002

**QUESTION ID**

## S81-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. hexagon

**GRADING NOTE** Name.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle with two equal sides is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

3003

QUESTION ID

**S81-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**isosceles**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A trapezoid has no parallel sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

3004

QUESTION ID

**S81-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** At least one pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4

# BACKWARD 3

Parts of solids • Grade 6 • Geometry • 6.G.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

3006

QUESTION ID

**S81-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the shape of the base of a cone?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

3007

QUESTION ID

**S81-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**A circle**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How many edges meet at each vertex of a cube?

**A.** 6

**B.** 2

**C.** 4

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

3008

**QUESTION ID**

**S81-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 3**

**GRADING NOTE** Corner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

3008

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A cylinder has two congruent circular bases.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

3009

QUESTION ID

**S81-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A sphere has one edge.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

3010

QUESTION ID

**S81-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** No edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSG-GMD.B.4

# FORWARD 1

Areas of cross-sections • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A cube has edge 4. Find the area of the diagonal cross-section through two opposite edges.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

3012

**QUESTION ID**
**S81-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**16√2**

**GRADING NOTE** Form required: exact value. 4 × 4√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

3012

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone has radius 6 and height 12. Find the area of the cross-section parallel to the base at height 4 (measured up from the base).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

3013

**QUESTION ID**
**S81-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**16π**

**GRADING NOTE** Form required: exact value. Radius at that level = 6 · (12 − 4)/12 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

**True or false?**

A sphere of radius 5 cut 3 units from its center gives a circle of radius 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

3014

QUESTION ID

## S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** 3-4-5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

A cube can have a cross-section that is a regular

**A.** octagon

**B.** pentagon

**C.** heptagon

**D.** hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

3015

QUESTION ID

**S81-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. hexagon**

**GRADING NOTE** Plane through the midpoints of six edges.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A square pyramid has base side 10 and height 15. Find the area of the cross-section parallel to the base 5 units below the apex.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

3016

**QUESTION ID**
**S81-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**100/9 (about 11.1)**

**GRADING NOTE** Form required: exact value or rounding to 0.1. Side 10 · 5/15 = 10/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

# FORWARD 2

Use cross-section areas to compare volumes (Cavalieri) • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Two solids have height 10 and every horizontal cross-section of each has area 7. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

3018

QUESTION ID

**S81-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Equal: 70 each**

**GRADING NOTE** Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A hemisphere has radius 5. Find the area of its cross-section 3 units above the base.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

3019

QUESTION ID

**S81-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**16π**

GRADING NOTE Form required: exact value. Radius 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

**True or false?**

Cavalieri's principle can compare two solids of different heights.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

3020

QUESTION ID

## S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The solids must have equal heights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

At height y, a hemisphere of radius r has cross-section area

**A.** $\pi(r^2 - y^2)$

**B.** $\pi(r - y)^2$

**C.** $\pi r^2$

**D.** $\pi y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

3021

QUESTION ID

**S81-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. $\pi(r^2 - y^2)$**

**GRADING NOTE** Pythagorean Theorem.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S81-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why cross-section areas are useful for finding volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

3022

**QUESTION ID**

## S81-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Volume can be built from thin slices; equal slice areas at each level give equal volumes.**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

# Identify three-dimensional objects generated by rotations of two-dimensional objects

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.B.4** |
| **BACKWARD 1** | Rotation of a point about a center | **HSG-CO.A.4** |
| **BACKWARD 2** | Cross-sections perpendicular to the axis | **HSG-GMD.B.4** |
| **BACKWARD 3** | Names of solids | **6.G.A.4** |
| **FORWARD 1** | Volumes of solids of rotation | **HSG-GMD.B.4** |
| **FORWARD 2** | Design an object as a solid of rotation | **HSG-MG.A.3** |

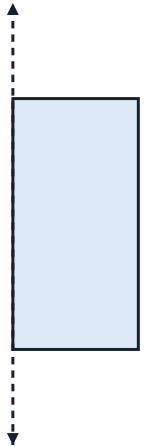
SET 82 • HSG-GMD.B.4

# MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

3025

**QUESTION ID**

## S82-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**A cylinder**

**GRADING NOTE** Any equivalent answer.

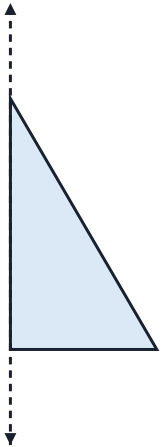
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The right triangle is rotated 360° about the dashed line. Name the solid formed.

Solid: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

3026

**QUESTION ID**

## S82-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

## A cone

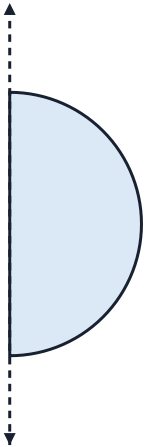
**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The semicircle is rotated 360° about the dashed line. Name the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

3027

QUESTION ID

**S82-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**A sphere**

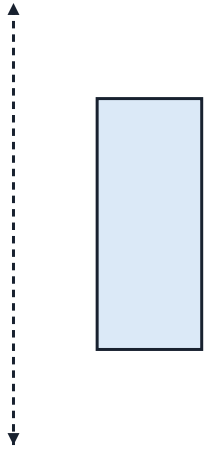
**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The rectangle is rotated 360° about the dashed line, which does not touch it. Describe the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

3028

**QUESTION ID**

## S82-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**A hollow cylinder (a cylinder with a cylindrical hole through the middle, like a pipe)**

**GRADING NOTE** Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

3028

A right triangle is rotated about its hypotenuse. The solid formed is

**A.** a sphere

**B.** one cone

**C.** two cones joined at their bases

**D.** a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

3029

**QUESTION ID**

## S82-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. two cones joined at their bases**

**GRADING NOTE** Each leg sweeps a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

SET 82 • HSG-GMD.B.4

# BACKWARD 1

Rotation of a point about a center • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

**True or false?**

Rotating a point 360° about a center traces a circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

3031

QUESTION ID

**S82-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Equal distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A point 3 cm from an axis is rotated about the axis. What does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

3032

QUESTION ID

**S82-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**A circle of radius 3 cm**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A point ON the axis of rotation

**A.** traces a line

**B.** stays fixed

**C.** traces a circle

**D.** disappears

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

3033

QUESTION ID

**S82-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. stays fixed**

GRADING NOTE Distance 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A segment parallel to the axis, 2 units away, is rotated about it. What surface does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

3034

QUESTION ID

**S82-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**The curved side of a cylinder**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

**True or false?**

Points farther from the axis trace smaller circles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

3035

QUESTION ID

**S82-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Larger circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

# BACKWARD 2

Cross-sections perpendicular to the axis • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

**True or false?**

A solid of rotation cut perpendicular to its axis has a circular cross-section (or a ring).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

3037

QUESTION ID

## S82-B2-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Every point traces a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cone is cut perpendicular to its axis. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

3038

QUESTION ID

**S82-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**A circle**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pipe cut perpendicular to its axis gives a

**A.** rectangle

**B.** ring (annulus)

**C.** circle

**D.** triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

3039

QUESTION ID

**S82-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. ring (annulus)**

**GRADING NOTE** Hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere is cut through its center. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

3040

QUESTION ID

**S82-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**A circle (a great circle)**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

**True or false?**

A cylinder cut perpendicular to its axis gives a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

3041

**QUESTION ID**

## S82-B2-Q5

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** A circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4

# BACKWARD 3

Names of solids • Grade 6 • Geometry • 6.G.A.4

Name the solid with two parallel circular bases and a curved side.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

3043

**QUESTION ID**

## S82-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

## Cylinder

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name the solid with one circular base and an apex.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

3044

QUESTION ID

**S82-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**Cone**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

All points equidistant from a center in space form a

**A.** sphere

**B.** cone

**C.** circle

**D.** cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

3045

**QUESTION ID**

**S82-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. sphere**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

**True or false?**

A cone has one flat face.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

3046

QUESTION ID

**S82-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Its base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

**True or false?**

A sphere has a flat base.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

**Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5**

3047

QUESTION ID

## S82-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

**Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5**

SET 82 • HSG-GMD.B.4

# FORWARD 1

Volumes of solids of rotation • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 5-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

3049

**QUESTION ID**
**S82-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**45π**

**GRADING NOTE** Form required: exact value. r = 3, h = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 3-by-5 rectangle is rotated about its 3-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

3050

**QUESTION ID**
**S82-F1-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**75π**

**GRADING NOTE** Form required: exact value. r = 5, h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

**True or false?**

Rotating one rectangle about two different sides can give solids with different volumes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

3051

QUESTION ID

## S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** A 3-by-5 rectangle gives 45π about its 5-unit side and 75π about its 3-unit side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle with legs 3 (on the axis) and 4 is rotated about the leg of length 3. Volume?

**A.** 48π

**B.** 9π

**C.** 12π

**D.** 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

3052

QUESTION ID

**S82-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 16π**

**GRADING NOTE** r = 4, h = 3: ⅓π · 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A semicircle of radius 3 is rotated about its diameter. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

3053

**QUESTION ID**
**S82-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**36π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

# FORWARD 2

Design an object as a solid of rotation • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A lathe turns a 2 cm by 10 cm rectangle about its 10 cm side to make a wooden dowel. Find the exact volume of wood.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

3055

**QUESTION ID**

## S82-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$40\pi$ cm$^3$**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A designer wants a cone-shaped lampshade by rotating a right triangle about a 30 cm leg. What other leg length gives a base of diameter 40 cm?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

3056

**QUESTION ID**

## S82-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**20 cm**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

**True or false?**

A hemispherical bowl can be modeled by rotating a right triangle about one of its legs.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

3057

QUESTION ID

## S82-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** That gives a cone; rotate a quarter-circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

To make a ring-shaped washer by rotation, rotate a rectangle about a line that

**A.** is a diagonal

**B.** contains one side

**C.** does not touch the rectangle

**D.** passes through its center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

3058

**QUESTION ID**

## S82-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. does not touch the rectangle**

**GRADING NOTE** Gap gives the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A vase is made by rotating a rectangle 6 cm wide and 20 cm tall about a line 2 cm from its long side (outside the rectangle). Find the exact volume of material.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

3059

**QUESTION ID**

## S82-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**1,200π cm³**

**GRADING NOTE** Form required: exact value. Outer radius 8, inner radius 2: π(64 − 4)(20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSG-MG.A.1 • MODELING WITH GEOMETRY

# Model objects with geometric shapes

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-MG.A.1** |
| **BACKWARD 1** | Volume formulas | **HSG-GMD.A.3** |
| **BACKWARD 2** | Radius from a circumference | **7.G.B.4** |
| **BACKWARD 3** | Convert cubic units | **6.RP.A.3.d** |
| **FORWARD 1** | Judge a model against a measurement | **HSG-MG.A.1** |
| **FORWARD 2** | Use the model to estimate mass with density | **HSG-MG.A.2** |

SET 83 • HSG-MG.A.1

# MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A tree trunk is about 6 m tall and has circumference 2.2 m. Model it as a cylinder and estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

3062

**QUESTION ID**

## S83-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**About 2.3 m³**

**GRADING NOTE** REQUIRED WORK: Find the radius from the circumference, then use $V = \pi r^2 h$.

Rounding to 0.1 required. $r = 2.2/(2\pi) \approx 0.35$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pile of sand is shaped like a cone with base diameter 8 m and height 3 m. Estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

3063

**QUESTION ID**

## S83-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**50.3 $m^3$**

**GRADING NOTE** Rounding to 0.1 required. 16π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which shape is the best model for a soup can?

**A.** Cone

**B.** Cylinder

**C.** Square pyramid

**D.** Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

3064

**QUESTION ID**

**S83-M-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. Cylinder**

**GRADING NOTE** Two circular bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pool is modeled as a rectangular prism 25 m long, 10 m wide, and 1.8 m deep. How many liters of water does it hold? (1 $m^3$ = 1,000 L)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

3065

**QUESTION ID**

**S83-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**450,000 L**

**GRADING NOTE** Any equivalent form. 450 $m^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pine tree is modeled as a cone. Name one measurement you would take and one way the model differs from the real tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

3066

**QUESTION ID**
**S83-M-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**For example: height and the width at the bottom of the branches; the real tree has gaps, a trunk, and an uneven outline.**

**GRADING NOTE** Any reasonable measurement and limitation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

SET 83 • HSG-MG.A.1

# BACKWARD 1

Volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 2 and h = 5?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

3068

QUESTION ID

**S83-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**20π**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

3068

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact volume of a cone with r = 3 and h = 4?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

3069

**QUESTION ID**

## S83-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**12π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 4 prism?

**A.** 52

**B.** 9

**C.** 26

**D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

3070

**QUESTION ID**

**S83-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 24**

**GRADING NOTE** Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

**True or false?**

$V = \frac{1}{3}\pi r^2 h$ is the cone formula.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

3071

QUESTION ID

## S83-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

**True or false?**

A sphere's volume is $\pi r^2 h$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

3072

QUESTION ID

**S83-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $4/3\ \pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

# BACKWARD 2

Radius from a circumference • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has circumference 10π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

3074

QUESTION ID

**S83-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has circumference 2.2 m. Radius to the nearest hundredth?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

3075

**QUESTION ID**
**S83-B2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**0.35 m**

**GRADING NOTE** Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

r =

**A.** C/(2π)

**B.** πC

**C.** 2πC

**D.** C/π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

3076

QUESTION ID

**S83-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. C/(2π)**

**GRADING NOTE** C = 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

**True or false?**

A tape measure around a trunk gives its circumference.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

3077

QUESTION ID

**S83-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Distance around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

**True or false?**

The radius is C/2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5**

3078

QUESTION ID

## S83-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** C/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

**Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5**

SET 83 • HSG-MG.A.1

# BACKWARD 3

Convert cubic units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

**ANSWER KEY** Reduced copy of the question page, with its answer

How many liters are in 2 $m^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

3080

**QUESTION ID**

**S83-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**2,000 L**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How many cubic centimeters are in 1 L?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

3081

**QUESTION ID**

## S83-B3-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**1,000 $cm^3$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

450 $m^3$ = ___ L

**A.** 45,000

**B.** 0.45

**C.** 450,000

**D.** 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

3082

**QUESTION ID**

**S83-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 450,000**

**GRADING NOTE** × 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$1\ mL = 1\ cm^3$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

3083

QUESTION ID

## S83-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$1\ m^3 = 100\ L$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

3084

QUESTION ID

**S83-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 1,000 L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSG-MG.A.1

# FORWARD 1

Judge a model against a measurement • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A water bottle is modeled as a cylinder with radius 3.8 cm and height 17 cm. The label says 750 mL. Find the model volume to the nearest mL and the percent error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

3086

**QUESTION ID**

## S83-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**About 771 mL; about 2.8% too high**

**GRADING NOTE** Form required: both (rounding to 1 mL and 0.1%). Percent error = (model − actual)/actual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A silo is modeled as a cylinder (r = 3 m, h = 12 m) with a hemisphere on top. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

3087

**QUESTION ID**

**S83-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**126π m³**

**GRADING NOTE** Form required: exact value. 108π + 18π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

**True or false?**

A model that overestimates by 3% may still be useful for planning.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

3088

QUESTION ID

## S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A football (pointed ends) is best modeled by

**A.** two cones joined at their bases

**B.** a cube

**C.** a sphere

**D.** a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

3089

QUESTION ID

**S83-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. two cones joined at their bases**

**GRADING NOTE** Pointed ends; other answers with reasons may be discussed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

Why might a cylinder model overestimate the volume of a bottle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

3090

**QUESTION ID**
**S83-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**The real bottle narrows at the neck and has thick walls, so it holds less than the full cylinder.**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

# FORWARD 2

Use the model to estimate mass with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A tree trunk modeled as a cylinder has volume about 2.3 m³. The wood has density 600 kg/m³. Estimate the trunk's mass to the nearest 10 kg.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

3092

**QUESTION ID**

**S83-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**About 1390 kg**

**GRADING NOTE** Rounding to the nearest 10 kg required; using 2.3 m³ gives 1,380 kg, also accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sand pile modeled as a cone has volume $16\pi$ m³. The sand has density 1,600 kg/m³. Estimate its mass in metric tons (1 t = 1,000 kg), to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3093

**QUESTION ID**
**S83-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**80 t**

**GRADING NOTE** Rounding to the nearest ton required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

3093

**True or false?**

A more accurate volume model gives a more accurate mass estimate.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

3094

**QUESTION ID**

## S83-F2-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pool holds 450 m³ of water (1,000 kg/m³). Mass?

**A.** 450,000 kg

**B.** 450 kg

**C.** 45,000 kg

**D.** 4,500 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

3095

**GRADING NOTE** Product.

**QUESTION ID**

## S83-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 450,000 kg**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

Explain how a geometric model and a density work together to estimate the mass of a log.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

3096

**QUESTION ID**

## S83-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The model gives the volume; mass = density × volume.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSG-MG.A.2 • MODELING WITH GEOMETRY

# Apply density based on area

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-MG.A.2** |
| **BACKWARD 1** | Unit rates | **6.RP.A.3.b** |
| **BACKWARD 2** | Areas of rectangles and circles | **7.G.B.6** |
| **BACKWARD 3** | Compare ratios | **7.RP.A.2** |
| **FORWARD 1** | Convert area units in density problems | **HSG-MG.A.2** |
| **FORWARD 2** | Plan capacity from an area density | **HSG-MG.A.3** |

SET 84 • HSG-MG.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A city has 840,000 people and an area of 120 square miles. Find its population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

3099

**QUESTION ID**

## S84-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**7,000 people per square mile**

**GRADING NOTE** Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

A circular forest region has radius 3 km and 4,500 trees. Find the tree density to the nearest tree per square kilometer.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

3100

**QUESTION ID**

## S84-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**159 trees per km²**

**GRADING NOTE** Rounding to the nearest whole number required. Area $9\pi \approx 28.27$ km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

Town A has 12,000 people on 30 km². Town B has 9,000 people on 20 km². Which town is more densely populated? Show the densities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

3101

**QUESTION ID**

## S84-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**Town B (450 per km² vs 400 per km²)**

**GRADING NOTE** Form required: the conclusion with both densities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A region has a population density of 250 people per km² and is a 4 km by 6 km rectangle. Estimate its population.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

3102

QUESTION ID

**S84-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**6,000 people**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is a unit of population density based on area?

**A.** people per mile

**B.** people per square mile

**C.** square miles per person per year

**D.** people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

3103

QUESTION ID

**S84-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. people per square mile**

GRADING NOTE Count ÷ area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

SET 84 • HSG-MG.A.2

# BACKWARD 1

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

**ANSWER KEY** Reduced copy of the question page, with its answer

840 items in 12 boxes: items per box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

3105

QUESTION ID

**S84-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**70**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$45 for 9 m² of tile: dollars per square meter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

3106

QUESTION ID

**S84-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$5 per m²**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Unit rate for 300 people on 4 km²?

**A.** 0.013 km² per person

**B.** 1,200 people per km²

**C.** 296 people per km²

**D.** 75 people per km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

3107

**QUESTION ID**

## S84-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 75 people per km²**

**GRADING NOTE** 300 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

**True or false?**

A unit rate compares a quantity to one unit of another.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

3108

QUESTION ID

**S84-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

**True or false?**

600 people on 3 km² is 1,800 people per km².

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

3109

QUESTION ID

**S84-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

# BACKWARD 2

Areas of rectangles and circles • Grade 7 • Geometry • 7.G.B.6

**ANSWER KEY** Reduced copy of the question page, with its answer

Area of a 4 km by 6 km rectangle?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

3111

QUESTION ID

**S84-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**24 km²**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of a circle with radius 3 km?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

3112

QUESTION ID

**S84-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**9π km²**

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A square region 5 mi on a side has area

**A.** 20 mi²

**B.** 10 mi²

**C.** 5 mi²

**D.** 25 mi²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3113

**QUESTION ID**

**S84-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 25 mi²**

**GRADING NOTE** $5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

3113

**True or false?**

Area units are square units.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

**Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5**

3114

QUESTION ID

## S84-B2-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Two dimensions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

**Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5**

**True or false?**

A circle with radius 3 has area 6π.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

**Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5**

3115

QUESTION ID

**S84-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

**Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5**

SET 84 • HSG-MG.A.2

# BACKWARD 3

Compare ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is greater: 12,000/30 or 9,000/20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

3117

QUESTION ID

**S84-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**9,000/20 (450 > 400)**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

**True or false?**

To compare densities, compare the unit rates.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

3118

QUESTION ID

**S84-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Same units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which rate is greatest?

**A.** 80 per 4 km²

**B.** 50 per 2 km²

**C.** 60 per 3 km²

**D.** 20 per 1 km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

3119

QUESTION ID

**S84-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 50 per 2 km²**

GRADING NOTE 25, 20, 20, 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A has 100 trees on 4 acres; B has 120 trees on 6 acres. Which is denser?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

3120

QUESTION ID

**S84-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**A (25 per acre vs 20)**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The town with more people is always denser.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

3121

QUESTION ID

**S84-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Depends on area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSG-MG.A.2

# FORWARD 1

Convert area units in density problems • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

1 square mile is about 2.59 km². Convert 7,000 people per square mile to people per km² (nearest whole number).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

3123

**QUESTION ID**
**S84-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**2703 people per km²**

**GRADING NOTE** Rounding to the nearest whole number required. Divide by 2.59.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A lawn of 400 m² needs grass seed at 35 g per m². How many kilograms of seed are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

3124

QUESTION ID

**S84-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**14 kg**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A density in people per km² is a larger number than the same density in people per mi².

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

3125

QUESTION ID

**S84-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE A km² is smaller, so fewer people per km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A farm plants 2 trees per 10 $m^2$. How many trees on 1 hectare (10,000 $m^2$)?

**A.** 2,000

**B.** 200

**C.** 5,000

**D.** 20,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

3126

**QUESTION ID**

**S84-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 2,000**

**GRADING NOTE** 0.2 per $m^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A park of 3 hectares (1 ha = 10,000 $m^2$) has 5,400 visitors. Visitors per 100 $m^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

3127

QUESTION ID

**S84-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**18**

GRADING NOTE Any equivalent form. 300 units of 100 $m^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

# FORWARD 2

Plan capacity from an area density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Safety rules allow 2 people per $m^2$ in a plaza. The plaza is a 30 m by 40 m rectangle. Maximum crowd?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

3129

**QUESTION ID**
**S84-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**2,400 people**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A festival expects 5,000 people and wants at most 1.5 people per m². What is the smallest area needed (nearest m²)?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

3130

QUESTION ID

**S84-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**3334 m²**

**GRADING NOTE** Rounding up required: 3,334 m² (3,333.3... rounded up so the limit is met).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

**True or false?**

When planning for a safety limit, round the capacity up.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

3131

QUESTION ID

**S84-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Round capacity down (and area up) to stay within the limit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

A circular dance floor of radius 5 m allows 3 people per m². Capacity is about

**A.** 235 people

**B.** 471 people

**C.** 75 people

**D.** 47 people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

3132

**QUESTION ID**

## S84-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. 235 people

**GRADING NOTE** $75\pi \approx 235.6$, round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

A circular dance floor of radius 5 m allows at most 3 people per m², and $3 \cdot 25\pi \approx 235.6$. Explain why the capacity is 235 people, not 236.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

3133

**QUESTION ID**

## S84-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Rounding up would exceed the allowed density.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

SET 85 • HSG-MG.A.2 • MODELING WITH GEOMETRY

# Apply density based on volume

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-MG.A.2** |
| **BACKWARD 1** | Volume of prisms and cylinders | **HSG-GMD.A.3** |
| **BACKWARD 2** | Unit rates | **6.RP.A.3.b** |
| **BACKWARD 3** | Products of numbers in scientific notation | **8.EE.A.4** |
| **FORWARD 1** | Use density to identify materials and find volumes | **HSG-MG.A.2** |
| **FORWARD 2** | Design within a mass limit | **HSG-MG.A.3** |

SET 85 • HSG-MG.A.2

# MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

An aluminum block measures 2 cm by 3 cm by 5 cm and has mass 81 g. Find its density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

3136

**QUESTION ID**
**S85-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**2.7 g/cm³**

**GRADING NOTE** Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylindrical tank (radius 0.5 m, height 2 m) is full of water (1,000 kg/m³). Find the mass of the water to the nearest kilogram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

3137

**QUESTION ID**

**S85-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**1571 kg**

**GRADING NOTE** Rounding to the nearest kilogram required. 0.5π m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A heater must supply 20 BTU per cubic foot of room. A room is 12 ft by 15 ft with an 8 ft ceiling. How many BTU are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

3138

**QUESTION ID**

**S85-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**28,800 BTU**

**GRADING NOTE** Any equivalent form. 1,440 $ft^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A sample has $3 \times 10^6$ bacteria per mL. How many bacteria are in 250 mL?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

3139

**QUESTION ID**

## S85-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**$7.5 \times 10^8$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

Water has density 1 g/cm³. Which object floats in water?

**A.** A 1,000 g brick with volume 400 cm³

**B.** A 600 g block with volume 500 cm³

**C.** A 100 g ball with volume 50 cm³

**D.** A 400 g block with volume 500 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

3140

**QUESTION ID**

## S85-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. A 400 g block with volume 500 cm³**

**GRADING NOTE** Density less than 1 g/cm³ (0.8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

SET 85 • HSG-MG.A.2

# BACKWARD 1

Volume of prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Volume of a 2 × 3 × 5 cm block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

3142

**QUESTION ID**

**S85-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**30 cm³**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact volume of a cylinder with r = 0.5 m, h = 2 m?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

3143

**QUESTION ID**

## S85-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$0.5\pi$ m³**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Volume of a 12 × 15 × 8 ft room?

**A.** 1,440 ft³

**B.** 180 ft³

**C.** 2,880 ft³

**D.** 35 ft³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

3144

QUESTION ID

**S85-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 1,440 ft³**

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

**True or false?**

Volume is needed before a volume density can be used.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

3145

QUESTION ID

## S85-B1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Density = amount ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A 1 m cube has volume 3 m³.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

3146

QUESTION ID

**S85-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 1 m³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

# BACKWARD 2

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

**ANSWER KEY** Reduced copy of the question page, with its answer

81 g in 30 cm³: grams per cm³?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

3148

QUESTION ID

**S85-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2.7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

20 BTU per $ft^3$ for 100 $ft^3$: total?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

3149

QUESTION ID

**S85-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**2,000 BTU**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

500 g in 250 $cm^3$ is

**A.** 250 $g/cm^3$

**B.** 2 $g/cm^3$

**C.** 0.5 $g/cm^3$

**D.** 750 $g/cm^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

3150

QUESTION ID

**S85-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 2 $g/cm^3$**

**GRADING NOTE** Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

**True or false?**

Amount = rate × quantity.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

3151

QUESTION ID

**S85-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Rate use.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

**True or false?**

Density = volume ÷ mass.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

3152

QUESTION ID

**S85-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Mass ÷ volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSG-MG.A.2

# BACKWARD 3

Products of numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

$(3 \times 10^6) \times 250$ in scientific notation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

3154

QUESTION ID

**S85-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**$7.5 \times 10^8$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write 2,400,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

3155

QUESTION ID

**S85-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**$2.4 \times 10^6$**

GRADING NOTE Form required: scientific notation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$(2 \times 10^3)(4 \times 10^2) =$

**A.** $6 \times 10^5$

**B.** $8 \times 10^5$

**C.** $8 \times 10^6$

**D.** $8 \times 10^1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

3156

**QUESTION ID**

**S85-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $8 \times 10^5$**

**GRADING NOTE** Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$10^3 \times 10^2 = 10^5$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

3157

QUESTION ID

**S85-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Exponent rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$5 \times 10^2 \times 3 = 15 \times 10^6$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

3158

QUESTION ID

**S85-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $1.5 \times 10^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSG-MG.A.2

# FORWARD 1

Use density to identify materials and find volumes • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A ring has mass 9.65 g and volume 0.5 cm³. Gold has density 19.3 g/cm³. Could the ring be pure gold?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

3160

QUESTION ID

**S85-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Yes:**

**9.65 ÷ 0.5 = 19.3 g/cm³**

GRADING NOTE Form required: the conclusion with the density.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 1,350 g piece of aluminum (2.7 g/cm³) is melted into a cube. Find the edge of the cube, to the nearest tenth of a centimeter.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

3161

**QUESTION ID**

**S85-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**7.9 cm**

**GRADING NOTE** Rounding to 0.1 required. Volume 500 cm³; edge $\sqrt[3]{500}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

**True or false?**

Two objects with equal mass but different volumes have different densities.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

3162

QUESTION ID

**S85-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Density = m/V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Oil (0.9 g/cm³) in a 2,000 cm³ jug has mass

**A.** 1,800 g

**B.** 2,000 g

**C.** 2,222 g

**D.** 1,100 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

3163

**QUESTION ID**

**S85-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 1,800 g**

**GRADING NOTE** 0.9 × 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A balloon holds 0.02 $m^3$ of helium (0.18 $kg/m^3$). Mass of the helium in grams?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

3164

**QUESTION ID**
**S85-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**3.6 g**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

# FORWARD 2

Design within a mass limit • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A truck can carry 12,000 kg. Gravel has density 1,500 kg/m³. What is the largest volume of gravel it can carry?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

3166

**QUESTION ID**

**S85-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8 m³**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

A shelf holds at most 40 kg. Books have density 0.8 g/cm³. How many cubic centimeters of books can it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

3167

**QUESTION ID**
**S85-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**50,000 cm³**

**GRADING NOTE** Any equivalent form. 40,000 g ÷ 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

**True or false?**

Mass limits become volume limits by multiplying by the density.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

3168

QUESTION ID

**S85-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Divide by the density: V = m/d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

A cylindrical water tank must weigh at most 3,140 kg when full (water 1,000 kg/m³, ignore the tank). With radius 1 m, its height can be at most about

**A.** 10 m

**B.** 3.14 m

**C.** 1 m

**D.** 0.32 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

3169

**QUESTION ID**

**S85-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 1 m**

**GRADING NOTE** 3.14 m³ ÷ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A concrete block must be under 25 kg. Concrete is 2,400 kg/m³. Will a 20 cm × 20 cm × 30 cm block work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

3170

QUESTION ID

**S85-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**No: 0.012 m³ × 2,400 kg/m³ = 28.8 kg, which exceeds 25 kg.**

GRADING NOTE Form required: the conclusion with the mass.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

SET 86 • HSG-MG.A.3 • MODELING WITH GEOMETRY

# Design to satisfy physical constraints

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-MG.A.3** |
| **BACKWARD 1** | Volume of a rectangular prism | **7.G.B.6** |
| **BACKWARD 2** | Use the Pythagorean Theorem | **8.G.B.7** |
| **BACKWARD 3** | Ratios and rates | **7.RP.A.2** |
| **BACKWARD 4** | Inequalities as limits | **7.EE.B.4** |
| **FORWARD 1** | Satisfy several constraints at once | **HSG-MG.A.3** |
| **FORWARD 2** | Find the best design with a function graph | **HSF-IF.C.7.c** |

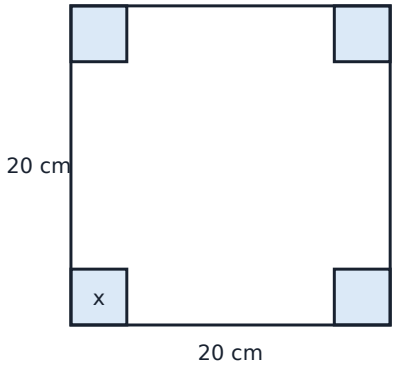
SET 86 • HSG-MG.A.3

# MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Compare the volumes for x = 2, 3, and 4. Which gives the largest box?




Not drawn to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

3173

QUESTION ID

**S86-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**x = 2: 512 cm³; x = 3: 588 cm³; x = 4: 576 cm³; x = 3 is largest**

GRADING NOTE Form required: the three volumes and the choice. $V = x(20 - 2x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Accessibility rules allow a ramp slope of at most 1 : 12 (1 unit of rise per 12 units of run). A ramp must rise 30 in. What is the shortest horizontal run allowed, in feet?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

3174

**QUESTION ID**
**S86-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**30 ft**

**GRADING NOTE** Any equivalent form. 30 × 12 = 360 in.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

A rectangular pen will use 40 m of fencing. Compare a 10 m by 10 m pen with a 5 m by 15 m pen. Which encloses more area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

3175

**QUESTION ID**

## S86-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**The 10 m by 10 m pen (100 m² vs 75 m²)**

**GRADING NOTE** Form required: the choice with both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylindrical tank must fit in a closet 2 m wide, 2 m deep, and 3 m tall. What is the largest possible volume, to the nearest tenth of a cubic meter?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

3176

**QUESTION ID**

## S86-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**9.4 m³**

**GRADING NOTE** Rounding to 0.1 required. Radius 1 m, height 3 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

A 13 ft ladder leans on a wall. For safety its base must be at least 5 ft from the wall. The highest point it can reach is

**A.** 18 ft

**B.** 12 ft

**C.** 13 ft

**D.** 8 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

3177

QUESTION ID

**S86-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. 12 ft**

GRADING NOTE √(169 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

SET 86 • HSG-MG.A.3

# BACKWARD 1

Volume of a rectangular prism • Grade 7 • Geometry • 7.G.B.6

**ANSWER KEY** Reduced copy of the question page, with its answer

Volume of a box 14 cm by 14 cm by 3 cm?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

3179

QUESTION ID

**S86-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**588 cm³**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 20 cm side with 3 cm cut from each end leaves

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

3180

QUESTION ID

**S86-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**14 cm**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Volume of a box 12 by 12 by 4?

**A.** 28

**B.** 576

**C.** 48

**D.** 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

3181

**QUESTION ID**

**S86-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 576**

**GRADING NOTE** Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

**True or false?**

Cutting larger corners makes the box taller but its base smaller.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

3182

QUESTION ID

**S86-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Trade-off.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

**True or false?**

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner. The box's base side is 20 − x.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

3183

**QUESTION ID**
**S86-B1-Q5**

**TYPE**
True or false

**CORRECT ANSWER**
**B. False**

**GRADING NOTE** 20 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

# BACKWARD 2

Use the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

Hypotenuse 13, leg 5: other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

3185

QUESTION ID

**S86-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Legs 30 in. and 360 in.: hypotenuse to the nearest inch?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

3186

**QUESTION ID**

## S86-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**361 in.**

**GRADING NOTE** Rounding to the nearest inch required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

A ladder, wall, and ground form a

**A.** right triangle

**B.** circle

**C.** square

**D.** equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

3187

**QUESTION ID**

## S86-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. right triangle**

**GRADING NOTE** Wall ⊥ ground.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

**True or false?**

A ladder leans against a vertical wall on level ground. The ladder is the hypotenuse of the right triangle formed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

**Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5**

3188

**QUESTION ID**

## S86-B2-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Opposite the right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

**Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5**

**True or false?**

The ladder's height on the wall is 13 + 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

3189

QUESTION ID

**S86-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Use the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q5/5

SET 86 • HSG-MG.A.3

# BACKWARD 3

Ratios and rates • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A ratio of 1 : 12 for a rise of 2 in. gives what run?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

3191

**QUESTION ID**

## S86-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**24 in.**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Is a slope of 3 : 30 steeper than 1 : 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

3192

QUESTION ID

**S86-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**Yes: 3 : 30 = 1 : 10, and 1/10 > 1/12.**

GRADING NOTE Form required: yes/no with the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

360 in. =

**A.** 30 ft

**B.** 36 ft

**C.** 12 ft

**D.** 3.6 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

3193

**GRADING NOTE** ÷ 12.

**QUESTION ID**

## S86-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 30 ft**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A ratio of 1 : 12 means 1 inch up for every 12 inches across.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

3194

QUESTION ID

**S86-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

**True or false?**

A 1 : 8 ramp is less steep than a 1 : 12 ramp.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

3195

QUESTION ID

## S86-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Steeper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

SET 86 • HSG-MG.A.3

# BACKWARD 4

Inequalities as limits • Grade 7 • Expressions & Equations • 7.EE.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

The run r must satisfy r ≥ 360. Is r = 300 allowed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

3197

QUESTION ID

**S86-B4-Q1**

TYPE

Short answer

CORRECT ANSWER

**No**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A diameter d must satisfy d ≤ 2. Largest radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

3198

**QUESTION ID**

## S86-B4-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**1**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

"At most 3 m" means

**A.** ≥ 3

**B.** < 3 only

**C.** ≤ 3

**D.** = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

3199

**QUESTION ID**

**S86-B4-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. ≤ 3**

**GRADING NOTE** Inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

"At least 5 ft" allows exactly 5 ft.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

3200

QUESTION ID

**S86-B4-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE ≥.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

3200

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

"At most 12" excludes 12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

3201

QUESTION ID

**S86-B4-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Includes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

SET 86 • HSG-MG.A.3

# FORWARD 1

Satisfy several constraints at once • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Which whole-number values of x give a volume of at least 500 cm³ AND a height of at most 3 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

3203

QUESTION ID

**S86-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**x = 2 and x = 3**

**GRADING NOTE** Form required: both values. Volumes 512 and 588; x = 4, 5 are too tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A ramp must rise 2 ft and fit within 20 ft of horizontal space. Can it meet the 1 : 12 limit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

3204

QUESTION ID

**S86-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**No: it needs 24 ft of run.**

GRADING NOTE Form required: the conclusion with the run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

**True or false?**

A design that satisfies one constraint may fail another.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

3205

QUESTION ID

**S86-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Check all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangular garden needs area 48 m² and width at most 6 m. Which works?

**A.** 6 m wide by 8 m long

**B.** 4 m wide by 10 m long

**C.** 7 m wide by 7 m long

**D.** 8 m wide by 6 m long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

3206

QUESTION ID

**S86-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 6 m wide by 8 m long**

**GRADING NOTE** Width 6, area 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A cylinder 3 m tall must hold at least 10 m³ and stand on a 2 m by 2 m floor space. Is that possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

3207

QUESTION ID

**S86-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**No: the largest is $r = 1$ m, giving $3\pi \approx 9.4$ m³ < 10 m³.**

GRADING NOTE Form required: the conclusion with the volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

# FORWARD 2

Find the best design with a function graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

**ANSWER KEY** Reduced copy of the question page, with its answer

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Write its volume as a function V(x) and state the domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

3209

**QUESTION ID**

## S86-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$V(x) = x(20 - 2x)^2$, $0 < x < 10$**

**GRADING NOTE** Form required: the function and the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An open box made from a 20 cm by 20 cm sheet with x-cm corner squares cut out has volume $V(x) = x(20 - 2x)^2$. Use a graph or table to estimate the x that gives the largest volume, to the nearest tenth, and that volume to the nearest $cm^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

3210

**QUESTION ID**

## S86-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$x \approx 3.3$; $V \approx 593\ cm^3$**

**GRADING NOTE** Form required: both. (Exact: x = 10/3, V = 16,000/27 ≈ 592.6.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

**True or false?**

$V(x) = x(20 - 2x)^2$ has zeros at $x = 0$ and $x = 20$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

3211

QUESTION ID

## S86-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The zeros are $x = 0$ and $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet, $V(x) = x(20 - 2x)^2$. The graph of V on $0 < x < 10$ rises and then falls. Its highest point gives

**A.** the cut that makes no box

**B.** the base area

**C.** the largest volume

**D.** the smallest volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

3212

QUESTION ID
**S86-F2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**C. the largest volume**

**GRADING NOTE** Maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For an open box made from a 20 cm by 20 cm sheet with x-cm corner cuts, $V(x) = x(20 - 2x)^2$ and V is largest near x = 3.3. Why is x = 5 not the best cut even though it is halfway between 0 and 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

3213

QUESTION ID
**S86-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**V(5) = 500 < 593; the maximum of a cubic need not be at the middle.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

SET 87 • HSG-MG.A.3 • MODELING WITH GEOMETRY

# Design to minimize cost and work with typographic grids

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-MG.A.3** |
| **BACKWARD 1** | Surface area of prisms and cylinders | **7.G.B.6** |
| **BACKWARD 2** | Proportional reasoning with ratios | **7.RP.A.2** |
| **BACKWARD 3** | Unit prices and rounding up purchases | **7.EE.B.3** |
| **FORWARD 1** | Compare designs with equal volume for least material | **HSG-MG.A.3** |
| **FORWARD 2** | Decide between designs using expected cost | **HSS-MD.B.7** |

SET 87 • HSG-MG.A.3

# MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A can with radius 4 cm and height 10 cm is made of metal costing \$0.002 per cm². Find the area of metal (top, bottom, and side) and the cost, to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

3216

**QUESTION ID**

## S87-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**112π cm² ≈ 351.9 cm²; cost \$0.70**

**GRADING NOTE** Form required: the area and the cost. $2\pi r^2 + 2\pi rh = 32\pi + 80\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Can A has r = 4 cm, h = 10 cm. Can B has r = 5 cm, h = 6.4 cm. Show that they hold the same volume, then decide which uses less metal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

3217

**QUESTION ID**

## S87-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Both hold 160π cm³; A uses 112π cm², B uses 114π cm², so A uses less metal.**

**GRADING NOTE** Form required: both volumes, both areas, and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A wall 12 m by 3 m has a window 1.5 m by 1.2 m. Paint covers 10 m² per liter and is sold only in whole liters at $8 each. Find the cost to paint the wall once.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

3218

**QUESTION ID**
**S87-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**$32**

**GRADING NOTE** Any equivalent form. 34.2 m² needs 3.42 L, so buy 4 L.

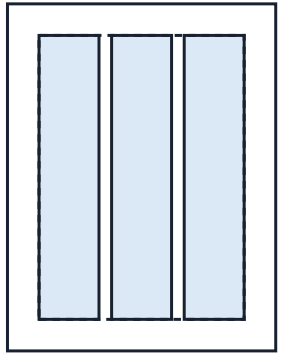
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A page is 8.5 in. wide with 1 in. margins on each side. The text area has 3 equal columns separated by 2 gutters of 0.2 in. Find the width of each column to the nearest hundredth of an inch.

Width: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

3219

**QUESTION ID**

**S87-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**2.03 in.**

**GRADING NOTE** Rounding to 0.01 required. (6.5 − 0.4) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A poster grid uses the golden ratio (about 1.618 : 1) for height : width. A poster is 60 cm wide. Its height is about

**A.** 120 cm

**B.** 97.1 cm

**C.** 37.1 cm

**D.** 61.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

3220

**QUESTION ID**
**S87-M-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. 97.1 cm**

**GRADING NOTE** 60 × 1.618.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

SET 87 • HSG-MG.A.3

# BACKWARD 1

Surface area of prisms and cylinders • Grade 7 • Geometry • 7.G.B.6

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of the two circular ends of a can with radius 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

3222

**QUESTION ID**

## S87-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**32π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Exact area of the side of a can with radius 4 and height 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

3223

QUESTION ID

**S87-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**80π**

GRADING NOTE Form required: exact value. 2πrh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

Surface area of a 2 × 3 × 4 box?

**A.** 24

**B.** 48

**C.** 52

**D.** 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

3224

**QUESTION ID**

**S87-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 52**

**GRADING NOTE** 2(6 + 8 + 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

**True or false?**

The side of a cylinder unrolls into a rectangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

3225

QUESTION ID

**S87-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Net.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

**True or false?**

A cylinder's surface area is $\pi r^2 h$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

3226

QUESTION ID

**S87-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE That is volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

# BACKWARD 2

Proportional reasoning with ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Height : width = 1.618 : 1 and width = 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q1/5

3228

QUESTION ID

**S87-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**16.18**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Split 6.1 in. into 3 equal columns. Width of each (nearest hundredth)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q2/5

3229

**QUESTION ID**

**S87-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**2.03 in.**

**GRADING NOTE** Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 2 : 3 rectangle 30 cm wide (width = 2 parts) is

**A.** 45 cm tall

**B.** 60 cm tall

**C.** 20 cm tall

**D.** 33 cm tall

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q3/5

3230

**QUESTION ID**

**S87-B2-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 45 cm tall**

**GRADING NOTE** 15 per part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q3/5

**True or false?**

Equal columns divide the available width equally.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q4/5

3231

QUESTION ID

**S87-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A 1 : 2 ratio means the second is half the first.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q5/5

3232

QUESTION ID

**S87-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q5/5

SET 87 • HSG-MG.A.3

# BACKWARD 3

Unit prices and rounding up purchases • Grade 7 • Expressions & Equations • 7.EE.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Paint needed is 3.42 L, sold in whole liters. How many liters must be bought?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

3234

QUESTION ID

**S87-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

3234

**ANSWER KEY** Reduced copy of the question page, with its answer

Metal costs $0.002 per cm². Cost of 350 cm²?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

3235

**QUESTION ID**

**S87-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$0.70**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Fabric is $6 per yard, sold in whole yards. You need 4.2 yd. Cost?

**A.** $30

**B.** $25.20

**C.** $24

**D.** $26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

3236

**GRADING NOTE** Buy 5 yd.

**QUESTION ID**

**S87-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. $30**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

3236

**True or false?**

When items are sold in whole units, round the amount needed up.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

3237

QUESTION ID

**S87-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Enough material.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

3.1 cans of paint can be bought as 3 cans.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

3238

QUESTION ID

**S87-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Need 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

SET 87 • HSG-MG.A.3

# FORWARD 1

Compare designs with equal volume for least material • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A box with a square base must hold 1,000 cm³. Compare the surface areas for base sides 8, 10, and 12 cm. Which uses the least material?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

3240

**QUESTION ID**

**S87-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8 cm: 628 cm²; 10 cm: 600 cm²; 12 cm: about 621 cm²; the 10 cm cube uses the least**

**GRADING NOTE** Form required: the three areas and the choice. h = 1000/s².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 12-column grid on a 6.5 in. text area has 11 gutters of 0.125 in. Find the width of each column to the nearest thousandth of an inch.

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

3241

**QUESTION ID**

## S87-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**0.427 in.**

**GRADING NOTE** Rounding to 0.001 required. (6.5 − 1.375) ÷ 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

**True or false?**

Among boxes with a square base and a fixed volume, the cube has the least surface area.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

3242

QUESTION ID

**S87-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Classic result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 12-column grid can also be used as

**A.** only 12 columns

**B.** 7 equal columns

**C.** 5 equal columns

**D.** 4 columns of 3, 3 of 4, or 2 of 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

3243

**QUESTION ID**
**S87-F1-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. 4 columns of 3, 3 of 4, or 2 of 6**

**GRADING NOTE** Divisors of 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

Why do designers often choose 12-column grids?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

3244

QUESTION ID

**S87-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**12 divides evenly into 2, 3, 4, and 6, giving many layout options.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

# FORWARD 2

Decide between designs using expected cost • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Packaging A costs $1.20 per item and 2% of items break. Packaging B costs $0.90 and 8% break. A break costs $20. Find the expected cost per item for each and choose.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

3246

QUESTION ID

**S87-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**A: $1.20 + 0.02 × $20 = $1.60; B: $0.90 + 0.08 × $20 = $2.50; choose A**

**GRADING NOTE** Form required: both expected costs and the choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A design uses a cheaper metal that fails 5% of the time, with replacement cost $40. How much cheaper must it be per can to be worth it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

3247

**QUESTION ID**
**S87-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**More than $2.00 cheaper (0.05 × $40)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

**True or false?**

The cheapest material always gives the cheapest design.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

3248

QUESTION ID
**S87-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Failures and replacement costs can make it more expensive overall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

Expected cost =

**A.** upfront cost × probability

**B.** upfront cost + probability of failure × cost of failure

**C.** cost of failure only

**D.** upfront cost − probability

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

3249

QUESTION ID
**S87-F2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. upfront cost + probability of failure × cost of failure**

**GRADING NOTE** Weighted cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name one cost besides material that a can designer might consider.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

3250

QUESTION ID
**S87-F2-Q5**

TYPE
Short answer

CORRECT ANSWER
**For example shipping weight, labor, breakage, or recycling.**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S87-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9 • ADVANCED EXTENSION (+)

# Derive and use the formula A = ½ab sin C

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.D.9** |
| **BACKWARD 1** | The sine ratio in a right triangle | **HSG-SRT.C.6** |
| **BACKWARD 2** | Area of a triangle from base and height | **6.G.A.1** |
| **BACKWARD 3** | The included angle of two sides | **HSG-CO.B.8** |
| **FORWARD 1** | Solve for a side or an angle from an area | **HSG-SRT.D.9** |
| **FORWARD 2** | From equal areas to the Law of Sines | **HSG-SRT.D.10** |

SET 88 • HSG-SRT.D.9

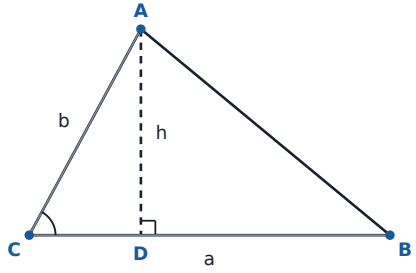
# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, the altitude from A to side CB has length h. Derive the formula Area = ½ab sin C.

Work space




Conclusion: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

3253

**QUESTION ID**

## S88-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**h = b sin C, so Area = ½ah = ½ab sin C.**

**GRADING NOTE** REQUIRED WORK: Express h using sin C in right triangle ADC, then use Area = ½ · base · height.

Any valid proof (paragraph, two-column, or flow). Essential: in right triangle ADC, sin C = h/b, so h = b sin C; Area = ½ · a · h; substitute. Alternative: an altitude from B gives the same result with h = a sin C.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-M-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has sides a = 8 and b = 11 with included angle C = 40°. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

3254

**QUESTION ID**

## S88-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**28.3**

**GRADING NOTE** Rounding to 0.1 required. ½ · 8 · 11 · sin 40°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-M-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5**

Two sides of a triangle are 6 and 10, and the angle between them is 150°. Find the exact area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

3255

**QUESTION ID**

**S88-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Any equivalent form. sin 150° = ½.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A parallelogram has sides 7 cm and 12 cm with an angle of 55° between them. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

3256

**QUESTION ID**

## S88-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**68.8 cm²**

**GRADING NOTE** Rounding to 0.1 required. Two congruent triangles: 7 · 12 · sin 55°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-M-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5**

A triangle has sides 5 and 9 with an included angle of 30°. Its area is

**A.** 45

**B.** 22.5

**C.** 11.25

**D.** 19.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

3257

**QUESTION ID**

**S88-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 11.25**

**GRADING NOTE** ½ · 45 · ½.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-M-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5**

SET 88 • HSG-SRT.D.9

# BACKWARD 1

The sine ratio in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

In a right triangle, sin of an acute angle = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

3259

QUESTION ID

**S88-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**opposite ÷ hypotenuse**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Hypotenuse 10, angle 30°. Find the side opposite the angle.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

3260

QUESTION ID

**S88-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If sin C = h/b, then h =

**A.** b + sin C

**B.** sin C/b

**C.** b/sin C

**D.** b sin C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

3261

**QUESTION ID**

**S88-B1-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. b sin C**

**GRADING NOTE** Multiply by b.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-B1-Q3**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

sin 90° = 1

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

3262

QUESTION ID

**S88-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

sin C = adjacent ÷ hypotenuse

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

3263

QUESTION ID
**S88-B1-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE That is cos C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

# BACKWARD 2

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Base 8, height 5: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

3265

**QUESTION ID**

**S88-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**20**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

**True or false?**

Area = ½ · base · height

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

**Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5**

3266

QUESTION ID

**S88-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

**Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5**

The height must be ___ to the base.

**A.** equal

**B.** adjacent

**C.** parallel

**D.** perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

3267

**QUESTION ID**

## S88-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. perpendicular**

**GRADING NOTE** Altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Base a and height b sin C: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

3268

QUESTION ID

**S88-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**½ab sin C**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A slanted side can be used as the height.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

3269

QUESTION ID

**S88-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9

# BACKWARD 3

The included angle of two sides • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In triangle ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

3271

**QUESTION ID**

**S88-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

$\angle A$

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

**True or false?**

SAS uses the angle between the two given sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

3272

QUESTION ID

**S88-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

For sides a (BC) and b (CA), the included angle is

**A.** C

**B.** none

**C.** A

**D.** B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

3273

QUESTION ID

**S88-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. C**

**GRADING NOTE** Shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Why must the angle be included for ½ab sin C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

3274

QUESTION ID

**S88-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**The altitude is built from side b and the angle between b and the base a.**

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

**True or false?**

Any angle of the triangle works in ½ab sin C.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

3275

QUESTION ID

**S88-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Only the included one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

SET 88 • HSG-SRT.D.9

# FORWARD 1

Solve for a side or an angle from an area • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has sides 10 and 12 and area 30. Find all possible measures of the included angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

3277

QUESTION ID

**S88-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**30° or 150°**

**GRADING NOTE** Form required: both. sin C = ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has area 50, one side 20, and an included angle of 45°. Find the other side to the nearest tenth.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

3278

**QUESTION ID**

## S88-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**7.1**

**GRADING NOTE** Rounding to 0.1 required. 50 = ½ · 20 · b · sin 45°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5**

**True or false?**

Two different triangles can have the same two sides and the same area.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

3279

QUESTION ID

## S88-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** C and 180° − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

The largest area of a triangle with sides 6 and 8 occurs when the included angle is

**A.** 60°

**B.** 180°

**C.** 90°

**D.** 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

3280

**QUESTION ID**

**S88-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 90°**

**GRADING NOTE** sin C = 1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F1-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the area of an equilateral triangle with side 6 using ½ab sin C (exact).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

3281

**QUESTION ID**

## S88-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**$9\sqrt{3}$**

**GRADING NOTE** Form required: exact value.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5**

SET 88 • HSG-SRT.D.9

# FORWARD 2

From equal areas to the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

The area of triangle ABC can be written as ½bc sin A, ½ac sin B, or ½ab sin C. Use this to show that sin A/a = sin B/b = sin C/c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

3283

**QUESTION ID**

**S88-F2-Q1**

**TYPE**

Answer with required work

**CORRECT ANSWER**

**½bc sin A = ½ac sin B = ½ab sin C; dividing by ½abc gives sin A/a = sin B/b = sin C/c.**

**GRADING NOTE** REQUIRED WORK: Set the three expressions equal and divide by ½abc.

Any valid proof (paragraph, two-column, or flow). Essential: three area expressions equal; division by ½abc; result.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Use sin A/a = sin B/b: A = 30°, a = 5, b = 8. Find sin B.

sin B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

3284

**QUESTION ID**

**S88-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**0.8**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F2-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5**

**True or false?**

The Law of Sines can be derived from the area formula.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

3285

QUESTION ID
**S88-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** As above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

Dividing ½bc sin A = ½ac sin B by ½abc gives

**A.** a sin A = b sin B

**B.** sin A/a = sin B/b

**C.** sin A = sin B

**D.** a/b = sin A sin B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

3286

**QUESTION ID**

**S88-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. sin A/a = sin B/b**

**GRADING NOTE** Cancel.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Why is it allowed to divide by ½abc?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S88-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

3287

**QUESTION ID**

## S88-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**Side lengths are positive, so ½abc ≠ 0.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S88-F2-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5**

SET 89 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

# Prove the Law of Sines

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.D.10** |
| **BACKWARD 1** | Use trigonometry in right triangles | **HSG-SRT.C.8** |
| **BACKWARD 2** | The area formula ½ab sin C | **HSG-SRT.D.9** |
| **BACKWARD 3** | Proportions and cross-multiplying | **7.RP.A.2** |
| **FORWARD 1** | Use the Law of Sines (AAS and ASA) | **HSG-SRT.D.10** |
| **FORWARD 2** | Surveying with the Law of Sines | **HSG-SRT.D.11** |

SET 89 • HSG-SRT.D.10

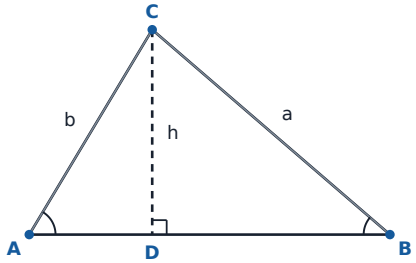
# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, CD is the altitude to AB. Prove that a/sin A = b/sin B.

Work space




Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

3290

**QUESTION ID**

## S89-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**h = b sin A and h = a sin B, so b sin A = a sin B and a/sin A = b/sin B.**

**GRADING NOTE** REQUIRED WORK: Write h two ways, using right triangles ADC and BDC.

Any valid proof (paragraph, two-column, or flow). Essential: sin A = h/b and sin B = h/a from the two right triangles; equate h; divide by sin A sin B. Repeating with another altitude gives c/sin C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has an obtuse angle at A, so the altitude from C meets line AB extended past A at D. Prove that a/sin A = b/sin B in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

3291

QUESTION ID

## S89-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

**h = b sin(180° − A) = b sin A and h = a sin B, so a/sin A = b/sin B.**

GRADING NOTE

REQUIRED WORK: Use the right triangle with angle 180° − A at A, and sin(180° − A) = sin A.

Any valid proof (paragraph, two-column, or flow). Essential: the exterior angle at A is 180° − A; h = b sin(180° − A); the identity sin(180° − A) = sin A; equate with h = a sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

Explain how the area formula ½bc sin A = ½ac sin B gives the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

3292

**QUESTION ID**

## S89-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**Dividing both sides by ½abc gives sin A/a = sin B/b.**

**GRADING NOTE** Any equivalent explanation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5**

In the proof of the Law of Sines, the altitude from C to AB has length h and forms right triangle ADC. Which fact gives h = b sin A?

**A.** The triangle angle sum

**B.** sin A = opposite/hypotenuse in right triangle ADC

**C.** cos A = h/b

**D.** The Pythagorean Theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

3293

**QUESTION ID**
**S89-M-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. sin A = opposite/hypotenuse in right triangle ADC**

**GRADING NOTE** Right triangle ADC.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-M-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5**

3293

**True or false?**

The Law of Sines works only in right triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

3294

QUESTION ID

## S89-M-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It works in every triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

SET 89 • HSG-SRT.D.10

# BACKWARD 1

Use trigonometry in right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangle: hypotenuse 12, angle 40°. Opposite side to the nearest tenth?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

3296

QUESTION ID

**S89-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**7.7**

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

An altitude splits a triangle into two right triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

3297

QUESTION ID
**S89-B1-Q2**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If sin B = h/a, then h =

**A.** h sin B

**B.** a/sin B

**C.** a sin B

**D.** sin B/a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

3298

QUESTION ID

**S89-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. a sin B**

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Right triangle: opposite 5, hypotenuse 10. Angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

3299

QUESTION ID

**S89-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**30°**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-B1-Q4**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5**

**True or false?**

sin is defined only for angles in right triangles, so it cannot be used after drawing an altitude.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

3300

QUESTION ID

## S89-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The altitude creates right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

# BACKWARD 2

The area formula ½ab sin C • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

**ANSWER KEY** Reduced copy of the question page, with its answer

Triangle ABC has b = 4, c = 5, and A = 90°. What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

3302

QUESTION ID

**S89-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**10**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

**True or false?**

½bc sin A, ½ac sin B, and ½ab sin C all give the same area.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

3303

QUESTION ID

## S89-B2-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Same triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Area of a triangle with sides 6, 6 and included 30°?

**A.** 3

**B.** 9

**C.** 36

**D.** 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

3304

**GRADING NOTE** ½ · 36 · ½.

**QUESTION ID**

## S89-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 9**

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-B2-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the area of triangle ABC using sides a and c.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

3305

QUESTION ID

**S89-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**½ac sin B**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

½ab sin C uses angle A.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

3306

QUESTION ID

## S89-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Angle C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10

# BACKWARD 3

Proportions and cross-multiplying • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve x/6 = 10/4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

3308

QUESTION ID

**S89-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

**True or false?**

If b sin A = a sin B, then a/sin A = b/sin B (when the sines are not 0).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

3309

QUESTION ID

**S89-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Divide by sin A sin B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From 3x = 4y, x/y =

**A.** 3/4

**B.** 4/3

**C.** 12

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

3310

QUESTION ID

**S89-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 4/3**

**GRADING NOTE** Divide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rewrite p · q = r · s as a proportion with p and r on top.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

3311

**QUESTION ID**

**S89-B3-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**p/r = s/q (or p/s = r/q)**

**GRADING NOTE** Any correct proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

**True or false?**

$a/b = c/d$ means $a + d = b + c$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

3312

QUESTION ID

**S89-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $ad = bc$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

SET 89 • HSG-SRT.D.10

# FORWARD 1

Use the Law of Sines (AAS and ASA) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, B = 60°, and a = 10. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

3314

**QUESTION ID**
**S89-F1-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**13.5**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, B = 60°, and a = 10. Find C, and find c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

3315

**QUESTION ID**

## S89-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**C = 80°; c ≈ 15.3**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5**

**True or false?**

Knowing two angles and any side determines the triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

3316

QUESTION ID

## S89-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** AAS/ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

A = 30°, B = 45°, a = 6. b =

**A.** 3√2 ≈ 4.2

**B.** 6√2 ≈ 8.5

**C.** 6

**D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

3317

QUESTION ID

**S89-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 6√2 ≈ 8.5**

**GRADING NOTE** 6 sin 45°/sin 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

Why must the third angle be found before using the Law of Sines for c in an ASA problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

3318

**QUESTION ID**

## S89-F1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**The law pairs each side with its opposite angle, so C is needed for c.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5**

SET 89 • HSG-SRT.D.10

# FORWARD 2

Surveying with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

Two observers 500 m apart sight a tree across a river. The angles at the observers (between the baseline and the tree) are 70° and 60°. How far is the tree from the observer at the 70° angle (nearest meter)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

3320

**QUESTION ID**
**S89-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**565 m**

**GRADING NOTE** Rounding to the nearest meter required. Third angle 50°; side opposite 60°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A surveyor measures a baseline of 200 m and angles of 52° and 71° to a peak. Find the distance from the 52° end to the peak (nearest meter).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

3321

**QUESTION ID**
**S89-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**225 m**

**GRADING NOTE** Rounding to the nearest meter required. Third angle 57°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F2-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5**

**True or false?**

Surveyors use a measured baseline and two angles to find distances they cannot walk.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

3322

QUESTION ID

**S89-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Triangulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

In a surveying triangle, the side opposite the largest angle is

**A.** equal to the others

**B.** the shortest

**C.** the longest

**D.** the baseline always

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

3323

**QUESTION ID**

**S89-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. the longest**

**GRADING NOTE** Law of Sines.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5**

Explain why measuring angles is easier than measuring the distance across the river directly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S89-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

3324

**QUESTION ID**
## S89-F2-Q5

**TYPE**
Short answer

**CORRECT ANSWER**
**The far side cannot be reached; angles can be measured from one bank.**

**GRADING NOTE** Any reasonable answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S89-F2-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5**

SET 90 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

# Prove the Law of Cosines

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.D.10** |
| **BACKWARD 1** | The Pythagorean Theorem | **8.G.B.7** |
| **BACKWARD 2** | The distance formula | **8.G.B.8** |
| **BACKWARD 3** | Sine and cosine in a right triangle | **HSG-SRT.C.6** |
| **BACKWARD 4** | Square a binomial | **HSA-APR.A.1** |
| **FORWARD 1** | Use the Law of Cosines (SAS and SSS) | **HSG-SRT.D.10** |
| **FORWARD 2** | Distances between moving objects | **HSG-SRT.D.11** |

SET 90 • HSG-SRT.D.10

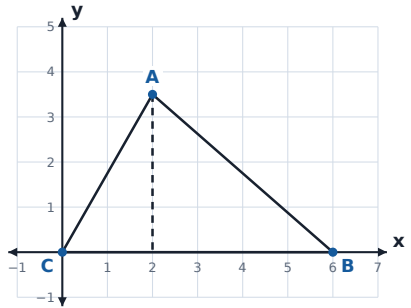
# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Place triangle ABC with C at the origin and B at (a, 0), so A = (b cos C, b sin C). Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space




Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

3327

**QUESTION ID**

## S90-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$c^2 = (b \cos C - a)^2 + (b \sin C)^2 = a^2 + b^2 - 2ab \cos C$**

**GRADING NOTE** REQUIRED WORK: Use the distance formula for AB and the identity $\sin^2 C + \cos^2 C = 1$.

Any valid proof (paragraph, two-column, or flow). Essential: A = (b cos C, b sin C) because CA = b; $c^2 = (b \cos C - a)^2 + b^2 \sin^2 C$; expand: $b^2\cos^2 C - 2ab \cos C + a^2 + b^2 \sin^2 C$; $\sin^2 C + \cos^2 C = 1$ gives $a^2 + b^2 - 2ab \cos C$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In acute triangle ABC, the altitude from A meets BC at D. Let CD = x. Use the two right triangles to prove $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

3328

**QUESTION ID**

## S90-M-Q2

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$h^2 = b^2 - x^2 = c^2 - (a - x)^2$, so $c^2 = a^2 + b^2 - 2ax = a^2 + b^2 - 2ab \cos C$.**

**GRADING NOTE** REQUIRED WORK: Write $h^2$ twice with the Pythagorean Theorem, then use $x = b \cos C$.

Any valid proof (paragraph, two-column, or flow). Essential: $BD = a - x$; $h^2 = b^2 - x^2$ and $h^2 = c^2 - (a - x)^2$; solve for $c^2$; substitute $x = b \cos C$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-M-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Show that the Law of Cosines becomes the Pythagorean Theorem when C = 90°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

3329

QUESTION ID

**S90-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**cos 90° = 0, so $c^2 = a^2 + b^2$.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5**

If angle C is obtuse, then

**A.** $c = a + b$

**B.** $c^2 = a^2 + b^2$

**C.** $c^2 < a^2 + b^2$

**D.** $c^2 > a^2 + b^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

3330

**QUESTION ID**

## S90-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $c^2 > a^2 + b^2$**

**GRADING NOTE** cos C < 0, so −2ab cos C > 0.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-M-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5**

**True or false?**

The Law of Cosines applies only to acute triangles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

3331

QUESTION ID

## S90-M-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It holds for every triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

SET 90 • HSG-SRT.D.10

# BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Legs 5 and 12: hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

3333

QUESTION ID

**S90-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**13**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Hypotenuse b, leg x: the other leg squared?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

3334

QUESTION ID

**S90-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$b^2 - x^2$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, the altitude from A to BC has length h and foot D, with CD = x and BC = a. In right triangle ABD, $h^2 + (a - x)^2 =$

**A.** $a^2$

**B.** $c^2$

**C.** $x^2$

**D.** $b^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

3335

QUESTION ID

**S90-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. $c^2$**

GRADING NOTE Right triangle ABD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

**True or false?**

The Pythagorean Theorem applies to the right triangles made by an altitude.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

3336

QUESTION ID

**S90-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Right angles at D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

**True or false?**

$a^2 + b^2 = c^2$ in every triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

3337

QUESTION ID

**S90-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Right triangles only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

# BACKWARD 2

The distance formula • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (0, 0) to (3, 4)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

3339

QUESTION ID

**S90-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the squared distance from (p, q) to (a, 0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

3340

QUESTION ID

**S90-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$(p - a)^2 + q^2$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

Squared distance from (1, 2) to (4, 6)?

**A.** 5

**B.** 25

**C.** 49

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

3341

QUESTION ID

**S90-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 25**

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

**True or false?**

The distance formula comes from the Pythagorean Theorem.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

3342

**QUESTION ID**

## S90-B2-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

**True or false?**

Distance from (2, 0) to (5, 0) is 7.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

3343

QUESTION ID

**S90-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10

# BACKWARD 3

Sine and cosine in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

Hypotenuse b and angle C: adjacent side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

3345

QUESTION ID

**S90-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**b cos C**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-B3-Q1**

**High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Hypotenuse b and angle C: opposite side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

3346

QUESTION ID

**S90-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**b sin C**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For a 3-4-5 triangle, $(\sin)^2 + (\cos)^2$ of an acute angle =

**A.** 25

**B.** 1

**C.** 12/25

**D.** 7/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

3347

**QUESTION ID**

## S90-B3-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 1**

**GRADING NOTE** 9/25 + 16/25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

**True or false?**

$\sin^2 C + \cos^2 C = 1$ for acute angles C.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

3348

QUESTION ID

## S90-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Pythagorean Theorem divided by the hypotenuse squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

cos C = opposite ÷ hypotenuse

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

3349

QUESTION ID

**S90-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Adjacent ÷ hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

SET 90 • HSG-SRT.D.10

# BACKWARD 4

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(b \cos C - a)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

3351

QUESTION ID
**S90-B4-Q1**

TYPE
Short answer

CORRECT ANSWER
**$b^2\cos^2 C - 2ab\cos C + a^2$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(a - x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

3352

QUESTION ID
**S90-B4-Q2**

TYPE
Short answer

CORRECT ANSWER
**$a^2 - 2ax + x^2$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$(p - q)^2 =$

**A.** $p^2 - 2pq + q^2$

**B.** $p^2 - pq + q^2$

**C.** $p^2 - q^2$

**D.** $p^2 + q^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

3353

QUESTION ID
**S90-B4-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**A. $p^2 - 2pq + q^2$**

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(a - x)^2 - x^2 = a^2 - 2ax$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

3354

QUESTION ID
**S90-B4-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE $x^2$ cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(a - x)^2 = a^2 - x^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

3355

QUESTION ID
**S90-B4-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE Missing −2ax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

SET 90 • HSG-SRT.D.10

# FORWARD 1

Use the Law of Cosines (SAS and SSS) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, a = 7, b = 9, and C = 50°. Find c to the nearest tenth.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

3357

**QUESTION ID**

**S90-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**7.0**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 7.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

3358

QUESTION ID
**S90-F1-Q2**

TYPE
Short answer

CORRECT ANSWER
**60°**

**GRADING NOTE** Any equivalent form. cos = (25 + 64 − 49)/80 = ½.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5**

**True or false?**

The Law of Cosines can find an angle when all three sides are known.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

3359

QUESTION ID

## S90-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** SSS.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F1-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5**

Sides 3, 4, 6. The angle opposite 6 is

**A.** obtuse

**B.** acute

**C.** right

**D.** impossible to tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

3360

**QUESTION ID**

## S90-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. obtuse

**GRADING NOTE** 36 > 9 + 16.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F1-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the largest angle of the triangle with sides 4, 5, and 6, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

3361

**QUESTION ID**
**S90-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**82.8°**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5**

SET 90 • HSG-SRT.D.10

# FORWARD 2

Distances between moving objects • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

Two ships leave port at the same time on courses 70° apart. One travels 20 km and the other 30 km. How far apart are they, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

3363

**QUESTION ID**

## S90-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**29.8 km**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F2-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A golfer's ball is 150 yd from the tee; the hole is 180 yd from the tee; the angle between the two directions is 12°. How far is the ball from the hole (nearest yard)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

3364

**QUESTION ID**
**S90-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**46 yd**

**GRADING NOTE** Rounding to the nearest yard required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

**True or false?**

The Law of Cosines can be used only when the angle between the two known sides is a right angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

3365

QUESTION ID

## S90-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It works for any included angle; with 90° it becomes the Pythagorean Theorem.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F2-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5**

Two hikers walk 3 km and 5 km from camp in directions 60° apart. Distance between them?

**A.** 8 km

**B.** √34 ≈ 5.8 km

**C.** 2 km

**D.** √19 ≈ 4.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

3366

**QUESTION ID**

## S90-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. √19 ≈ 4.4 km**

**GRADING NOTE** 9 + 25 − 15.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5**

Two ships leave port on courses 70° apart. Why can't the Pythagorean Theorem alone give the distance between them?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

3367

**QUESTION ID**
**S90-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**The angle between their courses is not 90°.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S90-F2-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5**

SET 91 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

# Apply the Law of Sines, including the ambiguous case

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.D.11** |
| **BACKWARD 1** | The Law of Sines | **HSG-SRT.D.10** |
| **BACKWARD 2** | Inverse sine | **HSG-SRT.C.8** |
| **BACKWARD 3** | Triangle angle sum | **HSG-CO.C.10** |
| **BACKWARD 4** | Sine of supplementary angles | **HSG-SRT.D.9** |
| **FORWARD 1** | Solve both triangles in the ambiguous case | **HSG-SRT.D.11** |
| **FORWARD 2** | Bearings and navigation with the Law of Sines | **HSG-MG.A.1** |

SET 91 • HSG-SRT.D.11

# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

Points A and B are 120 m apart on one bank of a river. A tree C is on the other bank. ∠A = 72° and ∠B = 63°. Find AC to the nearest tenth of a meter.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

3370

**QUESTION ID**

## S91-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**151.2 m**

**GRADING NOTE** Rounding to 0.1 required. ∠C = 45°; AC is opposite ∠B.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-M-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, A = 30°, a = 6, and b = 10. How many triangles are possible? Find all possible measures of B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

3371

**QUESTION ID**

## S91-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Two: B ≈ 56.4° or B ≈ 123.6°**

**GRADING NOTE** Form required: the number and both angles. sin B = 10 sin 30°/6 ≈ 0.833.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-M-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, A = 40°, a = 12, and b = 10. How many triangles are possible? Find B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

3372

**QUESTION ID**

## S91-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**One: B ≈ 32.4°**

**GRADING NOTE** Form required: the number and the angle. The supplement 147.6° is too large (with A = 40° the sum exceeds 180°).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5**

In triangle ABC, A = 50°, a = 5, and b = 10. How many triangles are possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

3373

**QUESTION ID**

## S91-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**None: sin B = 10 sin 50°/5 ≈ 1.53 > 1.**

**GRADING NOTE** Form required: the number with the reason.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-M-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5**

A is acute and b sin A < a < b. How many triangles have these measurements?

**A.** Infinitely many

**B.** One

**C.** Two

**D.** None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

3374

QUESTION ID

**S91-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. Two**

**GRADING NOTE** Side a can swing to two positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

SET 91 • HSG-SRT.D.11

# BACKWARD 1

The Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

3376

QUESTION ID

**S91-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**a/sin A = b/sin B = c/sin C**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A = 30°, B = 90°, a = 5. b =

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

3377

**QUESTION ID**

## S91-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-B1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solving a/sin A = b/sin B for sin B gives

**A.** b sin A/a

**B.** a/(b sin A)

**C.** sin A

**D.** a sin A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

3378

QUESTION ID
**S91-B1-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**A. b sin A/a**

**GRADING NOTE** Cross-multiply.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-B1-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5**

**True or false?**

Each side in the Law of Sines is paired with its opposite angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

3379

QUESTION ID

**S91-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$a/\sin B = b/\sin A$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

3380

QUESTION ID

## S91-B1-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Pairs are opposite.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

# BACKWARD 2

Inverse sine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

sin B = 0.5, B acute. B =

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

3382

QUESTION ID

**S91-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**30°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin B = 0.8. Find the acute B to the nearest tenth.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

3383

QUESTION ID

**S91-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**53.1°**

**GRADING NOTE** Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$\sin^{-1}(1) =$

**A.** 0°

**B.** 180°

**C.** 45°

**D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

3384

**QUESTION ID**

## S91-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 90°**

**GRADING NOTE** Value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

**True or false?**

The calculator's $\sin^{-1}$ always gives an angle between $-90°$ and $90°$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

3385

QUESTION ID

**S91-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sin^{-1}(1.2)$ is an angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

3386

QUESTION ID

**S91-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Sine cannot exceed 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11

# BACKWARD 3

Triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Two angles are 72° and 63°. Third angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

3388

QUESTION ID

**S91-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**45°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A triangle cannot have two obtuse angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

3389

QUESTION ID

**S91-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Sum would exceed 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A = 40° and B = 147.6°. Is this a triangle?

**A.** Only if C is 0°

**B.** No: the sum exceeds 180°

**C.** Yes

**D.** Only if right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

3390

QUESTION ID

**S91-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. No: the sum exceeds 180°**

GRADING NOTE 187.6°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A = 30°, B = 123.6°. Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

3391

QUESTION ID

**S91-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**26.4°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

**True or false?**

The angles of a triangle add to 360°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

3392

QUESTION ID

**S91-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

SET 91 • HSG-SRT.D.11

# BACKWARD 4

Sine of supplementary angles • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sin 150° = \sin 30°$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

3394

QUESTION ID

## S91-B4-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $\sin(180° - x) = \sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name an obtuse angle with the same sine as 56.4°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

3395

**QUESTION ID**

**S91-B4-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**123.6°**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-B4-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

sin 120° equals

**A.** cos 120°

**B.** sin 60°

**C.** sin 30°

**D.** −sin 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

3396

**QUESTION ID**

## S91-B4-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. sin 60°

**GRADING NOTE** Supplement.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-B4-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Why might sin B = 0.83 give two possible triangles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

3397

QUESTION ID

**S91-B4-Q4**

TYPE

Short answer

CORRECT ANSWER

**Both B and 180° − B have that sine.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

**True or false?**

An obtuse angle has a negative sine.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

3398

QUESTION ID

## S91-B4-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

SET 91 • HSG-SRT.D.11

# FORWARD 1

Solve both triangles in the ambiguous case • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

A = 30°, a = 6, b = 10. For the triangle with acute B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

3400

**QUESTION ID**

## S91-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**C ≈ 93.6°; c ≈ 12.0**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5**

In triangle ABC, A = 30°, a = 6, and b = 10, so B ≈ 56.4° or B ≈ 123.6°. For the triangle with obtuse B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

3401

**QUESTION ID**

## S91-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**C ≈ 26.4°; c ≈ 5.3**

**GRADING NOTE** Form required: both. B ≈ 123.6°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5**

**True or false?**

In the ambiguous case the two triangles have different third sides.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

3402

QUESTION ID

## S91-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Different shapes.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-F1-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5**

A = 30°, b = 10, and a = 5. How many triangles?

**A.** Three

**B.** Two

**C.** One (a right triangle)

**D.** None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

3403

QUESTION ID

**S91-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. One (a right triangle)**

**GRADING NOTE** a = b sin A.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S91-F1-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A = 120°, a = 8, b = 10. How many triangles? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

3404

QUESTION ID

**S91-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**None: with A obtuse, a must be the longest side, but a < b.**

**GRADING NOTE** Form required: the number with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

# FORWARD 2

Bearings and navigation with the Law of Sines • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A lighthouse is spotted from a ship at an angle of 35° to its course. After the ship sails 8 km farther, the angle is 70°. How far is the ship from the lighthouse at the second sighting (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3406

**QUESTION ID**

**S91-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**8.0 km**

**GRADING NOTE** Rounding to 0.1 required. The triangle has angles 35°, 110°, 35°, so it is isosceles: 8.0 km.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

3406

**ANSWER KEY** Reduced copy of the question page, with its answer

A forest ranger at A sees a fire at an angle of 48° from the line to a second tower B, 10 km away. From B, the angle is 61°. How far is the fire from A (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

3407

**QUESTION ID**

## S91-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**9.3 km**

**GRADING NOTE** Rounding to 0.1 required. Third angle 71°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

**True or false?**

Bearings and sightings are modeled as triangles with a known side and two angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

3408

QUESTION ID

**S91-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A ship sights a lighthouse at 35° to its course; after sailing 8 km farther along the course, the angle is 70°. In the triangle formed by the two ship positions and the lighthouse, the angle at the lighthouse is

**A.** 75°

**B.** 110°

**C.** 70°

**D.** 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

3409

**QUESTION ID**

**S91-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 35°**

**GRADING NOTE** 70° − 35° (exterior angle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Name one reason a navigation model using a flat triangle might be inaccurate over very long distances.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

3410

QUESTION ID

**S91-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**Earth's surface is curved.**

**GRADING NOTE** Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

# Apply the Law of Cosines, including resultant forces

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-SRT.D.11** |
| **BACKWARD 1** | The Law of Cosines | **HSG-SRT.D.10** |
| **BACKWARD 2** | Inverse cosine | **HSG-SRT.C.8** |
| **BACKWARD 3** | Angles in a parallelogram | **HSG-CO.C.11** |
| **FORWARD 1** | Solve a triangle with both laws | **HSG-SRT.D.11** |
| **FORWARD 2** | Velocity vectors: airspeed and wind | **HSN-VM.A.3** |

SET 92 • HSG-SRT.D.11

# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

To find the length AB across a lake, a surveyor at C measures CA = 350 m, CB = 420 m, and ∠C = 68°. Find AB to the nearest tenth of a meter.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

3413

**QUESTION ID**

## S92-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**434.5 m**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-M-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5**

3413

**ANSWER KEY** Reduced copy of the question page, with its answer

Two forces of 40 N and 60 N act on an object with an angle of 50° between them. Find the magnitude of the resultant force to the nearest tenth.

Resultant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

3414

**QUESTION ID**

## S92-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**91.0 N**

**GRADING NOTE** Rounding to 0.1 required. In the force parallelogram the angle opposite the resultant is 130°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-M-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A triangle has sides 8, 10, and 13. Find its largest angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

3415

**QUESTION ID**

## S92-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**91.8°**

**GRADING NOTE** Rounding to 0.1 required. Opposite 13.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane flies 200 km, turns 25° to the right, and flies 150 km. How far is it from its start, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

3416

**QUESTION ID**

## S92-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**341.9 km**

**GRADING NOTE** Rounding to 0.1 required. The angle between the legs inside the triangle is 155°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-M-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5**

Which law should be used first to solve a triangle given all three sides?

**A.** The Law of Sines

**B.** The Pythagorean Theorem

**C.** The angle sum

**D.** The Law of Cosines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

3417

**QUESTION ID**

## S92-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. The Law of Cosines

**GRADING NOTE** SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

SET 92 • HSG-SRT.D.11

# BACKWARD 1

The Law of Cosines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the Law of Cosines for side c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

3419

QUESTION ID

**S92-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$c^2 = a^2 + b^2 - 2ab \cos C$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

a = 3, b = 4, C = 90°: c?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

3420

QUESTION ID

**S92-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-B1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5**

Solved for cos C:

**A.** $(a^2 + b^2 - c^2)/(2ab)$

**B.** $(c^2 - a^2 - b^2)/(2ab)$

**C.** $2ab/(a^2 + b^2 - c^2)$

**D.** $(a^2 + b^2 + c^2)/(2ab)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

3421

**QUESTION ID**

## S92-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A.**

**$(a^2 + b^2 - c^2)/(2ab)$**

**GRADING NOTE** Rearrange.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-B1-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5**

**True or false?**

The angle in the formula is the one opposite the side being found.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

3422

**QUESTION ID**

## S92-B1-Q4

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** C opposite c.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-B1-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5**

**True or false?**

$c^2 = a^2 + b^2 + 2ab \cos C$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

3423

QUESTION ID

**S92-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Minus.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-B1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5**

SET 92 • HSG-SRT.D.11

# BACKWARD 2

Inverse cosine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

cos C = 0.5. C =

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

3425

**QUESTION ID**

## S92-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**60°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

cos C = −0.03125. C to the nearest tenth?

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

3426

**QUESTION ID**

## S92-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**91.8°**

**GRADING NOTE** Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

A negative cosine means the angle is

**A.** acute

**B.** right

**C.** obtuse

**D.** zero

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

3427

QUESTION ID

**S92-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. obtuse**

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

**True or false?**

$\cos^{-1}$ gives angles from 0° to 180°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

3428

QUESTION ID

**S92-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\cos^{-1}(0) = 0°$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

3429

QUESTION ID

**S92-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11

# BACKWARD 3

Angles in a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Consecutive angles of a parallelogram are supplementary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

3431

QUESTION ID

**S92-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

One angle of a parallelogram is 50°. The angle next to it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

3432

QUESTION ID

**S92-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**130°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Forces at 50° form a parallelogram; the angle opposite the resultant in the triangle is

**A.** 130°

**B.** 25°

**C.** 50°

**D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

3433

QUESTION ID

**S92-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 130°**

GRADING NOTE Supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane turns 25°. The angle inside the triangle at the turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

3434

QUESTION ID

**S92-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**155°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

**True or false?**

The angle between two forces is the angle used in the Law of Cosines for the resultant.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3435

QUESTION ID

**S92-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Use its supplement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

3435

SET 92 • HSG-SRT.D.11

# FORWARD 1

Solve a triangle with both laws • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, A = 40°. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

3437

**QUESTION ID**

**S92-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**9.7**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

In triangle ABC, b = 12, c = 15, and A = 40°. Find B to the nearest tenth of a degree (use the Law of Sines or Cosines).

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

3438

**QUESTION ID**

## S92-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**53.0°**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5**

**True or false?**

After the Law of Cosines, it is safest to find the smaller remaining angle with the Law of Sines.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

3439

QUESTION ID

## S92-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Avoids the ambiguous obtuse case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

Given two sides and the included angle, use first

**A.** $\sin^{-1}$

**B.** the Law of Sines

**C.** the Law of Cosines

**D.** the angle sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

3440

QUESTION ID

**S92-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. the Law of Cosines**

**GRADING NOTE** SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use Heron's formula or ½ab sin C to find the area of the triangle with sides 8, 10, 13 (nearest tenth).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

3441

**QUESTION ID**

**S92-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**40.0**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-F1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5**

SET 92 • HSG-SRT.D.11

# FORWARD 2

Velocity vectors: airspeed and wind • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane flies due east at 300 km/h; a wind blows due north at 50 km/h. Find its ground speed (nearest tenth) and direction (degrees north of east, nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

3443

**QUESTION ID**
**S92-F2-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**304.1 km/h at 9.5° north of east**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A boat heads across a river at 8 km/h; the current is 3 km/h at 90° to its heading. Find its actual speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

3444

QUESTION ID
**S92-F2-Q2**

TYPE
Short answer

CORRECT ANSWER
**8.5 km/h**

**GRADING NOTE** Rounding to 0.1 required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-F2-Q2**

**High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The resultant of two velocities is always found by adding their speeds.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

3445

QUESTION ID

## S92-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Only when they point the same way; otherwise use the parallelogram rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Velocities of 10 and 10 with 120° between them have a resultant of

**A.** 0

**B.** 17.3

**C.** 10

**D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

3446

**QUESTION ID**

## S92-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 10**

**GRADING NOTE** Law of Cosines with 60°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-F2-Q4**

**High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A plane's velocity is 400 km/h; a 60 km/h wind blows at 45° to its heading. Find the ground speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S92-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

3447

**QUESTION ID**

## S92-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

## 444.5 km/h

**GRADING NOTE** Rounding to 0.1 required. Angle in the triangle 135°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S92-F2-Q5**

**High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5**

SET 93 • HSG-C.A.4 • ADVANCED EXTENSION (+)

# Construct a tangent line from a point outside a circle

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-C.A.4** |
| **BACKWARD 1** | Construct the midpoint of a segment | **HSG-CO.D.12** |
| **BACKWARD 2** | Inscribed angles on a diameter | **HSG-C.A.2** |
| **BACKWARD 3** | Tangent is perpendicular to the radius | **HSG-C.A.2** |
| **FORWARD 1** | Prove that the two tangent segments are congruent and compute… | **HSG-C.A.4** |
| **FORWARD 2** | Lines of sight to the horizon | **HSG-MG.A.1** |

SET 93 • HSG-C.A.4

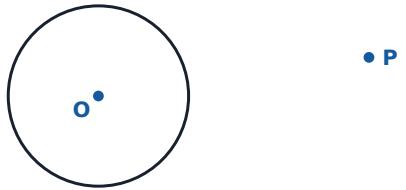
# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a line through P tangent to the circle with center O.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

3450

**QUESTION ID**

## S93-M-Q1

**TYPE**

Drawing / plotting

**CORRECT ANSWER**

**Draw OP; construct its midpoint M (perpendicular bisector); draw the circle with center M through O; it meets circle O at $T_1$ and $T_2$; draw $PT_1$ (and $PT_2$).**

**GRADING NOTE** Grade the construction marks, not only the final figure: segment OP, the perpendicular bisector arcs locating M, the circle on diameter OP, and the tangent through a crossing point T.

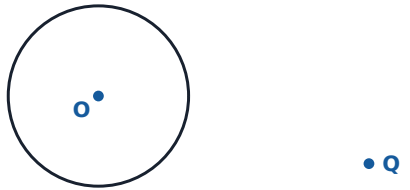
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct BOTH tangent lines from Q to the circle with center O.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

3451

QUESTION ID

## S93-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Same construction; both intersection points $T_1$ and $T_2$ give the two tangents $QT_1$ and $QT_2$.**

**GRADING NOTE** Grade the construction marks, not only the final figure: all construction marks and both tangents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the steps for constructing a tangent from an external point P to circle O.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

3452

**QUESTION ID**

## S93-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**1. Draw OP. 2. Bisect OP to find its midpoint M. 3. Draw the circle with center M and radius MO. 4. Label a point T where it meets circle O. 5. Draw line PT.**

**GRADING NOTE** Any equivalent steps.

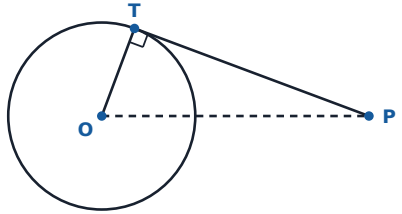
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why line PT in the construction is tangent to circle O.

Work space




Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

3453

QUESTION ID

## S93-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

**T is on the circle with diameter OP, so ∠OTP is inscribed in a semicircle and equals 90°; a line perpendicular to a radius at its endpoint on the circle is tangent.**

GRADING NOTE

REQUIRED WORK: Use the circle on diameter OP and the tangent-radius theorem.

Any valid explanation. Essential: ∠OTP = 90° (inscribed angle on a diameter); PT ⊥ OT at T on circle O; converse of the tangent-radius theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

The two tangent segments from P to circle O are

**A.** perpendicular

**B.** congruent

**C.** parallel

**D.** always equal to the radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

3454

QUESTION ID

**S93-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. congruent**

**GRADING NOTE** HL congruence of triangles OTP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

SET 93 • HSG-C.A.4

# BACKWARD 1

Construct the midpoint of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the midpoint of OP?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

3456

**QUESTION ID**

## S93-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**The perpendicular bisector of OP**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

**True or false?**

The perpendicular bisector crosses the segment at its midpoint.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

3457

QUESTION ID

**S93-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

The perpendicular bisector construction uses arcs

**A.** of any radius

**B.** centered at the midpoint

**C.** of equal radius from both endpoints

**D.** from one endpoint only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

3458

**QUESTION ID**

## S93-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. of equal radius from both endpoints**

**GRADING NOTE** Equal radii.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

To construct a tangent from a point P outside circle O, why is the midpoint of OP needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

3459

**QUESTION ID**

## S93-B1-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**It is the center of the circle with diameter OP.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The midpoint can be found by eye in a construction.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

3460

QUESTION ID

**S93-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It must be constructed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

SET 93 • HSG-C.A.4

# BACKWARD 2

Inscribed angles on a diameter • High School — Geometry • Geometry: Circles • HSG-C.A.2

**True or false?**

An angle inscribed in a semicircle is a right angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

3462

QUESTION ID

**S93-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A point T is on the circle with diameter OP. m∠OTP = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

3463

**QUESTION ID**

**S93-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**90°**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The intercepted arc of an angle inscribed in a semicircle is

**A.** 180°

**B.** 360°

**C.** 90°

**D.** 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

3464

QUESTION ID

**S93-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 180°**

GRADING NOTE Half of 180 is 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, why is a circle with diameter OP drawn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

3465

QUESTION ID

**S93-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**Points on it make a right angle with O and P.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

**True or false?**

An inscribed angle on a diameter can be 60°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

3466

QUESTION ID

**S93-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Always 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

SET 93 • HSG-C.A.4

# BACKWARD 3

Tangent is perpendicular to the radius • High School — Geometry • Geometry: Circles • HSG-C.A.2

**True or false?**

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

3468

QUESTION ID

**S93-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A tangent meets the radius at what angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

3469

QUESTION ID

**S93-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**90°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

If PT ⊥ OT and T is on circle O, then PT is

**A.** a secant

**B.** a diameter

**C.** a chord

**D.** tangent to the circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

3470

**QUESTION ID**

**S93-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. tangent to the circle**

**GRADING NOTE** Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

OT = 5 and OP = 13 with PT tangent at T. PT = ?

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

3471

QUESTION ID

**S93-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

**True or false?**

A tangent passes through the center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

3472

QUESTION ID

**S93-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It touches at one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

SET 93 • HSG-C.A.4

# FORWARD 1

Prove that the two tangent segments are congruent and compute lengths • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

PA and PB are tangent to circle O at A and B. Prove that PA = PB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

3474

**QUESTION ID**

## S93-F1-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**△OAP ≅ △OBP by HL, so PA = PB.**

**GRADING NOTE** REQUIRED WORK: Use the right triangles OAP and OBP.

Any valid proof (paragraph, two-column, or flow). Essential: right angles at A and B (tangent ⊥ radius); OA = OB (radii); OP common hypotenuse; HL; CPCTC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle has radius 5 and P is 13 from its center. Find the length of each tangent segment from P.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

3475

QUESTION ID

**S93-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**12**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

**True or false?**

OP bisects the angle between the two tangents.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

3476

QUESTION ID

**S93-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Tangents from P make a 60° angle. The angle AOB is

**A.** 30°

**B.** 120°

**C.** 60°

**D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

3477

**QUESTION ID**

**S93-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 120°**

**GRADING NOTE** Quadrilateral with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Radius 6, tangent segment 8. Distance from P to the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

3478

**QUESTION ID**

**S93-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

SET 93 • HSG-C.A.4

# FORWARD 2

Lines of sight to the horizon • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Model Earth as a sphere of radius 6,371 km. A person's eyes are 2 m (0.002 km) above the sea. How far is the horizon (tangent distance), to the nearest tenth of a km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

3480

**QUESTION ID**

**S93-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**5.0 km**

**GRADING NOTE** Rounding to 0.1 required. $d^2 = (6371.002)^2 - 6371^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

From the top of a 100 m tower (0.1 km), how far is the horizon, to the nearest km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

3481

**QUESTION ID**

## S93-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**36 km**

**GRADING NOTE** Rounding to the nearest km required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The line of sight to the horizon passes through Earth's center.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3482

QUESTION ID

**S93-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It is tangent, perpendicular to the radius at the horizon point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

3482

Raising the eye height by a factor of 4 multiplies the horizon distance by about

**A.** 16

**B.** 8

**C.** 2

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

3483

**QUESTION ID**

**S93-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 2**

**GRADING NOTE** $d \approx \sqrt{(2Rh)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In the model of the distance to the horizon from a point above Earth, why is the right angle of the triangle at the horizon point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

3484

**QUESTION ID**

**S93-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**The tangent line of sight is perpendicular to the radius at the point of tangency.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

3484

SET 94 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

# Derive the equation of an ellipse from its foci

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.A.3** |
| **BACKWARD 1** | The distance formula | **8.G.B.8** |
| **BACKWARD 2** | Derive a circle equation | **HSG-GPE.A.1** |
| **BACKWARD 3** | Expand squares | **HSA-APR.A.1** |
| **FORWARD 1** | Read foci and axes from an ellipse equation | **HSG-GPE.A.3** |
| **FORWARD 2** | Design with the reflective property of ellipses | **HSG-MG.A.3** |

SET 94 • HSG-GPE.A.3

# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

An ellipse is the set of points whose distances to the foci $(-3, 0)$ and $(3, 0)$ add to 10. Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

3487

**QUESTION ID**

## S94-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

$$x^2/25 + y^2/16 = 1$$

**GRADING NOTE** REQUIRED WORK: Write the distance sum, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{((x + 3)^2 + y^2)} + \sqrt{((x - 3)^2 + y^2)} = 10$; isolate and square; simplify to $3x - 25 = -5\sqrt{((x - 3)^2 + y^2)}$ (or equivalent); square again; $16x^2 + 25y^2 = 400$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

Write the equation of the ellipse with foci (0, −4) and (0, 4) whose distances to the foci add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

3488

**QUESTION ID**

## S94-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2/9 + y^2/25 = 1$**

**GRADING NOTE** Form required: an equation. a = 5 (vertical), c = 4, $b^2 = 9$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-M-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5**

Write the equation of the ellipse with foci (±5, 0) and vertices (±13, 0).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

3489

**QUESTION ID**

## S94-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2/169 + y^2/144 = 1$**

**GRADING NOTE** Form required: an equation. $b^2 = 169 - 25$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5**

Write the equation of the ellipse with foci (1, 2) and (7, 2) whose distances to the foci add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

3490

**QUESTION ID**

## S94-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**$(x - 4)^2/25 + (y - 2)^2/16 = 1$**

**GRADING NOTE** Form required: an equation. Center (4, 2), c = 3, a = 5, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

For any point on $x^2/36 + y^2/20 = 1$, the sum of its distances to the foci is

**A.** 12

**B.** $2\sqrt{20}$

**C.** 36

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

3491

**QUESTION ID**

**S94-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 12**

**GRADING NOTE** 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

SET 94 • HSG-GPE.A.3

# BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (x, y) to (3, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

3493

QUESTION ID

**S94-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

$\sqrt{((x - 3)^2 + y^2)}$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (0, 4) to (3, 0)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

3494

QUESTION ID

**S94-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

Distance from (5, 0) to (−3, 0)?

**A.** √34

**B.** 64

**C.** 8

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

3495

QUESTION ID

**S94-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 8**

**GRADING NOTE** Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

**True or false?**

The distance formula has a square root.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

3496

QUESTION ID

**S94-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE √.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sqrt{(a^2 + b^2)} = a + b$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

3497

QUESTION ID

**S94-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Not in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

# BACKWARD 2

Derive a circle equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Equation of the circle with center (0, 0) and radius 5?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

3499

**QUESTION ID**

## S94-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

$x^2 + y^2 = 25$

**GRADING NOTE** Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

A circle is an ellipse whose two foci coincide.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

3500

QUESTION ID

## S94-B2-Q2

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Special case.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$x^2/25 + y^2/25 = 1$ is

**A.** an ellipse with different axes

**B.** a parabola

**C.** a circle of radius 5

**D.** a hyperbola

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

3501

**QUESTION ID**

## S94-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. a circle of radius 5

**GRADING NOTE** Equal denominators.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-B2-Q3**

**High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A circle is the set of points at a fixed distance from

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

3502

QUESTION ID

**S94-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**one point (the center)**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

An ellipse has one focus.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

3503

QUESTION ID

**S94-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3

# BACKWARD 3

Expand squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(x + 3)^2 - (x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

3505

QUESTION ID
**S94-B3-Q1**

TYPE
Short answer

CORRECT ANSWER
**12x**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

3506

QUESTION ID
**S94-B3-Q2**

TYPE
Short answer

CORRECT ANSWER
**$x^2 - 6x + 9$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$(3x - 25)^2 =$

**A.** $9x^2 + 625$

**B.** $9x^2 - 75x + 625$

**C.** $3x^2 - 150x + 625$

**D.** $9x^2 - 150x + 625$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

3507

QUESTION ID
**S94-B3-Q3**

TYPE
Multiple choice

CORRECT ANSWER
**D. $9x^2 - 150x + 625$**

**GRADING NOTE** Middle term $2 \cdot 3x \cdot 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(a - b)^2 = a^2 - 2ab + b^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

3508

QUESTION ID
**S94-B3-Q4**

TYPE
True or false

CORRECT ANSWER
**A. True**

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(x + 3)^2 = x^2 + 9$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

3509

QUESTION ID
**S94-B3-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE Missing 6x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-B3-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

SET 94 • HSG-GPE.A.3

# FORWARD 1

Read foci and axes from an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the foci of $x^2/25 + y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

3511

QUESTION ID

**S94-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(±4, 0)**

**GRADING NOTE** Form required: both points. $c^2 = 25 - 9$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the vertices and foci of $x^2/16 + y^2/25 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

3512

QUESTION ID

**S94-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Vertices (0, ±5); foci (0, ±3)**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5**

**True or false?**

For an ellipse, $c^2 = a^2 - b^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

3513

**QUESTION ID**

## S94-F1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Relation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-F1-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5**

Which ellipse is closest to a circle?

**A.** $x^2/100 + y^2/9 = 1$

**B.** $x^2/25 + y^2/4 = 1$

**C.** $x^2/25 + y^2/24 = 1$

**D.** $x^2/25 + y^2/1 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

3514

**QUESTION ID**

## S94-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $x^2/25 + y^2/24 = 1$**

**GRADING NOTE** Nearly equal denominators.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-F1-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the length of the major axis of $x^2/49 + y^2/36 = 1$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

3515

**QUESTION ID**

**S94-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**14**

**GRADING NOTE** Any equivalent form. 2a.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S94-F1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5**

SET 94 • HSG-GPE.A.3

# FORWARD 2

Design with the reflective property of ellipses • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A whispering gallery is an ellipse 40 m long and 16 m wide. How far from the center should two people stand (at the foci), to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

3517

**QUESTION ID**

**S94-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**18.3 m**

**GRADING NOTE** Rounding to 0.1 required. a = 20, b = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A medical device places a kidney stone at one focus of an ellipse $x^2/100 + y^2/64 = 1$ (cm). Where is the emitter placed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

3518

**QUESTION ID**

## S94-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**At the other focus, 6 cm from the center: (−6, 0) if the stone is at (6, 0).**

**GRADING NOTE** Any equivalent answer. c = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

**True or false?**

Sound from one focus of an elliptical room reflects back to the same focus.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

3519

QUESTION ID

**S94-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** It reflects to the other focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

A garden ellipse is drawn with a 10 m rope tied to stakes 6 m apart. Its width (minor axis) is

**A.** 8 m

**B.** 10 m

**C.** 4 m

**D.** 6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

3520

**QUESTION ID**

## S94-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 8 m**

**GRADING NOTE** a = 5, c = 3, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain how the string-and-stakes method uses the definition of an ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

3521

QUESTION ID

**S94-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**The string length is the constant sum of the distances to the two stakes (foci).**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

# Derive the equation of a hyperbola from its foci

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GPE.A.3** |
| **BACKWARD 1** | The distance formula | **8.G.B.8** |
| **BACKWARD 2** | Derive an ellipse equation | **HSG-GPE.A.3** |
| **BACKWARD 3** | Absolute value as distance | **6.NS.C.7** |
| **FORWARD 1** | Read foci and asymptotes from a hyperbola equation | **HSG-GPE.A.3** |
| **FORWARD 2** | Locate a position from a time difference | **HSG-MG.A.3** |

SET 95 • HSG-GPE.A.3

# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

A hyperbola is the set of points whose distances to the foci (−5, 0) and (5, 0) differ by 6 (in absolute value). Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

3524

**QUESTION ID**

## S95-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**$x^2/9 - y^2/16 = 1$**

**GRADING NOTE** REQUIRED WORK: Write the distance difference, isolate one radical, square twice, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $|\sqrt{((x + 5)^2 + y^2)} - \sqrt{((x - 5)^2 + y^2)}| = 6$; isolate and square; simplify to $5x - 9 = \pm 3\sqrt{((x - 5)^2 + y^2)}$ (or equivalent); square again; $16x^2 - 9y^2 = 144$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-M-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5**

Write the equation of the hyperbola with foci (0, ±13) whose distances to the foci differ by 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

3525

**QUESTION ID**

## S95-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$y^2/25 - x^2/144 = 1$**

**GRADING NOTE** Form required: an equation. $a = 5$, $c = 13$, $b^2 = 144$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

Write the equation of the hyperbola with foci (±10, 0) and vertices (±6, 0).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

3526

**QUESTION ID**

## S95-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2/36 - y^2/64 = 1$**

**GRADING NOTE** Form required: an equation. $b^2 = 100 - 36$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-M-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5**

Find the asymptotes of $x^2/9 - y^2/16 = 1$.

Asymptotes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

3527

**QUESTION ID**

## S95-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**$y = \pm(4/3)x$**

**GRADING NOTE** Form required: both lines.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-M-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5**

For any point on $x^2/16 - y^2/9 = 1$, the difference of its distances to the foci is

**A.** 8

**B.** 7

**C.** 10

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

3528

**QUESTION ID**

**S95-M-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 8**

**GRADING NOTE** 2a.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-M-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5**

SET 95 • HSG-GPE.A.3

# BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (x, y) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

3530

QUESTION ID

**S95-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**√((x + 5)² + y²)**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (3, 0) to (5, 0) minus distance from (3, 0) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

3531

QUESTION ID

**S95-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**−6 (difference 6 in absolute value)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

3531

**ANSWER KEY** Reduced copy of the question page, with its answer

Distance from (6, 0) to (−10, 0)?

**A.** 4

**B.** 60

**C.** 16

**D.** −16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

3532

QUESTION ID

**S95-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 16**

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Distances are never negative.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

3533

QUESTION ID

**S95-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

**True or false?**

Distance from (0, 13) to (0, −13) is 13.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

3534

QUESTION ID

**S95-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

# BACKWARD 2

Derive an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

**True or false?**

An ellipse uses a constant SUM of distances to the foci.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

3536

QUESTION ID

**S95-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Ellipse with foci (±3, 0) and sum 10: equation?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

3537

QUESTION ID

**S95-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$x^2/25 + y^2/16 = 1$**

**GRADING NOTE** Form required: an equation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-B2-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

For an ellipse, $b^2 =$

**A.** $ac$

**B.** $a^2 - c^2$

**C.** $a^2 + c^2$

**D.** $c^2 - a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

3538

**QUESTION ID**

## S95-B2-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $a^2 - c^2$**

**GRADING NOTE** Relation.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-B2-Q3**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5**

What changes in the definition to get a hyperbola?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

3539

**QUESTION ID**

## S95-B2-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**The difference (not the sum) of the distances is constant.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-B2-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5**

**True or false?**

An ellipse equation has a minus sign between the squared terms.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

3540

QUESTION ID

## S95-B2-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Plus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3

# BACKWARD 3

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

**ANSWER KEY** Reduced copy of the question page, with its answer

$|4 - 10| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

3542

**QUESTION ID**

## S95-B3-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$|10 - 4| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

3543

**QUESTION ID**

**S95-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

|a − b| means

**A.** the size of the difference

**B.** a + b

**C.** a negative number

**D.** always a − b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

3544

QUESTION ID

**S95-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. the size of the difference**

**GRADING NOTE** Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

**True or false?**

$|x| = 6$ has two solutions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

3545

QUESTION ID

**S95-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE ±6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$|-3| = -3$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

3546

QUESTION ID

**S95-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

SET 95 • HSG-GPE.A.3

# FORWARD 1

Read foci and asymptotes from a hyperbola equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the foci of $x^2/16 - y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

3548

QUESTION ID

**S95-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(±5, 0)**

**GRADING NOTE** Form required: both points. $c^2 = 16 + 9$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5**

3548

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the vertices and asymptotes of $y^2/4 - x^2/9 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

3549

QUESTION ID

**S95-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Vertices (0, ±2); asymptotes y = ±(2/3)x**

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-F1-Q2**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

For a hyperbola, $c^2 = a^2 + b^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

3550

QUESTION ID

**S95-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which equation is a hyperbola?

**A.** $x^2/4 + y^2/9 = 1$

**B.** $y = x^2/4$

**C.** $x^2/4 - y^2/9 = 1$

**D.** $x^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

3551

**QUESTION ID**

**S95-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $x^2/4 - y^2/9 = 1$**

**GRADING NOTE** Minus sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

Explain how to tell an ellipse from a hyperbola by its equation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

3552

**QUESTION ID**
**S95-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**An ellipse has a plus sign between the squared terms; a hyperbola has a minus sign.**

**GRADING NOTE** Any equivalent answer.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S95-F1-Q5**

**High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5**

SET 95 • HSG-GPE.A.3

# FORWARD 2

Locate a position from a time difference • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Two radio stations are 100 km apart at (±50, 0). A ship receives signals showing it is 60 km closer to one station than the other. Write the equation of the hyperbola it lies on.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

3554

**QUESTION ID**

## S95-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2/900 - y^2/1600 = 1$**

**GRADING NOTE** Form required: an equation. $2a = 60$, $c = 50$, $b^2 = 2500 - 900$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Sound travels 0.34 km/s. Two microphones 2 km apart hear a sound 3 s apart. What is the difference in distances to the source?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

3555

QUESTION ID

**S95-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**1.02 km**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

**True or false?**

A single time difference places the source on one branch of a hyperbola, not at a single point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3556

QUESTION ID

**S95-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A second pair of stations is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

3556

Radio stations are at (−50, 0) and (50, 0) (km). A ship on the x-axis, to the right of the origin, is 60 km closer to the right station than to the left one. Its position is

**A.** (30, 0)

**B.** (40, 0)

**C.** (50, 0)

**D.** (60, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

3557

**QUESTION ID**

**S95-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. (30, 0)**

**GRADING NOTE** Vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

Why do navigation systems use at least two pairs of stations?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

3558

QUESTION ID

**S95-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**Each pair gives a hyperbola; the intersection pins down the position.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2 • ADVANCED EXTENSION (+)

# Use Cavalieri's principle to justify the volume of a sphere

| | | |
|---|---|---|
| **MAIN** | Geometry main question | **HSG-GMD.A.2** |
| **BACKWARD 1** | Cavalieri's principle for prisms and cylinders | **HSG-GMD.A.1** |
| **BACKWARD 2** | Cross-section radius with the Pythagorean Theorem | **8.G.B.7** |
| **BACKWARD 3** | Volumes of cylinders and cones | **HSG-GMD.A.3** |
| **BACKWARD 4** | Area of a ring | **7.G.B.4** |
| **FORWARD 1** | Use Cavalieri's principle for other solids | **HSG-GMD.A.2** |
| **FORWARD 2** | Model a dome as a hemispherical shell | **HSG-MG.A.2** |

SET 96 • HSG-GMD.A.2

# MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A hemisphere of radius r and a cylinder of radius r and height r with a cone removed (as shown) sit on the same plane. Use Cavalieri's principle to show that the hemisphere's volume is ⅔πr³, and conclude that a sphere has volume 4/3 πr³.

Work space

hemisphere, radius r  cylinder minus cone

Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION  ID S96-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

3561

**QUESTION ID**

## S96-M-Q1

**TYPE**

Answer with required work

**CORRECT ANSWER**

**Both cross-sections have area π(r² − y²), so the volumes are equal: πr³ − ⅓πr³ = ⅔πr³; the sphere is two hemispheres: 4/3 πr³.**

**GRADING NOTE** REQUIRED WORK: Compare the cross-section areas at height y.

Any valid explanation. Essential: hemisphere slice radius √(r² − y²) (Pythagorean), area π(r² − y²); the other slice is a ring with outer radius r and inner radius y (the cone radius at height y), area πr² − πy²; equal areas at every height and equal heights; Cavalieri; volume cylinder minus cone = ⅔πr³; double.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S96-M-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Let r = 5 and y = 3. Find the cross-section areas of the hemisphere and of the cylinder-minus-cone at height y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

3562

**QUESTION ID**

## S96-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**Both 16π**

**GRADING NOTE** Form required: both. 25π − 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Let r = 6. Find the volume of the cylinder-minus-cone and compare it with $\frac{2}{3}\pi r^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

3563

QUESTION ID

**S96-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**$216\pi - 72\pi = 144\pi = \frac{2}{3}\pi(216)$**

**GRADING NOTE** Form required: the computation and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Explain why a stack of coins pushed into a slanted stack keeps its volume, using Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

3564

QUESTION ID
**S96-M-Q4**

TYPE
Short answer

CORRECT ANSWER
**Each coin (cross-section) has the same area at every height, so the volumes are equal.**

**GRADING NOTE** Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Cavalieri's principle requires

**A.** equal surface areas

**B.** congruent bases only

**C.** the same shape at every level

**D.** equal heights and equal cross-section areas at every level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

3565

**QUESTION ID**
## S96-M-Q5

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. equal heights and equal cross-section areas at every level**

**GRADING NOTE** Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

SET 96 • HSG-GMD.A.2

# BACKWARD 1

Cavalieri's principle for prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

**True or false?**

Solids with equal heights and equal cross-section areas at every level have equal volumes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

3567

QUESTION ID

## S96-B1-Q1

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Principle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An oblique cylinder has radius 2 and height 5. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

3568

**QUESTION ID**

**S96-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**20π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A prism and a cylinder both have base area 30 and height 4. Volumes?

**A.** Cannot tell

**B.** 30 and 4

**C.** 120 and 40

**D.** Both 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

3569

**QUESTION ID**

## S96-B1-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. Both 120

**GRADING NOTE** Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

State what must be equal for Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

3570

QUESTION ID

**S96-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**The heights and the cross-section areas at every level.**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

3570

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Cavalieri's principle needs the cross-sections to be the same shape.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

3571

QUESTION ID
**S96-B1-Q5**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Only equal areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

# BACKWARD 2

Cross-section radius with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

A sphere of radius 5 is sliced 3 units from its center. Radius of the slice?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

3573

QUESTION ID

**S96-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Radius r, height y: slice radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

3574

QUESTION ID

**S96-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(r^2 - y^2)}$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

Radius 10, height 6: slice radius

**A.** 4

**B.** 8

**C.** 16

**D.** √136

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

3575

QUESTION ID

**S96-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 8**

GRADING NOTE 100 − 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

**True or false?**

The slice radius, the height, and the sphere radius form a right triangle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

3576

QUESTION ID

**S96-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Right angle at the slice center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

**True or false?**

Slices of a hemisphere get bigger as y increases.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

3577

QUESTION ID

**S96-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2

# BACKWARD 3

Volumes of cylinders and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Cylinder r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

3579

**QUESTION ID**

**S96-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**216π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Cone r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

3580

**QUESTION ID**

**S96-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**72π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Cylinder minus cone (same r and h = r) =

**A.** $4/3\ \pi r^3$

**B.** $\pi r^3$

**C.** $\frac{2}{3}\pi r^3$

**D.** $\frac{1}{3}\pi r^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

3581

**QUESTION ID**

**S96-B3-Q3**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $\frac{2}{3}\pi r^3$**

**GRADING NOTE** $\pi r^3 - \frac{1}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

**True or false?**

A cone is ⅓ of the cylinder with the same base and height.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

3582

QUESTION ID

## S96-B3-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\pi r^3 - \frac{1}{3}\pi r^3 = \frac{1}{3}\pi r^3$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

3583

QUESTION ID

## S96-B3-Q5

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** $\frac{2}{3}\pi r^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

SET 96 • HSG-GMD.A.2

# BACKWARD 4

Area of a ring • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Ring with outer radius 5 and inner radius 3: exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

3585

**QUESTION ID**

**S96-B4-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**16π**

**GRADING NOTE** Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

**True or false?**

Ring area = $\pi(R^2 - r^2)$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5**

3586

**QUESTION ID**

## S96-B4-Q2

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q2

**Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q2/5**

Ring with outer 4 and inner 1:

**A.** 3π

**B.** 15π

**C.** 9π

**D.** 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

3587

QUESTION ID

**S96-B4-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 15π**

GRADING NOTE 16 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Ring with outer r and inner y: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

3588

QUESTION ID

**S96-B4-Q4**

TYPE

Short answer

CORRECT ANSWER

**$\pi(r^2 - y^2)$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

**True or false?**

Ring area = $\pi(R - r)^2$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

3589

QUESTION ID

**S96-B4-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $\pi(R^2 - r^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

SET 96 • HSG-GMD.A.2

# FORWARD 1

Use Cavalieri's principle for other solids • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A twisted tower and a straight prism both have height 50 m and every horizontal slice is a 10 m by 10 m square. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

3591

QUESTION ID

**S96-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**Equal: 5,000 m³ each**

**GRADING NOTE** Form required: the comparison with the value.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S96-F1-Q1**

**High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A pyramid with base area 36 and height 10 and a cone with base area 36 and height 10: compare volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

3592

QUESTION ID
**S96-F1-Q2**

TYPE
Short answer

CORRECT ANSWER
**Equal: 120 each**

**GRADING NOTE** Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

**True or false?**

Cavalieri's principle shows that every pyramid with base area B and height h has volume ⅓Bh, once one such pyramid is known.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

3593

**QUESTION ID**

## S96-F1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Equal slices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

Two solids of height 8 have slice areas 5 and 5 at every level. Volumes?

**A.** 40 and 13.3

**B.** 5 each

**C.** Both 40

**D.** Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

3594

**QUESTION ID**

## S96-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. Both 40

**GRADING NOTE** 5 × 8.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S96-F1-Q4**

**High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Two solids have equal heights, but at one level their slice areas differ. Can Cavalieri's principle be used? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

3595

QUESTION ID

**S96-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**No: the areas must be equal at every level.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

# FORWARD 2

Model a dome as a hemispherical shell • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Find the volume of concrete to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

3597

**QUESTION ID**

## S96-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**298.7 m³**

**GRADING NOTE** Rounding to 0.1 required. $\frac{2}{3}\pi(10^3 - 9.5^3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Concrete has density 2.4 t/m³. Estimate the dome's mass to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

3598

**QUESTION ID**
**S96-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**717 t**

**GRADING NOTE** Rounding to the nearest ton required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

**True or false?**

A shell's volume is the outer volume minus the inner volume.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

3599

QUESTION ID

**S96-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A hemisphere of radius 3 has volume

**A.** 9π

**B.** 27π

**C.** 36π

**D.** 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3600

**QUESTION ID**

**S96-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 18π**

**GRADING NOTE** $\frac{2}{3}\pi \cdot 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

3600

**ANSWER KEY** Reduced copy of the question page, with its answer

A hemispherical shell has outer radius 10 m and inner radius 9.5 m. Why is its volume NOT $\frac{2}{3}\pi(10 - 9.5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5

3601

**QUESTION ID**
**S96-F2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**Volumes subtract, not radii: the shell is the difference of two hemispheres.**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5